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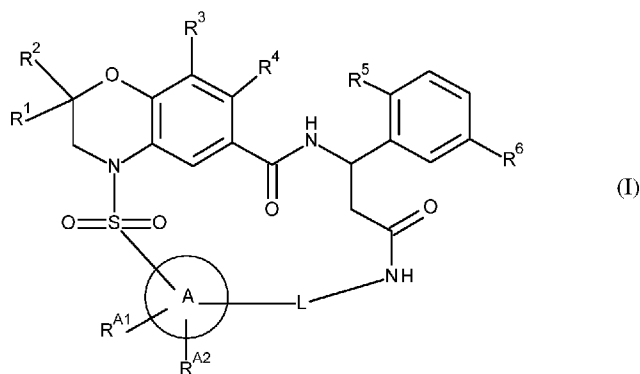
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(54) Title: DIHYDRO BENZOXAZINE DERIVATIVES

(57) Abstract: The invention provides new heterocyclic compounds having the general formula (I) wherein A, R¹, R², R³, R⁴, R⁵, R⁶, R^{A1}, R^{A2} and L are as described herein, compositions including the compounds and methods of using the compounds.

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DIHYDRO BENZOXAZINE DERIVATIVES

Field of the Invention

The present invention relates to organic compounds useful for the treatment and/or prophylaxis in a human, in particular to selected dihydro benzoxazine derivatives as inhibitors of the MET
5 receptor tyrosine kinase for the treatment and/or prophylaxis of MET mediated disease or disorder, in particular cancer.

Background of the Invention

MET is a transmembrane receptor tyrosine kinase. Binding of the hepatocyte growth factor (HGF) induces MET receptor dimerization and activation of the receptor. While HGF is mainly
10 expressed and secreted by mesenchymal cells, such as fibroblasts, MET is wider expressed by epithelial cells of a variety of tissues, but also found on endothelial cells, neurons, hepatocytes, and hematopoietic cells, and has been demonstrated to be involved in a variety of physiological processes during embryonic development and in tissues repair in adults. In particular, MET activates multiple signal transduction pathways, including the RAS-mitogen-activated protein
15 kinase (MAPK) cascade, the PI3K-AKT pathway, the Signal Transducer and Activator of Transcription (STAT) and NF- κ B pathway, involved in cell proliferation, differentiation, cell motility, invasion and survival. Moreover, by regulating cell-matrix adhesion and promoting cytoskeletal changes and cell migration, MET plays a key role in epithelial to mesenchymal transition (EMT). The pleiotropic biological effects of MET activation also rely on crosstalk and
20 signaling cooperation with various surface membrane proteins, including the MET homologue RON, and other TKs, such as EGFR, ERBB3, ROR1, CD44, integrins and plexin B1.

MET constitutive activation is a common event in several cancer types and can be the result of various mechanisms, including excessive autocrine or paracrine production of HGF, MET overexpression or genetic abnormalities of *MET*, such as gene copy-number amplification and
25 point mutations. Other mechanisms include inadequate internalization and degradation and receptor crosstalk. Accordingly, several therapeutic strategies have been developed to target MET signaling, including anti-HGF or anti-MET antibodies and small molecule TKIs.

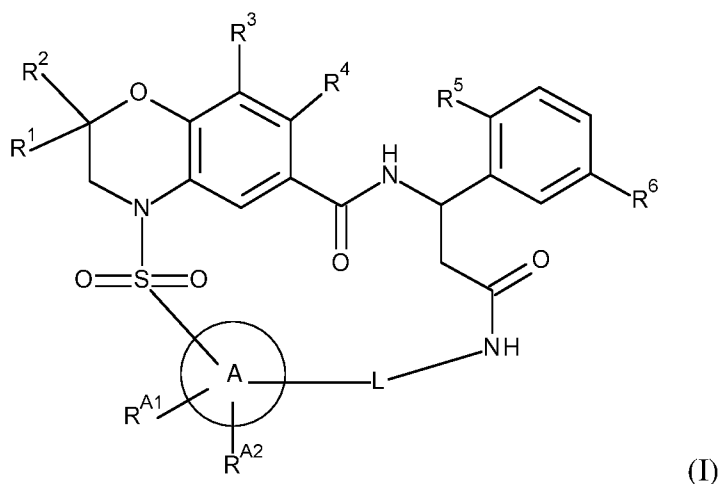
Lung cancer is the leading cause of cancer-related mortality worldwide, and non-small-cell lung cancer (NSCLC) represents the major histological subtypes of the disease. Approximately 3% of
30 NSCLC patients harbor *MET* exon 14 skipping alterations (*METex14*), resulting in decreased

MET turnover and extended oncogenic downstream signaling pathways. Today, there are a few highly selective and potent MET TKIs approved for the treatment of NSCLC patients harboring *MET* exon 14 skipping alterations (*METex14*). However, intrinsic or acquired resistance to TKIs has been a major challenge for these targeted therapies that impairs their clinical efficacies.

- 5 In conclusion, suppressing the action of MET receptor is a promising new therapeutic strategy for the treatment or prevention of various diseases and disorders, and there continues to be a high unmet medical need for new MET inhibitors.

Summary of the Invention

In a first aspect, the present invention provides a compound of formula (I)



wherein A , R^1 , R^2 , R^3 , R^4 , R^5 , R^6 , R^{A1} , R^{A2} and L are as described herein.

In further aspects, the present invention provides pharmaceutical compositions comprising the compounds of formula (I), as well as methods of using the compounds of formula (I) in the treatment or prophylaxis of disease or disorder mediated by MET kinase, in particular cancer.

Detailed Description of the Invention

Definitions

Features, integers, characteristics, compounds, chemical moieties or groups described in conjunction with a particular aspect, embodiment or example of the invention are to be understood to be applicable to any other aspect, embodiment or example described herein, unless incompatible therewith. All of the features disclosed in this specification (including any

accompanying claims, abstract and drawings), and/or all of the steps of any method or process so disclosed, may be combined in any combination, except combinations where at least some of such features and/or steps are mutually exclusive. The invention is not restricted to the details of any foregoing embodiments. The invention extends to any novel one, or any novel combination, of the features disclosed in this specification (including any accompanying claims, abstract and drawings), or to any novel one, or any novel combination, of the steps of any method or process so disclosed.

The term “acid” refers to a compound capable of giving proton of Broensted's definition, dissociating into proton and counter ion in water at 25°C and giving a solution having neutral pH or below. Concrete examples of the acid are phosphoric acid (orthophosphoric acid), sulfuric acid, nitric acid, phosphinic acid, phosphonic acid, diphosphonic acid, hydrochloric acid, pyrophosphoric acid, metaphosphoric acid and nitrous acid. These acids may be used in the form of metal salts, ammonium salts or the like; particularly acid means hydrochloric acid.

The term “alkyl” refers to a monovalent linear or branched saturated hydrocarbon group of 1 to 6 carbon atoms, e.g., 1, 2, 3, 4, 5, or 6 carbon atoms, having the number of carbon atoms designated (“C₁₋₆-alkyl”). In some embodiments, the alkyl group contains 1 to 3 carbon atoms (“C₁₋₃-alkyl”). Some non-limiting examples of alkyl include methyl, ethyl, n-propyl, isopropyl, n-butyl, iso-butyl, sec-butyl, tert-butyl, homologs and isomers of, for example, n-pentyl, n-hexyl, and the like. Particularly, yet non-limiting example of alkyl is methyl.

The term “alkylene” refers to a bivalent linear or branched hydrocarbon group of 1 to 3 carbon atoms, in particular linear hydrocarbon group of 1 to 3 carbon atoms. Examples of alkylene include methylene (-CH₂-), 1,2-ethylene (-CH₂-CH₂-), 1,1-ethylene (-CH₂(CH₃)-) or 1,3-propylene (-CH₂-CH₂-CH₂-), 1,2-propylene (-CH₂(CH₃)-CH₂-). Particular examples of alkylene are methylene (-CH₂-), 1,2-ethylene (-CH₂-CH₂-), or 1,3-propylene (-CH₂-CH₂-CH₂-). Particularly, yet non-limiting examples of alkylene is 1,2-ethylene (-CH₂-CH₂-), or 1,3-propylene (-CH₂-CH₂-CH₂-).

The term “C₁₋₆-alkoxy” refers to an alkyl group, as previously defined, attached to the parent molecular moiety via an oxygen atom. Some non-limiting examples of alkoxy groups include methoxy, ethoxy, n-propoxy, isopropoxy, n-butoxy, isobutoxy and tert-butoxy. Particularly, yet non-limiting example of alkoxy is methoxy.

The term “amino” alone or in combination with other groups, refers to -NH₂.

The term “aminoalkyl” refers to an alkyl group, having the number of carbon atoms designated (“amino-C₁₋₆-alkyl”), wherein at least one of the hydrogen atoms of the alkyl group has been replaced by amino group. Examples of aminoalkyl include aminomethyl, 2-aminoethyl, 2-aminopropyl, 3-aminopropyl, 2-aminobutyl, 3-aminobutyl, 4-aminobutyl, 2,3-diaminopropyl, 2-amino-1-aminomethylethyl, 2,3-diaminobutyl, or 3,4-diaminobutyl. Particularly, yet non-limiting example of aminoalkyl is aminomethyl.

The term “aryl” refers to a monovalent, monocyclic or bicyclic carbocyclic ring system of 6 to 10 ring members (“C₆₋₁₀-aryl”), wherein at least one ring in the system is aromatic. Some non-limiting examples of aryl include phenyl or naphthyl. Particularly, yet non-limiting example of aryl is phenyl.

The term “cancer” refers to a generic name of malignant tumor (or malignant neoplasma), including not only solid tumors such as tumors developed on epithelial tissues or sarcomas developed on connective tissues (bone, muscle, cartilage, sinew, ligament, adipose, and blood vessel), but also all non-solid tumors such as leukemia that are malignant tumors in the blood system and lymphomas. Thus, target diseases to which a pharmaceutical composition for treatment of cancers of the present invention can be applied include, but are not limited to, cervical cancer, esophageal cancer, gastric cancer, pancreatic cancer, liver cancer, lung cancer, non-small-cell lung cancer, parotid cancer, salivary gland cancer, colon cancer, breast cancer, renal cancer, prostate cancer, brain cancer, skin cancer, adrenal cancer, oral cancer, rectal cancer, endometrial cancer, thyroid cancer, ovarian cancer, laryngeal cancer, leukemia, and malignant lymphoma. In particular cancer refers to lung cancer or non-small-cell lung cancer.

The term “cycloalkyl” refers to a saturated monocyclic hydrocarbon group of 3 to 10 ring carbon atoms, having the number of carbon atoms designated. Particular, “C₃₋₁₀-cycloalkyl” refers to cycloalkyl having 3 to 10 carbon atoms; “C₅₋₆-cycloalkyl” refers to cycloalkyl having 5 to 6 carbon atoms; “C₃₋₄-cycloalkyl” refers to cycloalkyl having 3 to 4 carbon atoms. Examples of cycloalkyl include cyclopropyl, cyclobutyl, cyclopentyl, cyclohexyl. Particularly, yet non-limiting example of cycloalkyl is cyclopropyl. Particularly, yet non-limiting example of cycloalkyl is cyclopentyl.

The term “halo” or “halogen” refers to fluoro, chloro, bromo or iodo, particularly chloro or fluoro.

The term “haloalkyl” refers to an alkyl group, having the number of carbon atoms designated (“halo-C₁₋₆-alkyl”), wherein at least one of the hydrogen atoms of the alkyl group has been

replaced by same or different halogen atoms, particularly fluoro atoms. Examples of haloalkyl include monofluoro-, difluoro- or trifluoro-methyl, -ethyl or -propyl, for example 3,3,3-trifluoropropyl, 2-fluoroethyl, 2,2,2-trifluoroethyl, fluoromethyl, or trifluoromethyl. Particularly, yet non-limiting example of haloalkyl is trifluoromethyl.

- 5 The term “hydroxy”, alone or in combination with other groups, refers to -OH.

The term “hydroxyalkyl” refers to an alkyl group, having the number of carbon atoms designated (“hydroxy-C₁₋₆-alkyl”), wherein at least one of the hydrogen atoms of the alkyl group has been replaced by hydroxy group. Examples of hydroxyalkyl include hydroxymethyl, 2-hydroxyethyl, 2-hydroxypropyl, 3-hydroxypropyl, 2-hydroxybutyl, 3-hydroxybutyl, 4-hydroxybutyl, 10 2,3-dihydroxypropyl, 2-hydroxy-1-hydroxymethylethyl, 2,3-dihydroxybutyl, 3,4-dihydroxybutyl or 2-(hydroxymethyl)-3-hydroxypropyl. Particularly, yet non-limiting example of hydroxyalkyl is hydroxymethyl.

- The term “heteroaryl” refers to a multivalent, mono- or bicyclic ring system of 5 to 10 ring atoms, comprising 1, 2, or 3 heteroatoms independently selected from N, O and S, the 15 remaining ring atoms being carbon, wherein at least one ring in the ring system is aromatic and contains one or more heteroatoms. Particularly, heteroaryl is a 5- to 6-membered monocyclic heteroaryl comprising 1 or 2 heteroatoms independently selected from O, S and N, the remaining ring atoms being carbon. Examples of heteroaryl include pyridyl, quinoline, thienyl, pyrrolyl, furanyl, oxazolyl, imidazolyl. More particularly, yet non-limiting example of heteroaryl is 20 pyridyl.

- The terms “heterocyclyl” and “heterocycloalkyl” are used herein interchangeably and refer to a multivalent saturated mono- or bicyclic, particularly monocyclic ring system of 3 to 10 ring atoms, wherein 1, 2, or 3 of said ring atoms are heteroatoms independently selected from N, O and S, the remaining ring atoms being carbon. Particularly, “heterocyclyl” refers to monocyclic 25 ring system of 5 to 7 ring atoms, wherein 1 to 2 of said ring atoms are independently selected from N and O, and the remaining ring atoms being carbon. Some non-limiting examples of heterocyclyl include pyrrolidinyl, morpholinyl, piperazinyl, piperidyl, 2,5-diazabicyclo[2.2.1]heptanyl, 1,4-diazepanyl.

- The terms “MET”, “MET kinase”, and “MET receptor” are used herein interchangeably and 30 refer to a MET receptor tyrosine kinase.

“MET TKI” refers to a MET tyrosine kinase inhibitor.

The term “optionally substituted” means unsubstituted or substituted. Generally these substituents can be the same or different.

The term “pharmaceutically acceptable salt” refers to those salts which retain the biological effectiveness and properties of the free bases or free acids, which are not biologically or otherwise undesirable. The salts are formed with inorganic acids such as hydrochloric acid, hydrobromic acid, sulfuric acid, nitric acid, phosphoric acid and the like, in particular hydrochloric acid, and organic acids such as acetic acid, propionic acid, glycolic acid, pyruvic acid, oxalic acid, maleic acid, malonic acid, succinic acid, fumaric acid, tartaric acid, citric acid, benzoic acid, cinnamic acid, mandelic acid, methanesulfonic acid, ethanesulfonic acid, p-toluenesulfonic acid, salicylic acid, N-acetylcystein and the like. In addition, these salts may be prepared by addition of an inorganic base or an organic base to the free acid. Salts derived from an inorganic base include, but are not limited to, the sodium, potassium, lithium, ammonium, calcium, magnesium salts and the like. Salts derived from organic bases include, but are not limited to salts of primary, secondary, and tertiary amines, substituted amines including naturally occurring substituted amines, cyclic amines and basic ion exchange resins, such as isopropylamine, trimethylamine, diethylamine, triethylamine, tripropylamine, ethanolamine, lysine, arginine, N-ethylpiperidine, piperidine, polyimine resins and the like. Particular pharmaceutically acceptable salts of compounds of formula (I) are the salts of hydrochloric acid, hydrobromic acid, sulfuric acid, phosphoric acid, and methanesulfonic acid. More particularly, pharmaceutically acceptable salts of compounds of formula (I) are the salts of hydrochloric acid.

The term “protecting group” refers to the group, which selectively blocks a reactive site in a multifunctional compound such that a chemical reaction can be carried out selectively at another unprotected reactive site in the meaning conventionally associated with it in synthetic chemistry. Protective groups can be removed at the appropriate point. Exemplary protective groups are amino-protective groups, carboxy-protective groups or hydroxy-protective groups. Particular protective groups are the tert-butoxycarbonyl (Boc), benzyloxycarbonyl (Cbz), fluorenylmethoxycarbonyl (Fmoc), phthaloyl (Pht) and benzyl (Bn). Further particular protective groups are the tert-butoxycarbonyl (Boc), benzyloxycarbonyl (Cbz), or phthaloyl (Pht). More particular protective group is the tert-butoxycarbonyl (Boc). Exemplary protective groups and their application in organic synthesis are described, for example, in “Protective Groups in Organic Chemistry” by T. W. Greene and P. G. M. Wuts, 5th Ed., 2014, John Wiley & Sons, N.Y.

The term “prophylaxis” as used herein includes: preventing or delaying the appearance of clinical symptoms of the state, disorder or condition developing in a human that may be afflicted with or predisposed to the state, disorder or condition but does not yet experience or display clinical or subclinical symptoms of the state, disorder or condition.

- 5 The term “therapeutically effective amount” refers to an amount of a compound or molecule of the present invention that, when administered to a subject, (i) treats or prevents the particular disease, condition or disorder, (ii) attenuates, ameliorates or eliminates one or more symptoms of the particular disease, condition, or disorder, or (iii) prevents or delays the onset of one or more symptoms of the particular disease, condition or disorder described
10 herein. The therapeutically effective amount will vary depending on the compound, the disease state being treated, the severity of the disease treated, the age and relative health of the subject, the route and form of administration, the judgement of the attending medical or veterinary practitioner, and other factors.

By the term “treatment” or “treating” and grammatical variations thereof as used herein, is
15 meant therapeutic therapy. In reference to a particular condition, treating means: (1) to ameliorate the condition or one or more of the biological manifestations of the condition, (2) to interfere with (a) one or more points in the biological cascade that leads to or is responsible for the condition or (b) one or more of the biological manifestations of the condition, (3) to alleviate one or more of the symptoms, effects or side effects associated
20 with the condition or treatment thereof, or (4) to slow the progression of the condition or one or more of the biological manifestations of the condition. Prophylactic therapy using the methods and/or compositions of the invention is also contemplated. The skilled artisan will appreciate that “prevention” is not an absolute term. In medicine, “prevention” is understood to refer to the prophylactic administration of a drug to substantially diminish
25 the likelihood or severity of a condition or biological manifestation thereof, or to delay the onset of such condition or biological manifestation thereof. Prophylactic therapy is appropriate, for example, when a subject is considered at high risk for developing cancer, such as when a subject has a strong family history of cancer.

The compounds of formula (I) can contain several asymmetric centers and can be present in the
30 form of optically pure enantiomers, mixtures of enantiomers such as, for example, racemates, optically pure diastereoisomers, mixtures of diastereoisomers, diastereoisomeric racemates or mixtures of diastereoisomeric racemates. According to the Cahn-Ingold-Prelog Convention, the asymmetric carbon atom can be of the “R” or “S” configuration.

The term “treatment” as used herein includes: (1) inhibiting the state, disorder or condition (e.g. arresting, reducing or delaying the development of the disease, or a relapse thereof in case of maintenance treatment, of at least one clinical or subclinical symptom thereof); and/or (2) relieving the condition (i.e., causing regression of the state, disorder or condition or at least one of its clinical or subclinical symptoms). The benefit to a patient to be treated is either statistically significant or at least perceptible to the patient or to the physician. However, it will be appreciated that when a medicament is administered to a patient to treat a disease, the outcome may not always be effective treatment.

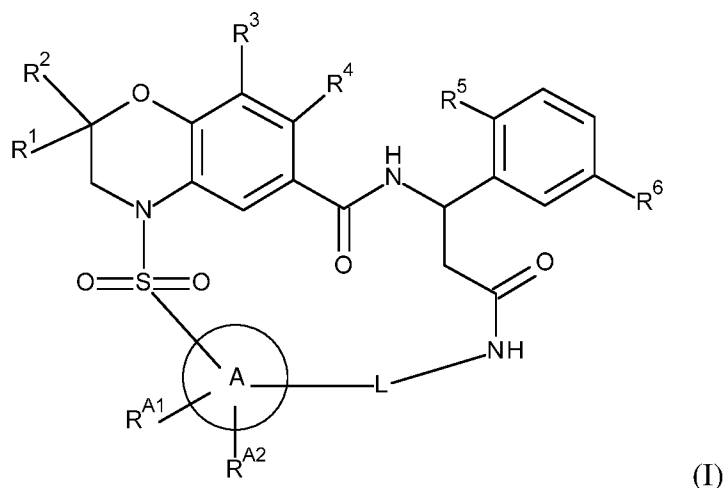
The following abbreviations are used in the present text:

- 10 Ac = acetyl, ACN = acetonitrile, °C = degrees celsius, µl = microliter, µm = micrometer, µmol = micromoles, ¹H=proton, Å=ångström, BBr₃ = Boron tribromide, BH₃·DMS = Borane dimethyl sulfide complex, BH₃·THF = Borane-tetrahydrofuran complex, Boc = tert-Butyloxycarbonyl, BSA = Bovine serum albumin, c=concentration, CAS=Chemical Abstracts Service registry number, Cbz = Carboxybenzyl, CH₃CN=acetonitrile, CDI = 1,1'-carbonyldiimidazole, CO₂ = carbon dioxide, Cs₂CO₃ = Cesium carbonate, CuI = Copper(I) iodide, DABCO = 1,4-Diazabicyclo[2.2.2]octane, dba = dibenzylideneacetone, dppf = 1,1'-bis(diphenylphosphino)ferrocene, DCC = N,N'-Dicyclohexylcarbodiimide, DCM = dichloromethane, DEAD = Diethyl Azodicarboxylate, DIAD = Diisopropyl Azodicarboxylat, DIPEA = N,N-Diisopropylethylamine, DMA = N,N-Dimethylacetamide, 20 DMAP = 4-Dimethylaminopyridine, DMF = N,N-Dimethylformamide, DMSO = dimethylsulfoxide, DMSO-d₆ = hexadeuterodimethylsulfoxide, DTG = Direct Tableting Grade, EC₅₀=half maximal effective concentration, EDTA = Ethylenediaminetetraacetic acid, EGTA = Ethylene glycol-bis(β-aminoethyl ether)-N,N,N',N'-tetraacetic acid, EMEM = Eagle's Minimum Essential Medium, eq = equivalent, ESI = electron spray ionization, 25 EtOAc = Ethyl acetate, Ex. = example, g = gram, g/L = gram per liter, Grubbs II catalyst = Benzyldiene[1,3-bis(2,4,6-trimethylphenyl)-2-imidazolidinylidene]dichloro(tricyclohexylphosphine)ruthenium, h = hour, HATU = hexafluorophosphate azabenzotriazole tetramethyl uranium, HBTU = hexafluorophosphate benzotriazole tetramethyl uronium, HCl = Hydrochloric acid, HEPES = 4-(2-hydroxyethyl)-1-piperazineethanesulfonic acid, HCOOH = formic acid, HOBt = 1-hydroxybenzotriazole, 30 HPLC = high performance liquid chromatography, IC₅₀ = half maximal inhibitory concentration, i-PrOH = Isopropanol, J=coupling constant, JCRB = Japanese Collection of Research Bioresources, kg = kilogram, K₂CO₃ = Potassium carbonate, KF = Potassium

fluoride, KOtBu = Potassium tert-butoxide, LC-MS = Liquid chromatography–mass spectrometry, LiAlH₄ = Lithium aluminium hydride, LiOH = Lithium hydroxide, M = molar, m/z=mass-to-charge ratio, MeOH=methanol, mg=milligram, MgSO₄=magnesium sulfate, MHz=megahertz, min=minute, ml=milliliter, mm=millimeter, mmol=millimole, 5 MS=mass spectrometry, N₂ = Nitrogen, Na₂SO₄ =sodium sulfate, NaBH₃CN = Sodium cyanoborohydride, NaBH₄ = Sodium borohydride, NaHMDS = Sodium bis(trimethylsilyl)amide, NaHCO₃ = sodium bicarbonate, NaOH = Sodium hydroxide, n-BuLi = n-Butyllithium, neg.= negative, NGS = Next-Generation Sequencing, NH₄Cl = ammonium chloride, nm = nanometer, NMI = N-Methylimidazole, NMP = N-Methyl-2-pyrrolidone, 10 NMR= nuclear magnetic resonance spectroscopy, PAM = Protospacer adjacent motif, PET = Positron Emission Tomography, Pd/C = Palladium on carbon, Pd(dppf)Cl₂ = [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium(II), Pd(OH)₂ = Palladium hydroxide, Pd(PPh₃)₄ = Tetrakis(triphenylphosphine)palladium(0), Pht = Phthaloyl, pos.=positive, psi = pounds per square inch, RH = Relative humidity, RP = reverse phase, 15 RT = room temperature, Rt = retention time, s = second, SFC = supercritical fluid chromatography, sgRNA = Single guide RNA, ssODN = single-stranded oligodeoxynucleotide, T3P = Propylphosphonic Anhydride, TBME = tert-Butyl methyl ether, TBTU = N,N,N',N'-tetramethyl-O-(benzotriazol-1-yl)uranium tetrafluoroborate, TFA = trifluoroacetic acid, TCFH = tetrachlorofluoroformamidinium hexafluorophosphate, TCFH- 20 NMI = A combination of tetrachlorofluoroformamidinium hexafluorophosphate and N-Methylimidazole, TEA = Triethylamine, THF = Tetrahydrofuran, TLC = thin layer chromatography, TR-FRET = Time-resolved fluorescence energy transfer, Xantphos = (9,9-Dimethyl-9H-xanthene-4,5-diyl)bis(diphenylphosphane), δ = chemical shift in parts per million

Compounds of the Invention

In a first aspect, the present invention provides a compound of formula (I)



or a pharmaceutically acceptable salt thereof, wherein:

- 5 ring A is C₆₋₁₀-aryl, 5- to 10-membered heteroaryl comprising 1 to 3 heteroatoms independently selected from N, O, S and the remaining atoms being carbon, or 3- to 10-membered heterocyclyl comprising 1 to 3 heteroatoms independently selected from N, O, S and the remaining atoms being carbon;
- 10 R^{A1} and R^{A2} are independently hydrogen, halogen, C₁₋₆-alkyl, cyano, C₁₋₆-alkoxy, halo-C₁₋₆-alkyl;
- R¹ is hydrogen, halogen, C₁₋₆-alkyl, halo-C₁₋₆-alkyl, or hydroxy-C₁₋₆-alkyl; and
- R² is hydrogen or C₁₋₆-alkyl; or
- R¹ and R² together with the carbon to which they are attached form C₃₋₄-cycloalkyl optionally substituted with hydroxy or halogen;
- 15 R³ is hydrogen or halogen;
- R⁴ is hydrogen, halogen, or C₁₋₆-alkoxy;
- R⁵ is halogen, C₁₋₆-alkyl, or halo-C₁₋₆-alkyl;
- R⁶ is halogen or halo-C₁₋₆-alkyl;
- L is
- 20 (i) a group *-X¹-L¹¹-Y¹-L¹²-; wherein:
- X¹ is -O-, -NR⁷-;
- Y¹ is -CH₂-, -O-, -NR⁸-, or -CH=CH-, -C≡C-;
- L¹¹ is C₁₋₃-alkylene optionally substituted with R⁹;

L^{12} is C_{1-3} -alkylene optionally substituted with R^{10} ;

R^7 and R^8 are independently hydrogen, C_{1-6} -alkyl, C_{1-6} -alkyl-C(O)-

R^9 and R^{10} are independently hydroxy, amino, halogen, C_{1-6} -alkyl, hydroxy- C_{1-6} -alkyl, or amino- C_{1-6} -alkyl; and

5 the asterisk indicates the point of attachment of L to ring A; or

(ii) a group $*-X^2-L^{21}-Y^2-L^{22}-$; wherein:

X^2 is $-C\equiv C-$, $-CH=CH-$

Y^2 is $-O-$, $-NR^{11}-$;

10 L^{21} and L^{22} are independently C_{1-3} -alkylene optionally substituted with hydroxy, amino, halogen, C_{1-6} -alkyl, hydroxy- C_{1-6} -alkyl, or amino- C_{1-6} -alkyl;

R^{11} is hydrogen, C_{1-6} -alkyl, or C_{1-6} -alkyl-C(O)-; and

the asterisk indicates the point of attachment of L to ring A; or

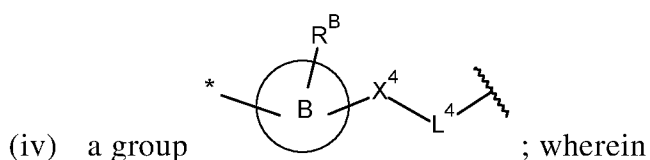
(iii) a group $*-L^{31}-X^3-L^{32}-$; wherein:

X^3 is $-O-$, $-NR^{12}-$;

15 L^{31} and L^{32} are independently C_{1-3} -alkylene optionally substituted with hydroxy, amino, halogen, C_{1-6} -alkyl, hydroxy- C_{1-6} -alkyl, or amino- C_{1-6} -alkyl;

R^{12} is hydrogen, C_{1-6} -alkyl, or C_{1-6} -alkyl-C(O)-; and

the asterisk indicates the point of attachment of L to ring A; or



20 X^4 is a covalent bond, $-O-$, $-NR^{13}-$, or $-CH(OH)-$, $-CH(NH_2)-$,

L^4 is C_{1-3} -alkylene;

R^{13} is hydrogen, C_{1-6} -alkyl, or C_{1-6} -alkyl-C(O)-;

R^B is hydrogen, hydroxy, amino, halogen, C_{1-6} -alkyl;

25 B is C_{3-10} -cycloalkyl, or 3- to 10-membered heterocyclyl comprising 1 to 3 heteroatoms independently selected from N, O, S and the remaining atoms being carbon; and

the asterisk indicates the point of attachment of L to ring A.

In one embodiment, the present invention provides a compound of formula (I) as described

herein, or a pharmaceutically acceptable salt thereof, wherein ring A is phenyl, 5- to 6-membered

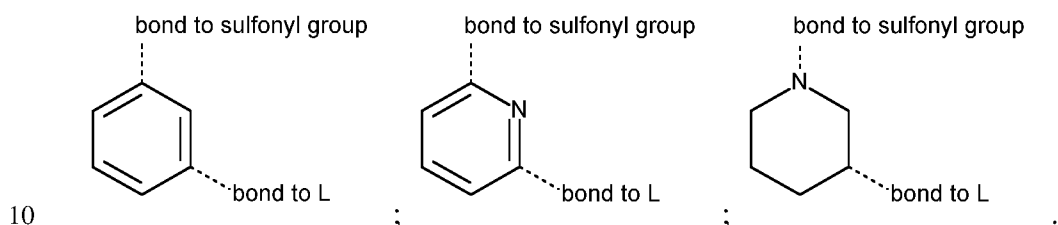
30 heteroaryl comprising 1 to 3 heteroatoms independently selected from N, O, S and the remaining

atoms being carbon, or 5- to 6-membered heterocyclyl comprising 1 to 3 heteroatoms independently selected from N, O, S and the remaining atoms being carbon.

In a particular embodiment, the present invention provides a compound of formula (I) as described herein, or a pharmaceutically acceptable salt thereof, wherein ring A is C₆-C₁₀-aryl.

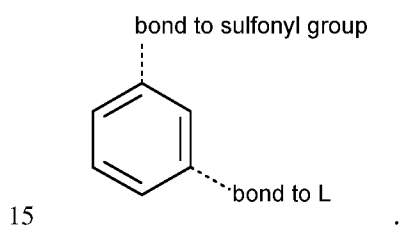
- 5 In a particular embodiment, the present invention provides a compound of formula (I) as described herein, or a pharmaceutically acceptable salt thereof, wherein ring A is phenyl, pyridyl, or piperidyl.

In a particular embodiment, the present invention provides a compound of formula (I) as described herein, or a pharmaceutically acceptable salt thereof, wherein ring A is:



In a more particular embodiment, the present invention provides a compound of formula (I) as described herein, or a pharmaceutically acceptable salt thereof, wherein ring A is phenyl.

In a particular embodiment, the present invention provides a compound of formula (I) as described herein, or a pharmaceutically acceptable salt thereof, wherein ring A is:



In one embodiment, the present invention provides a compound of formula (I) as described herein, or a pharmaceutically acceptable salt thereof, wherein R^{A1} and R^{A2} are independently hydrogen, halogen, C₁₋₆-alkyl.

- 20 In a particular embodiment, the present invention provides a compound of formula (I) as described herein, or a pharmaceutically acceptable salt thereof, wherein R^{A1} is hydrogen, halogen, C₁₋₆-alkyl; and R^{A2} is hydrogen.

In a more particular embodiment, the present invention provides a compound of formula (I) as described herein, or a pharmaceutically acceptable salt thereof, wherein R^{A1} is hydrogen, fluoro, or methyl; and R^{A2} is hydrogen.

5 In a more particular embodiment, the present invention provides a compound of formula (I) as described herein, or a pharmaceutically acceptable salt thereof, wherein R^{A1} is hydrogen or methyl; and R^{A2} is hydrogen or fluoro.

In a more particular embodiment, the present invention provides a compound of formula (I) as described herein, or a pharmaceutically acceptable salt thereof, wherein R^{A1} and R^{A2} are hydrogen.

10 In one embodiment, the present invention provides a compound of formula (I) as described herein, or a pharmaceutically acceptable salt thereof, wherein R^1 is hydrogen, halogen, C_{1-6} -alkyl, halo- C_{1-6} -alkyl, or hydroxy- C_{1-6} -alkyl; and R^2 is hydrogen; or R^1 and R^2 together with the carbon to which they are attached form C_{3-4} -cycloalkyl optionally substituted with hydroxy or halogen.

15 In a particular embodiment, the present invention provides a compound of formula (I) as described herein, or a pharmaceutically acceptable salt thereof, wherein R^1 is hydrogen, C_{1-6} -alkyl, or halo- C_{1-6} -alkyl; and R^2 is hydrogen.

In a particular embodiment, the present invention provides a compound of formula (I) as described herein, or a pharmaceutically acceptable salt thereof, wherein R^1 is hydrogen, fluoro, methyl, hydroxymethyl, or fluoromethyl, and R^2 is hydrogen; or R^1 and R^2 together with the
20 carbon to which they are attached form cyclopropyl or fluorocyclopropyl.

In a more particular embodiment, the present invention provides a compound of formula (I) as described herein, or a pharmaceutically acceptable salt thereof, wherein R^1 is hydrogen, methyl or fluoromethyl; and R^2 is hydrogen.

25 In one embodiment, the present invention provides a compound of formula (I) as described herein, or a pharmaceutically acceptable salt thereof, wherein R^3 is hydrogen or fluoro.

In one embodiment, the present invention provides a compound of formula (I) as described herein, or a pharmaceutically acceptable salt thereof, wherein R^4 is hydrogen or halogen.

30 In one embodiment, the present invention provides a compound of formula (I) as described herein, or a pharmaceutically acceptable salt thereof, wherein R^4 is hydrogen, fluoro, or methoxy.

In a particular embodiment, the present invention provides a compound of formula (I) as described herein, or a pharmaceutically acceptable salt thereof, wherein R^4 is hydrogen or fluoro.

In a particular embodiment, the present invention provides a compound of formula (I) as described herein, or a pharmaceutically acceptable salt thereof, wherein:

- 5 R^3 is hydrogen or fluoro;
 R^4 is hydrogen, fluoro, or methoxy.

In a particular embodiment, the present invention provides a compound of formula (I) as described herein, or a pharmaceutically acceptable salt thereof, wherein:

- R^3 is hydrogen or fluoro;
10 R^4 is hydrogen or fluoro.

In one embodiment, the present invention provides a compound of formula (I) as described herein, or a pharmaceutically acceptable salt thereof, wherein R^5 is halogen or C_{1-6} -alkyl.

- In one embodiment, the present invention provides a compound of formula (I) as described herein, or a pharmaceutically acceptable salt thereof, wherein R^5 is chloro, fluoro, methyl, or -
15 CF_3 .

In a particular embodiment, the present invention provides a compound of formula (I) as described herein, or a pharmaceutically acceptable salt thereof, wherein R^5 is chloro or methyl.

In one embodiment, the present invention provides a compound of formula (I) as described herein, or a pharmaceutically acceptable salt thereof, wherein R^6 is halo- C_{1-6} -alkyl.

- 20 In one embodiment, the present invention provides a compound of formula (I) as described herein, or a pharmaceutically acceptable salt thereof, wherein R^6 is chloro or $-CF_3$.

In a particular embodiment, the present invention provides a compound of formula (I) as described herein, or a pharmaceutically acceptable salt thereof, wherein R^6 is $-CF_3$.

- In a particular embodiment, the present invention provides a compound of formula (I) as
25 described herein, or a pharmaceutically acceptable salt thereof, wherein:

- R^5 is halogen or C_{1-6} -alkyl;
 R^6 is halo- C_{1-6} -alkyl.

In a particular embodiment, the present invention provides a compound of formula (I) as described herein, or a pharmaceutically acceptable salt thereof, wherein:

R^5 is chloro, fluoro, methyl, or $-CF_3$;

R^6 is chloro or $-CF_3$.

5 In a particular embodiment, the present invention provides a compound of formula (I) as described herein, or a pharmaceutically acceptable salt thereof, wherein:

R^5 is chloro or methyl;

R^6 is $-CF_3$.

In one embodiment, the present invention provides a compound of formula (I) as described
10 herein, or a pharmaceutically acceptable salt thereof, wherein L is

(i) a group $*-X^1-L^{11}-Y^1-L^{12}-$; wherein:

X^1 is $-O-$, $-NR^7-$;

Y^1 is $-CH_2-$, $-O-$, $-NR^8-$, or $-CH=CH-$;

L^{11} is C_{1-3} -alkylene optionally substituted with R^9 ;

15 L^{12} is C_{1-3} -alkylene optionally substituted with R^{10} ;

R^7 and R^8 are independently hydrogen, C_{1-6} -alkyl;

R^9 and R^{10} are independently hydroxy, amino, C_{1-6} -alkyl, or hydroxy- C_{1-6} -alkyl;

and the asterisk indicates the point of attachment of L to ring A; or

(ii) a group $*-X^2-L^{21}-Y^2-L^{22}-$; wherein:

20 X^2 is $-C\equiv C-$;

Y^2 is $-O-$, $-NR^{11}-$;

L^{21} and L^{22} are independently C_{1-3} -alkylene;

R^{11} is hydrogen, or C_{1-6} -alkyl- $C(O)-$; and

the asterisk indicates the point of attachment of L to ring A; or

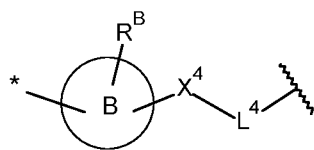
25 (iii) a group $*-L^{31}-X^3-L^{32}-$; wherein:

X^3 is $-O-$, $-NR^{12}-$;

L^{31} and L^{32} are independently C_{1-3} -alkylene;

R^{12} is hydrogen, C_{1-6} -alkyl; and

and the asterisk indicates the point of attachment of L to ring A; or



(iv) a group ; wherein

X^4 is a covalent bond, $-O-$, $-NR^{13}-$, or $-CH(OH)-$;

L^4 is C_{1-3} -alkylene;

R^{13} is hydrogen;

5 R^B is hydrogen or hydroxy;

B is C_{5-6} -cycloalkyl, or 5- to 7-membered heterocyclyl comprising 1 to 2 heteroatoms independently selected from N and O and the remaining atoms being carbon; and

the asterisk indicates the point of attachment of L to ring A.

10 In a particular embodiment, the present invention provides a compound of formula (I) as described herein, or a pharmaceutically acceptable salt thereof, wherein L is

(i) a group $*-X^1-L^{11}-Y^1-L^{12}-$; wherein:

X^1 is $-O-$, $-NR^7-$;

Y^1 is $-CH_2-$, $-O-$, $-NR^8-$, or $-CH=CH-$;

15 L^{11} is C_{1-3} -alkylene optionally substituted with R^9 ;

L^{12} is C_{1-3} -alkylene optionally substituted with R^{10} ;

R^7 and R^8 are independently hydrogen, C_{1-6} -alkyl;

R^9 and R^{10} are independently hydroxy, amino, C_{1-6} -alkyl, or hydroxy- C_{1-6} -alkyl;

and the asterisk indicates the point of attachment of L to ring A; or

20 (ii) a group $*-X^2-L^{21}-Y^2-L^{22}-$; wherein:

X^2 is $-C\equiv C-$;

Y^2 is $-O-$, $-NR^{11}-$;

L^{21} and L^{22} are independently C_{1-3} -alkylene;

R^{11} is hydrogen, or C_{1-6} -alkyl- $C(O)-$; and

25 the asterisk indicates the point of attachment of L to ring A; or

(iii) a group $*-L^{31}-X^3-L^{32}-$; wherein:

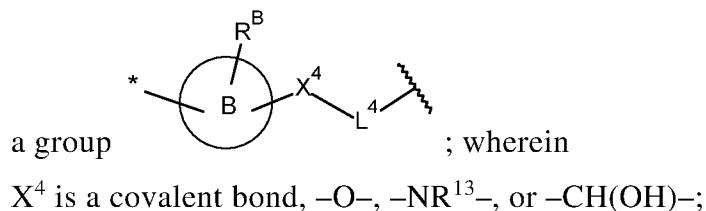
X^3 is $-O-$, $-NR^{12}-$;

L^{31} and L^{32} are independently C_{1-3} -alkylene;

R^{12} is hydrogen, C_{1-6} -alkyl; and

30 and the asterisk indicates the point of attachment of L to ring A.

In a particular embodiment, the present invention provides a compound of formula (I) as described herein, or a pharmaceutically acceptable salt thereof, wherein L is



L^4 is C_{1-3} -alkylene;

R^{13} is hydrogen;

R^B is hydrogen or hydroxy;

B is C_{5-6} -cycloalkyl, or 5- to 7-membered heterocyclyl comprising 1 to 2 heteroatoms independently selected from N and O and the remaining atoms being carbon; and

the asterisk indicates the point of attachment of L to ring A.

In a particular embodiment, the present invention provides a compound of formula (I) as described herein, or a pharmaceutically acceptable salt thereof, wherein L is

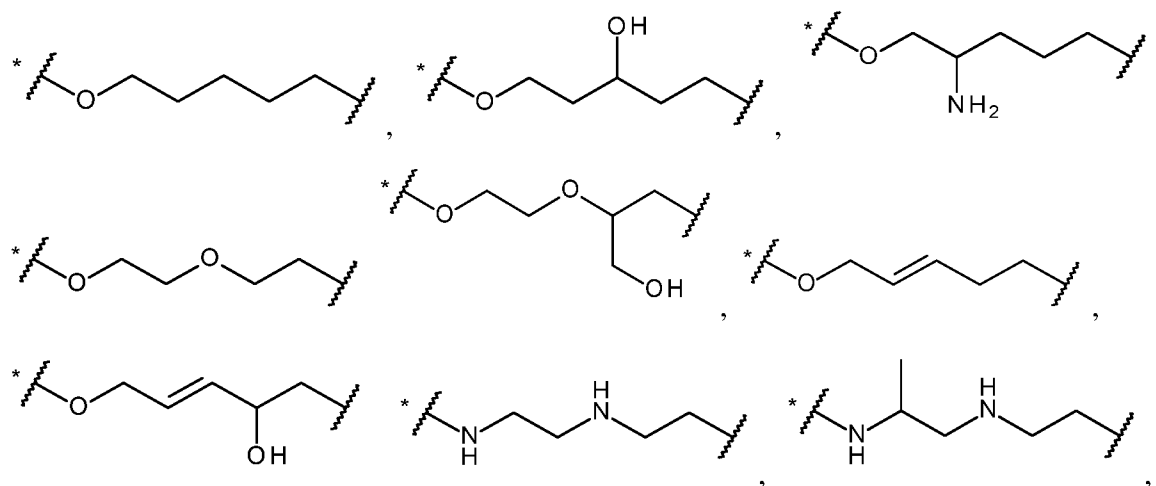
a group $*-L^{31}-X^3-L^{32}-$; wherein:

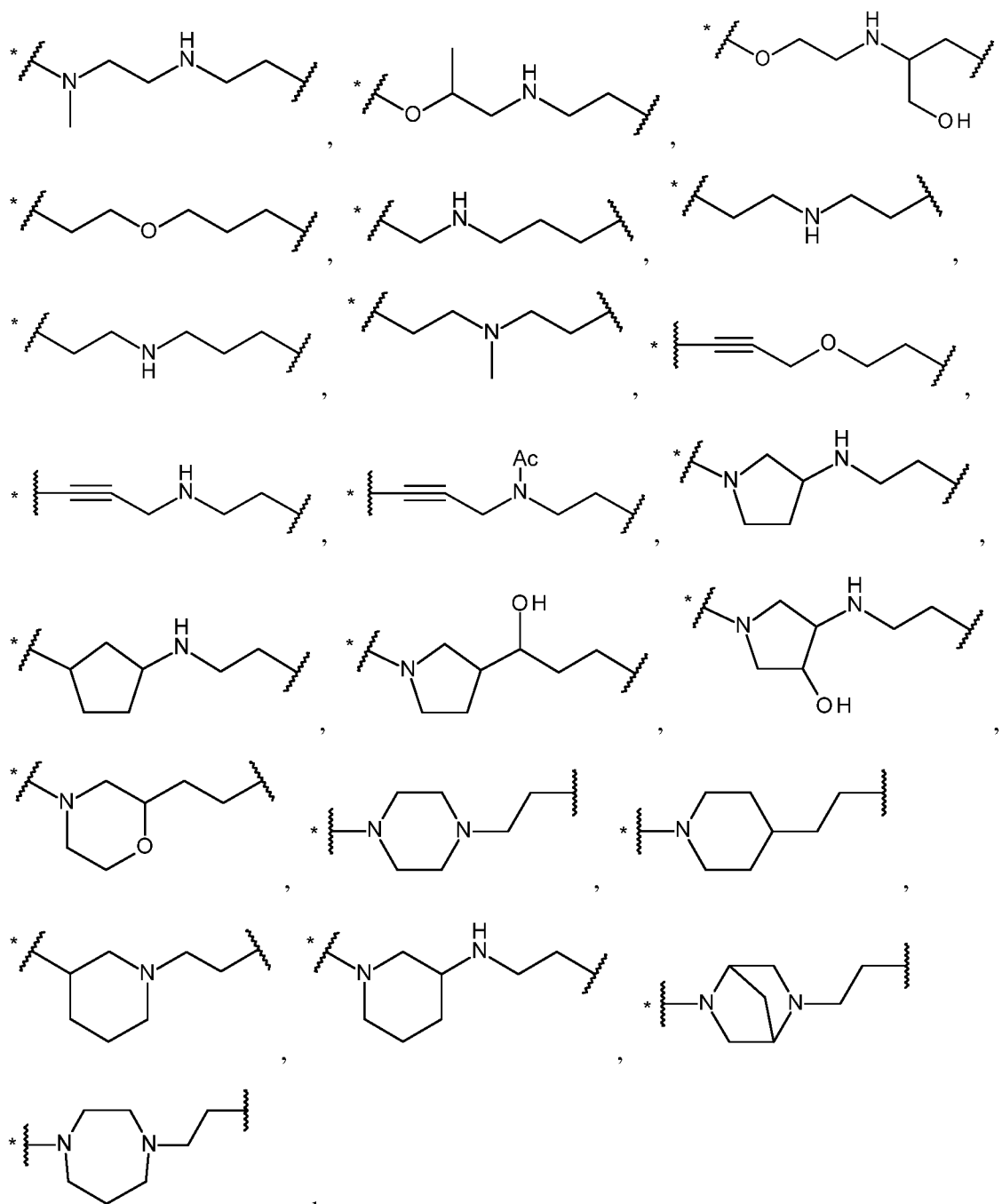
X^3 is $-NH-$;

L^{31} and L^{32} are independently C_{1-3} -alkylene;

the asterisk indicates the point of attachment of L to ring A.

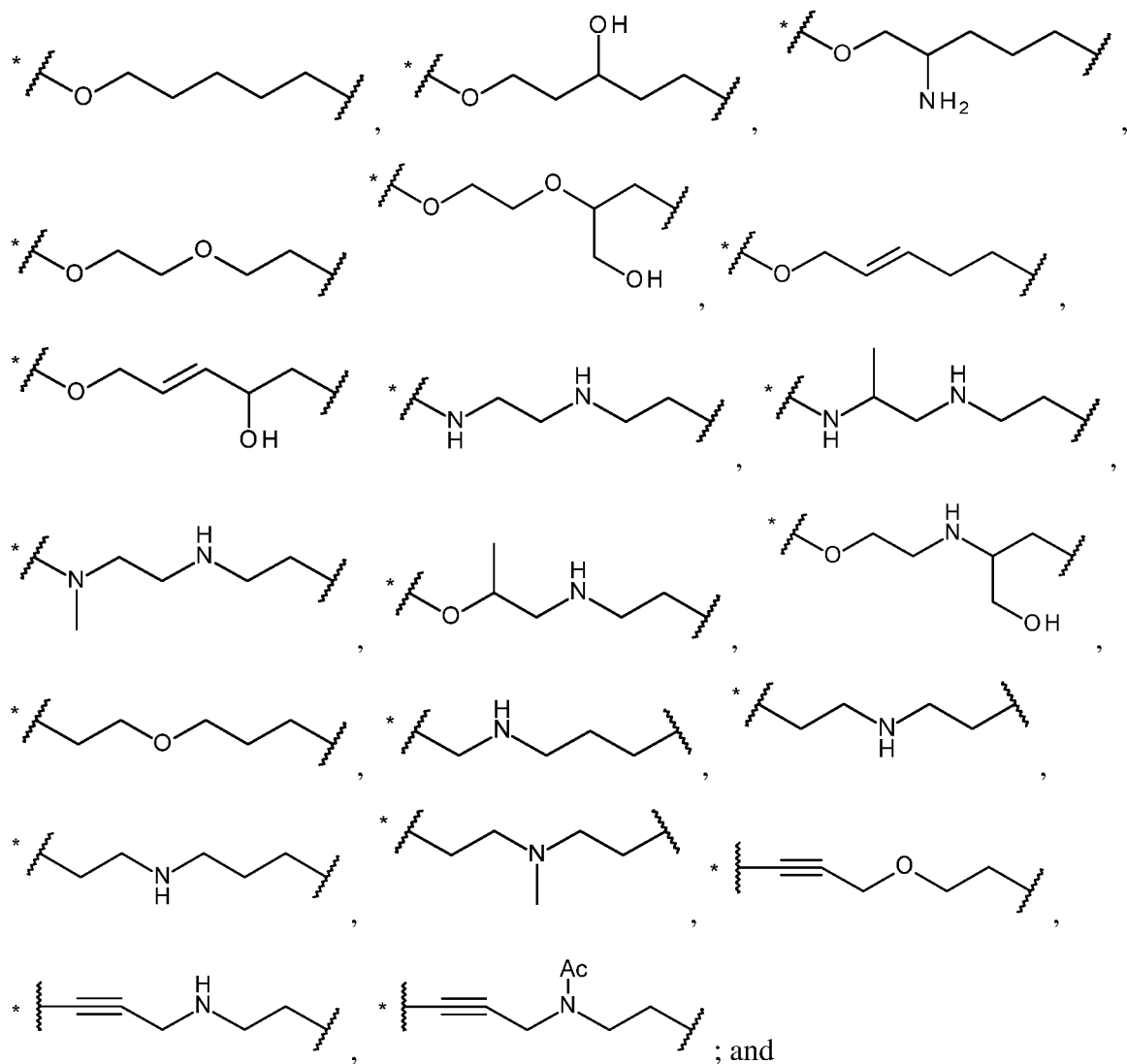
In one embodiment, the present invention provides a compound of formula (I) as described herein, or a pharmaceutically acceptable salt thereof, wherein L is





the asterisk indicates the point of attachment of L to ring A.

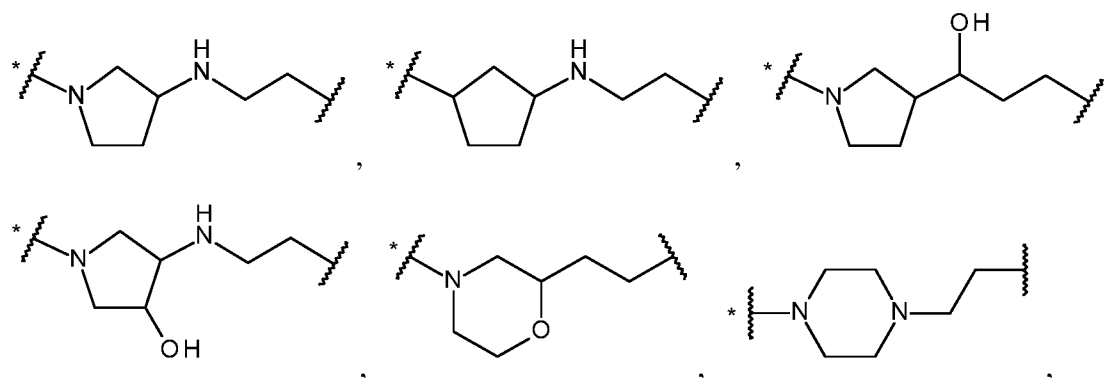
- 10 In a particular embodiment, the present invention provides a compound of formula (I) as described herein, or a pharmaceutically acceptable salt thereof, wherein L is

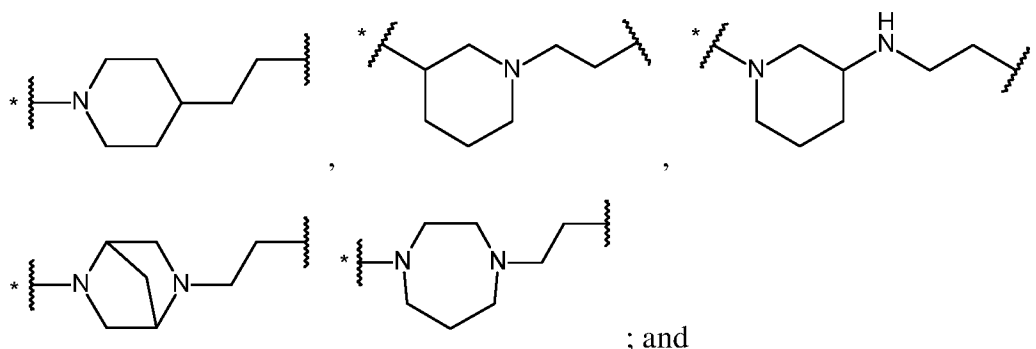


the asterisk indicates the point of attachment of L to ring A.

In a particular embodiment, the present invention provides a compound of formula (I) as

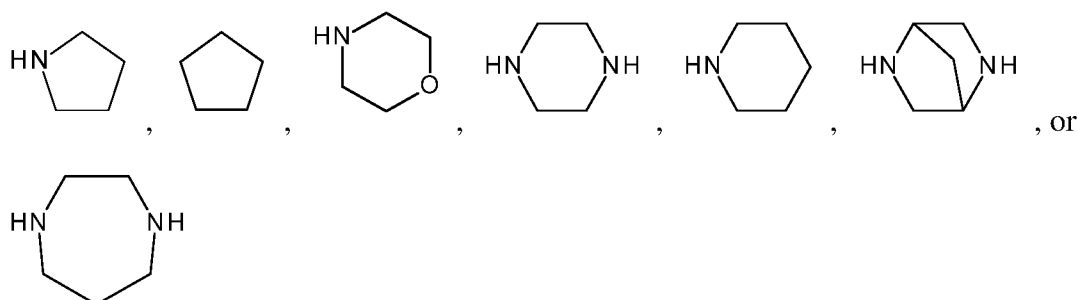
described herein, or a pharmaceutically acceptable salt thereof, wherein L is





the asterisk indicates the point of attachment of L to ring A.

In a particular embodiment, the present invention provides a compound of formula (I) as
 5 described herein, or a pharmaceutically acceptable salt thereof, wherein ring B is:



In a more particular embodiment, the present invention provides a compound of formula (I) as
 described herein, or a pharmaceutically acceptable salt thereof, wherein L is



In one embodiment, the present invention provides a compound of formula (I) as described
 herein, or a pharmaceutically acceptable salt thereof, wherein:

- ring A is phenyl, 5- to 6-membered heteroaryl comprising 1 to 3 heteroatoms
 independently selected from N, O, S and the remaining atoms being carbon, or 5- to
 15 6-membered heterocyclyl comprising 1 to 3 heteroatoms independently selected from
 N, O, S and the remaining atoms being carbon;
 R^{A1} and R^{A2} are independently hydrogen, halogen, C_{1-6} -alkyl;
 R^1 is hydrogen, halogen, C_{1-6} -alkyl, halo- C_{1-6} -alkyl, or hydroxy- C_{1-6} -alkyl; and
 R^2 is hydrogen; or R^1 and R^2 together with the carbon to which they are attached form
 20 C_{3-4} -cycloalkyl optionally substituted with hydroxy or halogen;

R^3 is hydrogen or halogen;

R^4 is hydrogen, halogen, or C_{1-6} -alkoxy;

R^5 is halogen, C_{1-6} -alkyl, or halo- C_{1-6} -alkyl;

R^6 is halogen or halo- C_{1-6} -alkyl;

5 L is

(i) a group $*-X^1-L^{11}-Y^1-L^{12}-$; wherein:

X^1 is $-O-$, $-NR^7-$;

Y^1 is $-CH_2-$, $-O-$, $-NR^8-$, or $-CH=CH-$;

L^{11} is C_{1-3} -alkylene optionally substituted with R^9 ;

10 L^{12} is C_{1-3} -alkylene optionally substituted with R^{10} ;

R^7 and R^8 are independently hydrogen, C_{1-6} -alkyl;

R^9 and R^{10} are independently hydroxy, amino, C_{1-6} -alkyl, or hydroxy- C_{1-6} -alkyl;

and

the asterisk indicates the point of attachment of L to ring A; or

15 (ii) a group $*-X^2-L^{21}-Y^2-L^{22}-$; wherein:

X^2 is $-C\equiv C-$;

Y^2 is $-O-$, $-NR^{11}-$;

L^{21} and L^{22} are independently C_{1-3} -alkylene;

R^{11} is hydrogen, or C_{1-6} -alkyl- $C(O)-$; and

20 the asterisk indicates the point of attachment of L to ring A; or

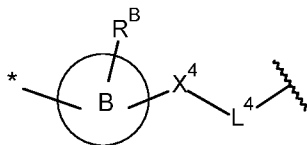
(iii) a group $*-L^{31}-X^3-L^{32}-$; wherein:

X^3 is $-O-$, $-NR^{12}-$;

L^{31} and L^{32} are independently C_{1-3} -alkylene;

R^{12} is hydrogen, C_{1-6} -alkyl; and

25 and the asterisk indicates the point of attachment of L to ring A; or

(iv) a group ; wherein

X^4 is a covalent bond, $-O-$, $-NR^{13}-$, or $-CH(OH)-$;

L^4 is C_{1-3} -alkylene;

R^{13} is hydrogen;

30 R^B is hydrogen or hydroxy;

B is C₅₋₆-cycloalkyl, or 5- to 7-membered heterocyclyl comprising 1 to 2 heteroatoms independently selected from N and O and the remaining atoms being carbon; and

the asterisk indicates the point of attachment of L to ring A.

- 5 In a particular embodiment, the present invention provides a compound of formula (I) as described herein, or a pharmaceutically acceptable salt thereof, wherein:

ring A is phenyl;

R^{A1} and R^{A2} are hydrogen;

R¹ is hydrogen, C₁₋₆-alkyl, or halo-C₁₋₆-alkyl; and

10 R² is hydrogen;

R³ is hydrogen or halogen;

R⁴ is hydrogen or halogen;

R⁵ is halogen or C₁₋₆-alkyl;

R⁶ is halo-C₁₋₆-alkyl;

15 L is a group *-L³¹-X³-L³²-; wherein:

X³ is-NH-;

L³¹ and L³² are independently C₁₋₃-alkylene; and

the asterisk indicates the point of attachment of L to ring A.

- In a particular embodiment, the present invention provides a compound of formula (I) as
20 described herein, or a pharmaceutically acceptable salt thereof, wherein:

ring A is phenyl, pyridyl, or piperidyl;

R^{A1} and R^{A2} are hydrogen;

R¹ is hydrogen, fluoro, methyl, hydroxymethyl, or fluoromethyl, and R² is hydrogen; or

25 R¹ and R² together with the carbon to which they are attached form cyclopropyl or fluorocyclopropyl;

R³ is hydrogen or fluoro;

R⁴ is hydrogen, fluoro, or methoxy;

R⁵ is chloro, fluoro, methyl, or -CF₃;

R⁶ is chloro or -CF₃;

30 L is



; and

the asterisk indicates the point of attachment of L to ring A.

In a more particular embodiment, the present invention provides a compound of formula (I) as described herein, or a pharmaceutically acceptable salt thereof, wherein:

ring A is phenyl;

5 R^{A1} and R^{A2} are hydrogen;

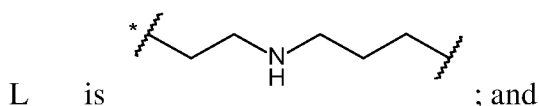
R^1 is hydrogen, methyl or fluoromethyl; and R^2 is hydrogen;

R^3 is hydrogen or fluoro;

R^4 is hydrogen or fluoro;

R^5 is chloro or methyl;

10 R^6 is $-CF_3$;



the asterisk indicates the point of attachment of L to ring A.

In a particular embodiment, the present invention provides a compound of formula (I) as described herein, or a pharmaceutically acceptable salt thereof, wherein

15 In one embodiment, the present invention provides a compound of formula (I) as described herein, or a pharmaceutically acceptable salt thereof, wherein said compound of formula (I) is:

(17S)-17-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-diketo-8,11,24-trioxa-2 λ^6 -thia-1,14,18-triazatetracyclo[18.6.2.13,7.023,27]nonacosa-3(29),4,6,20,22,27-hexaene-15,19-quinone;

20 (17S)-17-[2-fluoro-5-(trifluoromethyl)phenyl]-2,2-dioxo-8,11,24-trioxa-2 λ^6 -thia-1,14,18-triazatetracyclo[18.6.2.13,7.023,27]nonacosa-3,5,7(29),20(28),21,23(27)-hexaene-15,19-dione;

(17S)-17-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-spiro[8,11,24-trioxa-2 λ^6 -thia-1,14,18-triazatetracyclo[18.6.2.13,7.023,27]nonacosa-3,5,7(29),20(28),21,23(27)-hexaene-25,1'-cyclopropane]-15,19-dione;

25

(17S)-17-[2-chloro-5-(trifluoromethyl)phenyl]-5-methyl-2,2-dioxo-8,11,24-trioxa-2 λ^6 -thia-1,14,18-triazatetracyclo[18.6.2.13,7.023,27]nonacosa-3,5,7(29),20(28),21,23(27)-hexaene-15,19-dione;

- (10R,17S)-10-Amino-17-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-spiro[8,24-dioxa-2 λ^6 -thia-1,14,18-triazatetracyclo[18.6.2.13,7.023,27]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-25,1'-cyclopropane]-15,19-dione;
- 5 (18S)-18-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-25-oxa-2 λ^6 -thia-1,8,12,15,19-pentazapentacyclo[19.6.2.28,12.13,7.024,28]dotriaconta-3,5,7(32),21(29),22,24(28)-hexaene-16,20-dione;
- (17S)-17-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-24-oxa-2 λ^6 -thia-1,8,11,14,18-pentazapentacyclo[18.6.2.28,11.13,7.023,27]hentriaconta-3,5,7(31),20(28),21,23(27)-hexaene-15,19-dione;
- 10 (10R,12R,20S)-20-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-spiro[27-oxa-2 λ^6 -thia-1,8,14,17,21-pentazahexacyclo[21.6.2.13,7.08,12.010,14.026,30]dotriaconta-3(32),4,6,23(31),24,26(30)-hexaene-28,1'-cyclopropane]-18,22-dione;
- (17S)-17-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-spiro[24-oxa-2 λ^6 -thia-1,8,14,18-tetrazapentacyclo[18.6.2.28,11.13,7.023,27]hentriaconta-3,5,7(31),20(28),21,23(27)-hexaene-25,1'-cyclopropane]-15,19-dione;
- 15 (11S,18S)-18-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-25-oxa-2 λ^6 -thia-1,8,12,15,19-pentazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-3,5,7(31),21(29),22,24(28)-hexaene-16,20-dione;
- (11S,18S)-18-[2-chloro-5-(trifluoromethyl)phenyl]-5-methyl-2,2-dioxo-25-oxa-2 λ^6 -thia-1,8,12,15,19-pentazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-3,5,7(31),21(29),22,24(28)-hexaene-16,20-dione;
- 20 (11S,18S,26R)-18-[2-chloro-5-(trifluoromethyl)phenyl]-26-methyl-2,2-dioxo-25-oxa-2 λ^6 -thia-1,8,12,15,19-pentazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-3(31),4,6,21(29),22,24(28)-hexaene-16,20-dione;
- (11S,18S)-18-[2-chloro-5-(trifluoromethyl)phenyl]-5-fluoro-2,2-dioxo-25-oxa-2 λ^6 -thia-1,8,12,15,19-pentazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-3,5,7(31),21(29),22,24(28)-hexaene-16,20-dione;
- 25 (11S,18S)-18-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-spiro[25-oxa-2 λ^6 -thia-1,8,12,15,19-pentazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-3(31),4,6,21(29),22,24(28)-hexaene-26,1'-cyclopropane]-16,20-dione;
- 30

- (12S,19S)-19-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-26-oxa-2 λ^6 -thia-1,8,13,16,20-pentazapentacyclo[20.6.2.13,7.18,12.025,29]dotriaconta-3,5,7(32),22(30),23,25(29)-hexaene-17,21-dione;
- 5 (17S)-17-[2-chloro-5-(trifluoromethyl)phenyl]-8-methyl-2,2-dioxo-24-oxa-2 λ^6 -thia-1,8,11,14,18-pentazatetracyclo[18.6.2.13,7.023,27]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-15,19-dione;
- (12R,18S)-18-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-spiro[11,25-dioxa-2 λ^6 -thia-1,8,15,19-tetrazapentacyclo[19.6.2.13,7.18,12.024,28]hentriaconta-3(31),4,6,21(29),22,24(28)-hexaene-26,1'-cyclopropane]-16,20-dione;
- 10 (12S,18S)-18-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-spiro[11,25-dioxa-2 λ^6 -thia-1,8,15,19-tetrazapentacyclo[19.6.2.13,7.18,12.024,28]hentriaconta-3(31),4,6,21(29),22,24(28)-hexaene-26,1'-cyclopropane]-16,20-dione;
- (12R,18S)-18-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-11,25-dioxa-2 λ^6 -thia-1,8,15,19-tetrazapentacyclo[19.6.2.13,7.18,12.024,28]hentriaconta-3(31),4,6,21(29),22,24(28)-hexaene-16,20-dione;
- 15 (12S,18S)-18-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-11,25-dioxa-2 λ^6 -thia-1,8,15,19-tetrazapentacyclo[19.6.2.13,7.18,12.024,28]hentriaconta-3(31),4,6,21(29),22,24(28)-hexaene-16,20-dione;
- (17S)-17-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-11,24-dioxa-2 λ^6 -thia-1,14,18-triazatetracyclo[18.6.2.13,7.023,27]nonacosa-3,5,7(29),20(28),21,23(27)-hexaen-8-yne-15,19-dione;
- 20 rac-(11S,18S)-18-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-25-oxa-2 λ^6 -thia-1,3,8,12,15,19-hexazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-21(29),22,24(28)-triene-16,20-dione;
- rac-(10E,17S)-17-[2-chloro-5-(trifluoromethyl)phenyl]-12-hydroxy-2,2-dioxo-8,24-dioxa-2 λ^6 -thia-1,14,18-triazatetracyclo[18.6.2.13,7.023,27]nonacosa-3(29),4,6,10,20(28),21,23(27)-heptaene-15,19-dione;
- 30

(17S)-17-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-8,24-dioxa-2 λ^6 -thia-1,11,14,18-tetrazatetracyclo[18.6.2.13,7.023,27]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-15,19-dione;

(17S)-17-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-8,24-dioxa-2 λ^6 -thia-1,14,18-triazatetracyclo[18.6.2.13,7.023,27]nonacosa-3,5,7(29),20(28),21,23(27)-hexaene-15,19-dione;

(17S)-17-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-8,24-dioxa-2 λ^6 -thia-1,14,18-triazatetracyclo[18.6.2.13,7.023,27]nonacosa-3,5,7(29),10,20(28),21,23(27)-heptaene-15,19-dione;

(9R,17S)-17-[2-chloro-5-(trifluoromethyl)phenyl]-9-methyl-2,2-dioxo-spiro[8,24-dioxa-2 λ^6 -thia-1,11,14,18-tetrazatetracyclo[18.6.2.13,7.023,27]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-25,1'-cyclopropane]-15,19-dione;

(9S,17S)-17-[2-chloro-5-(trifluoromethyl)phenyl]-9-methyl-2,2-dioxo-spiro[8,24-dioxa-2 λ^6 -thia-1,11,14,18-tetrazatetracyclo[18.6.2.13,7.023,27]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-25,1'-cyclopropane]-15,19-dione;

(12S,17S)-17-[2-chloro-5-(trifluoromethyl)phenyl]-12-(hydroxymethyl)-2,2-dioxo-8,11,24-trioxa-2 λ^6 -thia-1,14,18-triazatetracyclo[18.6.2.13,7.023,27]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-15,19-dione;

(11S,17S)-17-[2-chloro-5-(trifluoromethyl)phenyl]-11-hydroxy-2,2-dioxo-8,24-dioxa-2 λ^6 -thia-1,14,18-triazatetracyclo[18.6.2.13,7.023,27]nonacosa-3,5,7(29),20(28),21,23(27)-hexaene-15,19-dione;

(17S)-17-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-spiro[24-oxa-2 λ^6 -thia-1,11,14,18-tetrazatetracyclo[18.6.2.13,7.023,27]nonacosa-3,5,7(29),20(28),21,23(27)-hexaen-8-yne-25,1'-cyclopropane]-15,19-dione;

(17S)-11-acetyl-17-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-spiro[24-oxa-2 λ^6 -thia-1,11,14,18-tetrazatetracyclo[18.6.2.13,7.023,27]nonacosa-3,5,7(29),20(28),21,23(27)-hexaen-8-yne-25,1'-cyclopropane]-15,19-dione;

(19S)-19-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-26-oxa-2 λ^6 -thia-1,12,16,20-tetrazapentacyclo[20.6.2.13,7.18,12.025,29]dotriaconta-3(32),4,6,22(30),23,25(29)-hexaene-17,21-dione;

- (11S,18S,26R)-18-[2-chloro-5-(trifluoromethyl)phenyl]-26-fluoro-2,2-dioxo-25-oxa-2 λ^6 -thia-1,8,12,15,19-pentazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-3(31),4,6,21(29),22,24(28)-hexaene-16,20-dione;
- 5 (12S,18S)-18-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-spiro[11,25-dioxa-2 λ^6 -thia-1,8,15,19-tetrazapentacyclo[19.6.2.13,7.18,12.024,28]hentriaconta-3(31),4,6,21(29),22,24(28)-hexaene-26,1'-cyclopropane]-16,20-dione;
- (11S,18S,26S)-18-[2-chloro-5-(trifluoromethyl)phenyl]-26-fluoro-2,2-dioxo-25-oxa-2 λ^6 -thia-1,8,12,15,19-pentazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-3(31),4,6,21(29),22,24(28)-hexaene-16,20-dione;
- 10 (10R,11R,18S)-18-[2-chloro-5-(trifluoromethyl)phenyl]-10-hydroxy-2,2-dioxo-25-oxa-2 λ^6 -thia-1,8,12,15,19-pentazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-3(31),4,6,21(29),22,24(28)-hexaene-16,20-dione;
- (11S,12R,18S)-18-[2-chloro-5-(trifluoromethyl)phenyl]-12-hydroxy-2,2-dioxo-25-oxa-2 λ^6 -thia-1,8,15,19-tetrazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-3(31),4,6,21(29),22,24(28)-hexaene-16,20-dione;
- 15 (11S,12S,18S)-18-[2-chloro-5-(trifluoromethyl)phenyl]-12-hydroxy-2,2-dioxo-25-oxa-2 λ^6 -thia-1,8,15,19-tetrazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-3(31),4,6,21(29),22,24(28)-hexaene-16,20-dione;
- (11R,12R,18S)-18-[2-chloro-5-(trifluoromethyl)phenyl]-12-hydroxy-2,2-dioxo-25-oxa-2 λ^6 -thia-1,8,15,19-tetrazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-3(31),4,6,21(29),22,24(28)-hexaene-16,20-dione;
- 20 (11R,12S,18S)-18-[2-chloro-5-(trifluoromethyl)phenyl]-12-hydroxy-2,2-dioxo-25-oxa-2 λ^6 -thia-1,8,15,19-tetrazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-3(31),4,6,21(29),22,24(28)-hexaene-16,20-dione;
- (18S)-18-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-25-oxa-2 λ^6 -thia-1,12,15,19-tetrazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-3(31),4,6,21(29),22,24(28)-hexaene-16,20-dione;
- 25 (8S,11S,18S)-18-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-25-oxa-2 λ^6 -thia-1,12,15,19-tetrazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-3(31),4,6,21(29),22,24(28)-hexaene-16,20-dione;
- 30

(8R,11S,18S)-18-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-25-oxa-2 λ^6 -thia-1,12,15,19-tetrazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-3(31),4,6,21(29),22,24(28)-hexaene-16,20-dione;

5 (8S,11R,18S)-18-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-25-oxa-2 λ^6 -thia-1,12,15,19-tetrazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-3(31),4,6,21(29),22,24(28)-hexaene-16,20-dione;

(8R,11R,18S)-18-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-25-oxa-2 λ^6 -thia-1,12,15,19-tetrazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-3(31),4,6,21(29),22,24(28)-hexaene-16,20-dione;

10 (17S)-17-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-24-oxa-2 λ^6 -thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.023,27]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-15,19-dione;

15 (16S)-16-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-23-oxa-2 λ^6 -thia-1,10,13,17-tetrazatetracyclo[17.6.2.13,7.022,26]octacosa-3(28),4,6,19(27),20,22(26)-hexaene-14,18-dione;

(17S)-17-[2,5-bis(trifluoromethyl)phenyl]-2,2-dioxo-24-oxa-2 λ^6 -thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.023,27]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-15,19-dione;

20 (17S)-17-[2-chloro-5-(trifluoromethyl)phenyl]-10-methyl-2,2-dioxo-24-oxa-2 λ^6 -thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.023,27]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-15,19-dione;

(17S,25R)-17-[2-chloro-5-(trifluoromethyl)phenyl]-22-fluoro-25-methyl-2,2-dioxo-24-oxa-2 λ^6 -thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.023,27]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-15,19-dione;

25 (17S)-17-[2-methyl-5-(trifluoromethyl)phenyl]-2,2-dioxo-24-oxa-2 λ^6 -thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.023,27]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-15,19-dione;

30 (17S)-17-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-24-oxa-2 λ^6 -thia-1,10,14,18,29-pentazatetracyclo[18.6.2.13,7.023,27]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-15,19-dione;

(17S)-17-[2-chloro-5-(trifluoromethyl)phenyl]-21-methoxy-2,2-dioxo-24-oxa-2 λ^6 -thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.023,27]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-15,19-dione;

5 (17S)-17-[2-chloro-5-(trifluoromethyl)phenyl]-21-fluoro-2,2-dioxo-24-oxa-2 λ^6 -thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.023,27]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-15,19-dione;

(17S)-17-(2,5-dichlorophenyl)-2,2-dioxo-24-oxa-2 λ^6 -thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.023,27]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-15,19-dione;

10 (17S)-17-[2-chloro-5-(trifluoromethyl)phenyl]-2'-fluoro-2,2-dioxo-spiro[24-oxa-2 λ^6 -thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.023,27]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-25,1'-cyclopropane]-15,19-dione;

15 (17S)-17-[2-chloro-5-(trifluoromethyl)phenyl]-2'-fluoro-2,2-dioxo-spiro[24-oxa-2 λ^6 -thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.023,27]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-25,1'-cyclopropane]-15,19-dione;

(17S,25S)-17-[2-chloro-5-(trifluoromethyl)phenyl]-25-(hydroxymethyl)-2,2-dioxo-24-oxa-2 λ^6 -thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.023,27]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-15,19-dione;

20 (17S,25S)-17-[2-chloro-5-(trifluoromethyl)phenyl]-25-(fluoromethyl)-2,2-dioxo-24-oxa-2 λ^6 -thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.023,27]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-15,19-dione;

(17S)-17-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-10,24-dioxa-2 λ^6 -thia-1,14,18-triazatetracyclo[18.6.2.13,7.023,27]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-15,19-dione;

25 (18S)-18-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-9,25-dioxa-2 λ^6 -thia-1,12,15,19-tetrazatetracyclo[19.6.2.03,8.024,28]nonacosa-3(8),4,6,21(29),22,24(28)-hexaene-16,20-dione;

30 (16S)-16-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-23-oxa-2 λ^6 -thia-1,9,13,17-tetrazatetracyclo[17.6.2.13,7.022,26]octacosa-3(28),4,6,19(27),20,22(26)-hexaene-14,18-dione.

In a particular embodiment, the present invention provides a compound of formula (I) as described herein, or a pharmaceutically acceptable salt thereof, wherein said compound of formula (I) is:

(17S)-17-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-24-oxa-2 λ^6 -thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.023,27]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-15,19-dione;

(17S,25R)-17-[2-chloro-5-(trifluoromethyl)phenyl]-22-fluoro-25-methyl-2,2-dioxo-24-oxa-2 λ^6 -thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.023,27]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-15,19-dione;

(17S)-17-[2-methyl-5-(trifluoromethyl)phenyl]-2,2-dioxo-24-oxa-2 λ^6 -thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.023,27]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-15,19-dione;

(17S)-17-[2-chloro-5-(trifluoromethyl)phenyl]-21-fluoro-2,2-dioxo-24-oxa-2 λ^6 -thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.023,27]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-15,19-dione;

(17S,25S)-17-[2-chloro-5-(trifluoromethyl)phenyl]-25-(fluoromethyl)-2,2-dioxo-24-oxa-2 λ^6 -thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.023,27]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-15,19-dione;

In a particular embodiment, the present invention provides pharmaceutically acceptable salts of the compounds according to formula (I) as described herein. In a further particular embodiment, the present invention provides compounds according to formula (I) as described herein.

In some embodiments, the compounds of formula (I) are isotopically-labeled by having one or more atoms therein replaced by an atom having a different atomic mass or mass number. Such isotopically-labeled (i.e., radiolabeled) compounds of formula (I) are considered to be within the scope of this disclosure. Examples of isotopes that can be incorporated into the compounds of formula (I) include isotopes of hydrogen, carbon, nitrogen, oxygen, phosphorous, sulfur, fluorine, chlorine, and iodine, such as, but not limited to, ^2H , ^3H , ^{11}C , ^{13}C , ^{14}C , ^{13}N , ^{15}N , ^{15}O , ^{17}O , ^{18}O , ^{31}P , ^{32}P , ^{35}S , ^{18}F , ^{36}Cl , ^{123}I , and ^{125}I , respectively. Certain isotopically-labeled compounds of formula (I), for example, those incorporating a radioactive isotope, are useful in drug and/or substrate tissue distribution studies. The radioactive isotopes tritium, i.e. ^3H , and carbon-14, i.e., ^{14}C , are particularly useful for this purpose in view of their ease of incorporation and ready

means of detection. For example, a compound of formula (I) can be enriched with 1, 2, 5, 10, 25, 50, 75, 90, 95, or 99 percent of a given isotope.

Substitution with heavier isotopes such as deuterium, i.e. ^2H , may afford certain therapeutic advantages resulting from greater metabolic stability, for example, increased *in vivo* half-life or reduced dosage requirements. Thus, the present invention encompasses compounds of formula (I) wherein one or more hydrogen atoms have been replaced by deuterium.

Substitution with positron emitting isotopes, such as ^{11}C , ^{18}F , ^{15}O and ^{13}N , can be useful in Positron Emission Topography (PET) studies for examining substrate receptor occupancy. Isotopically-labeled compounds of formula (I) can generally be prepared by conventional techniques known to those skilled in the art or by processes analogous to those described in the Examples as set out below using an appropriate isotopically-labeled reagent in place of the non-labeled reagent previously employed.

Using the Compounds of the Invention

In one aspect, the present invention provides a compound of formula (I) as described herein, or a pharmaceutically acceptable salt thereof, for use as therapeutically active substance.

In a further aspect, the present invention provides the use of a compound of formula (I) as described herein, or a pharmaceutically acceptable salt thereof, for the treatment and/or prophylaxis of disease or disorder mediated by MET kinase.

In a further aspect, the present invention provides the use of a compound of formula (I) as described herein, or a pharmaceutically acceptable salt thereof, for the preparation of a medicament for the treatment and/or prophylaxis of disease or disorder mediated by MET kinase.

In a further aspect, the present invention provides a compound of formula (I) as described herein, or a pharmaceutically acceptable salt thereof, for use in the treatment and/or prophylaxis of disease or disorder mediated by MET kinase.

In a further aspect, the present invention provides a method for the treatment and/or prophylaxis of disease or disorder mediated by MET kinase, which method comprises administering a therapeutically effective amount of a compound of formula (I) as described herein to a human.

In some embodiments, said disease or disorder that is mediated by MET kinase is cancer.

In some embodiments, said cancer is lung cancer.

In some embodiments, said lung cancer is non-small-cell lung cancer (NSCLC).

Pharmaceutical Compositions and Administration

In one aspect, the present invention provides a pharmaceutical composition comprising a
5 compound of formula (I), or a pharmaceutically acceptable salt thereof, as described herein and a pharmaceutically acceptable excipient.

In one embodiment, said pharmaceutical composition further comprises an additional therapeutic agent.

The compounds of formula (I) and their pharmaceutically acceptable salts and esters can be used
10 as medicaments (e.g. in the form of pharmaceutical preparations). The pharmaceutical preparations can be administered internally, such as orally (e.g. in the form of tablets, coated tablets, dragées, hard and soft gelatin capsules, solutions, emulsions or suspensions), nasally (e.g. in the form of nasal sprays) or rectally (e.g. in the form of suppositories). However, the administration can also be effected parentally, such as intramuscularly or intravenously (e.g. in
15 the form of injection solutions).

The compounds of formula (I) and their pharmaceutically acceptable salts and esters can be processed with pharmaceutically inert, inorganic or organic adjuvants for the production of tablets, coated tablets, dragées and hard gelatin capsules. Lactose, corn starch or derivatives thereof, talc, stearic acid or its salts etc. can be used, for example, as such adjuvants for tablets,
20 dragées and hard gelatin capsules.

Suitable adjuvants for soft gelatin capsules are, for example, vegetable oils, waxes, fats, semi-solid substances and liquid polyols, etc.

Suitable adjuvants for the production of solutions and syrups are, for example, water, polyols, saccharose, invert sugar, glucose, etc.

25 Suitable adjuvants for injection solutions are, for example, water, alcohols, polyols, glycerol, vegetable oils, etc.

Suitable adjuvants for suppositories are, for example, natural or hardened oils, waxes, fats, semi-solid or liquid polyols, etc.

Moreover, the pharmaceutical preparations can contain preservatives, solubilizers, viscosity-increasing substances, stabilizers, wetting agents, emulsifiers, sweeteners, colorants, flavorants, salts for varying the osmotic pressure, buffers, masking agents or antioxidants. They can also contain still other therapeutically valuable substances.

- 5 The dosage can vary in wide limits and will, of course, be fitted to the individual requirements in each particular case. In general, in the case of oral administration a daily dosage of about 0.1 mg to 20 mg per kg body weight, particularly about 0.5 mg to 4 mg per kg body weight (e.g. about 300 mg per person), divided into particularly 1-3 individual doses, which can consist, for example, of the same amounts, should be appropriate. It will, however, be clear that the upper
10 limit given herein can be exceeded when this is shown to be indicated.

Tablet Formulation (Wet Granulation)

<u>Item</u>	<u>Ingredients</u>	<u>mg/tablet</u>			
1.	Compound of formula (I)	5	25	100	500
2.	Lactose Anhydrous DTG	125	105	30	150
15 3.	Sta-Rx 1500	6	6	6	30
4.	Microcrystalline Cellulose	30	30	30	150
5.	Magnesium Stearate	1	1	1	1
	Total	<u>167</u>	<u>167</u>	<u>167</u>	<u>831</u>

20 Manufacturing Procedure

1. Mix items 1, 2, 3 and 4 and granulate with purified water.
2. Dry the granules at 50°C.
3. Pass the granules through suitable milling equipment.
4. Add item 5 and mix for three minutes; compress on a suitable press.

25 Capsule Formulation

<u>Item</u>	<u>Ingredients</u>	<u>mg/capsule</u>			
1.	Compound of formula (I)	5	25	100	500
2.	Hydrous Lactose	159	123	148	---
3.	Corn Starch	25	35	40	70
30 4.	Talc	10	15	10	25

5.	Magnesium Stearate	1	2	2	5
	Total	<u>200</u>	<u>200</u>	<u>300</u>	<u>600</u>

Manufacturing Procedure

1. Mix items 1, 2 and 3 in a suitable mixer for 30 minutes.
- 5 2. Add items 4 and 5 and mix for 3 minutes.
3. Fill into a suitable capsule.

Processes of Manufacturing

Processes for the manufacture of compounds of formula (I), or pharmaceutically acceptable salt thereof, as described herein are also an object of the invention.

- 10 The preparation of compounds of formula (I) of the present invention may be carried out in sequential or convergent synthetic routes. Syntheses of the invention are shown in the following general schemes. The skills required for carrying out the reaction and purification of the resulting products are known to those persons skilled in the art. The substituents and indices used in the following description of the processes have the significance given herein, unless indicated to the
- 15 contrary.

If one of the starting materials, intermediates or compounds of formula (I) contain one or more functional groups which are not stable or are reactive under the reaction conditions of one or more reaction steps, appropriate protective groups (as described e.g., in “Protective Groups in Organic Chemistry” by T. W. Greene and P. G. M. Wutts, 5th Ed., 2014, John Wiley & Sons,

20 N.Y.) can be introduced before the critical step applying methods well known in the art. Such protective groups can be removed at a later stage of the synthesis using standard methods described in the literature.

If starting materials or intermediates contain stereogenic centers, compounds of formula (I) can be obtained as mixtures of diastereomers or enantiomers, which can be separated by methods

25 well known in the art e.g., chiral HPLC, chiral SFC or chiral crystallization. Racemic compounds can e.g., be separated into their antipodes via diastereomeric salts by crystallization with optically pure acids or by separation of the antipodes by specific chromatographic methods using either a chiral adsorbent or a chiral eluent. It is equally possible to separate starting materials and intermediates containing stereogenic centers to afford diastereomerically/enantiomerically

30 enriched starting materials and intermediates. Using such diastereomerically/enantiomerically

enriched starting materials and intermediates in the synthesis of compounds of formula (I) will typically lead to the respective diastereomerically/enantiomerically enriched compounds of formula (I).

The compounds described herein, including compounds of general Formula (I), can be readily prepared according to the following reaction schemes and Examples, or modifications thereof, using readily available starting materials, reagents, and conventional synthesis procedures. Many of the reactions can also be carried out under microwave conditions or using conventional heating or utilizing other technologies such as solid phase reagents/scavengers or flow chemistry. In these reactions, it is also possible to make use of variants which are themselves known to those skilled in the art but are not mentioned in greater detail. For Example, where specific acids, bases, reagents, coupling agents, solvents, etc. are mentioned, it is understood that other suitable acids, bases, reagents, coupling agents, solvents etc. may be used and are included within the scope of the present invention. Furthermore, other methods for preparing compounds of the invention will be readily apparent to a person of ordinary skill in the art in light of the following reaction schemes and Examples. In cases where synthetic intermediates and final products contain potentially reactive functional groups, for Example Amino, hydroxy, thiol and carboxylic acid groups that may interfere with the desired reaction, it may be advantageous to employ protected forms of the intermediate. Methods for the selection, introduction, and subsequent removal of protecting groups are well known to those skilled in the art. The compounds obtained by using the general reaction sequences may be of insufficient purity. The compounds can be purified by using any of the methods of purification of organic compounds, for Example, crystallization or silica gel, alumina or C18 column chromatography, using different solvents in suitable ratios. All possible stereoisomers are envisioned within the scope of the invention. In the discussion below variables have the meaning indicated above unless otherwise indicated.

A person skilled in the art will acknowledge that in the synthesis of compounds of formula (I) - insofar not desired otherwise - an “orthogonal protection group strategy” will be applied, allowing the cleavage of several protective groups one at a time each without affecting other protective groups in the molecule. The principle of orthogonal protection is well known in the art and has also been described in literature (e.g. Barany and R. B. Merrifield, *J. Am. Chem. Soc.* **1977**, 99, 7363; H. Waldmann et al., *Angew. Chem. Int. Ed. Engl.* **1996**, 35, 2056).

A person skilled in the art will acknowledge that the sequence of reactions may be varied depending on reactivity and nature of the intermediates.

Unless otherwise defined, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this invention belongs.

5 Although methods and materials similar or equivalent to those described herein can be used in the practice or testing of the invention, suitable methods and materials are described below.

All publications, patent applications, patents, and other references mentioned herein are incorporated by reference in their entirety.

10 The nomenclature used in this application is based on IUPAC systematic nomenclature, unless indicated otherwise. Any open valency appearing on a carbon, oxygen or nitrogen atom in the structures herein indicates the presence of a hydrogen atom.

In more detail, the compounds of formula (I) can be manufactured by the methods given below, by the methods given in the examples or by analogous methods. Appropriate reaction conditions for the individual reaction steps are known to a person skilled in the art. Also, for reaction
15 conditions described in literature affecting the described reactions see for example:

Comprehensive Organic Transformations: A Guide to Functional Group Preparations, 2nd Edition, Richard C. Larock. John Wiley & Sons, New York, NY. 1999). It was found convenient to carry out the reactions in the presence or absence of a solvent. There is no particular restriction on the nature of the solvent to be employed, provided that it has no adverse effect on
20 the reaction or the reagents involved and that it can dissolve the reagents, at least to some extent. The described reactions can take place over a wide range of temperatures, and the precise reaction temperature is not critical to the invention. It is convenient to carry out the described reactions in a temperature range between -78 °C to reflux. The time required for the reaction may also vary widely, depending on many factors, notably the reaction temperature and the nature of
25 the reagents. However, a period of from 0.5 hours to several days will usually suffice to yield the described intermediates and compounds. The reaction sequence is not limited to the one displayed in the schemes, however, depending on the starting materials and their respective reactivity, the sequence of reaction steps can be freely altered.

If starting materials or intermediates are not commercially available or their synthesis not
30 described in literature, they can be prepared in analogy to existing procedures for close analogues or as outlined in the experimental section.

All possible stereoisomers are envisioned within the scope of the invention. In the discussion below variables have the meaning indicated above unless otherwise indicated. All final compounds have been characterized using for example LC-MS, NMR and/or Specific Optical Rotation.

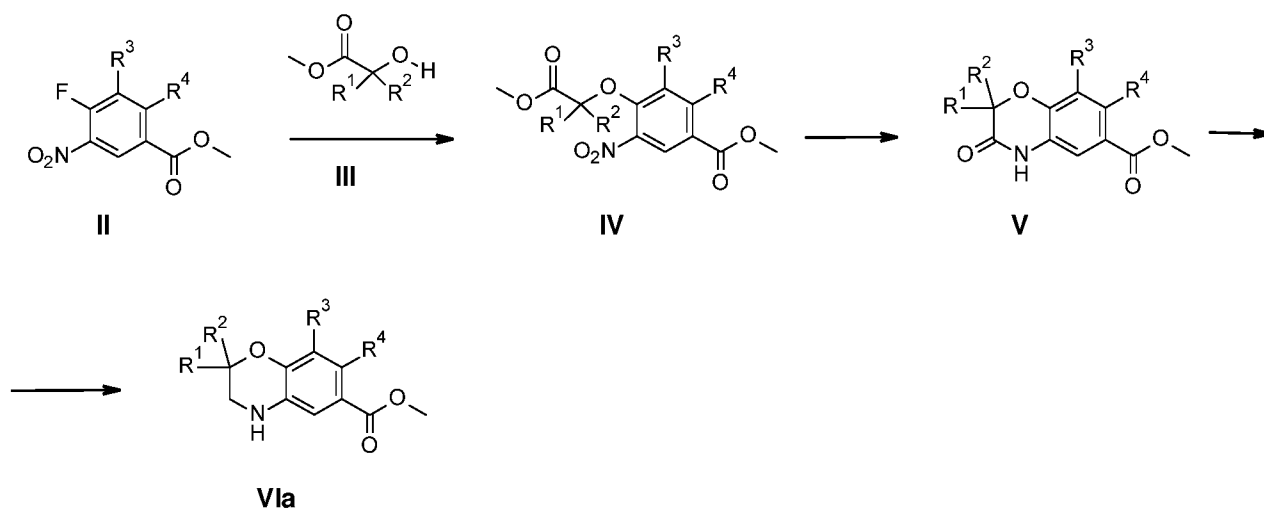
- 5 As depicted in the Examples below, in certain exemplary embodiments, compounds are prepared according to the following general procedures. It will be appreciated that, although the general methods depict the synthesis of certain compounds of the invention, the following general methods, and other methods known to one skilled in the art, can be applied to all compounds and subclasses and species of each of these compounds, as
10 described herein.

INTERMEDIATES

SYNTHESIS OF INTERMEDIATES A

- Intermediates A of general formula VI in which R₃, R₄ are described herein can be prepared by
15 methods well known in the art and as exemplified by the general synthetic routes outlined in schemes A to D.

Scheme A

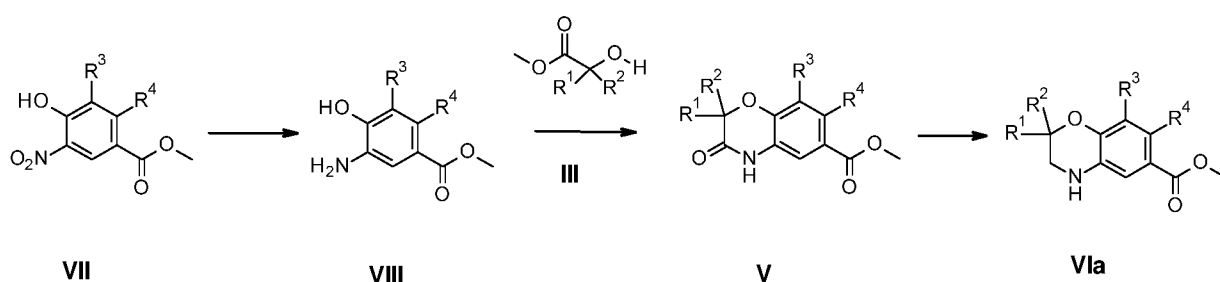


- 20 Intermediate II, either commercially available or prepared by methods known in the art, can react first with intermediates III using a suitable base such as sodium hydride or NaHMDS in a dipolar aprotic solvent such as e.g. N,N-dimethylformamide to form compounds IV by nucleophilic substitution.

Intermediates **IV** can be reduced using a metal such as iron or zinc in acetic acid to form in one step the cyclised intermediates **V** which gives compound **VI** after reduction of the amide bond using a borane complex e.g. $\text{BH}_3\cdot\text{THF}$ or $\text{BH}_3\cdot\text{DMS}$ in a solvent, commonly THF at temperatures from 0°C to room temperature (see Brown, H. C.; Krishnamurthy, S. "Selective Reductions. 20. Borane-Methyl Sulfide: A Highly Convenient, Stable, and Selective Reagent for the Reduction of Representative Organic Compounds." *J. Am. Chem. Soc.* 1972, 94, 7159-7161.)

Alternatively, intermediates **V**, can be obtained as described in [scheme B](#).

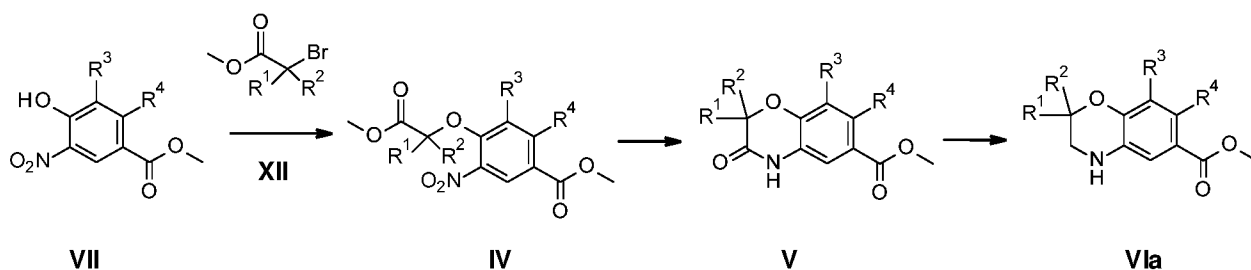
Scheme B



In this method, intermediates **VII** are first reduced with a metal such as iron or zinc in acetic acid using methods well known in the art and react after with intermediates **III** via a Mitsunobu reaction using an appropriate phosphine such as triphenylphosphine and a dialkyl azodicarboxylate such as DEAD or DIAD in a suitable solvent such as THF to give after concomitant cyclisation intermediates **V**. Mitsunobu reactions of that type are broadly described in literature (*e.g. Org. Chem. Front.* **2015**, 2, 739; *Chem. Rev.* **2009**, 109 (6), 2551).

Additionally, intermediates **IV**, can be obtained as described in [scheme C](#).

Scheme C

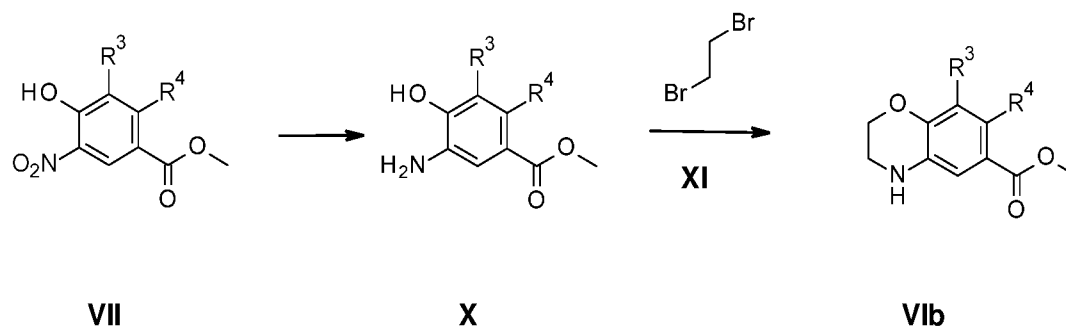


Alkylation of intermediates **VII** with intermediates **XII** in presence of a mineral base such as potassium carbonate or cesium carbonate in a solvent such as N,N-dimethylformamide or

acetonitrile at temperatures between 0°C and the boiling temperature of the solvent yields intermediates **IV**.

For the Intermediates **VIb**, the procedure is described in scheme D

Scheme D

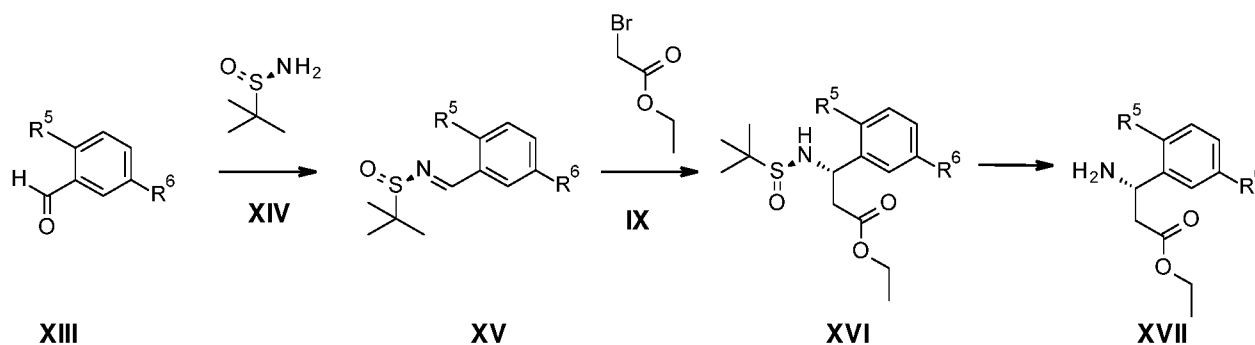


Intermediates **VII** react with dibromoethane **XI** in the presence of a mineral base such as potassium carbonate or cesium carbonate in a solvent such as N,N-dimethylformamide or acetonitrile at temperatures between 0°C and the boiling temperature of the solvent to give after concomitant cyclisation the compounds **VIb**.

SYNTHESIS OF INTERMEDIATES B

Intermediates **B** of general formula **XVII** in which R⁵ and R⁶ are described herein can be prepared by methods well known in the art and as exemplified by the general synthetic routes outlined in scheme E.

Scheme E



Chiral Intermediates **XVII** can be synthesized using a chiral auxiliary such as reagent **XIV** to form first the imines **XV** following the procedure described by Tang, T. P.; Ellman, J. A. The tert-

Butanesulfinyl Group: An Ideal Chiral Directing Group and Boc-Surrogate for the Asymmetric Synthesis and Applications of β -Amino Acids. J. Org. Chem. 1999, 64, 12–13.

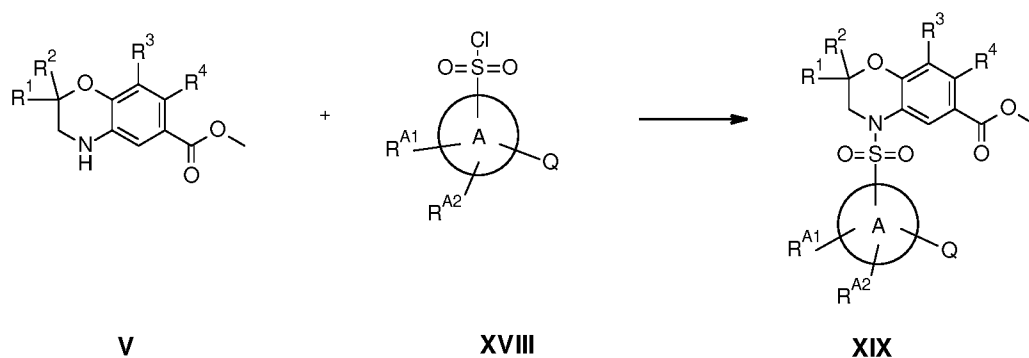
To construct the carbon-carbon bond, intermediates **XV** are subjected to a Reformatsky reaction using intermediate **IX** as alkylating reagent in presence of Zn. Reformatsky reactions of that type
 5 are broadly described in literature and in particular in Journal of Medicinal Chemistry, 2013, vol. 56, # 10, p. 3980 - 3995.

Removal of the auxiliary to obtained beta-amino esters **XVII** is done by acidic treatment.

SYNTHESIS OF INTERMEDIATES C

Intermediates **C** of general formula **XIX** in which A, R^1 , R^{A1} , R^{A2} , R^2 , R^3 , R^4 are described herein
 10 and Q is either an heteroatom or a functional group can be prepared by methods well known in the art and as exemplified by the general synthetic routes outlined in scheme F.

Scheme F

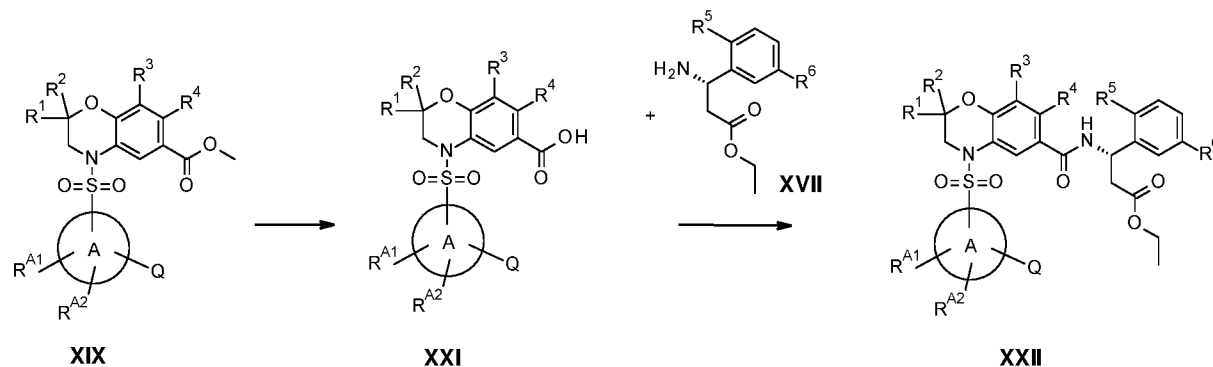


Intermediates **XIX** are synthesized by reaction of intermediates **V** with intermediates **XVIII**, either
 15 commercially available or prepared by methods known in the art, in presence of a base such as triethylamine or pyridine in a solvent such as dichloromethane or without the use of any additional solvent.

SYNTHESIS OF INTERMEDIATES D

Intermediates **D** of general formula **XXII** in which A, R^1 , R^{A1} , R^{A2} , R^2 , R^3 , R^4 , R^5 , R^6 are described
 20 herein Q is either an heteroatom or a functional group can be prepared by methods well known in the art and as exemplified by the general synthetic routes outlined in scheme G.

Scheme G



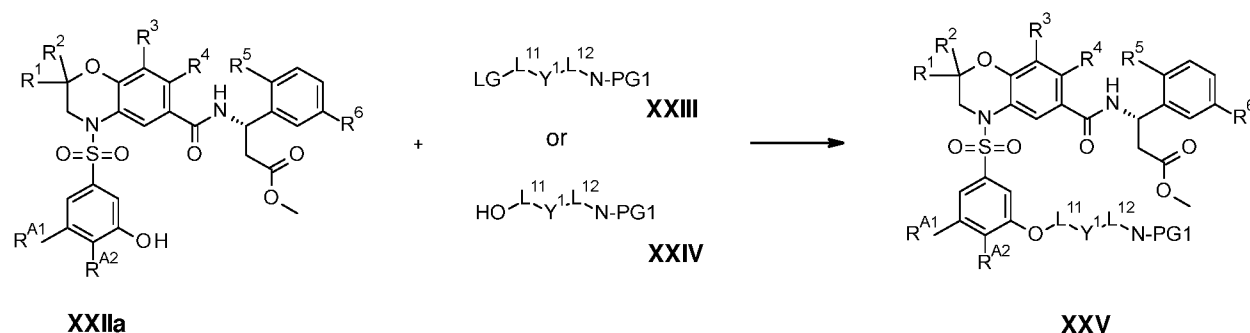
Saponification of the intermediates **XIX** by applying methods known in the art using NaOH or LiOH in a solvent or a mixture of solvents like THF, ethanol or water at temperatures between 0°C and room temperature gives the corresponding carboxylic acids **XXI**.

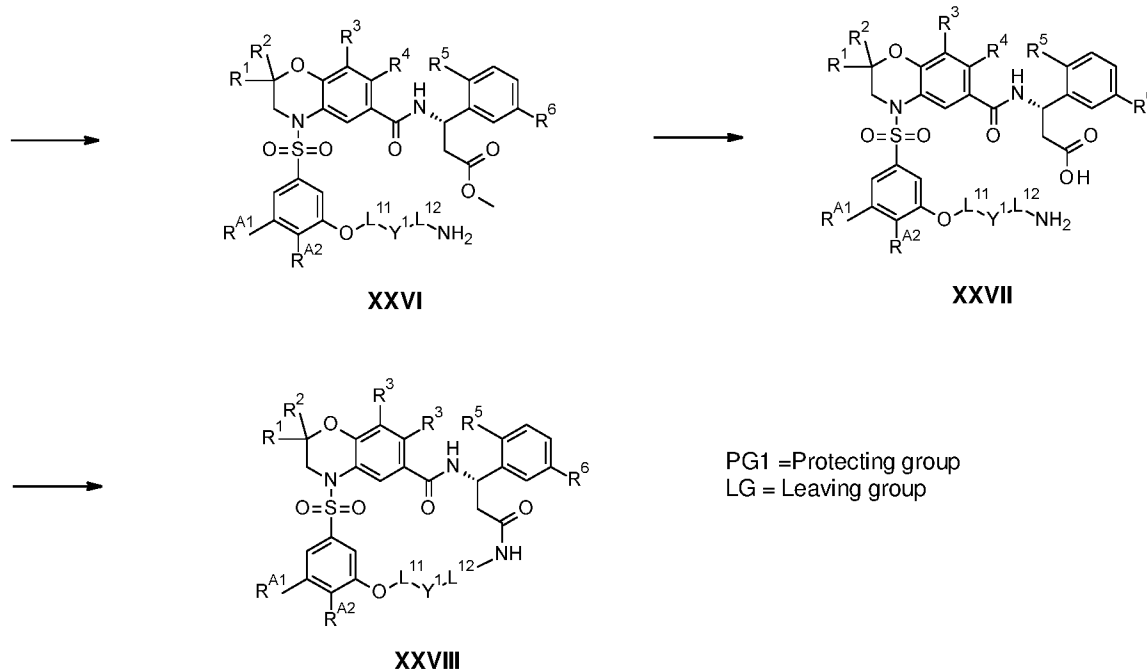
- 5 Next, the carboxylic acids can be activated and react with intermediates **XVII** to provide compounds **XXII**. Amide couplings of this type with carboxylic acids are widely described in the literature and can be accomplished by the usage of coupling reagents such as CDI, DCC, HATU, HBTU, HOBT, TBTU, T3P or combination TCFH-NMI (Org. Lett. 2018, 20, 4218-4222) in a suitable solvent e.g., DMF, DMA, DCM or dioxane, optionally in the presence of a base (e.g.,
- 10 TEA, DIPEA (Huenig's base) or DMAP).

SYNTHESIS OF THE MACROCYCLES

- In some embodiments, intermediates **I** are intermediates of type **XXVIII**. Intermediates **XXVIII** in which R^1 , R^{A1} , R^{A2} , R^2 , R^3 , R^4 , R^5 , R^6 , L^{11} , L^{12} , Y^1 are as described herein and Q is hydroxy can be prepared by methods well known in the art and as exemplified by the general synthetic
- 15 procedures outlined in Scheme H.

Scheme H





Alcohols of type **XXIIa** can be subjected to a Mitsunobu reaction with intermediates **XXIV**, either commercially available or prepared by methods known in the art, in which PG1 is a suitable protective group such as a Cbz, Boc, or Pht, using an appropriate phosphine such as triphenylphosphine and a dialkyl azodicarboxylate such as DEAD or DIAD in a suitable solvent such as THF to give intermediates **XXV**. Mitsunobu reactions of that type are broadly described in literature (*e.g. Org. Chem. Front.* **2015**, 2, 739; *Chem. Rev.* **2009**, 109 (6), 2551).

Alternatively, intermediates **XXV** can be obtained by alkylation using intermediates **XXIII** in which PG1 is a suitable protective group such as a Cbz, Boc, or Pht and LG is a leaving group such as bromine, chlorine or mesylate in presence of a base like potassium carbonate or cesium carbonate in an appropriate solvent such as DMF or ACN at temperatures between 0°C and the boiling temperature of the solvent.

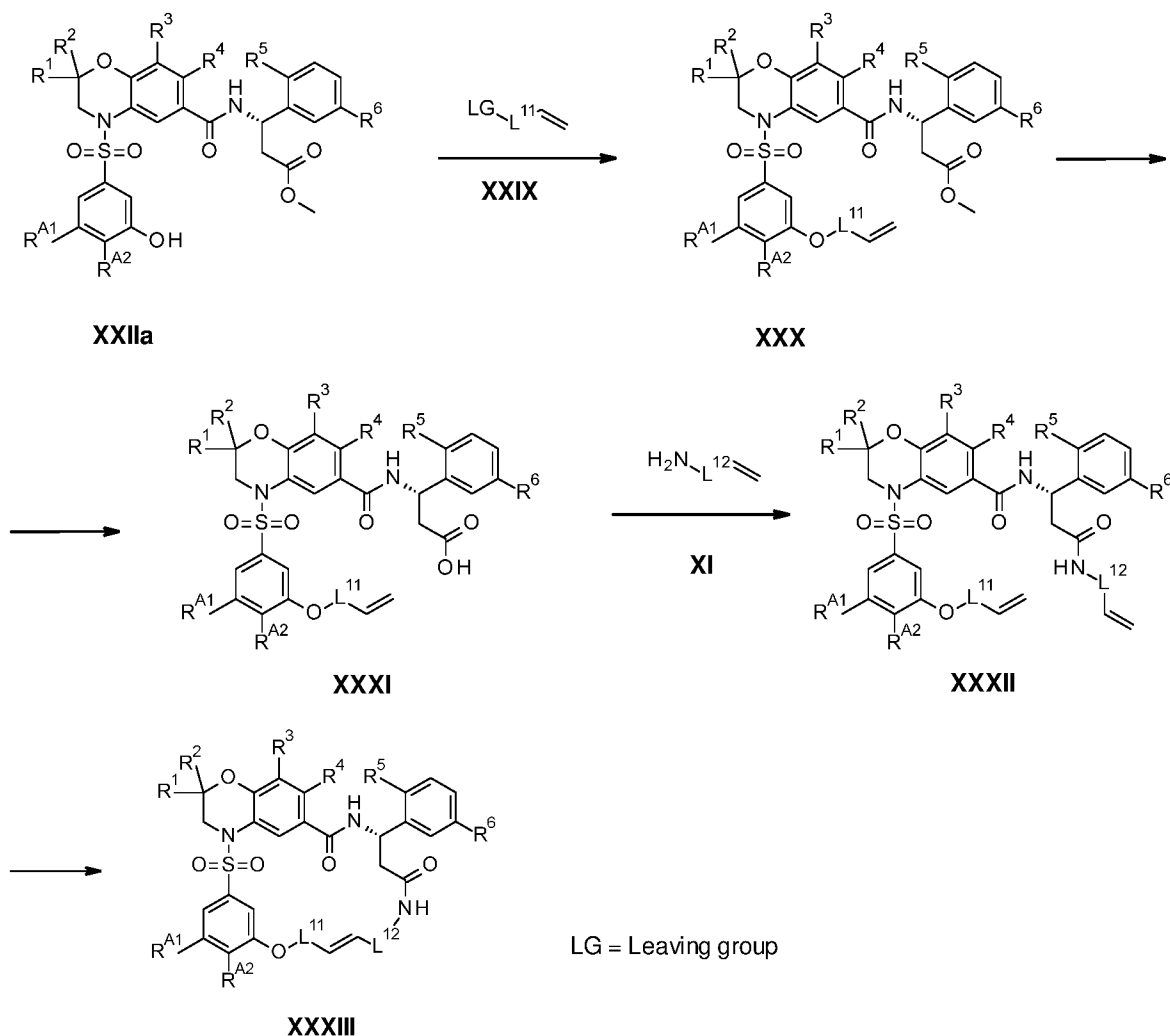
Next, removal of the protective group from intermediates **XXV** in which PG1 signifies a protective group such as tert-butyloxycarbonyl (BOC) or phthalimide (Pht) can be performed using methods well known in the art and as described (see “Protective Groups in Organic Chemistry” by T. W. Greene and P. G. M. Wutts, 5th Ed., 2014, John Wiley & Sons, New York), *e.g.*, using TFA or hydrochloric acid in a solvent like dichloromethane, 1,4-dioxane or THF preferable at room temperature. In the case of using phthalimide as PG1 group, the removal can be done under either acidic or basic conditions or using hydrazine in ethanol or THF.

Next, saponification of intermediates **XXVI** with a suitable base such as NaOH or LiOH in a solvent or a mixture of solvents like THF, ethanol or water gives the intermediates **XXVII**.

Accordingly, intermediates **XXVII** can be cyclized by activating the carboxylic acid group to provide compounds **XXVIII**. Amide couplings of this type with carboxylic acids are widely described in the literature and can be accomplished by the usage of coupling reagents such as CDI, DCC, HATU, HBTU, HOBT, TBTU, T3P or combination like TCFH-NMI (Org. Lett. 2018, 20, 4218-4222) in a suitable solvent e.g., DMF, DMA, DCM or dioxane, optionally in the presence of a base (e.g., TEA, DIPEA (Huenig's base) or DMAP).

In some embodiments, intermediates **I** are intermediates of type **XXXIII**. Intermediates **XXXIII** in which R^1 , R^{A1} , R^{A2} , R^2 , R^3 , R^4 , R^5 , R^6 , L^{11} , L^{12} are as described herein and Q is oxygen can be prepared by a methods well known in the art and as exemplified by the general synthetic procedures outlined in Scheme I

Scheme I



Intermediates **XXX** can be obtained by alkylation of intermediates **XXIIa** using intermediates **XXIX** in which LG is a leaving group such as bromine, chlorine or mesylate in presence of a base like potassium carbonate or cesium carbonate in an appropriate solvent such as DMF or ACN at temperatures between 0°C and the boiling temperature of the solvent.

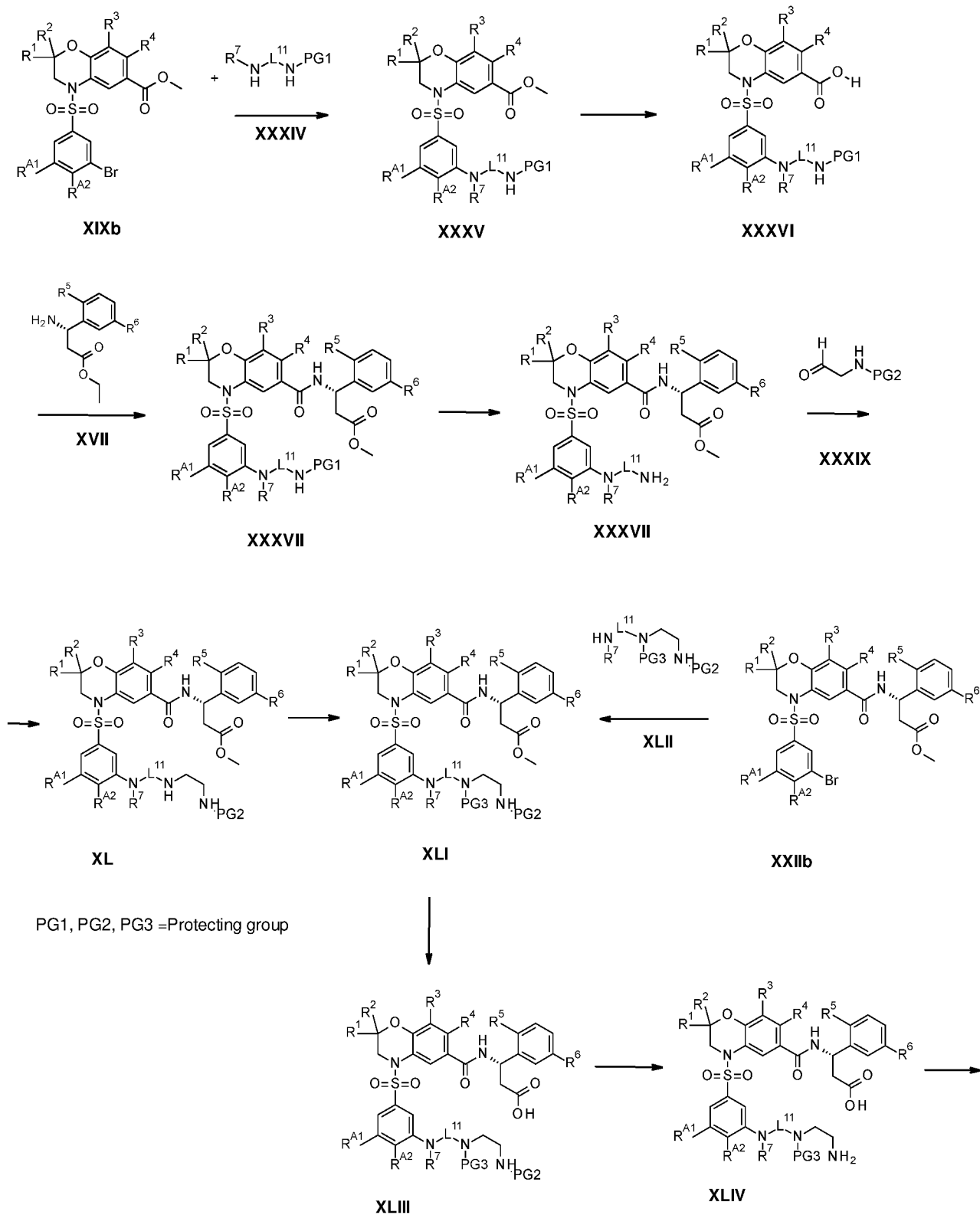
- 5 Saponification of the intermediates just formed by applying methods known in the art using NaOH or LiOH in a solvent or a mixture of solvents like THF, ethanol or water at temperatures between 0°C and room temperature gives the corresponding carboxylic acids **XXXI**.

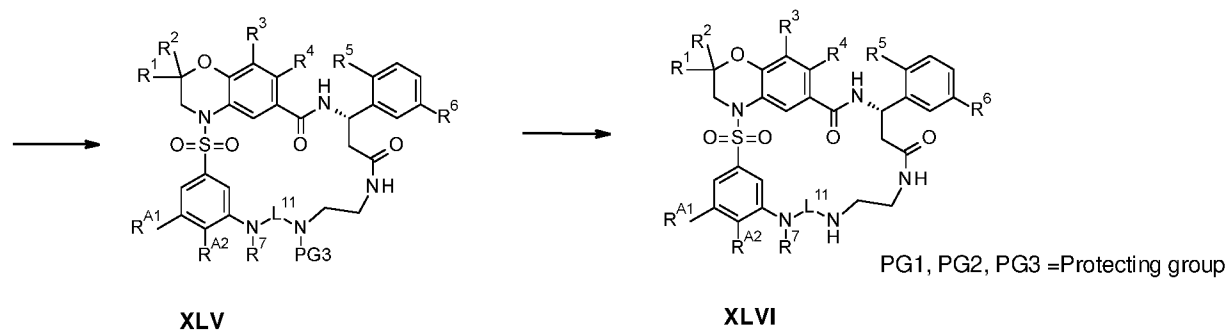
Next, the carboxylic acids can be activated and react with intermediates **XI** to provide intermediates **XXXII**. Amide couplings of this type with carboxylic acids are widely described in
10 the literature and can be accomplished by the usage of coupling reagents such as CDI, DCC, HATU, HBTU, HOBT, TBTU, T3P or combination TCFH-NMI (Org. Lett. 2018, 20, 4218-4222) in a suitable solvent e.g., DMF, DMA, DCM or dioxane, optionally in the presence of a base (e.g., TEA, DIPEA (Huenig's base) or DMAP).

Compounds **XXXIII** can be obtained by ring-closure metathesis using the appropriate catalyst,
15 particularly Grubbs II, and using methods well known in the art as described in R.H. Grubbs, S. Miller, G.C. Fu Acc. Chem. Res., 28 (1995), 446-452

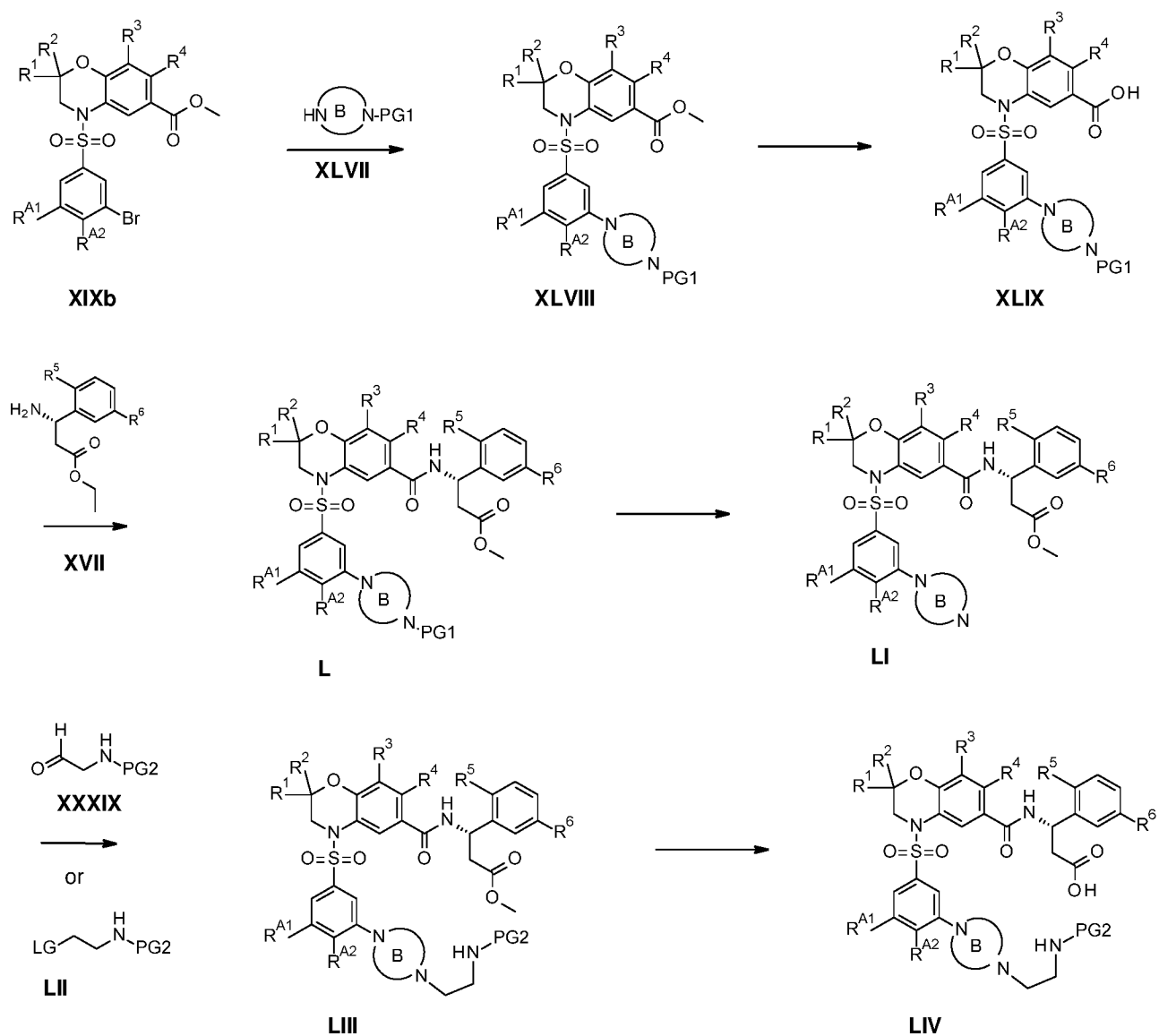
In some embodiments, intermediates **I** are intermediates of type **XLVI**, **LVI** and **LXVIII**. .
Intermediates **XLVI**, **LVI** and **LXVIII** in which B, R¹, R^{A1}, R^{A2}, R², R^B, R³, R⁴, R⁵, R⁶, R⁷ L¹¹,
L¹² are as described herein and Q is bromine can be prepared by methods well known in the art
20 and as exemplified by the general synthetic procedures outlined in Scheme J to Scheme L.

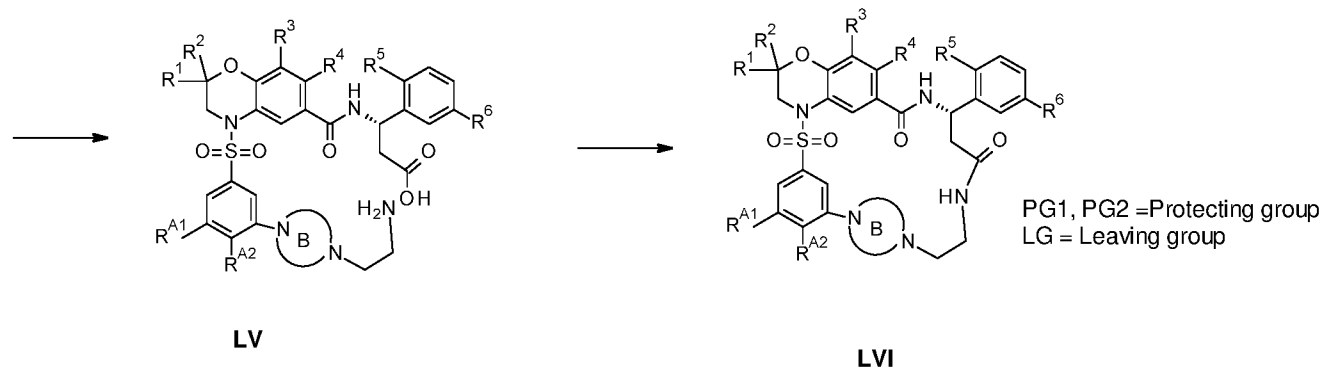
Scheme J



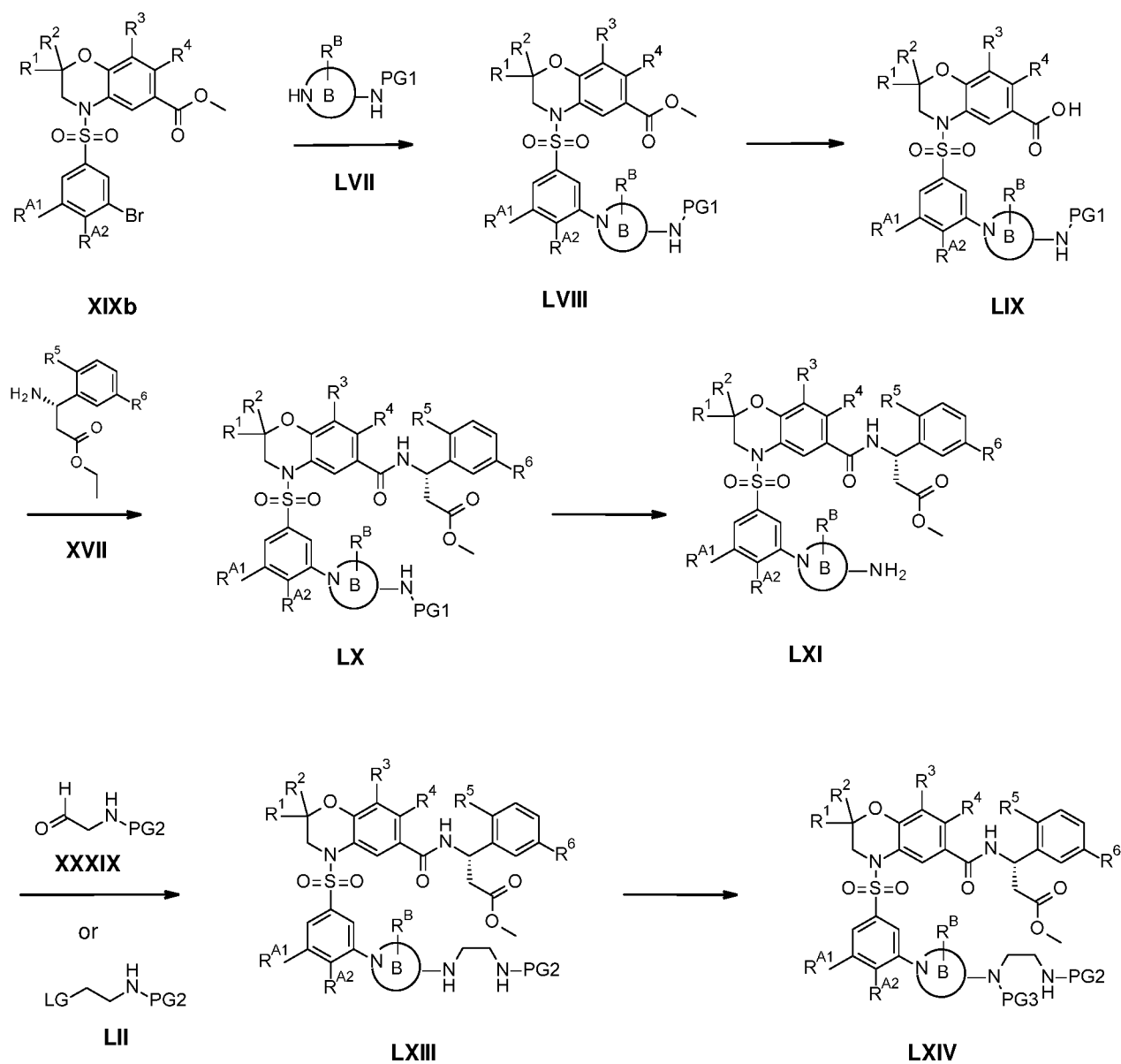


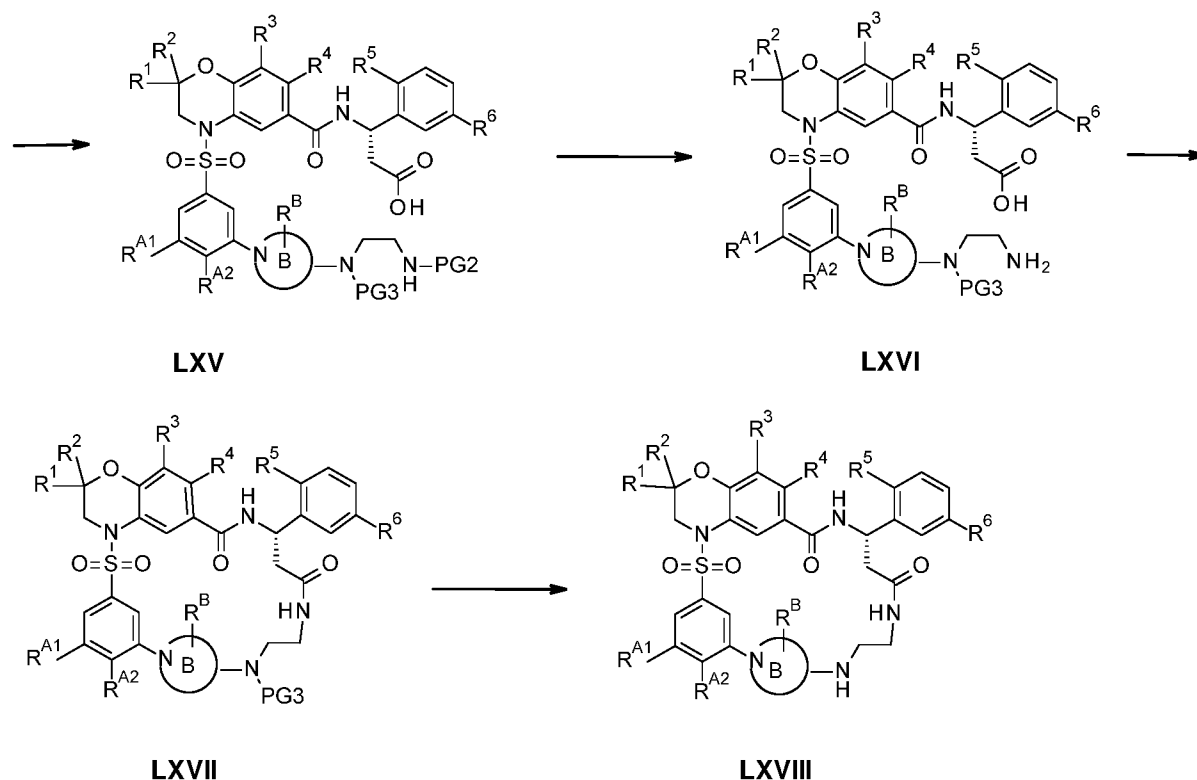
Scheme K





Scheme L





PG1, PG2, PG3 =Protecting group
LG = Leaving group

Intermediates **XIXb** can be reacted with intermediates **XXXIV** (Scheme J), **XLVII** (Scheme K), **LVII** (Scheme L) using a suitable catalyst and solvent such as, e.g. tris(dibenzylideneacetone)dipalladium (0) chloroform adduct and X-Phos or bis(diphenylphosphino)-1,1'-binaphtalene and palladium(II) acetate in tert-butanol or 1,4-dioxane and a suitable base such as Cs₂CO₃ or KOtBu at temperatures between room temperature and the boiling point of the solvent or solvent mixture to provide intermediates **XXXV** (Scheme J), **XLVIII** (Scheme K) and **LVIII** (Scheme L). Buchwald Hartwig reactions of that type are well known in the art and described in literature, e.g. *Angewandte Chemie*. **2019**, 58, 17118-17129.

Saponification of the intermediates just formed by applying methods known in the art using NaOH or LiOH in a solvent or a mixture of solvents like THF, ethanol or water at temperatures between 0°C and room temperature gives the corresponding carboxylic acids **XXXVI** (Scheme J), **XLIX** (Scheme K) and **LIX** (Scheme L).

Accordingly, the carboxylic acids can be activated and react with intermediates **XVII** to provide compounds **XXXVII** (Scheme J), **L** (Scheme K) and **LX** (Scheme L). Amide couplings of this type with carboxylic acids are widely described in the literature and can be accomplished by the usage of coupling reagents such as CDI, DCC, HATU, HBTU, HOBT, TBTU, T3P or combination

TCFH-NMI (Org. Lett. 2018, 20, 4218-4222) in a suitable solvent e.g., DMF, DMA, DCM or dioxane, optionally in the presence of a base (e.g., TEA, DIPEA (Huenig's base) or DMAP).

Removal of the protective group from the lastly prepared intermediates applying methods known in the art (e.g., a Boc group using TFA in DCM or 4M HCl in dioxane at temperatures between
5 0°C and room temperature, a Cbz group using hydrogen in the presence of a suitable catalyst such as Pd or Pd(OH)₂ on charcoal in a suitable solvent such as MeOH, EtOH, EtOAc or mixtures thereof and as described for example in "Protective Groups in Organic Chemistry" by T.W. Greene and P.G.M. Wuts, 4th Ed., 2006, Wiley N.Y.), furnishes intermediates **XXXVIII** (Scheme J), **LI** (Scheme K) and **LXI** (Scheme L).

- 10 Elongation of the chain for intermediates **XXXVIII**, **XLIX** and **LXI** may be achieved by reductive amination. Reductive aminations of carbonyl compounds are generally carried out by reacting an aldehyde or a ketone like intermediates **XXXIX** with suitable amines like intermediates **XXXVIII** (Scheme J), **LI** (Scheme K) and **LXI** (Scheme L) in the presence of a reducing agent e.g. hydrogen/Pd, NaBH₃CN or NaBH₄ in a suitable solvent e.g. methanol, ethanol or
15 dichloromethane (see, for example: "Reductive alkylation - Preparation of Amines" in Catalytic Hydrogenation over Platinum Metals, p. 291 -303 or Borch, Bernstein, J. Am. Chem. Soc, 1971, 93, 2897)

- Alternatively, intermediates **XLIX** and **LXI** can be alkylated with compound **LII** in which LG is a suitable leaving group such as chlorine, bromine, iodine, OSO₂alkyl (e.g. methanesulfonic acid),
20 OSO₂fluoroalkyl (e.g. trifluoromethanesulfonic acid) or OSO₂aryl (e.g. p-toluenesulfonic acid) using a suitable base such as K₂CO₃ or Cs₂CO₃ in an appropriate solvent such as DMF or ACN at temperatures between 0°C and the boiling temperature of the solvent.

- Intermediates **XL** (Scheme J) and **LXIII** (Scheme L) can be protected by applying procedures known in the art using e.g. BOC anhydride or CBz-Cl in the presence of a base (e.g., TEA, DIPEA
25 (Huenig's base) or DMAP) in solvent such as N,N-dimethylformamide or dichloromethane to get intermediates **XLI** (Scheme J) and **LXIV** (Scheme L).

Furthermore, intermediate **XLI** (scheme J) can be formed from **XXIIb** by Buchwald type reaction similar to step 1 (Scheme J) with compound **XLII** in which PG3 is a suitable protective group such as a Cbz, Boc, or Pht.

- 30 Saponification of the intermediates **XLI**, **LIII** and **LXIV** by applying methods known in the art using NaOH or LiOH in a solvent or a mixture of solvents like THF, ethanol or water at

temperatures between 0°C and room temperature then gives the corresponding carboxylic acids **XLIV** (Scheme J), **LIV** (Scheme K) and **LXV** (Scheme L).

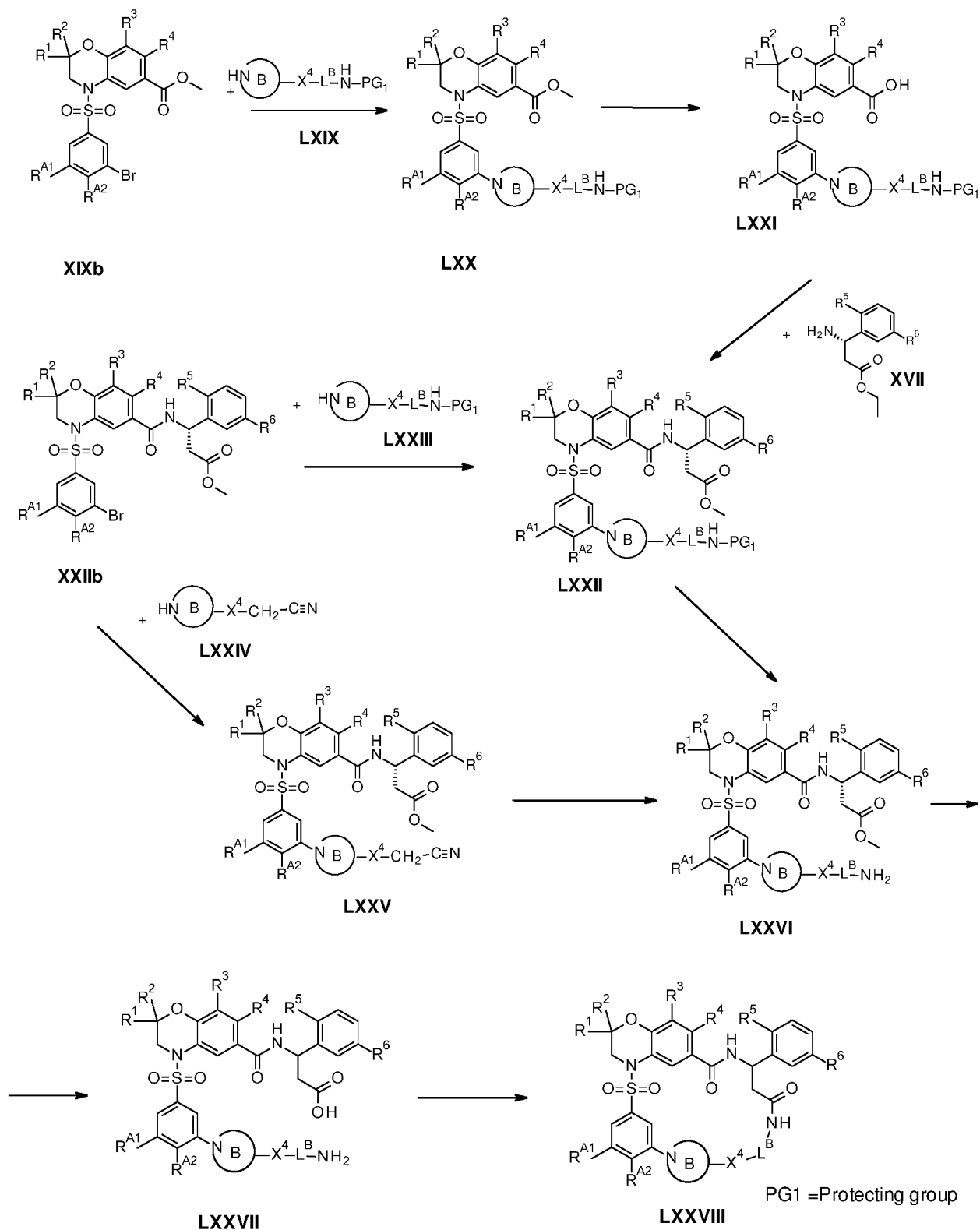
Next, the acids were subjected to selective N-deprotection, removal of PG2 which can be a protective group such as tert-butyloxycarbonyl (BOC) or phthalimide (Pht) can be performed using methods well known in the art and as described (see “Protective Groups in Organic Chemistry” by T. W. Greene and P. G. M. Wutts, 5th Ed., 2014, John Wiley & Sons, New York), e.g., using TFA or hydrochloric acid in a solvent like DCM, 1,4-dioxane or THF preferable at room temperature or in case of phthalimide using hydrazine in ethanol or THF.

Inter-molecular cyclisation to macrocycles **XLVI** (Scheme J), **LVI** (Scheme K) and **LXVIII** (Scheme L) can be accomplished by reaction of intermediates **XLIV** (Scheme J), **LV** (Scheme K) and **LXVI** (Scheme L) with coupling reagents such as CDI, DCC, HATU, HBTU, HOBT, TBTU, T3P or combination TCFH-NMI (Org. Lett. 2018, 20, 4218-4222) in a suitable solvent e.g., DMF, DMA, DCM or dioxane, optionally in the presence of a base (e.g., TEA, DIPEA (Huenig’s base) or DMAP).

The final N-deprotection for intermediates **XLV** (Scheme J) and **LXVII** (Scheme L) can be done using standard procedures (see “Protective Groups in Organic Chemistry” by T. W. Greene and P. G. M. Wutts, 5th Ed., 2014, John Wiley & Sons, New York)

In some embodiments, intermediates **I** are intermediates of type **LXXVIII** in which B, R¹, R^{A1}, R^{A2}, R², R³, R⁴, R⁵, R⁶, L^B, X⁴ are as described herein and Q is bromine can be prepared by methods well known in the art and as exemplified by the general synthetic procedures outlined in Scheme M.

Scheme M



Synthesis of intermediates **LXXII** can be done using the same general procedure as for intermediates **LI** (Scheme K).

- 5 Alternatively, the synthesis of intermediates **LXXII** can be done starting from intermediates **XXIIb** and intermediates **LXXIII**, using a suitable catalyst and solvent such as, e.g.

tris(dibenzylideneacetone)dipalladium (0) chloroform adduct and X-Phos or bis(diphenylphosphino)-1,1'-binaphtalene and palladium(II) acetate in tert-butanol or 1,4-dioxane and a suitable base such as Cs₂CO₃ or KOtBu at temperatures between room temperature and the boiling point of the solvent or solvent mixture. Buchwald Hartwig reactions of that type are well known in the art and described in literature, e.g. *Angewandte Chemie*. **2019**, 58, 17118-17129.

Removal of the protective group PG1 applying methods known in the art (e.g., a Boc group using TFA in DCM or 4M HCl in dioxane at temperatures between 0°C and room temperature or Pht using hydrazine in a suitable solvent such as MeOH, EtOH or THF and as described for example in “Protective Groups in Organic Chemistry” by T.W. Greene and P.G.M. Wuts, 4th Ed., 2006, Wiley N.Y.), furnishes intermediates **LXXVI**.

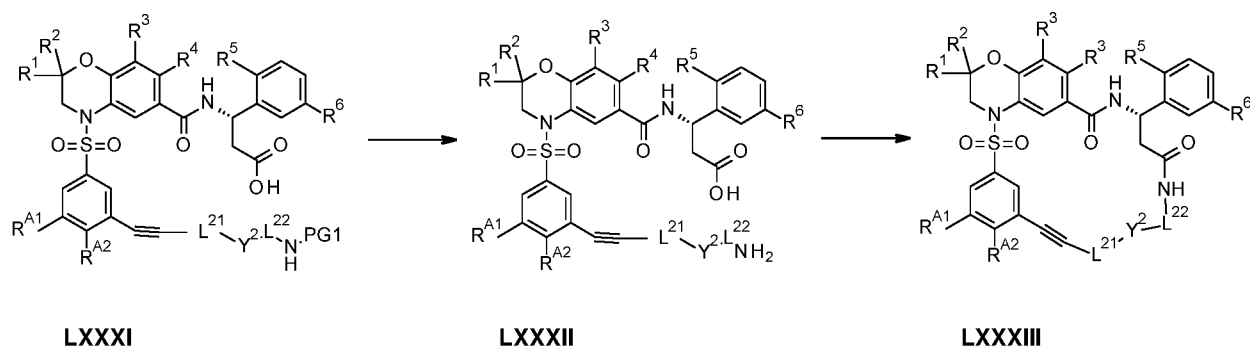
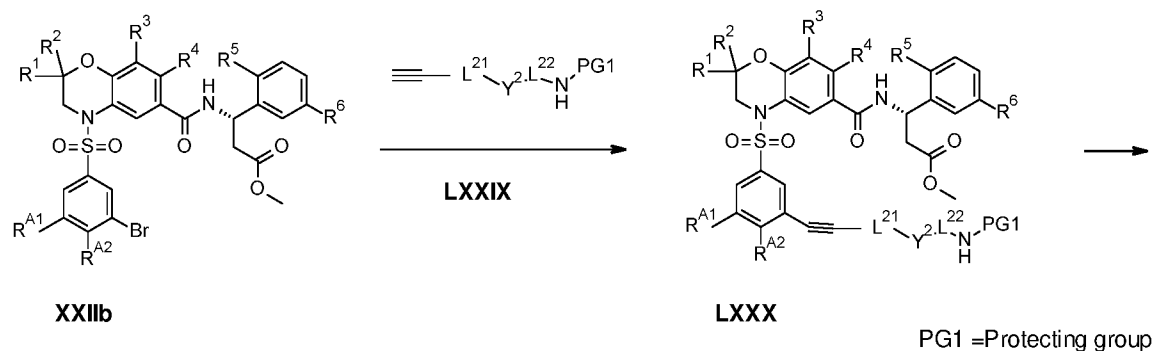
Another route to form intermediates **LXXVI**, starts from intermediates **XXIIIb** and intermediates **LXXIV**, either commercially available or prepared by literature methods following the standard Buchwald above. Reduction of the terminal nitrile can be done by catalytic hydrogenation of intermediates **LXXV** using a catalyst such as Pd/C or Raney Nickel in a solvent like THF, MeOH, EtOH, EtOAc, aqueous ammonia or a mixture thereof, particularly Raney Nickel in a mixture of methanol and ammonia under e.g. a pressure of hydrogen (15 psi).

Saponification of the intermediates **LXXVII** by applying methods known in the art using NaOH or LiOH in a solvent or a mixture of solvents like THF, ethanol or water at temperatures between 0°C and room temperature gives the corresponding carboxylic acids **LXXVII**.

Macrocyclisation to compounds **LXXVIII** can be accomplished by reaction of intermediates **LXXVIII** with coupling reagents such as CDI, DCC, HATU, HBTU, HOBT, TBTU, T3P or combination TCFH-NMI (Org. Lett. 2018, 20, 4218-4222) in a suitable solvent e.g., DMF, DMA, DCM or dioxane, optionally in the presence of a base (e.g., TEA, DIPEA (Huenig's base) or DMAP).

In some embodiments, intermediates **I** are intermediates of type **LXXXIII** in which R¹, R^{A1}, R^{A2}, R², R³, R⁴, R⁵, R⁶, L²¹, L²², Y² are as described herein and Q is bromine can be prepared by methods well known in the art and as exemplified by the general synthetic procedures outlined in Scheme N.

Scheme N



Alkynes **LXXIX**, either commercially available or prepared by methods known in the art and in which PG1 signifies a suitable protecting group such as, e.g. a Boc, Cbz or Pht protecting group, can be subjected to a Sonogashira reaction (e.g. Chem. Soc. Rev. 2011, 40, 5084) with aryl bromides **XXIIb**, and using a suitable catalyst, such as Pd(OAc)₂/PPh₃, Pd(PPh₃)₄ or PdCl₂(PPh₃)₂, particularly PdCl₂(PPh₃)₂ in presence of CuI and an appropriate base such as, e.g. K₂CO₃, Cs₂CO₃, TEA or DIPEA, particularly DIPEA, and suitable solvent such as, e.g. THF, DMSO, DMF, NMP, CH₃CN or dioxane, particularly dioxane and in a temperature range between room temperature and 100°C, particularly around 80°C to give intermediates **LXXX**.

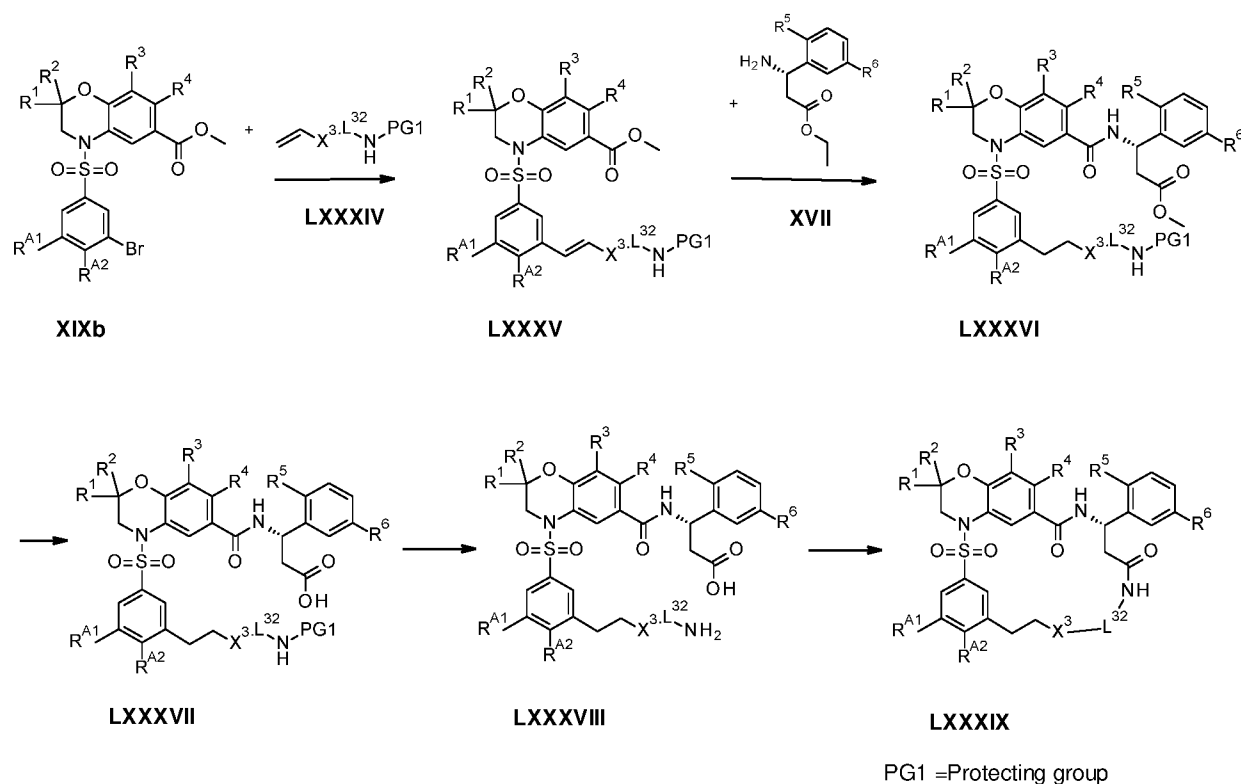
- 10 Saponification of the intermediates **LXXX** by applying methods known in the art using NaOH or LiOH in a solvent or a mixture of solvents like THF, ethanol or water at temperatures between 0°C and room temperature gives the corresponding carboxylic acids **LXXXI**.

Next, the acids are subjected to deprotection to give intermediates **LXXXII**. Removal of PG1 which can be a protective group such as tert-butyloxycarbonyl (BOC) or phtalimide (Pht) can be performed using methods well known in the art and as described (see “Protective Groups in Organic Chemistry” by T. W. Greene and P. G. M. Wuts, 5th Ed., 2014, John Wiley & Sons, New York), e.g., using TFA or hydrochloric acid in a solvent like DCM, 1,4-dioxane or THF preferable at room temperature or in case of phtalimide using hydrazine in ethanol or THF.

Final cyclisation to compounds **LXXXIII** can be accomplished by reaction of intermediates **LXXXII** with coupling reagents such as CDI, DCC, HATU, HBTU, HOBT, TBTU, T3P or combination as TCFH-NMI (Org. Lett. 2018, 20, 4218-4222) in a suitable solvent e.g., DMF, DMA, DCM or dioxane, optionally in the presence of a base (e.g., TEA, DIPEA (Huenig's base) or DMAP).

In some embodiments, intermediates **I** are intermediates of type **LXXXIX** in which R^1 , R^{A1} , R^{A2} , R^2 , R^3 , R^4 , R^5 , R^6 , L^{32} , X^3 , Y^2 are as described herein and Q is bromine can be prepared by methods well known in the art and as exemplified by the general synthetic procedures outlined in Scheme O.

10 Scheme O



Intermediate **XIXb** can be reacted with intermediates **LXXXIV** using a suitable catalyst like palladium(II) acetate or palladium(II) chloride and an additional ligand such as, e.g. triphenylphosphine or 1,1'-bis(diphenylphosphino)ferrocene (dppf) in a solvent such N,N dimethylformamide, NMP or toluene or 1,4-dioxane and a suitable base such as Na_2CO_3 or Et_3N at temperatures between room temperature and the boiling point of the solvent or solvent mixture to provide intermediates **LXXXV**. Heck reactions of that type are well known in the art and described in literature, e.g. Heck, R. F. (1968). "The Palladium-Catalyzed Arylation of Enol

Esters, Halides, and Dienes". Journal of the American Chemical Society. 90 (20): 5518–5526. doi:10.1021/ja01020a005.

Reduction of the double bond in intermediates **LXXXV** using, *e.g.* hydrogen in the presence of a suitable catalyst such as palladium on charcoal or Raney Nickel in an appropriate solvent or solvent
5 mixture such as EtOAc or MeOH or aqueous ammonia yields compounds **LXXXVI**.

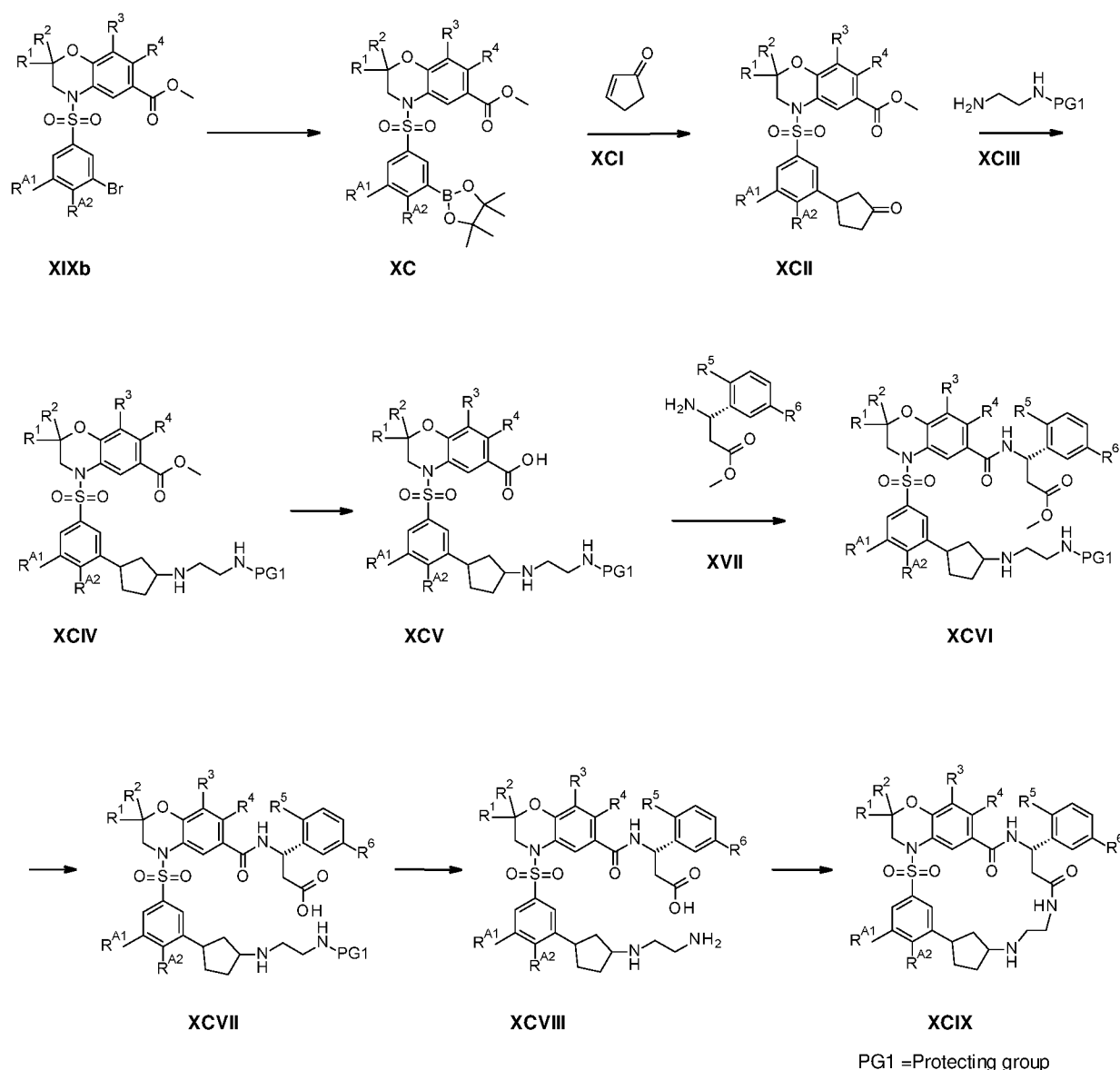
Saponification of the intermediates **LXXXVI** by applying methods known in the art using NaOH or LiOH in a solvent or a mixture of solvents like THF, ethanol or water at temperatures between 0°C and room temperature gives the corresponding carboxylic acids **LXXXVII**.

Next, the acids are subjected to deprotection to give intermediates **LXXXVIII**. Removal of PG1
10 which can be a protective group such as tert-butyloxycarbonyl (BOC) or phthalimide (Pht) can be performed using methods well known in the art and as described (see “Protective Groups in Organic Chemistry” by T. W. Greene and P. G. M. Wuts, 5th Ed., 2014, John Wiley & Sons, New York), *e.g.*, using TFA or hydrochloric acid in a solvent like DCM, 1,4-dioxane or THF preferable at room temperature or in case of phthalimide using hydrazine in ethanol or THF.

15 Final cyclisation to compounds **LXXXIX** can be accomplished by reaction of intermediates **LXXXII** with coupling reagents such as CDI, DCC, HATU, HBTU, HOBT, TBTU, T3P or combination as TCFH-NMI (Org. Lett. 2018, 20, 4218-4222) in a suitable solvent *e.g.*, DMF, DMA, DCM or dioxane, optionally in the presence of a base (*e.g.*, TEA, DIPEA (Huenig’s base) or DMAP).

20 In some embodiments, intermediates **I** are intermediates of type **XCIX** in which R¹, R^{A1}, R^{A2} R², R³, R⁴, R⁵, R⁶ are as described herein and Q is bromine can be prepared by methods well known in the art and as exemplified by the general synthetic procedures outlined in Scheme P.

Scheme P



Intermediates **XC** can be prepared from intermediates **XIXb**, using literature procedures as described for example in “Boronic Acids - Preparation and Applications in Organic Synthesis and Medicine” by Dennis G. Hall (ed.) 1st Ed., 2005, John Wiley & Sons, New York). Those intermediates can be subjected to cross coupling reaction with intermediates **XCI** using a suitable catalyst (e.g. dichloro[1,1'-bis(diphenylphosphino)-ferrocene]palladium(II) DCM adduct, tetrakis(triphenylphosphine)palladium(0) or palladium(II)acetate with triphenylphosphine) in an appropriate solvent (e.g. dioxane, dimethoxyethane, water, toluene, DMF or mixtures thereof) and a suitable base (e.g. Na_2CO_3 , NaHCO_3 , KF , K_2CO_3 or TEA) at temperatures between room temperature and the boiling point of the solvent or solvent mixture, to yield intermediates **XCII**. Suzuki reactions of this type are broadly described in literature (e.g. A. Suzuki, *Pure Appl. Chem.* **1991**, 63, 419-422; A. Suzuki, N. Miyaura, *Chem. Rev.* **1995**, 95, 2457-2483; A. Suzuki, *J.*

Organomet. Chem. **1999**, 576, 147-168; V. Polshettiwar *et al.*, *Chem. Sus. Chem.* **2010**, 3, 502-522) and are well known to those skilled in the art.

To attach the linker, a classical reductive amination reaction can be performed. Reductive aminations of carbonyl compounds are generally carried out by reacting an aldehyde or a ketone like intermediates **XCI** with suitable amines like intermediates **XCIII** in the presence of a reducing agent e.g. hydrogen/Pd, NaBH₃CN or NaBH₄ in a suitable solvent e.g. methanol, ethanol or dichloromethane (see, for example: "Reductive alkylation - Preparation of Amines" in *Catalytic Hydrogenation over Platinum Metals*, p. 291 -303 or Borch, Bernstein, J. Am. Chem. Soc, 1971, 93, 2897)

- 10 Saponification of the intermediates **XCIV** by applying methods known in the art using NaOH or LiOH in a solvent or a mixture of solvents like THF, ethanol or water at temperatures between 0°C and room temperature gives the corresponding carboxylic acids **XCIV**.

Accordingly, the carboxylic acids can be activated and react with intermediate **XVII** to provide compounds **XCVII**. Amide couplings of this type with carboxylic acids are widely described in the literature and can be accomplished by the usage of coupling reagents such as CDI, DCC, HATU, HBTU, HOBT, TBTU, T3P or combination TCFH-NMI (*Org. Lett.* 2018, 20, 4218-4222) in a suitable solvent e.g., DMF, DMA, DCM or dioxane, optionally in the presence of a base (e.g., TEA, DIPEA (Huenig's base) or DMAP).

- 20 Esters **XCVI** are converted to the corresponding acids **XCVII** by applying methods known in the art using NaOH or LiOH in a solvent or a mixture of solvents like THF, ethanol or water at temperatures between 0°C and room temperature.

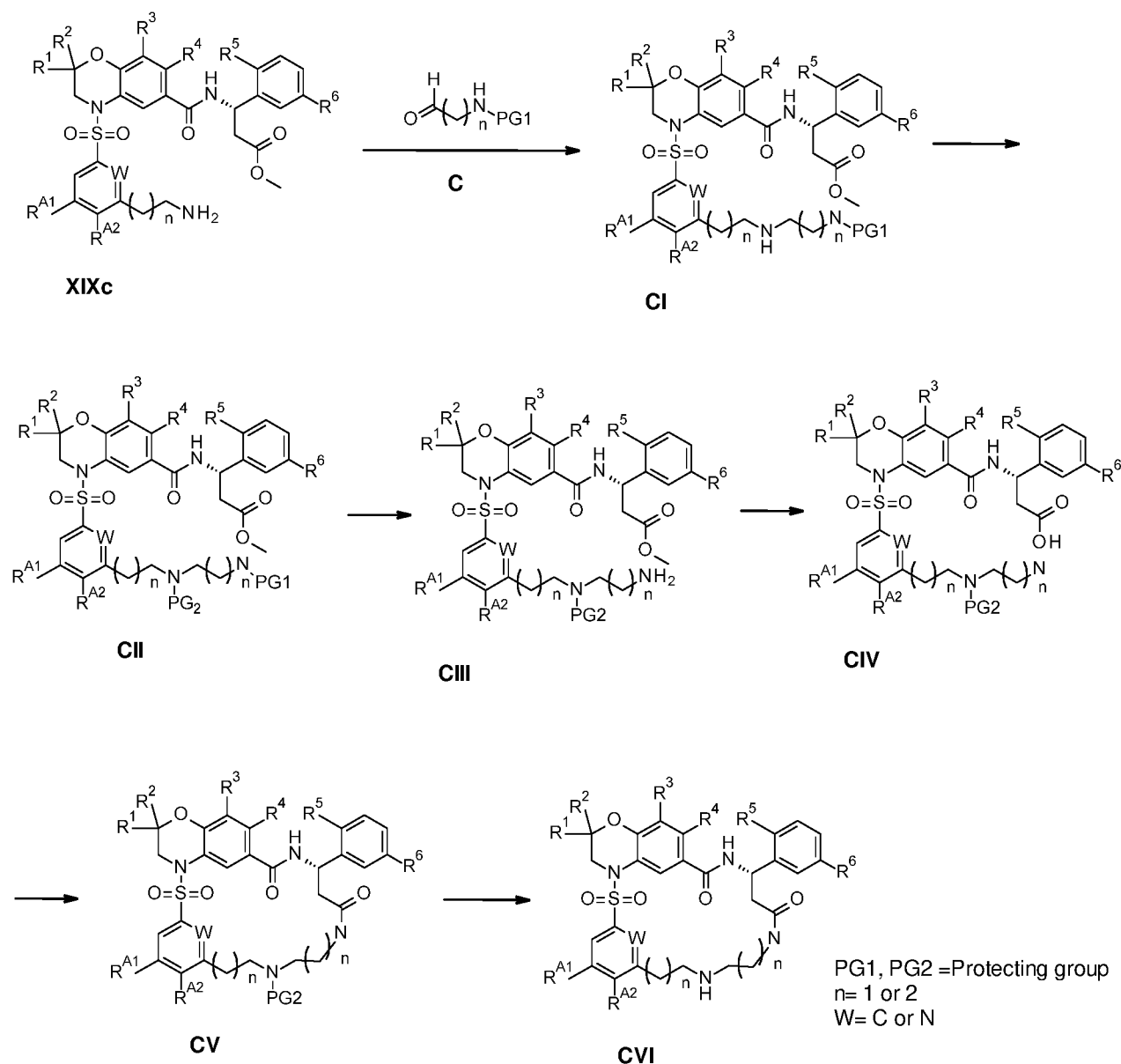
Removal of the protective group from the lastly prepared intermediates applying methods known in the art (e.g., a Boc group using TFA in DCM or 4M HCl in dioxane at temperatures between 0°C and room temperature, a Cbz group using hydrogen in the presence of a suitable catalyst such as Pd or Pd(OH)₂ on charcoal in a suitable solvent such as MeOH, EtOH, EtOAc or mixtures thereof and as described for example in "Protective Groups in Organic Chemistry" by T.W. Greene and P.G.M. Wuts, 4th Ed., 2006, Wiley N.Y.), furnishes intermediates XXXVIII (Scheme J), LI (Scheme K) and LXI (Scheme L).

- 30 Macrocyclisation to compounds **XCIX** can be accomplished by reaction of intermediates **XCVII** with coupling reagents such as CDI, DCC, HATU, HBTU, HOBT, TBTU, T3P or combination

TCFH-NMI (Org. Lett. 2018, 20, 4218-4222) in a suitable solvent e.g., DMF, DMA, DCM or dioxane, optionally in the presence of a base (e.g., TEA, DIPEA (Huenig's base) or DMAP).

In some embodiments, intermediates **I** are intermediates of type **CVI** in which R^1 , R^{A1} , R^{A2} , R^2 , R^3 , R^4 , R^5 , R^6 are as described herein and Q is $-(CH_2)_nNH_2$ ($n=1$ or 2) can be prepared by methods well known in the art and as exemplified by the general synthetic procedures outlined in Scheme Q.

Scheme Q



Amines **XIXc**, either commercially available or prepared by methods known in the art can be subjected to a reductive amination with aldehydes **C** in which PG1 signifies a suitable protecting group such as, e.g. a Boc, Cbz or Pht protecting group. Reductive aminations of carbonyl

compounds are generally carried out by reacting an aldehyde or a ketone like intermediates **XCI** with suitable amines like intermediates **XCIII** in the presence of a reducing agent e.g. hydrogen/Pd, NaBH₃CN or NaBH₄ in a suitable solvent e.g. methanol, ethanol or dichloromethane (see, for example: "Reductive alkylation - Preparation of Amines" in Catalytic Hydrogenation over
5 Platinum Metals, p. 291 -303 or Borch, Bernstein, J. Am. Chem. Soc, 1971, 93, 2897).

Introduction of a second protecting group PG2 can be done by applying procedures known in the art using e.g. BOC anhydride or CBz-Cl in the presence of a base (e.g., TEA, DIPEA (Huenig's base) or DMAP) in solvent such as N,N-dimethylformamide or dichloromethane to get intermediates **CII**.

10 Next, the intermediates **CII** were subjected to selective N-deprotection, removal of PG1 which can be a protective group such as tert-butyloxycarbonyl (BOC) or phthalimide (Pht) can be performed using methods well known in the art and as described (see "Protective Groups in Organic Chemistry" by T. W. Greene and P. G. M. Wuts, 5th Ed., 2014, John Wiley & Sons, New York), e.g., using TFA or hydrochloric acid in a solvent like DCM, 1,4-dioxane or THF preferable
15 at room temperature or in case of phthalimide using hydrazine in ethanol or THF.

Saponification of the intermediates **CIII** using NaOH or LiOH in a solvent or a mixture of solvents like THF, ethanol or water at temperatures between 0°C and room temperature gives the corresponding carboxylic acids **CIV**

Compounds **CVI** can be obtained by reaction of intermediates **CIV** with coupling reagents such
20 as CDI, DCC, HATU, HBTU, HOBT, TBTU, T3P or combination TCFH-NMI (Org. Lett. 2018, 20, 4218-4222) in a suitable solvent e.g., DMF, DMA, DCM or dioxane, optionally in the presence of a base (e.g., TEA, DIPEA (Huenig's base) or DMAP) and by final N-deprotection as using the same methods as above.

In some embodiments, intermediates **I** are intermediates of type **CXIV** in which R¹, R^{A1}, R^{A2} R²,
25 R³, R⁴, R⁵, R⁶ are as described herein, Q is -(CH₂)_nNH₂ (n=1 or 2) and W is C or N can be prepared by methods well known in the art and as exemplified by the general synthetic procedures outlined in Scheme R.

Amines **XIXc**, either commercially available or prepared by methods known in the art can be subjected to reductive amination with aldehydes **C** in which PG1 signifies a suitable protecting
30 group such as, e.g. a Boc, Cbz or Pht protecting group. Reductive aminations of carbonyl compounds are generally carried out by reacting an aldehyde or a ketone like intermediates **XCI**

with suitable amines like intermediates **XCIII** in the presence of a reducing agent e.g. hydrogen/Pd, NaBH₃CN or NaBH₄ in a suitable solvent e.g. methanol, ethanol or dichloromethane (see, for example: "Reductive alkylation - Preparation of Amines" in Catalytic Hydrogenation over Platinum Metals, p. 291 -303 or Borch, Bernstein, J. Am. Chem. Soc, 1971, 93, 2897)

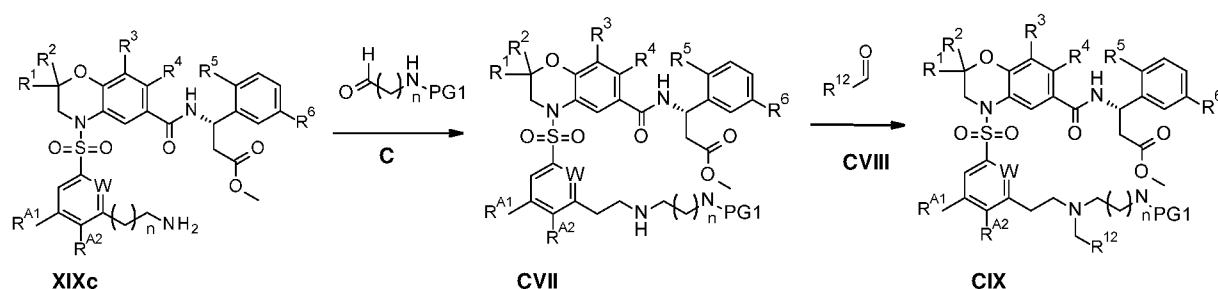
- 5 A second reductive amination with the same conditions as above using intermediates **CVII** as amines and reagents **CVIII** as aldehyde can be done to obtain intermediates **CIX**.

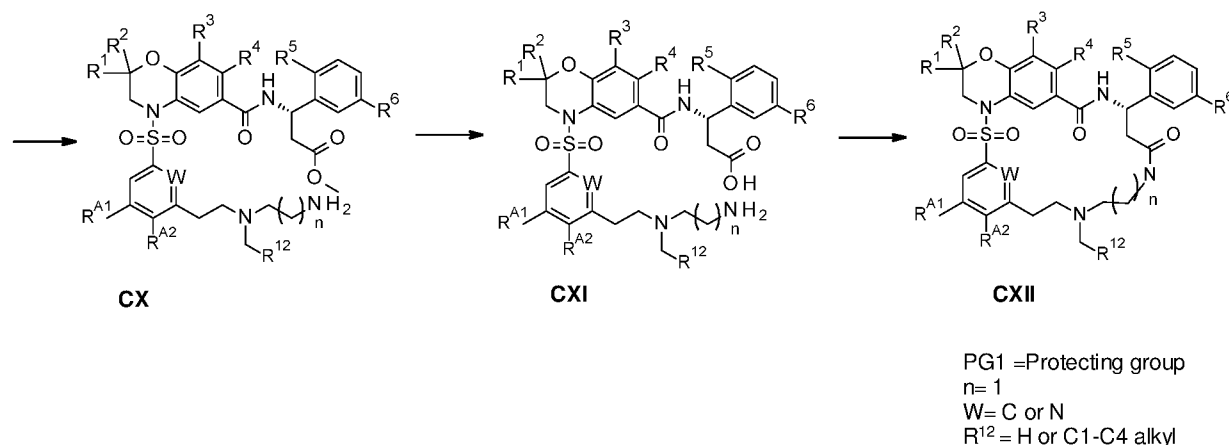
Next, the intermediates **CIX** were subjected to selective N-deprotection, removal of PG1 which can be a protective group such as tert-butyloxycarbonyl (BOC) or phthalimide (Pht) can be performed using methods well known in the art and as described (see "Protective Groups in
10 Organic Chemistry" by T. W. Greene and P. G. M. Wuts, 5th Ed., 2014, John Wiley & Sons, New York), e.g., using TFA or hydrochloric acid in a solvent like DCM, 1,4-dioxane or THF preferable at room temperature or in case of phthalimide using hydrazine in ethanol or THF.

Saponification of the intermediates **CX** using NaOH or LiOH in a solvent or a mixture of solvents like THF, ethanol or water at temperatures between 0°C and room temperature gives the
15 corresponding carboxylic acids **CXI**.

Finally, compounds **CXII** can be obtained by reaction of intermediates **CXI** with coupling reagents such as CDI, DCC, HATU, HBTU, HOBT, TBTU, T3P or combination TCFH-NMI (Org. Lett. 2018, 20, 4218-4222) in a suitable solvent e.g., DMF, DMA, DCM or dioxane, optionally in the presence of a base (e.g., TEA, DIPEA (Huenig's base) or DMAP).

20 Scheme R





Examples

The invention will be more fully understood by reference to the following examples. The claims should not, however, be construed as limited to the scope of the examples.

- 5 In case the preparative examples are obtained as a mixture of enantiomers, the pure enantiomers can be separated by methods described herein or by methods known to the person skilled in the art, such as e.g., chiral chromatography (e.g., chiral SFC) or crystallization.

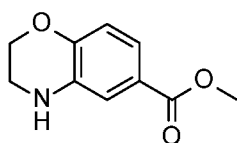
All reaction examples and intermediates were prepared under an argon atmosphere if not specified otherwise.

- 10 The compounds disclosed and described herein have been named using the IUPAC naming function of Biovia Draw 22.1. If there is a discrepancy between a depicted structure and a name given to that structure, then the depicted structure controls.

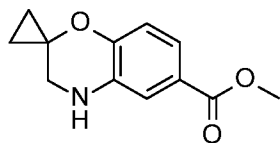
The following intermediates were prepared according to the procedures given, are commercially available or can be prepared according to literature procedures.

15 SYNTHESIS OF INTERMEDIATES A

Intermediate A1: Methyl 3,4-dihydro-2*H*-1,4-benzoxazine-6-carboxylate



Methyl 3,4-dihydro-2*H*-1,4-benzoxazine-6-carboxylate (CAS 758684-29-6) is commercially available.

Intermediate A2: Methyl spiro[3,4-dihydro-1,4-benzoxazine-2,1'-cyclopropane]-6-carboxylate**Step 1:** Methyl 4-(1-methoxycarbonylcyclopropoxy)-3-nitrobenzoate

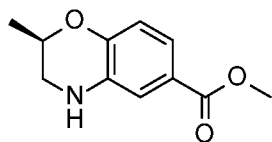
To a stirred solution of methyl 1-hydroxy-1-cyclopropane carboxylate (CAS 33689-29-1, 7 g, 50.2 mmol) in N,N-dimethylformamide (100 ml), was added sodium hydride (2.41 g, 60.3 mmol) at 0 °C under a nitrogen atmosphere. The reaction mixture was stirred at 0 °C for 30 minutes and then methyl 4-fluoro-3-nitrobenzoate (CAS 329-59-9, 10.0 g, 50.2 mmol) was added. After stirring at 25 °C for 16 hours under a nitrogen atmosphere, the reaction mixture was slowly poured into saturated aqueous NH₄Cl (500 ml) and stirred at 0 °C for 10 minutes, after which time the mixture was extracted with ethyl acetate three times. The organic layer was washed with brine twice, dried over Na₂SO₄, and concentrated *in vacuo*. The crude product was purified by flash chromatography (silica gel, ethyl acetate in petroleum ether, 1-30 %) to afford methyl 4-(1-methoxycarbonylcyclopropoxy)-3-nitrobenzoate (13 g, 88 % yield) as an orange oil.

¹H NMR (400 MHz, CHLOROFORM-d) δ = 8.50 (d, J = 2.0 Hz, 1H), 8.16 (dd, J = 2.0, 8.8 Hz, 1H), 7.18 (d, J = 8.8 Hz, 1H), 3.93 (s, 3H), 3.73 (s, 3H), 1.77 - 1.69 (m, 2H), 1.49 - 1.41 (m, 2H)

Step 2: Methyl 3-oxospiro[4H-1,4-benzoxazine-2,1'-cyclopropane]-6-carboxylate

To a solution of methyl 4-(1-methoxycarbonylcyclopropoxy)-3-nitro-benzoate (13.0 g, 44 mmol) in acetic acid (100 ml) was added iron (7.4 g, 132 mmol). The resulting mixture was stirred at 90 °C for 2 hours. The mixture was filtered through diatomaceous earth and washed with ethyl acetate. The filtrate was poured into water and the mixture was extracted with ethyl acetate three times. The combined organic phase was washed with brine three times, dried over Na₂SO₄ and concentrated *in vacuo*. The crude product was purified by flash chromatography (silica gel, methanol in dichloromethane, 10-20%) to afford methyl 3-oxospiro[4H-1,4-benzoxazine-2,1'-cyclopropane]-6-carboxylate (56.0 g, 79% yield) as a yellow solid. MS m/z: 234.0 [M+H]⁺, ESI pos.

Intermediate A3: Methyl (2R)-2-methyl-3,4-dihydro-2H-1,4-benzoxazine-6-carboxylate



Step 1: Methyl 4-[(1R)-2-methoxy-1-methyl-2-oxo-ethoxy]-3-nitrobenzoate

To a solution of methyl (*R*)-(+)-lactate (CAS 17392-83-5, 2.35 g, 22.6 mmol) in tetrahydrofuran (40 ml), sodium hydride 60 % (903 mg, 22.6 mmol) was added in portions at 0-10 °C. The mixture
5 was stirred at 0-10 °C for 30 minutes. Methyl 4-fluoro-3-nitrobenzoate (CAS 329-59-9, 3.0 g, 15.1 mmol) was added to the mixture at 0-10 °C in portions, and the mixture was stirred at 40 °C for 16 hours. The reaction mixture was added to a saturated solution of NH₄Cl and extracted twice with ethyl acetate. The combined organic phase was washed twice with brine, dried over Na₂SO₄, filtered and concentrated *in vacuo*. The crude product was purified by flash chromatography (silica
10 gel, ethyl acetate in petroleum ether, 0-50 %) to afford methyl 4-[(1R)-2-methoxy-1-methyl-2-oxo-ethoxy]-3-nitrobenzoate (3.4 g, 80 % yield) as a yellow oil.

¹H NMR (400 MHz, CHLOROFORM-*d*) δ = 8.49 (d, *J* = 2.0 Hz, 1H), 8.15 (dd, *J* = 2.0, 8.8 Hz, 1H), 6.95 (d, *J* = 8.8 Hz, 1H), 4.94 (q, *J* = 6.8 Hz, 1H), 3.93 (s, 3H), 3.77 (s, 3H), 1.72 (d, *J* = 6.8 Hz, 3H).

15 Step 2: Methyl (2R)-2-methyl-3-oxo-4H-1,4-benzoxazine-6-carboxylate

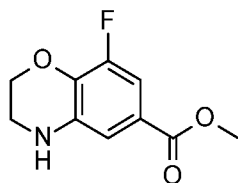
To a solution of methyl 4-[(1R)-2-methoxy-1-methyl-2-oxo-ethoxy]-3-nitrobenzoate (4 g, 14.1 mmol) in acetic acid (60 ml) was added iron (2.37 g, 42.4 mmol). The resulting suspension was stirred at 90 °C for 2 hours. The reaction mixture was concentrated *in vacuo*. The crude was diluted with ethyl acetate, filtered, and washed with water. The filtrate was then washed with a saturated
20 aqueous solution of sodium bicarbonate and brine twice each, dried over Na₂SO₄, and concentrated *in vacuo* to afford methyl (2R)-2-methyl-3-oxo-4H-1,4-benzoxazine-6-carboxylate (3.1 g, 99 % yield) as a white solid. MS *m/z*: 222.2 [M+H]⁺, ESI pos.

Step 3: Methyl (2R)-2-methyl-3,4-dihydro-2H-1,4-benzoxazine-6-carboxylate

To a solution of methyl (2R)-2-methyl-3-oxo-4H-1,4-benzoxazine-6-carboxylate (1.6 g, 7.23
25 mmol) in tetrahydrofuran (30 ml) was added dropwise a 10 M solution of borane dimethyl sulfide complex in tetrahydrofuran (3.62 ml, 36.2 mmol) at 20 °C. After stirring at 20 °C for 16 hours, the reaction mixture was added to a saturated solution of NH₄Cl and extracted twice with ethyl acetate. The combined organic phase was washed with brine twice, dried over Na₂SO₄, and concentrated

in vacuo. The crude product was purified by flash chromatography (silica gel, ethyl acetate in petroleum ether, 0-50 %) to afford methyl (2*R*)-2-methyl-3,4-dihydro-2*H*-1,4-benzoxazine-6-carboxylate (1.0 g, 62 % yield) as a white solid. MS *m/z*: 208.2 [M+H]⁺, ESI pos.

Intermediate A4: Methyl 8-fluoro-3,4-dihydro-2*H*-1,4-benzoxazine-6-carboxylate



5

Step 1: Methyl 3-fluoro-4-hydroxy-5-nitrobenzoate

To a cooled solution of methyl 3-fluoro-4-hydroxybenzoate (CAS 403-01-0, 5 g, 29.4 mmol) in a mixture of acetic acid (33 ml) and acetic anhydride (17 ml) was added at 0 °C a solution of fuming nitric acid (1.35 ml, 32.3 mmol) in acetic acid (20 ml) over 1 minute. The temperature of the light brown solution was allowed to rise to 22 °C and stirred for 15 minutes. The reaction mixture was diluted with water and left to settle for 30 minutes. The mixture was then filtered, and the filter cake was washed with water. The filtrate was extracted with ethyl acetate twice, washed with brine, dried over Na₂SO₄, filtered and concentrated *in vacuo*. The crude product was purified by flash chromatography (silica gel, ethyl acetate in petroleum ether, 0-50 %) to afford methyl 3-fluoro-4-hydroxy-5-nitrobenzoate (5.2 g, 82 % yield) as a yellow solid. MS *m/z*: 214.1 [M-H]⁻, ESI neg.

Step 2: Methyl 3-amino-5-fluoro-4-hydroxybenzoate

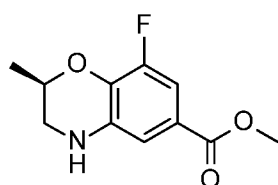
To a solution of methyl 3-fluoro-4-hydroxy-5-nitrobenzoate (1.0 g, 4.65 mmol) in ethanol (20 ml) and water (4 ml) were added iron (1.3 g, 23.2 mmol) and NH₄Cl (1.23 g, 23.3 mmol). After stirring at 80 °C for 2 hours, the mixture was concentrated *in vacuo*, filtered, and extracted with ethyl acetate three times. The combined organic phase was washed with water twice, dried over Na₂SO₄, filtered and concentrated *in vacuo*. The crude product was purified by flash chromatography (silica gel, ethyl acetate in petroleum ether, 0-50 %) to afford methyl 3-amino-5-fluoro-4-hydroxybenzoate (600 mg, 66 % yield) as a yellow solid. MS *m/z*: 184 [M-H]⁻, ESI neg.

Step 3: Methyl 8-fluoro-3,4-dihydro-2*H*-1,4-benzoxazine-6-carboxylate

To a solution of methyl 3-amino-5-fluoro-4-hydroxybenzoate (500 mg, 2.7 mmol) in N,N-dimethylformamide (10 ml) was added potassium carbonate (745 mg, 5.4 mmol). After stirring at

20 °C for 15 minutes, 1,2-dibromoethane (CAS 106-93-4, 558 mg, 2.97 mmol) was added and the reaction mixture was stirred at 120 °C for 16 hours. The mixture was then concentrated *in vacuo*, filtered, and purified by reverse phase flash chromatography (Flash Spherical C18, 20-35µm; 100 Å; 20 g, water with 2.25 % formic acid and acetonitrile) to afford methyl 8-fluoro-3,4-dihydro-2H-1,4-benzoxazine-6-carboxylate (130 mg, 22 % yield) as a white solid. MS m/z: 212.1 [M+H]⁺, ESI pos.

Intermediate A5: Methyl (2R)-8-fluoro-2-methyl-3,4-dihydro-2H-1,4-benzoxazine-6-carboxylate



10 Step 1: Methyl 3-fluoro-4-hydroxy-5-nitrobenzoate

To a cooled solution of methyl 3-fluoro-4-hydroxybenzoate (CAS 403-01-0, 5 g, 29.4 mmol) in a mixture of acetic acid (33 ml) and acetic anhydride (17 ml) was added a mixture of fuming nitric acid (1.35 ml, 32.3 mmol) in acetic acid (20 ml) over 1 minute at 0 °C. Upon completing the addition, the temperature of the light brown solution was allowed to rise to 22 °C and stirred for 15 minutes. The reaction mixture was diluted with water and allowed to settle for 30 minutes, after which it was filtered. The filter cake was washed with water and the filtrate was extracted with ethyl acetate twice, washed with brine, dried over sodium sulfate, and concentrated *in vacuo*. The crude product was purified by flash chromatography (silica gel, ethyl acetate in petroleum ether, 0-100 %) and concentrated *in vacuo* to afford methyl 3-fluoro-4-hydroxy-5-nitrobenzoate (5.2 g, 82 % yield) as a yellow solid. MS m/z: 214.1 [M-H]⁻, ESI neg.

Step 2: Methyl 3-fluoro-4-[(1R)-2-methoxy-1-methyl-2-oxo-ethoxy]-5-nitrobenzoate

To a solution of methyl 3-fluoro-4-hydroxy-5-nitrobenzoate (1g, 4.6 mmol) and methyl (S)-(-)-lactate (CAS 27871-49-4, 726 mg, 7 mmol) in tetrahydrofuran (20 ml) was added triphenylphosphine (2.44 g, 9.3 mmol) and diethyl azodicarboxylate (1.46 ml, 9.3 mmol) at 0 °C, under inert atmosphere. The mixture was stirred at 20 °C for 1 hour, then diluted with water and extracted with ethyl acetate three times. The combined organic phase was washed with brine twice, dried over sodium sulfate and concentrated *in vacuo*. The crude product was purified by flash

chromatography (silica gel, ethyl acetate in petroleum ether, 0-50 %) to afford methyl 3-fluoro-4-[(1*R*)-2-methoxy-1-methyl-2-oxo-ethoxy]-5-nitrobenzoate (1.6 g, >100 % yield) as a colorless oil.

¹H NMR (400 MHz, DMSO-*d*₆) δ = 8.25 (t, *J* = 1.6 Hz, 1H), 8.10 (dd, *J* = 2.0, 12.0 Hz, 1H), 5.22 (dq, *J* = 1.2, 6.8 Hz, 1H), 4.10 - 3.96 (m, 1H), 3.89 (s, 3H), 3.66 (s, 3H), 1.55 (d, *J* = 6.8 Hz, 3H)

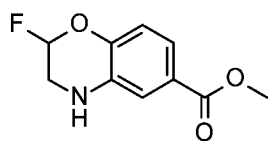
5 Step 3: Methyl (2*R*)-8-fluoro-2-methyl-3-oxo-4*H*-1,4-benzoxazine-6-carboxylate

To a solution of methyl 3-fluoro-4-[(1*R*)-2-methoxy-1-methyl-2-oxo-ethoxy]-5-nitrobenzoate (1.3 g, 4.32 mmol) in acetic acid (30 ml), was added iron (725 mg, 13.0 mmol) at 25 °C. After stirring at 90 °C for 2 hours, the mixture was concentrated *in vacuo*, filtered, and extracted with ethyl acetate three times. The pH of the combined organic phase was adjusted to pH = 7 by addition of
10 aqueous sodium hydrogen carbonate solution, then was washed with water twice, dried over Na₂SO₄, filtered, and concentrated *in vacuo* to give crude methyl (2*R*)-8-fluoro-2-methyl-3-oxo-4*H*-1,4-benzoxazine-6-carboxylate (1g, 96 % yield) as a white solid. MS *m/z*: 240.0 [M+H]⁺, ESI pos.

Step 4: Methyl (2*R*)-8-fluoro-2-methyl-3,4-dihydro-2*H*-1,4-benzoxazine-6-carboxylate

15 To a solution of methyl (2*R*)-8-fluoro-2-methyl-3-oxo-4*H*-1,4-benzoxazine-6-carboxylate (1.0 g, 4.18 mmol) in tetrahydrofuran (20 ml) was added dropwise under inert atmosphere a 10 M solution of borane dimethyl sulfide complex in tetrahydrofuran (4.18 ml, 41.8 mmol). After stirring at 20 °C for 18 hours, the reaction mixture was quenched by dropwise addition of methanol (20 ml) at 0 °C, followed by stirring at 20 °C for 30 minutes and at 50 °C for another
20 30 minutes. The solution was finally concentrated *in vacuo*. The crude product was purified by flash chromatography (silica gel, ethyl acetate in petroleum ether, 0-50 %) to afford methyl (2*R*)-8-fluoro-2-methyl-3,4-dihydro-2*H*-1,4-benzoxazine-6-carboxylate (1.0 g, >100 % yield) as a white solid. MS *m/z*: 225.9 [M+H]⁺, ESI pos.

Intermediate A6: Methyl 2-fluoro-3,4-dihydro-2*H*-1,4-benzoxazine-6-carboxylate



25

Step 1: Methyl 4-(2-ethoxy-1-fluoro-2-oxo-ethoxy)-3-nitrobenzoate

To a solution of methyl 4-hydroxy-3-nitrobenzoate (CAS 99-42-3, 2.0 g, 10.2 mmol) in N,N-dimethylformamide (30 ml) was added potassium carbonate (5.61 g, 40.6 mmol) and ethyl bromofluoroacetate (CAS 401-55-8, 3.75 g, 20.3 mmol). The reaction mixture was stirred at 80 °C for 16 hours. The mixture was poured into water and extracted three times with ethyl acetate. The combined organic layer was washed with brine, dried with Na₂SO₄, filtered and evaporated. The crude product was purified via flash chromatography (silica gel, ethyl acetate/petroleum ether 20-50 %) to give (methyl 4-(2-ethoxy-1-fluoro-2-oxo-ethoxy)-3-nitrobenzoate (1.8 g, 52 % yield) as a white solid.

¹H NMR (400 MHz, CHLOROFORM-d) δ = 8.55 (d, J = 2.2 Hz, 1H), 8.26 (dd, J = 2.1, 8.7 Hz, 1H), 7.43 (dd, J = 1.4, 8.8 Hz, 1H), 6.18 - 5.91 (m, 1H), 4.54 - 4.24 (m, 2H), 3.96 (s, 3H), 1.36 (t, J = 7.2 Hz, 3H)

Step 2: Methyl 2-fluoro-3-oxo-4H-1,4-benzoxazine-6-carboxylate

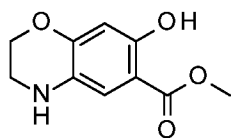
To a solution of methyl 4-(2-ethoxy-1-fluoro-2-oxo-ethoxy)-3-nitro-benzoate (1.3 g, 4.32 mmol) in acetic acid (15 ml) was added iron (723 mg, 13 mmol). The mixture was stirred at 90 °C for 2 hours. The reaction mixture was evaporated and filtered. The filter cake was washed with ethyl acetate. The filtrate obtained was treated with a 5% aqueous solution of NaHCO₃ until pH > 7 was obtained. The organic layer was separated and was washed twice with water. The combined organic layer was washed with brine, dried with Na₂SO₄, filtered and evaporated. The crude product was purified via flash chromatography (silica gel, ethyl acetate/petroleum ether 30-60 %) to give (methyl 2-fluoro-3-oxo-4H-1,4-benzoxazine-6-carboxylate (700 mg, 71 % yield) as a white solid. MS m/z: 226.1 [M+H]⁺, ESI pos.

Step 3: Methyl 2-fluoro-3,4-dihydro-2H-1,4-benzoxazine-6-carboxylate

To a solution of methyl 2-fluoro-3-oxo-4H-1,4-benzoxazine-6-carboxylate (650 mg, 2.89 mmol) in tetrahydrofuran (20 ml) was added dropwise a 1 M solution of BH₃·THF in tetrahydrofuran (8.66 ml, 8.66 mmol) at 0 °C under nitrogen. The mixture was stirred at 0 °C for 1 hour under nitrogen. The reaction was quenched by a slow addition of methanol (30 ml) at 0 °C under nitrogen. The mixture obtained was stirred for 30 minutes then the cooling bath was removed and the solution was stirred at 25 °C for 30 minutes then 50 °C for 30 minutes. The solvents were evaporated and the residue was purified via flash chromatography (silica gel, ethyl acetate/petroleum ether 20-50 %) to give methyl 2-fluoro-3,4-dihydro-2H-1,4-benzoxazine-6-carboxylate (540 mg, 71 % yield) as a white solid.

^1H NMR (400 MHz, DMSO- d_6) δ = 7.29 (d, J = 2.0 Hz, 1H), 7.20 (dd, J = 2.2, 8.4 Hz, 1H), 6.89 (d, J = 8.4 Hz, 1H), 6.37 - 6.09 (m, 2H), 3.79 (s, 3H), 3.48 (br s, 1H), 3.30 - 3.17 (m, 1H)

Intermediate A7: Methyl 7-hydroxy-3,4-dihydro-2*H*-1,4-benzoxazine-6-carboxylate



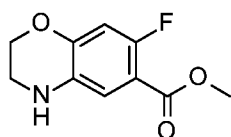
5 Step 1: Methyl 5-amino-2,4-dihydroxybenzoate

To a solution of methyl 2,4-dihydroxy-5-nitrobenzoate (CAS 271261-71-3, 2.0 g, 9.38 mmol) in methanol (40 ml) was added Pd/C (800 mg) and the reaction mixture was stirred at 30 °C for 2 hours under a hydrogen atmosphere (3 bar). The reaction mixture was filtered and the filter cake was washed three times with methanol. The filtrate was concentrated to give methyl 5-amino-2,4-dihydroxybenzoate (1.8 g, 89 % yield) as a yellow solid. MS m/z : 182.1 $[\text{M}-\text{H}]^-$, ESI neg.

Step 2: Methyl 7-hydroxy-3,4-dihydro-2*H*-1,4-benzoxazine-6-carboxylate

The title compound was prepared in analogy to **Intermediate A4, step 1** starting from methyl 5-amino-2,4-dihydroxybenzoate instead of methyl 3-amino-5-fluoro-4-hydroxybenzoate and stirring at 125 °C for 2 hours instead of 120 °C for 16 hours. Black oil. MS m/z : 210.1 $[\text{M}+\text{H}]^+$, ESI pos.

Intermediate A8: Methyl 7-fluoro-3,4-dihydro-2*H*-1,4-benzoxazine-6-carboxylate



Step 1: Methyl 2-fluoro-4-hydroxy-5-nitrobenzoate

To a cooled solution of methyl 2-fluoro-4-hydroxybenzoate (CAS 197507-22-5, 5 g, 29.4 mmol) in acetic acid (50 ml) was added at 5-10 °C a solution of HNO_3 (3.05 ml, 44.1 mmol, 1.5 eq) in acetic acid (10 ml). The reaction mixture was warmed to 20 °C and stirred for 18 hours. The mixture was poured into water, stirred at 0 °C for 10 minutes and extracted three times with dichloromethane. The combined organic layer was dried with Na_2SO_4 , filtered and evaporated. The crude product was purified via flash chromatography (silica gel, ethyl acetate/petroleum

ether 0-100 %) to give as second fraction, methyl 2-fluoro-4-hydroxy-5-nitrobenzoate (1.7 g, 27 % yield) as a yellow solid. MS m/z: 215.9 [M+H]⁺, ESI pos.

Step 2: Methyl 4-(2-ethoxy-2-oxo-ethoxy)-2-fluoro-5-nitrobenzoate

To a solution of methyl 2-fluoro-4-hydroxy-5-nitrobenzoate (3.6 g, 16.7 mmol)
5 in tetrahydrofuran (55 ml) were added triphenylphosphine (5.26 g, 20.1 mmol) and ethyl glycolate (CAS 623-50-7, 2.61 g, 25.1 mmol) followed by diethyl azodicarboxylate (4.37 g, 25.1 mmol) at 0 °C (dropwise addition). The reaction mixture was allowed to reach 20 °C and was stirred for 16 hours. The mixture was poured into water and extracted three times with ethyl acetate. The combined organic layer was dried with Na₂SO₄, filtered and evaporated. The crude
10 product was purified via flash chromatography (silica gel, ethyl acetate/petroleum ether 0-30 %) to give a solid. The solid was triturated with ethyl acetate/petroleum ether 1:20, filtered and dried to afford methyl 4-(2-ethoxy-2-oxo-ethoxy)-2-fluoro-5-nitro-benzoate (5.3 g, > 100 % yield) as a white solid. MS m/z: 324.0 [M+Na]⁺, ESI pos.

Step 3: Methyl 7-fluoro-3-oxo-4*H*-1,4-benzoxazine-6-carboxylate

15 To a solution of methyl 4-(2-ethoxy-2-oxo-ethoxy)-2-fluoro-5-nitrobenzoate (4.8 g, 15.9 mmol) in acetic acid (50 ml) was added iron (2.68 g, 47.8 mmol) and the mixture was stirred at 90 °C for 2 hours. The reaction mixture was concentrated and ethyl acetate was added. After stirring overnight, the mixture was filtered. The filtrate was carefully poured into saturated aqueous NaHCO₃ (pH > 7) and extracted three times with ethyl acetate. The combined organic layer was
20 dried over Na₂SO₄ to give the product as a yellow solid (1.2 g, 33 % yield). The filter cake was treated with 1M HCl (60 mL) and stirred at 20 °C for 15 min, then the mixture was filtered. The filtrate was lyophilized to give methyl 7-fluoro-3-oxo-4*H*-1,4-benzoxazine-6-carboxylate (2.6 g, 72 % yield) as an off-white solid. MS m/z: 225.9 [M+H]⁺, ESI pos.

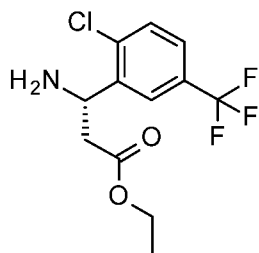
Step 4: Methyl 7-fluoro-3,4-dihydro-2*H*-1,4-benzoxazine-6-carboxylate

25 To a solution of methyl 7-fluoro-3-oxo-4*H*-1,4-benzoxazine-6-carboxylate (2.5 g, 11.1 mmol) in tetrahydrofuran (40 ml) was added dropwise a solution of 1M BH₃·THF (44.4 ml, 44.4 mmol) in tetrahydrofuran at 0 °C. The reaction mixture was stirred at 20 °C for 18 hours. The reaction mixture was carefully poured into saturated aqueous NH₄Cl (300 ml) and stirred at 20 °C for 1 hour, then the mixture was extracted with three times with ethyl acetate. The combined organic
30 layer was dried over Na₂SO₄ and evaporated. The crude product was purified via flash chromatography (silica gel, ethyl acetate/petroleum ether 0-30 %) to give methyl 7-fluoro-3,4-

dihydro-2H-1,4-benzoxazine-6-carboxylate (1.55 g, 65 % yield) as a yellow oil. MS m/z: 212 [M+H]⁺, ESI pos.

SYNTHESIS OF INTERMEDIATES B

Intermediate B1: Ethyl (3*S*)-3-amino-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate



5

Step 1: *N*-[[2-Chloro-5-(trifluoromethyl)phenyl]methylene]-2-methyl-propane-2-sulfonamide

To a clear solution of 2-chloro-5-(trifluoromethyl)benzaldehyde (CAS 82386-89-8, 30.6 g, 147 mmol) in dry tetrahydrofuran (306 ml) was added (*R*)-2-methylpropane-2-sulfonamide (+) (CAS 196929-78-9, 17.8 g, 146 mmol). The solution was cooled to 0 °C then titanium (IV) ethoxide (CAS 3087-36-3, 100 g, 93 ml, 440 mmol) was added dropwise maintaining the temperature below 5 °C. The mixture was allowed to warm to room temperature and was stirred for 1.5 hours. For the work-up, about 10 g of dicalite were added and the mixture was stirred for 5 minutes. The solution was poured into a stirred mixture of brine and ice. More dicalite was added and the suspension was stirred for further 30 minutes at room temperature. The solids were filtered, washed with tetrahydrofuran and water. The solvent was removed under reduced pressure and the residue was extracted with ethyl acetate three times. The combined organic layer was washed with water and brine, dried over MgSO₄, filtered and evaporated. The crude product was purified by flash chromatography (silica gel, ethyl acetate in heptane, 5-50 %) to afford (*NZ*)-*N*-[[2-chloro-5-(trifluoromethyl)phenyl]methylene]-2-methyl-propane-2-sulfonamide (36.7 g, 80 % yield) as a white waxy solid. MS m/z: 312.0 [M+H]⁺, ESI pos.

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Step 2: Ethyl (3*S*)-3-(*tert*-butylsulfonylamino)-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate

To a suspension of activated zinc (31.4 g, 480 mmol) in dry tetrahydrofuran (200 ml) was added cuprous chloride (CAS 7758-89-6, 9.55 g, 96.5 mmol) under nitrogen. The resulting suspension was stirred at 65 °C for 45 minutes then was cooled to 45 °C before ethyl bromoacetate (CAS 105-36-2, 17.9 ml, 160 mmol) was added dropwise so that the reaction. The mixture started to reflux again and the temperature was maintained between 49 °C - 53 °C (completed within 35

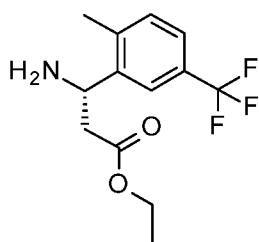
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minutes) The reaction mixture was stirred at 50 °C for further 15 minutes. The dark reaction mixture was cooled down to 0 °C before a solution of *N*-[[2-chloro-5-(trifluoromethyl)phenyl]methylene]-2-methyl-propane-2-sulfonamide (25 g, 80.2 mmol) in dry tetrahydrofuran (150 ml) was added within 25 minutes. After stirring at 0 °C for two additional hours, the mixture was poured into water. The suspension was stirred for 10 minutes, the solid was filtered off. The filtrate was evaporated and the residue was extracted three times with ethyl acetate. The combined organic layer was washed with sodium bicarbonate and brine, dried over MgSO₄ and evaporated. The crude product was purified by flash chromatography (silica gel, tert-butyl methyl ether in heptane, 10-60 %) to afford ethyl (3*S*)-3-(*tert*-butylsulfonylamino)-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate (27.3 g, 81 % yield) as a light yellow viscous oil. MS *m/z*: 400.1 [M+H]⁺, ESI pos.

Step 3: Ethyl (3*S*)-3-amino-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate

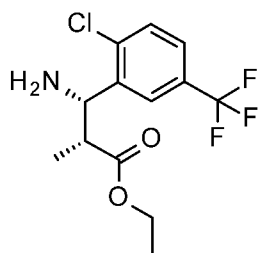
To a cooled solution of ethyl (3*S*)-3-(*tert*-butylsulfonylamino)-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate (15.2 g, 38 mmol) in tetrahydrofuran (180 ml) was added dropwise over 30 minutes at 0 °C a solution of 4 M hydrogen chloride in 1,4-dioxane (38 ml, 152 mmol). The reaction mixture was stirred for 2 hours at room temperature. The solvents were removed *in vacuo* and the residue was diluted with ethyl acetate and washed with a solution of saturated Na₂CO₃. The aqueous phase was then extracted twice with ethyl acetate and the combined organic layer was successively washed with water and brine, dried over MgSO₄, filtered and evaporated. The crude product was purified by flash chromatography (silica gel, tert-butyl methyl ether in heptane, 0-40 %) to afford ethyl (3*S*)-3-amino-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate (9.2 g, 78 % yield) as a colorless liquid. MS *m/z*: 297.1 [M+H]⁺, ESI pos.

Intermediate B2: Ethyl (3*S*)-3-amino-3-[2-methyl-5-(trifluoromethyl)phenyl]propanoate



The title compound was prepared in analogy to **Intermediate B1** starting from 2-methyl-5-(trifluoromethyl)benzaldehyde (CAS 886498-85-7) instead of 2-chloro-5-(trifluoromethyl)benzaldehyde in step 1. Light yellow liquid, MS *m/z*: 276.2 [M+H]⁺, ESI pos.

Intermediate B3: Ethyl (2*R*,3*S*)-3-amino-3-[2-chloro-5-(trifluoromethyl)phenyl]-2-methyl-propanoate



Step 1: (2*R*,3*S*)-3-(tert-Butylsulfinylamino)-3-[2-chloro-5-(trifluoromethyl)phenyl]-2-methyl-propionic acid ethyl ester

The title compound was prepared in analogy to **Intermediate B1, step2** using (*NZ*)-*N*-[[2-chloro-5-(trifluoromethyl)phenyl]methylene]-2-methyl-propane-2-sulfonamide instead of (*NZ*)-*N*-[[2-chloro-5-(trifluoromethyl)phenyl]methylene]-2-methyl-propane-2-sulfonamide and 2-bromopropionic acid ethyl ester instead of ethyl bromoacetate.

- 10 The diastereomers were separated by chiral SFC (column: chiral OZ-H, 5 μ m, 250 x 20 mm, co-solvent 5 % EtOH/Heptane 1/1)

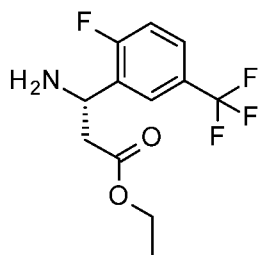
First eluting diastereomer: (2*R*,3*S*)-3-(tert-butylsulfinylamino)-3-[2-chloro-5-(trifluoromethyl)phenyl]-2-methyl-propionic acid ethyl ester. Colorless oil. MS *m/z*: 414.1 [M+H]⁺, ESI pos., final absolute assignment was made by later analysis of derivatives.

- 15 Second eluting diastereomer: (2*S*,3*S*)-3-(tert-butylsulfinylamino)-3-[2-chloro-5-(trifluoromethyl)phenyl]-2-methyl-propionic acid ethyl ester. Colorless oil. MS *m/z*: 414.2 [M+H]⁺, ESI pos., final absolute assignment was made by later analysis of derivatives.

Step 2: Ethyl (2*R*,3*S*)-3-amino-3-[2-chloro-5-(trifluoromethyl)phenyl]-2-methyl-propanoate

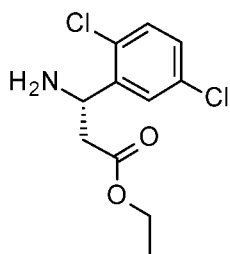
- 20 The title compound was prepared in analogy to **Intermediate B1, step 3** starting from (2*R*,3*S*)-3-(tert-butylsulfinylamino)-3-[2-chloro-5-(trifluoromethyl)phenyl]-2-methyl-propionic acid ethyl ester instead of (3*S*)-3-[[(*R*)-tert-butylsulfinyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid ethyl ester. Colorless oil. MS *m/z*: 309.6 [M+H]⁺, ESI pos.

Intermediate B4: Ethyl (3*S*)-3-amino-3-[2-fluoro-5-(trifluoromethyl)phenyl]propanoate



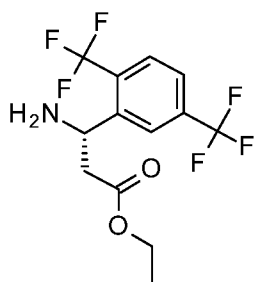
The title compound was prepared in analogy to **Intermediate B1** starting from 2-fluoro-5-(trifluoromethyl)benzaldehyde (146137-78-2) instead of 2-chloro-5-(trifluoromethyl)benzaldehyde in step 1. Light yellow oil. MS m/z: 280.1 [M+H]⁺, ESI pos.

5 **Intermediate B5:** Ethyl (3S)-3-amino-3-(2,5-dichlorophenyl)propanoate



The title compound was prepared in analogy to **Intermediate B1** starting from 2,5-dichlorobenzaldehyde (CAS 6361-23-5) instead of 2-chloro-5-(trifluoromethyl)benzaldehyde in step 1. Colorless oil. MS m/z: 263.1 [M+H]⁺, ESI pos.

10 **Intermediate B6:** Ethyl (3S)-3-amino-3-[2,5-bis(trifluoromethyl)phenyl]propanoate



Step 1: (R)-N-[[2,5-Bis(trifluoromethyl)phenyl]methylene]-2-methyl-propane-2-sulfinamide

To a solution of 2,5-bis(trifluoromethyl)benzaldehyde (CAS 395-64-2, 3.41 ml, 20.7 mmol) in dichloroethane (50 ml) was added (R)-(+)-2-methyl-2-propanesulfinamide (CAS 196929-78-9, 5.0 g, 41.3 mmol) and CuSO₄ (4.94 mg, 31 mmol). The reaction mixture was stirred at 50 °C for 64 hours. The reaction mixture was filtered, the filter cake was washed three times with dichloromethane. The filtrate was evaporated and was purified via flash chromatography (silica

gel, ethyl acetate/petroleum ether, 0-20 %) to give ((*NE,R*)-*N*-[[2,5-bis(trifluoromethyl)phenyl]methylene]-2-methyl-propane-2-sulfinamide (4.3 g, 57 % yield) as a yellow solid. MS *m/z*: 346.1 [M+H]⁺, ESI pos.

Step 2: Ethyl (3*S*)-3-[2,5-bis(trifluoromethyl)phenyl]-3-[[(*R*)-tert-

5 butylsulfinyl]amino]propanoate

To a suspension of activated zinc (8.14 g, 124 mmol) in tetrahydrofuran (80 ml) under nitrogen was slowly added chloro(trimethyl)silane (1.58 ml, 12.5 mmol) at 20 °C. After stirring at 50 °C for 1 hour, ethyl bromoacetate (2 ml, 18.7 mmol) was slowly added. The reaction mixture was stirred at 50 °C for an additional hour then was cooled down to 0 °C. A solution of *N*-[[2,5-bis(trifluoromethyl)phenyl]methylene]-2-methyl-propane-2-sulfinamide (2.86 ml, 12.4 mmol) in 10 tetrahydrofuran (20 ml) was slowly added. The reaction mixture was carefully poured into saturated aqueous NH₄Cl and was stirred at 0 °C for 15 minutes. The suspension was filtered and the filtrate was extracted three times with ethyl acetate. The combined organic layer was dried with Na₂SO₄, filtered and evaporated. The crude product was purified via flash chromatography 15 (silica gel, ethyl acetate/petroleum ether, 0-50 %) to give ethyl (3*S*)-3-[2,5-bis(trifluoromethyl)phenyl]-3-[[(*R*)-tert-butylsulfinyl]amino]propanoate (4.5 g, 75 % yield) as a light yellow oil. MS *m/z*: 434.2 [M+H]⁺, ESI pos.

Step 3: Ethyl (3*S*)-3-amino-3-[2,5-bis(trifluoromethyl)phenyl]propanoate hydrochloride salt

To a solution of ethyl (3*S*)-3-[2,5-bis(trifluoromethyl)phenyl]-3-[[(*R*)-tert-butylsulfinyl]amino]propanoate (800 mg, 1.85 mmol) in ethanol (4 ml) was added a solution of 20 2 M HCl in ethyl acetate (8.0 ml, 16.0 mmol) at 0 °C. The reaction mixture was stirred at 25 °C for 2 hours and was concentrated under reduced pressure to give ethyl (3*S*)-3-amino-3-[2,5-bis(trifluoromethyl)phenyl]propanoate hydrochloride salt (660 mg, 84% yield) as a colorless oil. MS *m/z*: 330.1 [M+H]⁺, ESI pos.

25 **SYNTHESIS OF INTERMEDIATES C**

The intermediates **C1** to **C20** were obtained from intermediates **A**.

General procedure for intermediates C

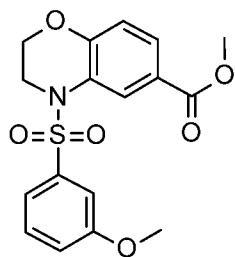
Step 1: Sulfonamide formation

Method a1: To a solution of **Intermediate C** in dichloromethane (0.2-0.3 M) were added 30 triethylamine (3 eq) and *N,N*-dimethylpyridin-4-amine (0.1 eq). The appropriate

benzenesulfonyl chloride (1.5 eq) dissolved in dichloromethane (0.2-0.3 M) was added dropwise at room temperature over 5 minutes. The reaction mixture was stirred for 2-16 hours, then was quenched by addition of water, and extracted twice with dichloromethane. The combined organic phase was washed with a solution of 1 M HCl and with brine, dried over MgSO_4 or Na_2SO_4 ,
 5 filtered and concentrated *in vacuo*. The crude product was purified by flash chromatography using silica gel with a gradient of ethyl acetate/heptane (usually ethyl acetate in heptane 0-50 %) to yield the targeted compound.

Method a2: To a solution of **Intermediate C** in dichloromethane (0.1-0.2 M) and pyridine (5-10 eq) was added dropwise the appropriate benzenesulfonyl chloride (1-1.5 eq) at 0 °C. The
 10 mixture was stirred with cooling or not for 2-16 hours, then was quenched by addition of water, and extracted with dichloromethane. The organic phase was washed with saturated aqueous sodium bicarbonate. The solution was dried over Na_2SO_4 , and concentrated *in vacuo*. When needed, the crude product was purified by flash chromatography using silica gel with a gradient of ethyl acetate/heptane (usually ethyl acetate in heptane 0-50 %) to yield the targeted
 15 compound.

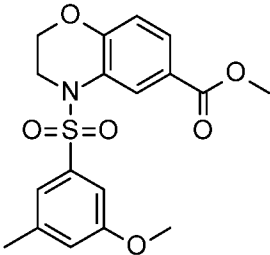
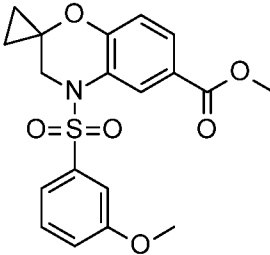
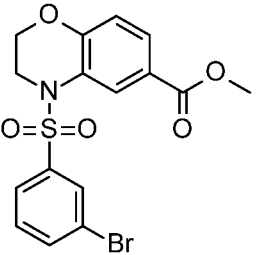
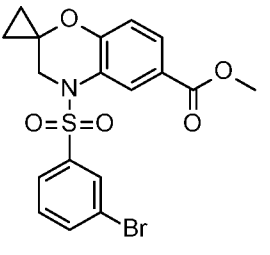
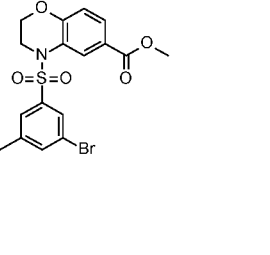
Intermediate C1: Methyl 4-(3-methoxyphenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate

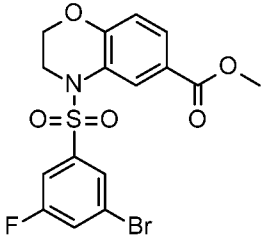
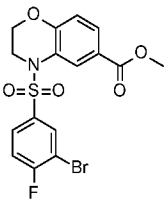
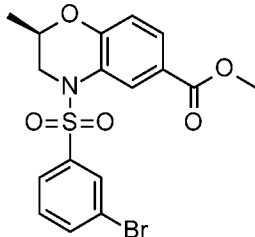
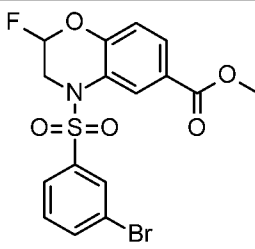
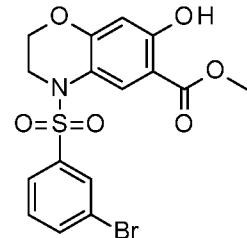


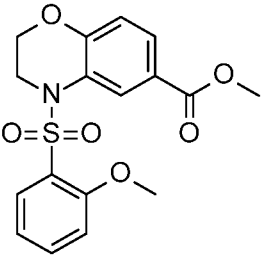
The title intermediate was prepared using **Method a1** starting with methyl 3,4-dihydro-2H-1,4-benzoxazine-6-carboxylate (**Intermediate A1**, 600 mg, 3.1 mmol) and 3-methoxybenzenesulfonyl chloride (CAS 10130-74-2, 1 g, 4.7 mmol). Methyl 4-(3-methoxyphenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate

(9.2 g, 78 % yield) was obtained as an off-white powder (947 mg, 82 % yield). MS m/z : 364.2 $[\text{M}+\text{H}]^+$, ESI pos.

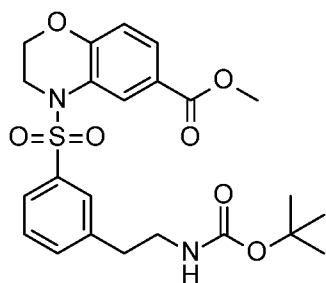
Interme diate	Name	Structure	Reac tant1	Reactant2	Methods

C2	methyl 4-(3-methoxy-5-methyl-phenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate		A1	3-methoxy-5-methylbenzenesulfonyl chloride (CAS 861560-21-6)	a2
C3	methyl 4-(3-methoxyphenyl)sulfonylspiro[3H-1,4-benzoxazine-2,1'-cyclopropane]-6-carboxylate		A2	3-methoxybenzenesulfonyl chloride (CAS 10130-74-2)	a2
C4	methyl 4-(3-bromophenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate		A1	3-bromobenzenesulfonyl chloride (CAS 2905-24-0)	a2
C5	methyl 4-(3-bromophenyl)sulfonylspiro[3H-1,4-benzoxazine-2,1'-cyclopropane]-6-carboxylate		A2	3-bromobenzenesulfonyl chloride (CAS 2905-24-0)	a2
C6	methyl 4-(3-bromo-5-methylphenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate		A1	3-bromo-5-methylbenzenesulfonyl chloride (CAS 885520-33-2)	a1

C7	methyl 4-(3-bromo-5-fluorophenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate		A1	3-bromo-5-fluorobenzene-1-sulfonyl chloride CAS (1214342-44-5)	a2
C8	methyl 4-(3-bromo-4-fluorophenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate		A1	3-bromo-4-fluorobenzenesulphonyl chloride (CAS 631912-19-1)	a2
C9	methyl (2<i>R</i>)-4-(3-bromophenyl)sulfonyl-2-methyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate		A3	3-bromobenzenesulfonyl chloride (CAS 2905-24-0)	a2
C10	methyl 4-(3-bromophenyl)sulfonyl-2-fluoro-2,3-dihydro-1,4-benzoxazine-6-carboxylate		A6	3-bromobenzenesulfonyl chloride (CAS 2905-24-0)	a2
C14	methyl 4-(3-bromophenyl)sulfonyl-7-hydroxy-2,3-dihydro-1,4-benzoxazine-6-carboxylate		A7	3-bromobenzenesulfonyl chloride (CAS 2905-24-0)	a2

C20	methyl 4-(2-methoxyphenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate		A1	2-methoxybenzenesulfonyl chloride (CAS 10130-87-7)	a2
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Intermediate C11: Methyl 4-[3-[2-(tert-butoxycarbonylamino)ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate



5 Step1: *tert*-Butyl *N*-[2-(3-benzylsulfonylphenyl)ethyl]carbamate

To a solution of *tert*-butyl 3-bromophenethylcarbamate (CAS 153732-25-3, 9.5 g, 31.65 mmol) in 1,4-dioxane (150 ml) were added Xantphos (1.83 g, 3.16 mmol), Pd₂(dba)₃ (1.45 g, 1.58 mmol), DIPEA (10.5 ml, 63.3 mmol) under nitrogen and finally benzyl mercaptan (5.57 ml, 47.5 mmol). The reaction mixture was stirred at 100 °C for 16 hours.

- 10 The mixture was concentrated under reduced pressure and the crude product was purified via flash column chromatography (silica gel, ethyl acetate/petroleum ether, 20-50 %) to give *tert*-butyl *N*-[2-(3-benzylsulfonylphenyl)ethyl]carbamate (11.5 g, 77 % yield) as an orange oil. MS *m/z*: 244.1 [M-100+H]⁺, ESI pos.

Step 2: *Tert*-Butyl *N*-[2-(3-chlorosulfonylphenyl)ethyl]carbamate

- 15 To a mixture of *tert*-butyl *N*-[2-(3-benzylsulfonylphenyl)ethyl]carbamate (1.5 g, 4.37 mmol) in a mixture of acetic acid (24 ml) and water (6 ml) was added portionwise *N*-chlorosuccinimide (2.34 g, 17.5 mmol) at 0 °C. The reaction mixture was stirred at 20 °C for 2 hours. The mixture was poured into water and extracted twice with ethyl acetate. The combined organic layer was dried with Na₂SO₄, filtered and evaporated. The crude product was purified via flash

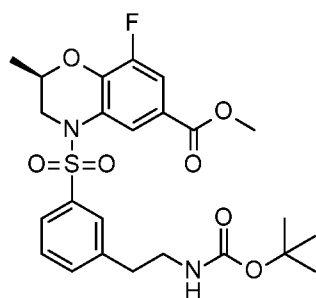
chromatography (silica gel, ethyl acetate/petroleum ether 0-50 %) to give *tert*-butyl *N*-[2-(3-chlorosulfonylphenyl)ethyl]carbamate (1.4 g, 100 % yield) as a colorless oil.

¹H NMR (400 MHz, CHLOROFORM-*d*) δ = 7.95 - 7.90 (m, 1H), 7.87 (s, 1H), 7.64 - 7.53 (m, 2H), 7.38 (d, *J* = 4.8 Hz, 1H), 3.42 (q, *J* = 6.8 Hz, 2H), 2.95 (br t, *J* = 6.8 Hz, 2H), 1.44 (s, 9H)

5 Step 3: Methyl 4-[3-[2-(*tert*-butoxycarbonylamino)ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate

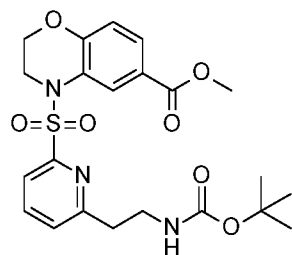
To a solution of methyl 3,4-dihydro-2*H*-1,4-benzoxazine-6-carboxylate (0.7 g, 3.62 mmol) and pyridine (1.47 ml, 18.1 mmol) in dichloromethane (20 ml) was added slowly a solution of *tert*-butyl *N*-[2-(3-chlorosulfonylphenyl)ethyl]carbamate (1.7 g, 5.32 mmol) in dichloromethane at
10 0 °C. The reaction mixture was stirred at 20 °C for 16 hours. The reaction mixture was concentrated, the residue was diluted with ethyl acetate (30 ml) and washed successively with a solution of 1 N HCl, saturated NaHCO₃ and brine, dried over Na₂SO₄, filtered and concentrated under reduced pressure. The crude product was purified via flash chromatography (silica gel, ethyl acetate/petroleum ether, 0-50 %) to give methyl 4-[3-[2-(*tert*-
15 butoxycarbonylamino)ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate (1.6 g, 91 % yield) as a yellow oil. MS *m/z*: 377.1 [M-BOC+H]⁺, ESI pos.

Intermediate C12: Methyl (2*R*)-4-[3-[2-(*tert*-butoxycarbonylamino)ethyl]phenyl]sulfonyl-8-fluoro-2-methyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate



20 The title compound was prepared in analogy to **Intermediate C11 step 3** using ethyl (2*R*)-8-fluoro-2-methyl-3,4-dihydro-2*H*-1,4-benzoxazine-6-carboxylate (**Intermediate A5**) instead of methyl 3,4-dihydro-2*H*-1,4-benzoxazine-6-carboxylate (**Intermediate A1**), yellow oil, MS *m/z*: 409.0 [M-BOC+H]⁺, ESI pos.

Intermediate C13: Methyl 4-[[6-[2-(*tert*-butoxycarbonylamino)ethyl]-2-pyridyl]sulfonyl]-2,3-dihydro-1,4-benzoxazine-6-carboxylate
25



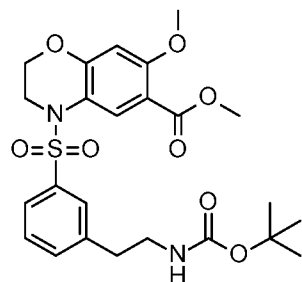
Step 1: Methyl 4-[(6-bromo-2-pyridyl)sulfonyl]-2,3-dihydro-1,4-benzoxazine-6-carboxylate

The title compound was prepared in analogy to **Intermediate C4** using 6-bromopyridine-2-sulfonyl chloride (CAS 912934-77-1) instead of 3-bromobenzenesulfonyl chloride (CAS 2905-24-0). Yellow solid, MS m/z: 415 [M+2+H]⁺, ESI pos.

Step 2: Methyl 4-[[6-[2-(tert-butoxycarbonylamino)ethyl]-2-pyridyl]sulfonyl]-2,3-dihydro-1,4-benzoxazine-6-carboxylate

To a solution of methyl 4-[(6-bromo-2-pyridyl)sulfonyl]-2,3-dihydro-1,4-benzoxazine-6-carboxylate (480 mg, 1.16 mmol) in 1,4-dioxane (10 ml) and water (1 ml) were added potassium carbonate (481 mg, 3.48 mmol), Pd(dppf)Cl₂ (85 mg, 0.12 mmol) and potassium (2-((tert-butoxycarbonyl)amino)ethyl)trifluoroborate (CAS 926280-83-3, 350 mg, 1.39 mmol). The reaction mixture was stirred at 100 °C for 4 hours under nitrogen. The solvents were evaporated and the residue was purified by flash chromatography (silica gel, ethyl acetate/petroleum ether 20-40%) to give methyl 4-[[6-[2-(tert-butoxycarbonylamino)ethyl]-2-pyridyl]sulfonyl]-2,3-dihydro-1,4-benzoxazine-6-carboxylate (650 mg, >100 % yield) as a yellow solid. MS m/z: 378.1[M-BOC+H]⁺, ESI pos.

Intermediate C15: Methyl 4-[3-[2-(tert-butoxycarbonylamino)ethyl]phenyl]sulfonyl-7-methoxy-2,3-dihydro-1,4-benzoxazine-6-carboxylate



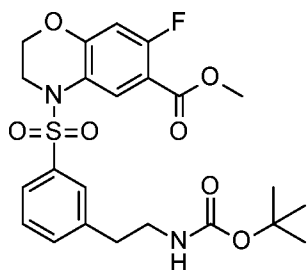
Step 1: Methyl 4-(3-bromophenyl)sulfonyl-7-methoxy-2,3-dihydro-1,4-benzoxazine-6-carboxylate

To a solution of methyl 4-(3-bromophenyl)sulfonyl-7-hydroxy-2,3-dihydro-1,4-benzoxazine-6-carboxylate (**Intermediate C14**, 270 mg, 0.63 mmol) in N,N-Dimethylformamide (10 ml) was added methyl iodide (134 mg, 0.95 mmol) and potassium carbonate (174 mg, 1.26 mmol). The reaction mixture was stirred at 25 °C for 16 hours. The mixture was poured into water and
5 extracted three times with ethyl acetate. The combined organic layer was dried with Na₂SO₄, filtered and evaporated. The crude product was purified via flash chromatography (silica gel, ethyl acetate/petroleum ether 0-50 %) to give methyl 4-(3-bromophenyl)sulfonyl-7-methoxy-2,3-dihydro-1,4-benzoxazine-6-carboxylate (300 mg, >100 % yield) as a white solid. MS m/z: 444.0 [M+H]⁺, ESI pos.

10 Step 2: Methyl 4-[3-[2-(tert-butoxycarbonylamino)ethyl]phenyl]sulfonyl-7-methoxy-2,3-dihydro-1,4-benzoxazine-6-carboxylate

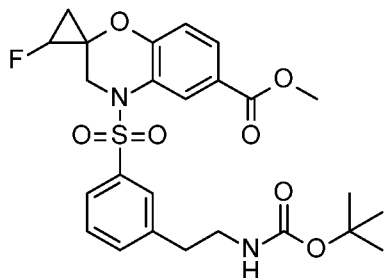
To a solution of methyl 4-(3-bromophenyl)sulfonyl-7-methoxy-2,3-dihydro-1,4-benzoxazine-6-carboxylate (260 mg, 0.59 mmol) in 1,4-dioxane (10 ml) and water (1 ml) were added potassium carbonate (243 mg, 1.76 mmol), Pd(dppf)Cl₂ (43.01 mg, 0.06 mmol) and potassium 2-(tert-
15 butoxycarbonylamino)ethyl-trifluoro-boranuide (CAS 926280-83-3, 221 mg, 0.88 mmol). The reaction mixture was stirred at 80 °C for 16 hours under nitrogen. The reaction mixture was filtered and the filtrate was concentrated under reduced pressure. The crude product was purified via flash chromatography (silica gel, ethyl acetate/petroleum ether 0-50 %) to give methyl 4-[3-[2-(tert-butoxycarbonylamino)ethyl]phenyl]sulfonyl-7-methoxy-2,3-dihydro-1,4-benzoxazine-6-
20 carboxylate (200 mg, 62 % yield) as a colorless oil. MS m/z: 529.2 [M+Na]⁺, ESI pos.

Intermediate C16: Methyl 4-[3-[2-(tert-butoxycarbonylamino)ethyl]phenyl]sulfonyl-7-fluoro-2,3-dihydro-1,4-benzoxazine-6-carboxylate



The title compound was prepared in analogy to **Intermediate C11 step 3** using methyl 7-fluoro-
25 3,4-dihydro-2H-1,4-benzoxazine-6-carboxylate (**Intermediate A8**) instead of methyl 3,4-dihydro-2H-1,4-benzoxazine-6-carboxylate (**Intermediate A1**), colorless oil. MS m/z: 395.0 [M-BOC+H]⁺, ESI pos.

Intermediate C17: Methyl 4-[3-[2-(tert-butoxycarbonylamino)ethyl]phenyl]sulfonyl-2'-fluoro-spiro[3H-1,4-benzoxazine-2,1'-cyclopropane]-6-carboxylate



Step 1: Methyl 4-hydroxy-3-(prop-2-ynylamino)benzoate

- 5 To a solution of methyl 3-amino-4-hydroxybenzoate (CAS 536-25-4, 5.0 g, 29.9 mmol) in *N,N*-dimethylformamide (50 ml) were added propargyl bromide (4.45 g, 29.9 mmol), potassium carbonate (8.25 g, 59.8 mmol) and potassium iodide (0.25 g, 1.5 mmol). The mixture was stirred at 25 °C for 16 hours. The mixture was poured into water and extracted three times with ethyl acetate. The combined organic layer was dried with Na₂SO₄, filtered and evaporated. The crude
- 10 product was purified via flash chromatography (silica gel, ethyl acetate/petroleum ether 20-30 %) to give methyl 4-hydroxy-3-(prop-2-ynylamino)benzoate (3.0 g, 47 % yield) as an orange solid. MS *m/z*: 206.1 [M+H]⁺, ESI pos.

Step 2: Methyl 2-methylene-3,4-dihydro-1,4-benzoxazine-6-carboxylate

- To a solution of methyl 4-hydroxy-3-(prop-2-ynylamino)benzoate (2.15 g, 10.5 mmol) in
- 15 trimethylamine (40.0 ml, 10.48 mmol) was added CuI (399 mg, 2.1 mmol). The reaction mixture was stirred at 20 °C for 30 minutes and at 100 °C for 2.5 hours under nitrogen. The mixture was poured into water and extracted three times with ethyl acetate. The combined organic layer was dried with Na₂SO₄, filtered and evaporated. The crude product was purified via flash
- 20 chromatography (silica gel, ethyl acetate/petroleum ether 20-40 %) to methyl 2-methylene-3,4-dihydro-1,4-benzoxazine-6-carboxylate (1.4 g, 64 % yield) as a yellow solid. MS *m/z*: 206.1 [M+H]⁺, ESI pos.

Step 3: Methyl 4-(3-bromophenyl)sulfonyl-2-methylene-3H-1,4-benzoxazine-6-carboxylate

- To a solution of methyl 2-methylene-3,4-dihydro-1,4-benzoxazine-6-carboxylate (1.4 g, 6.82 mmol) in dichloromethane (14 ml) and pyridine (2.76 ml, 34.1 mmol) was added 3-
- 25 bromobenzenesulfonyl chloride (2.61 g, 10.2 mmol) at 0-10°C. The mixture was stirred at 25 °C for 16 hours. The mixture was poured into water and extracted three times with dichloromethane.

The combined organic layer was dried with Na₂SO₄, filtered and evaporated. The crude product was purified via flash chromatography (silica gel, ethyl acetate/petroleum ether 20-50 %) to give methyl 4-(3-bromophenyl)sulfonyl-2-methylene-3*H*-1,4-benzoxazine-6-carboxylate (2.6 g, 90 % yield) as a white solid. MS m/z: 424 [M+H]⁺, ESI pos.

5 Step 4: Methyl 1'-bromo-4-(3-bromophenyl)sulfonyl-1'-fluoro-spiro[3*H*-1,4-benzoxazine-2,2'-cyclopropane]-6-carboxylate

To a solution of methyl 4-(3-bromophenyl)sulfonyl-2-methylene-3*H*-1,4-benzoxazine-6-carboxylate (2.0 g, 4.71 mmol) in dichloromethane (40 ml) were added benzyltriethylammonium chloride (0.52 g, 1.89 mmol), dibromofluoromethane (4.52 g, 23.6 mmol) and sodium hydroxide
10 (4.53 g, 56.6 mmol). The reaction mixture was stirred at 25 °C for 16 hours. The mixture was poured into water and extracted three times with ethyl acetate. The combined organic layer was washed with brine, dried with Na₂SO₄, filtered and evaporated. The crude product was purified by prep-HPLC (Phenomenex Luna C18, 150 x 25mm x 10 µm, gradient water with 2.25 % formic acid and acetonitrile) to give methyl 1'-bromo-4-(3-bromophenyl)sulfonyl-1'-fluoro-
15 spiro[3*H*-1,4-benzoxazine-2,2'-cyclopropane]-6-carboxylate (700 mg, 27 % yield) as a white solid. MS m/z: 536 [M+H]⁺, ESI pos.

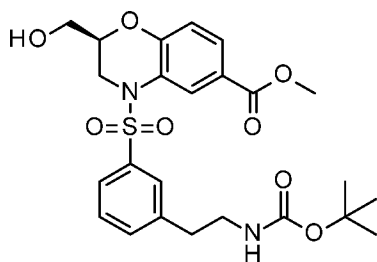
Step 5: Methyl 1'-bromo-4-[3-[2-(*tert*-butoxycarbonylamino)ethyl]phenyl]sulfonyl-1'-fluoro-spiro[3*H*-1,4-benzoxazine-2,2'-cyclopropane]-6-carboxylate

To a solution of methyl 1'-bromo-4-(3-bromophenyl)sulfonyl-1'-fluoro-spiro[3*H*-1,4-
20 benzoxazine-2,2'-cyclopropane]-6-carboxylate (600 mg, 1.12 mmol) in toluene (10 ml) and water (3 ml) were added Cs₂CO₃ (305 mg, 2.24 mmol), Pd(dppf)Cl₂ (82 mg, 0.11 mmol) and (2-((*tert*-butoxycarbonyl)amino)ethyl)trifluoroborate (CAS 926280-83-3, 422 mg, 1.68 mmol). The reaction mixture was stirred at 80 °C for 16 hours. The mixture was poured into water and extracted three times with ethyl acetate. The combined organic layer was dried with Na₂SO₄,
25 filtered and evaporated. The crude product was purified by flash chromatography (silica gel, ethyl acetate/petroleum ether 20-40 %) to give methyl 1'-bromo-4-[3-[2-(*tert*-butoxycarbonylamino)ethyl]phenyl]sulfonyl-1'-fluoro-spiro[3*H*-1,4-benzoxazine-2,2'-cyclopropane]-6-carboxylate (300 mg, 38 % yield) as a yellow solid. MS m/z: 501.1 [M+H]⁺, ESI pos.

30 Step 6: Methyl 4-[3-[2-(*tert*-butoxycarbonylamino)ethyl]phenyl]sulfonyl-2'-fluoro-spiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carboxylate

To a solution of methyl 1'-bromo-4-[3-[2-(tert-butoxycarbonylamino)ethyl]phenyl]sulfonyl-1'-fluoro-spiro[3*H*-1,4-benzoxazine-2,2'-cyclopropane]-6-carboxylate (560 mg, 0.93 mmol) in ethanol (11 ml) were added ammonium chloride (198 mg, 3.74 mmol) and zinc (1.19 g, 18.7 mmol). The reaction mixture was stirred at 25 °C for 16 hours. The reaction mixture was filtered, the filter cake was washed with ethanol and the filtrate was evaporated to give methyl 4-[3-[2-(tert-butoxycarbonylamino)ethyl]phenyl]sulfonyl-2'-fluoro-spiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carboxylate (600 mg, > 100 % yield) as a white solid. MS *m/z*: 421.1 [M-BOC+H]⁺, ESI pos.

- 10 **Intermediate C18:** Methyl (2*S*)-4-[3-[2-(*tert*-butoxycarbonylamino)ethyl]phenyl]sulfonyl-2-(hydroxymethyl)-2,3-dihydro-1,4-benzoxazine-6-carboxylate



Step1: Methyl 3-[[3-[2-(tert-butoxycarbonylamino)ethyl]phenyl]sulfonylamino]-4-fluoro-benzoate

- 15 To a mixture of methyl 3-amino-4-fluoro-benzoate (CAS 369-26-6, 400 mg, 2.4 mmol) and *tert*-butyl *N*-[2-(3-chlorosulfonylphenyl)ethyl]carbamate (1.13g, 3.5 mmol) in dichloromethane (10 ml) was added pyridine (0.96 ml, 11.8 mmol) and the mixture was stirred at 25 °C for 16 hours. The mixture was poured into water and extracted twice with ethyl acetate. The combined organic layer was dried with Na₂SO₄, filtered and evaporated. The crude product was purified via flash chromatography (silica gel, ethyl acetate/petroleum ether 0-60 %) to give methyl 3-[[3-[2-(tert-butoxycarbonylamino)ethyl]phenyl]sulfonylamino]-4-fluoro-benzoate (900 mg, 76 % yield as a light yellow solid. MS *m/z*: 475.2 [M+Na]⁺, ESI pos.

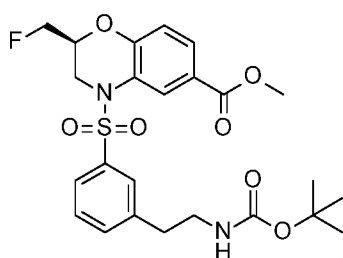
Step 2: Methyl (2*S*)-4-[3-[2-(tert-butoxycarbonylamino)ethyl]phenyl]sulfonyl-2-(hydroxymethyl)-2,3-dihydro-1,4-benzoxazine-6-carboxylate

- 25 To a solution of methyl 3-[[3-[2-(tert-butoxycarbonylamino)ethyl]phenyl]sulfonylamino]-4-fluoro-benzoate (900 mg, 1.99 mmol) in *N*-methyl-2-pyrrolidone (24 ml) was added (*R*)-(+)-glycidol (354 mg, 4.77 mmol) and potassium carbonate (412 mg, 2.98 mmol). The reaction

mixture was stirred at 140 °C for 2 hours. The mixture was poured into water and extracted four times with ethyl acetate. The combined organic layer was washed with brine, dried with Na₂SO₄, filtered and evaporated. The crude product was purified via flash chromatography (silica gel, ethyl acetate/petroleum ether 20-50 %) to give methyl (2*S*)-4-[3-[2-(tert-

5 butoxycarbonylamino)ethyl]phenyl]sulfonyl-2-(hydroxymethyl)-2,3-dihydro-1,4-benzoxazine-6-carboxylate (150 mg, 10 % yield) as a yellow oil. MS *m/z*: 529.2 [M+Na]⁺, ESI pos.

Intermediate C19: Methyl (2*S*)-4-[3-[2-(tert-butoxycarbonylamino)ethyl]phenyl]sulfonyl-2-(fluoromethyl)-2,3-dihydro-1,4-benzoxazine-6-carboxylate



10 Step 1: Methyl 3-[(3-bromophenyl)sulfonylamino]-4-fluorobenzoate

The title compound was prepared in analogy to **Intermediate C18, step 1** using 3-bromobenzenesulfonyl chloride (CAS 2905-24-0) instead of *tert*-butyl *N*-[2-(3-chlorosulfonylphenyl)ethyl]carbamate. Yellow solid, MS *m/z*: 390 [M+2+H]⁺, ESI pos.

15 Step 2: Methyl (2*S*)-4-(3-bromophenyl)sulfonyl-2-(hydroxymethyl)-2,3-dihydro-1,4-benzoxazine-6-carboxylate

The title compound was prepared in analogy to **Intermediate C18, step 2** using methyl 3-[(3-bromophenyl)sulfonylamino]-4-fluoro-benzoate instead of methyl 3-[[3-[2-(tert-butoxycarbonylamino)ethyl]phenyl]sulfonylamino]-4-fluoro-benzoate. Yellow oil. MS *m/z*: 466 [M+2+Na]⁺, ESI pos.

20 Step 3: methyl (2*S*)-4-(3-bromophenyl)sulfonyl-2-(fluoromethyl)-2,3-dihydro-1,4-benzoxazine-6-carboxylate

To a solution of methyl (2*S*)-4-(3-bromophenyl)sulfonyl-2-(hydroxymethyl)-2,3-dihydro-1,4-benzoxazine-6-carboxylate (1.9 g, 4.3 mmol) in dichloromethane (40 ml) was added diethylaminosulfur trifluoride (CAS 38078-09-0, 0.57 ml, 4.3 mmol) at -78 °C under nitrogen.

25 The reaction mixture was stirred at -78 °C for 1 hour then more diethylaminosulfur trifluoride (CAS 38078-09-0, 0.57 ml, 4.3 mmol) was added and the reaction mixture was stirred at -78 °C

for an additional hour and finally at 0 °C for 1 hour. The reaction mixture was poured into a cold saturated aqueous solution of NaHCO₃ (50ml) until the pH was above 7 at 0 °C and was extracted three times with dichloromethane. The combined organic layer was washed with brine, dried with Na₂SO₄, filtered and evaporated. The crude product was purified via flash chromatography (silica gel, ethyl acetate/petroleum ether 0-20%) to give methyl (2*S*)-4-(3-bromophenyl)sulfonyl-2-(fluoromethyl)-2,3-dihydro-1,4-benzoxazine-6-carboxylate (450 mg, 18 % yield) as a yellow oil. MS m/z: 446 [M+2+H]⁺, ESI pos.

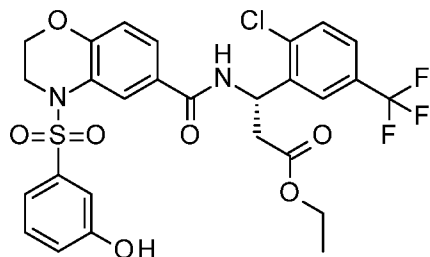
Step 4: Methyl (2*S*)-4-[3-[2-(*tert*-butoxycarbonylamino)ethyl]phenyl]sulfonyl-2-(fluoromethyl)-2,3-dihydro-1,4-benzoxazine-6-carboxylate

- 10 To a solution of methyl-(2*S*)-4-(3-bromophenyl)sulfonyl-2-(fluoromethyl)-2,3-dihydro-1,4-benzoxazine-6-carboxylate (450 mg, 1.01 mmol) in toluene (3 ml) and water (1 ml) were added Cs₂CO₃ (276 mg, 2.03 mmol), Pd(dppf)Cl₂ (74 mg, 0.1 mmol) and potassium 2-(*tert*-butoxycarbonylamino)ethyl-trifluoro-boranuide (280 mg, 1.11 mmol). The reaction mixture was stirred at 80° C for 16 hours under nitrogen. The reaction mixture was filtered and concentrated.
- 15 The crude product was purified via flash chromatography (silica gel, ethyl acetate/petroleum ether 20-60%) to give methyl-(2*S*)-4-[3-[2-(*tert*-butoxycarbonylamino)ethyl]phenyl]sulfonyl-2-(fluoromethyl)-2,3-dihydro-1,4-benzoxazine-6-carboxylate (340 mg, 54 % yield) as a yellow solid. MS m/z: 409.1 [M-BOC+H]⁺, ESI pos.

20 **SYNTHESIS OF INTERMEDIATES D**

The intermediates **D1** to **D21** using appropriate intermediates **A** and **C**.

Intermediate D1: Ethyl (3*S*)-3-[2-chloro-5-(trifluoromethyl)phenyl]-3-[[4-(3-hydroxyphenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate



- 25 Step 1: 4-(3-Methoxyphenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid

To a solution of 4-(3-methoxyphenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid methyl ester (**Intermediate C1**, 2.75 g, 7.57 mmol) in a mixture of methanol (15 ml) and tetrahydrofuran (29 ml) was added dropwise a solution of 4 M sodium hydroxide (5.68 ml, 22.7 mmol) at room temperature. The reaction was stirred overnight. The mixture was concentrated in *vacuo* and acidified with a solution of 4 M sulfuric acid. The free acid was recovered by filtration, and the remaining solvent was evaporated to dryness to afford 4-(3-methoxyphenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid (2.42 g, 89 % yield) as a white powder MS m/z: 350.1 [M+H]⁺, ESI pos.

Step 2: (3S)-3-[2-Chloro-5-(trifluoromethyl)phenyl]-3-[[4-(3-methoxyphenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propionic acid ethyl ester

To a solution of 4-(3-methoxyphenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid (800 mg, 2.29 mmol) in dry *N,N*-dimethylformamide (7.21 ml), HATU (958 mg, 2.52 mmol) was added. The mixture was left to stir for 10 minutes, after which (3S)-3-amino-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid ethyl ester (**Intermediate B1**, 745 mg, 2.52 mmol) and DIPEA (1.2 ml, 6.87 mmol) were added. The reaction was left to stir for 16 hours. LC/MS showed poor conversion, therefore additional HATU (958 mg, 2.52 mmol) and DIPEA (1.2 ml, 6.87 mmol) were added and the mixture was stirred for another 3 hours. The reaction mixture was extracted twice with ethyl acetate. The combined organics were washed with brine, dried over Na₂SO₄, filtered and concentrated in *vacuo*. The crude product was purified by flash chromatography (silica gel, 40 g, ethyl acetate in heptane, 0-50 %) to yield (3S)-3-[2-chloro-5-(trifluoromethyl)phenyl]-3-[[4-(3-methoxyphenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propionic acid ethyl ester (1.32 g, 92 %) as a white foam MS m/z: 627.2 [M+H]⁺, ESI pos.

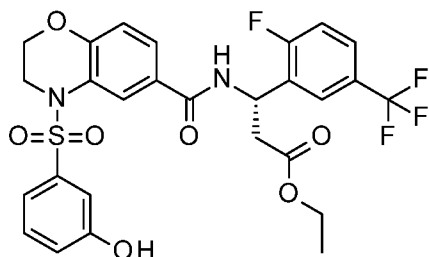
Step 3: (3S)-3-[2-Chloro-5-(trifluoromethyl)phenyl]-3-[[4-(3-hydroxyphenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propionic acid ethyl ester

To a solution of methyl 4-(3-methoxyphenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate (**Intermediate C1**, 1 g, 1.6 mmol) in dichloromethane (8.6 ml), a solution of 1 M BBr₃ in dichloromethane (8.0 ml, 8.0 mmol) was added dropwise at 0 °C. The reaction was stirred for 2 hours, and then was quenched by adding the reaction mixture dropwise on a 20 ml ice-cold saturated NaHCO₃ solution, with stirring. The mixture was extracted twice using ethyl acetate. The combined organic phase was washed with brine, dried over Na₂SO₄, filtered, and concentrated in *vacuo*. The crude product was purified via flash chromatography (silica gel, 40 g,

ethyl acetate in heptane 0-50 %), and concentrated *in vacuo* to yield ethyl (3*S*)-3-[2-chloro-5-(trifluoromethyl)phenyl]-3-[[4-(3-hydroxyphenyl)sulfonyl]-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate (282 mg, 86 %) as a white foam MS *m/z*: 613.1 [M+H]⁺, ESI pos.

Intermediate D2: Ethyl (3*S*)-3-[2-fluoro-5-(trifluoromethyl)phenyl]-3-[[4-(3-

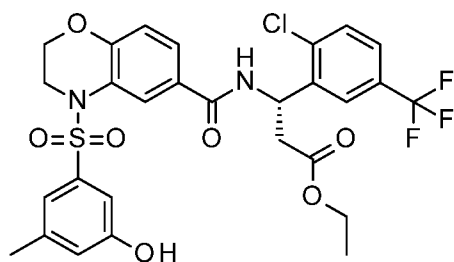
5 hydroxyphenyl)sulfonyl]-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate



The title compound was prepared in analogy to **Intermediate D1** using ethyl (3*S*)-3-amino-3-[2-fluoro-5-(trifluoromethyl)phenyl]propanoate (**Intermediate B4**) instead of ethyl (3*S*)-3-amino-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate (**Intermediate B1**)

10 in step 2, off-white powder, MS *m/z*: 597.2 [M+H]⁺, ESI pos.

Intermediate D3: Ethyl (3*S*)-3-[2-chloro-5-(trifluoromethyl)phenyl]-3-[[4-(3-hydroxy-5-methylphenyl)sulfonyl]-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate



Step 1: 4-(3-Methoxy-5-methyl-phenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid

15 To a mixture of methyl 4-(3-methoxy-5-methyl-phenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate (**Intermediate C2**, 950 mg, 2.5 mmol) in a mixture of tetrahydrofuran (5 ml), methanol (5 ml) and water (5 ml) was added lithium hydroxide monohydrate (317 mg, 7.6 mmol). The reaction mixture was stirred at 50 °C for 2 hours and was poured into water. A solution of 1N HCl was added until the pH was about 4 and the mixture was extracted three
20 times with ethyl acetate. The combined organic layer was dried over Na₂SO₄ and concentrated to

give crude 4-(3-methoxy-5-methyl-phenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid (900 mg, 89 % yield) as an off-white solid, MS m/z: 364.1 [M+Na+H]⁺, ESI pos.

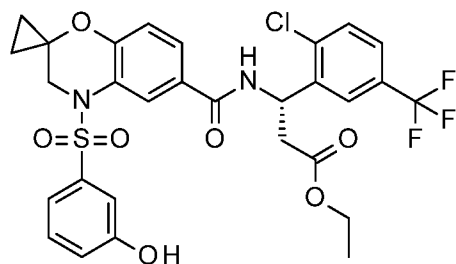
Step 2: 4-(3-hydroxy-5-methyl-phenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid

To a mixture of 4-(3-methoxy-5-methyl-phenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid (400 mg, 1.1 mmol) in dichloromethane (8 ml) was added dropwise at -70 °C a solution of 1 M BBr₃ in dichloromethane (0.53 ml, 5.5 mmol) in dichloromethane. The reaction mixture was stirred at -70 °C for 1 hour then was allowed to reach room temperature and stir for an additional hour. The reaction mixture was poured into water at 0 °C and was extracted with ethyl acetate three times. The combined organic layer was washed with brine and dried with Na₂SO₄ and evaporated. The crude product was purified by flash chromatography (silica gel, ethyl acetate/petroleum ether 30-100 %) to afford 4-(3-hydroxy-5-methyl-phenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid (370 mg, 81 % yield) as a yellow solid. MS m/z: 350.1 [M+H]⁺, ESI pos.

Step 3: Ethyl (3S)-3-[2-chloro-5-(trifluoromethyl)phenyl]-3-[[4-(3-hydroxy-5-methyl-phenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate

To a solution of ethyl (3S)-3-amino-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate (**Intermediate B1**, 313.2 mg, 1.06 mmol) and 4-(3-hydroxy-5-methyl-phenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid (370 mg, 1.06 mmol) in N,N-dimethylformamide (5 ml) was added DIPEA (0.55 ml, 3.18 mmol), then HATU (483 mg, 1.27 mmol) was added to the mixture. The reaction mixture was stirred at 20 °C for 16 hours. The reaction mixture was directly purified by reverse phase flash chromatography (Flash Spherical C18, 20-35µm; 100 Å; water-acetonitrile). The combined solution was concentrated and extracted three times with ethyl acetate. The combined organic layer was dried with Na₂SO₄, filtered and concentrated to afford ethyl (3S)-3-[2-chloro-5-(trifluoromethyl)phenyl]-3-[[4-(3-hydroxy-5-methyl-phenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate (320 mg, 47 % yield) as a white solid. MS m/z: 627.2 [M+H]⁺, ESI pos.

Intermediate D4: Ethyl (3S)-3-[2-chloro-5-(trifluoromethyl)phenyl]-3-[[4-(3-hydroxyphenyl)sulfonylspiro[3H-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]propanoate

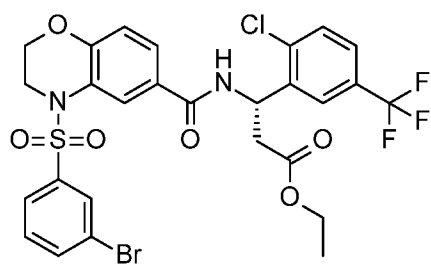


The title compound was prepared in analogy to **Intermediate D3** starting from methyl 4-(3-methoxyphenyl)sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carboxylate (

Intermediate C3) instead of methyl 4-(3-methoxy-5-methyl-phenyl)sulfonyl-2,3-dihydro-1,4-

- 5 benzoxazine-6-carboxylate (**Intermediate C2**) in step 1. Yellow solid, MS m/z : 639.1 $[M+H]^+$, ESI pos.

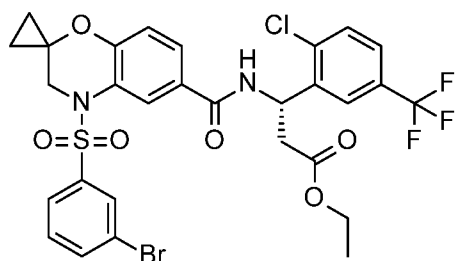
Intermediate D5: Ethyl (3*S*)-3-[[4-(3-bromophenyl)sulfonyl]-2,3-dihydro-1,4-benzoxazine-6-carbonylamino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate



- 10 The title compound was prepared in analogy to **Intermediate D1 step 1 and step 2** only, starting from methyl 4-(3-bromophenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate

(**Intermediate C4**) instead of methyl 4-(3-methoxy-5-methyl-phenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate (**Intermediate C2**) in step 1. Yellow solid, MS m/z : 677.0 $[M+H]^+$, ESI pos.

- 15 **Intermediate D6:** Ethyl (3*S*)-3-[[4-(3-bromophenyl)sulfonyl]spiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonylamino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate



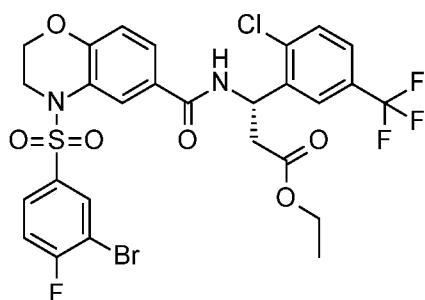
Step 1: 4-(3-Bromophenyl)sulfonylspiro[3H-1,4-benzoxazine-2,1'-cyclopropane]-6-carboxylic acid

To a solution of methyl 4-(3-bromophenyl)sulfonylspiro[3H-1,4-benzoxazine-2,1'-cyclopropane]-6-carboxylate (**Intermediate C5**, 2 g, 4.56 mmol) in methanol (9.25 ml) and tetrahydrofuran (18.5 ml), was added dropwise a solution of 4 M sodium hydroxide (3.42 ml, 13.7 mmol). The reaction was left to stir for 16 hours, after which LC/MS showed complete saponification. The mixture was acidified. The solid was filtered and dried to give 4-(3-bromophenyl)sulfonylspiro[3H-1,4-benzoxazine-2,1'-cyclopropane]-6-carboxylic acid (1.86 g, 96 % yield) as a white powder. MS m/z: 424.2 [M-H]⁻, ESI neg.

Step2: Ethyl (3S)-3-[[4-(3-bromophenyl)sulfonylspiro[3H-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate

To a solution of 4-(3-bromophenyl)sulfonylspiro[3H-1,4-benzoxazine-2,1'-cyclopropane]-6-carboxylic acid (1.66 g, 3.91 mmol) in dry N,N-dimethylformamide (15.65 ml) were added (3S)-3-amino-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid ethyl ester (**Intermediate B1**, 1.5 g, 5.09 mmol), DIPEA (2.02 g, 2.73 ml, 15.7 mmol), and HATU (2.23 g, 5.87 mmol). The reaction mixture was stirred for 3 hours, after which LC/MS showed complete conversion of the starting material. Water was added and the reaction mixture was extracted twice with ethyl acetate. The combined organic layer was washed with brine, dried over Na₂SO₄, and concentrated in vacuo. The crude product was purified by flash chromatography (silica gel, ethyl acetate in heptane 0-50 %) to yield ethyl (3S)-3-[[4-(3-bromophenyl)sulfonylspiro[3H-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate (2.88 g, 97 % yield) as an orange solid. MS m/z: 703.2 [M+H]⁺, ESI pos.

Intermediate D7: Ethyl (3S)-3-[[4-(3-bromo-4-fluoro-phenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate

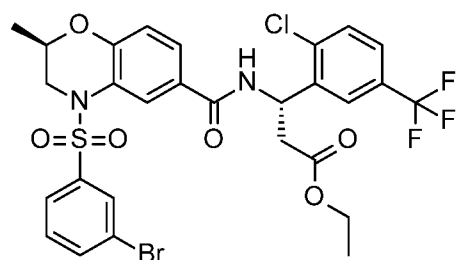


The title compound was prepared in analogy to **Intermediate D6** step 1 and 2 only, starting from methyl 4-(3-bromo-4-fluoro-phenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate

(**Intermediate C8**) instead of methyl 4-(3-methoxy-5-methyl-phenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate (**Intermediate C2**) in step 1. White solid, MS m/z: 695.1 [M+H]⁺,

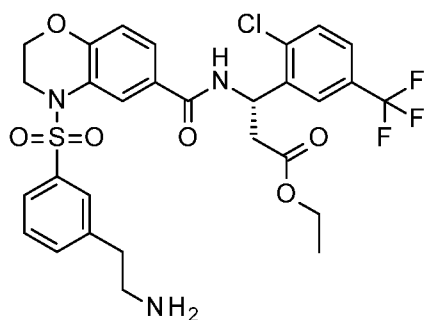
5 ESI pos.

Intermediate D8: Ethyl (3*S*)-3-[[*(2R)*-4-(3-bromophenyl)sulfonyl-2-methyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate



The title compound was prepared in analogy to **Intermediate D6**, starting from methyl (*2R*)-4-(3-bromophenyl)sulfonyl-2-methyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate (**Intermediate C9**) instead of methyl 4-(3-bromophenyl)sulfonylspiro[3H-1,4-benzoxazine-2,1'-cyclopropane]-6-carboxylate (**Intermediate C5**) in step 1. White solid, MS m/z: 691.0 [M+H]⁺, ESI pos.

Intermediate D9: Ethyl (3*S*)-3-[[4-[3-(2-aminoethyl)phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate



Step 1: 4-[3-[2-(*tert*-Butoxycarbonylamino)ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid

The title compound was prepared in analogy to **Intermediate D3**, step 1 using methyl 4-[3-[2-(*tert*-butoxycarbonylamino)ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate

20 (**Intermediate C11**) instead of methyl 4-(3-methoxy-5-methyl-phenyl)sulfonyl-2,3-dihydro-1,4-

benzoxazine-6-carboxylate (**Intermediate C2**) and stirring at 20 °C for 16 hours instead of 50 °C for 2 hours. Yellow oil. MS m/z: 363.1 [M-BOC+H]⁺, ESI pos.

Step 2: Ethyl (3S)-3-[[4-[3-[2-(tert-butoxycarbonylamino)ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate

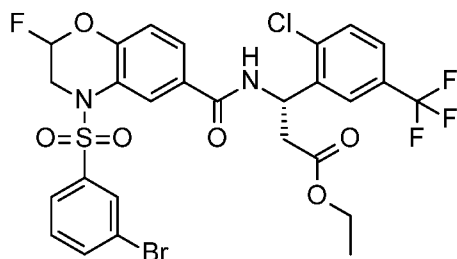
- 5 The title compound was prepared in analogy to **Intermediate D3, step 3** using 4-[3-[2-(tert-butoxycarbonylamino)ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid instead of 4-(3-hydroxy-5-methyl-phenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid. Yellow solid. MS m/z: 762.1 [M+H]⁺, ESI pos.

Step3: Ethyl -(3S)-3-[[4-[3-(2-aminoethyl)phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-

- 10 carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate 2,2,2-trifluoroacetic acid salt

To a solution of ethyl (3S)-3-[[4-[3-[2-(tert-butoxycarbonylamino)ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate (430.0 mg, 0.58 mmol) in dichloromethane (2 ml) was added 2,2,2-trifluoroacetic acid (2.0 ml, 0.58 mmol). The reaction mixture was stirred at 25 °C for 2 hours. The reaction mixture was concentrated and lyophilized to give (ethyl rac-(3S)-3-[[4-[3-(2-aminoethyl)phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate 2,2,2-trifluoroacetic acid salt. Yellow solid. MS m/z: 640.1 [M+H]⁺, ESI pos.

Intermediate D10: Ethyl (3S)-3-[[4-(3-bromophenyl)sulfonyl-2-fluoro-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate



Step 1: 4-(3-Bromophenyl)sulfonyl-2-fluoro-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid

The title compound was prepared in analogy to **Intermediate D3, step 1** using methyl 4-(3-bromophenyl)sulfonyl-2-fluoro-2,3-dihydro-1,4-benzoxazine-6-carboxylate (**Intermediate C10**) instead of methyl 4-(3-methoxy-5-methyl-phenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate (**Intermediate C2**) and stirring at 25 °C for 2 hours instead of 50 °C for 2 hours. Yellow oil.

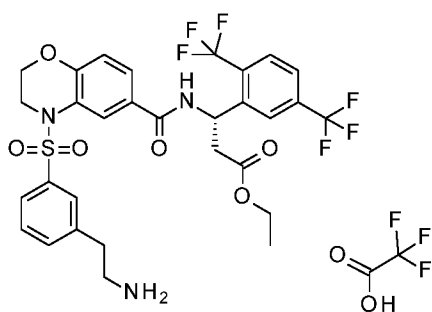
¹H NMR (400 MHz, DMSO-d₆) δ = 8.30 (d, *J* = 2.0 Hz, 1H), 7.95 - 7.83 (m, 2H), 7.71 (dt, *J* = 2.4, 5.4 Hz, 2H), 7.58 - 7.45 (m, 1H), 7.12 (d, *J* = 8.6 Hz, 1H), 6.47 - 6.24 (m, 1H), 4.61 (br dd, *J* = 3.2, 15.2 Hz, 1H), 3.88 - 3.68 (m, 2H)

Step 2: Ethyl (3*S*)-3-[[4-(3-bromophenyl)sulfonyl-2-fluoro-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate

To a solution of 4-(3-bromophenyl)sulfonyl-2-fluoro-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid (200 mg, 0.48 mmol) in N,N-dimethylformamide (4 ml) were added ethyl (3*S*)-3-amino-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate (**Intermediate A1**, 171 mg, 0.58 mmol), DIPEA (186 mg, 1.44 mmol) and HATU (219 mg, 0.58 mmol). The reaction mixture was stirred at 25 °C for 16 hours. The mixture was poured into water and extracted three times with ethyl acetate. The combined organic layer was dried with Na₂SO₄, filtered and evaporated. The crude product was purified via flash chromatography (silica gel, ethyl acetate/petroleum ether, 20-60 %) to give

(ethyl (3*S*)-3-[[4-(3-bromophenyl)sulfonyl-2-fluoro-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate (300 mg, 90 % yield) as a white solid. MS *m/z*: 694.9 [M+H]⁺, ESI pos.

Intermediate D11: Ethyl (3*S*)-3-[2,5-bis(trifluoromethyl)phenyl]-3-[[4-[3-[2-(tert-butoxycarbonylamino)ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate 2,2,2-trifluoroacetic acid salt



Step 1: 4-[3-[2-(tert-butoxycarbonylamino)ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid

The title compound was prepared in analogy to **Intermediate D3, step 1** using methyl 4-[3-[2-(tert-butoxycarbonylamino)ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate (**Intermediate C11**) instead of methyl 4-(3-methoxy-5-methyl-phenyl)sulfonyl-2,3-dihydro-1,4-

benzoxazine-6-carboxylate (**Intermediate C2**) and stirring at 20 °C for 16 hours instead of 50°C for 2 hours. Yellow oil. MS m/z: 363.1 [M-BOC+H]⁺, ESI pos.

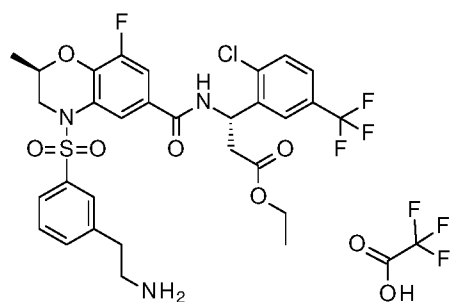
Step 2: Ethyl (3*S*)-3-[2,5-bis(trifluoromethyl)phenyl]-3-[[4-[3-[2-(tert-butoxycarbonylamino)ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate

The title compound was prepared in analogy to **Intermediate D3, step 3** using 4-[3-[2-(tert-butoxycarbonylamino)ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid and ethyl (3*S*)-3-amino-3-[2,5-bis(trifluoromethyl)phenyl]propanoate (**Intermediate B6**) instead of 4-(3-hydroxy-5-methyl-phenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid and ethyl (3*S*)-3-amino-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate (**Intermediate B1**). White solid. MS m/z: 774.5 [M+H]⁺, ESI pos.

Step 3: Ethyl (3*S*)-3-[[4-[3-(2-aminoethyl)phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2,5-bis(trifluoromethyl)phenyl]propanoate 2,2,2-trifluoroacetic acid

To a solution of ethyl (3*S*)-3-[2,5-bis(trifluoromethyl)phenyl]-3-[[4-[3-[2-(tert-butoxycarbonylamino)ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate (700 mg, 0.9 mmol) in dichloromethane (17.5 ml) was added 2,2,2-trifluoroacetic acid (2.5 ml, 0.9 mmol). The mixture was stirred at 25 °C for 1 hour. The reaction mixture was concentrated under reduced pressure and the residue was diluted with water and acetonitrile before lyophilization to give ethyl (3*S*)-3-[[4-[3-(2-aminoethyl)phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2,5-bis(trifluoromethyl)phenyl]propanoate 2,2,2-trifluoroacetic acid salt (800 mg, > 100 % yield) as a light yellow solid. MS m/z: 674.4 [M-BOC+H]⁺, ESI pos.

Intermediate D12: Ethyl (3*S*)-3-[(2*R*)-4-[3-(2-aminoethyl)phenyl]sulfonyl-8-fluoro-2-methyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate 2,2,2-trifluoroacetic acid salt



Step 1: (2R)-4-[3-[2-(tert-butoxycarbonylamino)ethyl]phenyl]sulfonyl-8-fluoro-2-methyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid

The title compound was prepared in analogy to **Intermediate D3, step 1** using methyl (2R)-4-[3-[2-(tert-butoxycarbonylamino)ethyl]phenyl]sulfonyl-8-fluoro-2-methyl-2,3-dihydro-1,4-

5 benzoxazine-6-carboxylate (**Intermediate C12**) instead of methyl 4-(3-methoxy-5-methyl-phenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate (**Intermediate C2**) and stirring at 25 °C for 3 hours instead of 50 °C for 2 hours. Yellow oil. MS m/z: 395.0 [M-BOC+H]⁺, ESI pos.

Step 2: Ethyl (3S)-3-[[4-[(2R)-4-[3-[2-(tert-butoxycarbonylamino)ethyl]phenyl]sulfonyl-2-methyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate

10

The title compound was prepared in analogy to **Intermediate D3, step 3** using (2R)-4-[3-[2-(tert-butoxycarbonylamino)ethyl]phenyl]sulfonyl-8-fluoro-2-methyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid instead of 4-(3-hydroxy-5-methyl-phenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid. Yellow solid. MS m/z: 672.2 [M-BOC+H]⁺, ESI pos.

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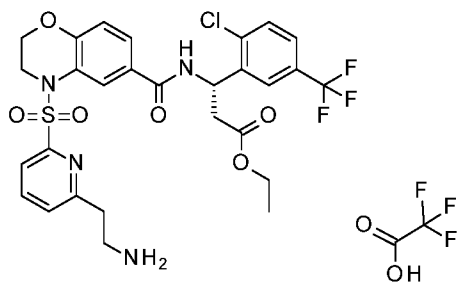
Step3: Ethyl (3S)-3-[[4-[3-(2-aminoethyl)phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2,5-bis(trifluoromethyl)phenyl]propanoate 2,2,2-trifluoroacetic acid

The title compound was prepared in analogy to **Intermediate D11, step 3** using ethyl (3S)-3-[[4-[(2R)-4-[3-[2-(tert-butoxycarbonylamino)ethyl]phenyl]sulfonyl-2-methyl-2,3-dihydro-1,4-

20 benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate instead of ethyl (3S)-3-[2,5-bis(trifluoromethyl)phenyl]-3-[[4-[3-[2-(tert-butoxycarbonylamino)ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate. Yellow solid. MS m/z: 672.1 [M+H]⁺, ESI pos.

Intermediate D13: Ethyl (3S)-3-[[4-[[6-(2-aminoethyl)-2-pyridyl]sulfonyl]-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate 2,2,2-trifluoroacetic acid salt

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Step 1: 4-[[6-[2-(*tert*-butoxycarbonylamino)ethyl]-2-pyridyl]sulfonyl]-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid

The title compound was prepared in analogy to **Intermediate D3, step 1** using methyl 4-[[6-[2-(*tert*-butoxycarbonylamino)ethyl]-2-pyridyl]sulfonyl]-2,3-dihydro-1,4-benzoxazine-6-carboxylate (**Intermediate C13**) instead of methyl 4-(3-methoxy-5-methyl-phenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate (**Intermediate C2**). White solid. MS *m/z*: 770.4 [M+H]⁺, ESI pos.

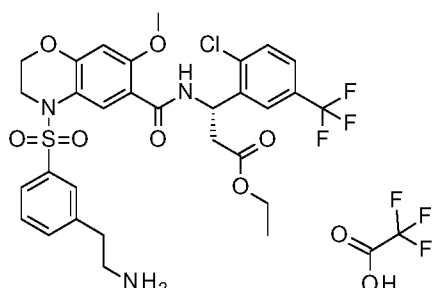
Step 2: Ethyl (3*S*)-3-[[4-[[6-[2-(*tert*-butoxycarbonylamino)ethyl]-2-pyridyl]sulfonyl]-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate

The title compound was prepared in analogy to **Intermediate D3, step 3** using 4-[[6-[2-(*tert*-butoxycarbonylamino)ethyl]-2-pyridyl]sulfonyl]-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid instead of 4-(3-hydroxy-5-methyl-phenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid and stirring at 25 °C for 1 hour instead of 50 °C for 2 hours. White solid. MS *m/z*: 641.1 [M-BOC+H]⁺, ESI pos.

Step3: Ethyl (3*S*)-3-[(2*S*)-4-[3-(2-aminoethyl)phenyl]sulfonyl-2-(hydroxymethyl)-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate 2,2,2-trifluoroacetic acid salt

The title compound was prepared in analogy to **Intermediate D11, step 3** using ethyl (3*S*)-3-[[4-[[6-[2-(*tert*-butoxycarbonylamino)ethyl]-2-pyridyl]sulfonyl]-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate instead of ethyl (3*S*)-3-[2,5-bis(trifluoromethyl)phenyl]-3-[[4-[3-[2-(*tert*-butoxycarbonylamino)ethyl]phenyl]sulfonyl]-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate. Yellow solid. MS *m/z*: 641.1 [M+H]⁺, ESI pos.

Intermediate D14: Ethyl (3*S*)-3-[[4-[3-(2-aminoethyl)phenyl]sulfonyl-7-methoxy-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate 2,2,2-trifluoroacetic acid salt



5 Step 1: 4-[3-[2-(*tert*-butoxycarbonylamino)ethyl]phenyl]sulfonyl-7-methoxy-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid

The title compound was prepared in analogy to **Intermediate D3, step 1** starting with methyl 4-[3-[2-(*tert*-butoxycarbonylamino)ethyl]phenyl]sulfonyl-7-methoxy-2,3-dihydro-1,4-benzoxazine-6-carboxylate (**Intermediate C15**) instead of methyl 4-(3-methoxy-5-methyl-phenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate (**Intermediate C2**) and stirring at
10 25°C for 18 hours instead of 50°C for 2 hours. Colorless oil. MS *m/z*: 515.3 [M+Na]⁺, ESI pos.

Step 2: Ethyl (3*S*)-3-[[4-[3-[2-(*tert*-butoxycarbonylamino)ethyl]phenyl]sulfonyl-7-methoxy-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate

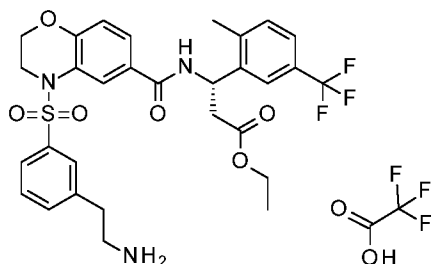
The title compound was prepared in analogy to **Intermediate D3, step 3** starting with 4-[3-[2-(*tert*-butoxycarbonylamino)ethyl]phenyl]sulfonyl-7-methoxy-2,3-dihydro-1,4-benzoxazine-6-
15 carboxylic acid instead of 4-(3-hydroxy-5-methyl-phenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid and stirring at 25°C for 2 hours instead of 20°C for 16 hours. White solid. MS *m/z*: 770.4 [M+H]⁺, ESI pos.

Step3: Ethyl (3*S*)-3-[[4-[3-(2-aminoethyl)phenyl]sulfonyl-7-methoxy-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate 2,2,2-trifluoroacetic acid salt

The title compound was prepared in analogy to **Intermediate D11, step 3** using ethyl (3*S*)-3-[[4-[3-[2-(*tert*-butoxycarbonylamino)ethyl]phenyl]sulfonyl-7-methoxy-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate instead of
25 ethyl (3*S*)-3-[2,5-bis(trifluoromethyl)phenyl]-3-[[4-[3-[2-(*tert*-

butoxycarbonylamino)ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate. Light yellow solid. MS m/z: 670.4 [M+H]⁺, ESI pos.

Intermediate D15: Ethyl (3*S*)-3-[[4-[3-(2-aminoethyl)phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-methyl-5-(trifluoromethyl)phenyl]propanoate 2,2,2-trifluoroacetic acid salt



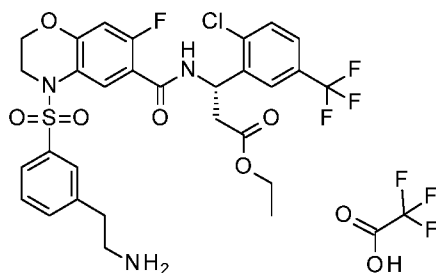
Step 1: Ethyl (3*S*)-3-[[4-[3-[2-(*tert*-butoxycarbonylamino)ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-methyl-5-(trifluoromethyl)phenyl]propanoate

The title compound was prepared in analogy to **Intermediate D3, step 3** using 4-[3-[2-(*tert*-butoxycarbonylamino)ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid and ethyl (3*S*)-3-amino-3-[2-methyl-5-(trifluoromethyl)phenyl]propanoate (**Intermediate B2**) instead of 4-(3-hydroxy-5-methyl-phenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid and ethyl (3*S*)-3-amino-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate (**Intermediate B1**). White solid. MS m/z: 774.5 [M+H]⁺, ESI pos.

Step 2: Ethyl (3*S*)-3-[[4-[3-(2-aminoethyl)phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-methyl-5-(trifluoromethyl)phenyl]propanoate 2,2,2-trifluoroacetic acid salt

The title compound was prepared in analogy to **Intermediate D11, step 3** using ethyl (3*S*)-3-[[4-[3-[2-(*tert*-butoxycarbonylamino)ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-methyl-5-(trifluoromethyl)phenyl]propanoate instead of ethyl (3*S*)-3-[2,5-bis(trifluoromethyl)phenyl]-3-[[4-[3-[2-(*tert*-butoxycarbonylamino)ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate. Light yellow solid. MS m/z: 620.2 [M+H]⁺, ESI pos.

Intermediate D16: Ethyl-(3*S*)-3-[[4-[3-(2-aminoethyl)phenyl]sulfonyl-7-fluoro-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate 2,2,2-trifluoroacetic acid salt



Step 1: 4-[3-[2-(*tert*-butoxycarbonylamino)ethyl]phenyl]sulfonyl-7-fluoro-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid

The title compound was prepared in analogy to **Intermediate D3, step 1** starting methyl 4-[3-[2-(*tert*-butoxycarbonylamino)ethyl]phenyl]sulfonyl-7-fluoro-2,3-dihydro-1,4-benzoxazine-6-carboxylate (**Intermediate C16**) instead of methyl 4-(3-methoxy-5-methyl-phenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate (**Intermediate C2**) and stirring at 25 °C for 2 hours instead of 50 °C for 2 hours. Colorless oil. MS m/z: 381.0 [M-BOC+H]⁺, ESI pos.

Step 2: Ethyl-(3*S*)-3-[[4-[3-[2-(*tert*-butoxycarbonylamino)ethyl]phenyl]sulfonyl-7-fluoro-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate

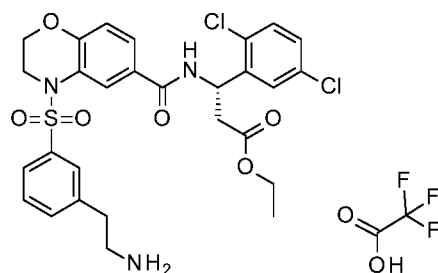
The title compound was prepared in analogy to **Intermediate D3, step 3** starting with 4-[3-[2-(*tert*-butoxycarbonylamino)ethyl]phenyl]sulfonyl-7-fluoro-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid instead of 4-(3-hydroxy-5-methyl-phenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid. White solid. MS m/z: 658.1 [M-BOC+H]⁺, ESI pos.

Step3: Ethyl-(3*S*)-3-[[4-[3-(2-aminoethyl)phenyl]sulfonyl-7-fluoro-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate 2,2,2-trifluoroacetic acid salt

The title compound was prepared in analogy to **Intermediate D11, step 3** using ethyl (3*S*)-3-[[4-[3-[2-(*tert*-butoxycarbonylamino)ethyl]phenyl]sulfonyl-7-fluoro-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate instead of ethyl (3*S*)-3-[2,5-bis(trifluoromethyl)phenyl]-3-[[4-[3-[2-(*tert*-butoxycarbonylamino)ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate. Yellow solid. MS m/z: 658.2 [M+H]⁺, ESI pos.

Intermediate D17: Ethyl -(3*S*)-3-[[4-[3-(2-aminoethyl)phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-(2,5-dichlorophenyl)propanoate 2,2,2-trifluoroacetic acid

salt



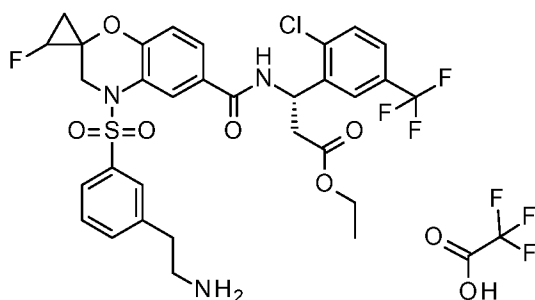
Step 1: Ethyl (3S)-3-[[4-[3-[2-(*tert*-butoxycarbonylamino)ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-methyl-5-(trifluoromethyl)phenyl]propanoate

- 5 The title compound was prepared in analogy to **Intermediate D3**, **step 3** using 4-[3-[2-(*tert*-butoxycarbonylamino)ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid and ethyl (3S)-3-amino-3-(2,5-dichlorophenyl)propanoate (**Intermediate B5**) instead of 4-(3-hydroxy-5-methyl-phenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid and ethyl (3S)-3-amino-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate (**Intermediate B1**). Yellow
10 solid. MS m/z: 706.3 [M+H]⁺, ESI pos.

Step 2: Ethyl-(3S)-3-[[4-[3-(2-aminoethyl)phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-(2,5-dichlorophenyl)propanoate 2,2,2-trifluoroacetic acid salt

- The title compound was prepared in analogy to **Intermediate D11**, **step 3** using ethyl (3S)-3-[[4-[3-[2-(*tert*-butoxycarbonylamino)ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-
15 carbonyl]amino]-3-[2-methyl-5-(trifluoromethyl)phenyl]propanoate instead of ethyl (3S)-3-[2,5-bis(trifluoromethyl)phenyl]-3-[[4-[3-[2-(*tert*-butoxycarbonylamino)ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate. Yellow solid. MS m/z: 606.2 [M+H]⁺, ESI pos.

- Intermediate D18:** Ethyl-(3S)-3-[[4-[3-(2-aminoethyl)phenyl]sulfonyl-2'-fluoro-spiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate 2,2,2-trifluoroacetic acid salt
20



Step 1: 4-[3-[2-(tert-butoxycarbonylamino)ethyl]phenyl]sulfonyl-2'-fluoro-spiro[3H-1,4-benzoxazine-2,1'-cyclopropane]-6-carboxylic acid

The title compound was prepared in analogy to **Intermediate D3, step 1** using methyl 4-[3-[2-(tert-butoxycarbonylamino)ethyl]phenyl]sulfonyl-2'-fluoro-spiro[3H-1,4-benzoxazine-2,1'-cyclopropane]-6-carboxylate (**Intermediate C17**) instead of methyl 4-(3-methoxy-5-methyl-phenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate (**Intermediate C2**). Yellow solid. MS m/z: 407.1 [M-BOC+H]⁺, ESI pos.

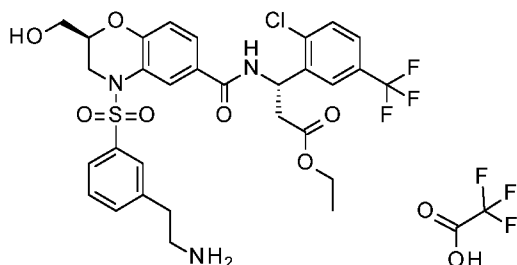
Step 5: Ethyl (3S)-3-[[4-[3-[2-(tert-butoxycarbonylamino)ethyl]phenyl]sulfonyl-2'-fluoro-spiro[3H-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate

The title compound was prepared in analogy to **Intermediate D3, step 3** using 4-[3-[2-(tert-butoxycarbonylamino)ethyl]phenyl]sulfonyl-2'-fluoro-spiro[3H-1,4-benzoxazine-2,1'-cyclopropane]-6-carboxylic acid instead of 4-(3-hydroxy-5-methyl-phenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid. White solid. MS m/z: 784.1 [M+H]⁺, ESI pos.

Step 6: Ethyl-(3S)-3-[[4-[3-(2-aminoethyl)phenyl]sulfonyl-2'-fluoro-spiro[3H-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate 2,2,2-trifluoroacetic acid salt

The title compound was prepared in analogy to **Intermediate D11, step 3** using ethyl (3S)-3-[[4-[3-[2-(tert-butoxycarbonylamino)ethyl]phenyl]sulfonyl-2'-fluoro-spiro[3H-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate instead of ethyl (3S)-3-[2,5-bis(trifluoromethyl)phenyl]-3-[[4-[3-[2-(tert-butoxycarbonylamino)ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate. Yellow solid. MS m/z: 684.1 [M+H]⁺, ESI pos.

Intermediate D19: Ethyl (3*S*)-3-[[*(2S)*-4-[3-(2-aminoethyl)phenyl]sulfonyl-2-(hydroxymethyl)-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate 2,2,2-trifluoroacetic acid salt



5 Step 1: (*2S*)-4-[3-[2-(*tert*-butoxycarbonylamino)ethyl]phenyl]sulfonyl-2-(hydroxymethyl)-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid

The title compound was prepared in analogy to **Intermediate D3, step 1** using methyl (*2S*)-4-[3-[2-(*tert*-butoxycarbonylamino)ethyl]phenyl]sulfonyl-2-(hydroxymethyl)-2,3-dihydro-1,4-benzoxazine-6-carboxylate (**Intermediate C18**) instead of methyl 4-(3-methoxy-5-methyl-phenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate (**Intermediate C2**) and stirring at
10 25 °C for 2 hours instead of 50 °C for 2 hours. White solid. MS *m/z*: 770.4 [M+H]⁺, ESI pos.

Step 2: Ethyl (3*S*)-3-[[*(2S)*-4-[3-[2-(*tert*-butoxycarbonylamino)ethyl]phenyl]sulfonyl-2-(hydroxymethyl)-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate

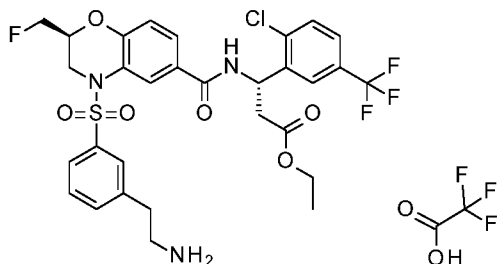
15 The title compound was prepared in analogy to **Intermediate D3, step 3** using (*2S*)-4-[3-[2-(*tert*-butoxycarbonylamino)ethyl]phenyl]sulfonyl-2-(hydroxymethyl)-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid instead of 4-(3-hydroxy-5-methyl-phenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid and stirring at 25 °C for 2 hours instead of 50 °C for 2 hours. Yellow solid. MS *m/z*: 672.2 [M-BOC+H]⁺, ESI pos.

20 Step 3: Ethyl (3*S*)-3-[[*(2S)*-4-[3-(2-aminoethyl)phenyl]sulfonyl-2-(hydroxymethyl)-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate 2,2,2-trifluoroacetic acid salt

The title compound was prepared in analogy to **Intermediate D11, step 3** using ethyl (3*S*)-3-[[*(2S)*-4-[3-[2-(*tert*-butoxycarbonylamino)ethyl]phenyl]sulfonyl-2-(hydroxymethyl)-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate instead of
25 ethyl (3*S*)-3-[2,5-bis(trifluoromethyl)phenyl]-3-[[4-[3-[2-(*tert*-

butoxycarbonylamino)ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate. White solid. MS m/z : 670.3 $[M+H]^+$, ESI pos.

Intermediate D20: Ethyl-(3*S*)-3-[2-chloro-5-(trifluoromethyl)phenyl]-3-[[*(2S)*-4-[3-(2-aminoethyl)phenyl]sulfonyl-2-(fluoromethyl)-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate 2,2,2-trifluoroacetic acid salt



Step 1: (*2S*)-4-[3-[2-(tert-butoxycarbonylamino)ethyl]phenyl]sulfonyl-2-(fluoromethyl)-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid

The title compound was prepared in analogy to **Intermediate D3, step 1** using methyl (*2S*)-4-[3-[2-(tert-butoxycarbonylamino)ethyl]phenyl]sulfonyl-2-(fluoromethyl)-2,3-dihydro-1,4-benzoxazine-6-carboxylate (**Intermediate C19**) instead of methyl 4-(3-methoxy-5-methylphenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate (**Intermediate C2**) and stirring at 25 °C for 16 hours instead of 50 °C for 2 hours. White solid. MS m/z : 395.1 $[M-BOC+H]^+$, ESI pos.

Step 2: Ethyl (3*S*)-3-[[*(2S)*-4-[3-[2-(tert-butoxycarbonylamino)ethyl]phenyl]sulfonyl-2-(fluoromethyl)-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate.

To a solution of (*2S*)-4-[3-[2-(tert-butoxycarbonylamino)ethyl]phenyl]sulfonyl-2-(fluoromethyl)-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid (300 mg, 0.61 mmol) in *N,N*-dimethylformamide (4 ml) were added DIPEA (0.24 ml, 1.82 mmol), ethyl (3*S*)-3-amino-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate (**Intermediate B1**, 197 mg, 0.67 mmol) and HATU (171 mg, 0.73 mmol). The reaction mixture was stirred at 25 °C for 2 hours.

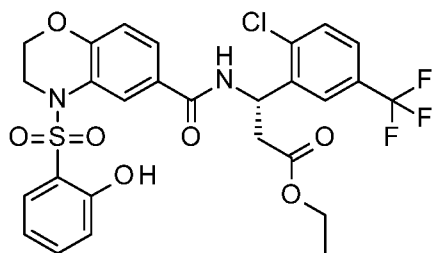
The reaction mixture was directly purified by reverse phase flash chromatography (Flash Spherical C18, 20-35 μ m; 100 Å; 40 g, gradient water with 0.1% TFA and acetonitrile) to ethyl (3*S*)-3-[[*(2S)*-4-[3-[2-(tert-butoxycarbonylamino)ethyl]phenyl]sulfonyl-2-(fluoromethyl)-2,3-

dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate (440 mg, 0.57 mmol, 92 % yield) as a yellow oil. MS m/z: 672.4 [M-BOC+H]⁺, ESI pos.

Step 3: Ethyl-(3*S*)-3-[2-chloro-5-(trifluoromethyl)phenyl]-3-[[2*S*)-4-[3-(2-aminoethyl)phenyl]sulfonyl-2-(fluoromethyl)-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate 2,2,2-trifluoroacetic acid salt

The title compound was prepared in analogy to **Intermediate D11, step 3** using ethyl (3*S*)-3-[[2*S*)-4-[3-[2-(tert-butoxycarbonylamino)ethyl]phenyl]sulfonyl-2-(fluoromethyl)-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate instead of ethyl (3*S*)-3-[2,5-bis(trifluoromethyl)phenyl]-3-[[4-[3-[2-(tert-butoxycarbonylamino)ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate. White solid. MS m/z: 672.4 [M+H]⁺, ESI pos.

Intermediate D21: Ethyl (3*S*)-3-[2-chloro-5-(trifluoromethyl)phenyl]-3-[[4-(2-hydroxyphenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate



Step 1: Methyl 4-(2-hydroxyphenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate

To a solution of methyl 4-(2-methoxyphenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate (200 mg, 0.55 mmol) in dichloromethane (2 ml) cooled to -70 °C was added dropwise a solution of 2 M BBr₃ in dichloromethane (1.38 ml, 2.75 mmol). The reaction mixture was stirred at -70 °C for 1 hour and was allowed to reach 0 °C and to stir for one additional hour. The reaction mixture was added dropwise to water at 0 °C and was stirred at 25 °C for 30 minutes. The mixture was poured into water and extracted three times with ethyl acetate. The combined organic layer was dried with Na₂SO₄, filtered and evaporated to give crude methyl 4-(2-hydroxyphenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate (200 mg, 77 % yield) as a yellow solid. MS m/z: 372.1 [M+Na+H]⁺, ESI pos.

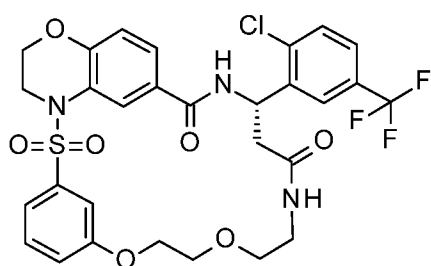
Step 2: 4-(2-hydroxyphenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid

The title compound was prepared in analogy to **Intermediate D3, step 1** using methyl 4-(2-hydroxyphenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate instead of methyl 4-(3-methoxy-5-methyl-phenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate (**Intermediate C2**) and stirring at 25 °C for 16 hours instead of 50 °C for 2 hours. White solid. MS m/z: 358 [M+Na]⁺, ESI pos.

Step 3: Ethyl (3*S*)-3-[2-chloro-5-(trifluoromethyl)phenyl]-3-[[4-(2-hydroxyphenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate

To a solution of 4-(2-hydroxyphenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid (300 mg, 0.89 mmol) in N,N-dimethylformamide (6 ml) were added ethyl (3*S*)-3-amino-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate (**Intermediate B1**, 317 mg, 1.07 mmol), DIPEA (231 mg, 1.79 mmol), HOBt (242 mg, 1.79 mmol) and finally 1-ethyl-3-(3-dimethylaminopropyl)carbodiimide (257 mg, 1.34 mmol). The reaction mixture was stirred at 25 °C for 1 hour. The reaction mixture was directly purified by reverse phase flash chromatography (Flash Spherical C18, 20-35µm; 100Å; gradient water with 0.05% formic acid and acetonitrile) to give ethyl (3*S*)-3-[2-chloro-5-(trifluoromethyl)phenyl]-3-[[4-(2-hydroxyphenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate (310 mg, 55 % yield) as a white solid. MS m/z: 613.1 [M+H]⁺, ESI pos.

Example 1: (17*S*)-17-[2-Chloro-5-(trifluoromethyl)phenyl]-2,2-diketo-8,11,24-trioxa-2λ⁶-thia-1,14,18-triazatetracyclo[18.6.2.13,7.0^{23,27}]nonacosa-3(29),4,6,20,22,27-hexaene-15,19-quinone



Step 1: (3*S*)-3-[[4-[3-[2-[2-(tert-Butoxycarbonylamino)ethoxy]ethoxy]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid ethyl ester

To a solution of ethyl (3*S*)-3-[2-chloro-5-(trifluoromethyl)phenyl]-3-[[4-(3-hydroxyphenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate (**Intermediate D1**, 279 mg, 455 µmol) in N,N-dimethylformamide (4.6 ml), potassium carbonate (186 mg, 1.4 mmol) and *N*-[2-(2-bromoethoxy)ethyl]carbamic acid *tert*-butyl ester

(CAS 165963-71-3, 254 mg, 948 μ mol) were added. The vial was sealed and the mixture stirred at 80 °C for 18 hours. The reaction mixture was extracted twice with TBME. The combined organics were washed with brine, dried over Na₂SO₄, and concentrated *in vacuo*. The crude product was purified by flash chromatography (silica gel, 25 g, ethyl acetate in heptane 0-50 %) to give (3*S*)-3-[[4-[3-[2-[2-(tert-butoxycarbonylamino)ethoxy]ethoxy]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid ethyl ester (227 mg, 57 % yield) as a yellow waxy solid. MS *m/z*: 700.3 [M+H-BOC]⁺, ESI pos.

Step 2: Ethyl (3*S*)-3-[[4-[3-[2-(2-aminoethoxy)ethoxy]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate hydrochloride salt.

To a solution of (3*S*)-3-[[4-[3-[2-[2-(tert-butoxycarbonylamino)ethoxy]ethoxy]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid ethyl ester (227 mg, 278 μ mol) in ethyl acetate (1.9 ml), was added dropwise a solution of 4M HCl in dioxane (826 μ l, 3.3 mmol) at 0 °C. The mixture was allowed to warm to room temperature and was stirred for 16 hours. The mixture was concentrated *in vacuo* to give crude (3*S*)-3-[[4-[3-[2-(2-aminoethoxy)ethoxy]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid ethyl ester (218 mg > 100 % yield) as a yellow foam. MS *m/z*: 700.4 [M+H]⁺, ESI pos.

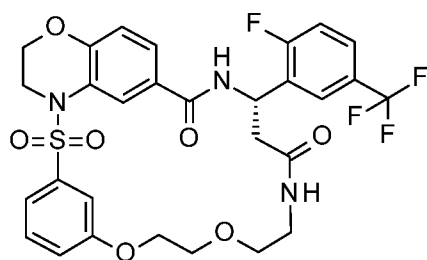
Step 3: (3*S*)-3-[[4-[3-[2-(2-Aminoethoxy)ethoxy]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid sodium salt.

To a solution of crude (3*S*)-3-[[4-[3-[2-(2-aminoethoxy)ethoxy]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-fluoro-5-(trifluoromethyl)phenyl]propionic acid ethyl ester (218 mg, 311 μ mol) in methanol (622 μ l) was added an aqueous solution of 4 M sodium hydroxide (311 μ l, 1.24 mmol) at room temperature. After stirring for 30 minutes, the mixture was concentrated *in vacuo* and residual water was removed by coevaporation with toluene (2 ml) to give (3*S*)-3-[[4-[3-[2-(2-aminoethoxy)ethoxy]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid sodium salt (265 mg, > 100 % yield) as an off-white solid. MS *m/z*: 672.2 [M+H]⁺, ESI pos.

Step 4: (17*S*)-17-[2-Chloro-5-(trifluoromethyl)phenyl]-2,2-diketo-8,11,24-trioxo-2 λ^6 -thia-1,14,18-triazatetracyclo[18.6.2.13,7.0.23,27]nonacosa-3(29),4,6,20,22,27-hexaene-15,19-quinone

To a solution of HATU (599.25 mg, 1.58 mmol) in dry N,N-dimethylformamide (4.0 ml) was added dropwise a suspension of crude (3*S*)-3-[[4-[3-[2-(2-aminoethoxy)ethoxy]phenyl]sulfonyl]-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid (265 mg, 394 μ mol) in dry N,N-dimethylformamide (0.9 ml) followed by DIPEA (274 μ l, 1.58 mmol). The reaction was stirred at room temperature for 18 hours, then was directly purified with preparative HPLC (YMC-Triart C18, 12 nm, 5 μ m, 100 x 30 mm, acetonitrile/water with 0.1 % formic acid) to yield (17*S*)-17-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-diketo-8,11,24-trioxa-2 λ^6 -thia-1,14,18-triazatetracyclo[18.6.2.13,7.0^{23,27}]nonacosa-3(29),4,6,20,22,27-hexaene-15,19-quinone (39.4 mg, 15 % yield) as a white lyophilized powder. MS *m/z*: 698.4 [M+H]⁺, ESI pos

Example 2: (17*S*)-17-[2-Fluoro-5-(trifluoromethyl)phenyl]-2,2-dioxo-8,11,24-trioxa-2 λ^6 -thia-1,14,18-triazatetracyclo[18.6.2.13,7.0^{23,27}]nonacosa-3,5,7(29),20(28),21,23(27)-hexaene-15,19-dione



Step 1: (3*S*)-3-[[4-[3-[2-[2-(tert-Butoxycarbonylamino)ethoxy]ethoxy]phenyl]sulfonyl]-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-fluoro-5-(trifluoromethyl)phenyl]propionic acid ethyl ester

The title compound was prepared in analogy to **Example 1, step 1** starting from ethyl (3*S*)-3-[2-fluoro-5-(trifluoromethyl)phenyl]-3-[[4-(3-hydroxyphenyl)sulfonyl]-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate (**Intermediate D2**) instead of ethyl (3*S*)-3-[2-chloro-5-(trifluoromethyl)phenyl]-3-[[4-(3-hydroxyphenyl)sulfonyl]-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate (**Intermediate D1**). White foam, MS *m/z*: 684.4 [M+H-BOC]⁺, ESI pos.

Step 2: (3*S*)-3-[[4-[3-[2-(2-Aminoethoxy)ethoxy]phenyl]sulfonyl]-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-fluoro-5-(trifluoromethyl)phenyl]propionic acid ethyl ester

The title compound was prepared in analogy to **Example 1, step 2** starting from (3*S*)-3-[[4-[3-[2-[2-(tert-butoxycarbonylamino)ethoxy]ethoxy]phenyl]sulfonyl]-2,3-dihydro-1,4-benzoxazine-6-

carbonyl]amino]-3-[2-fluoro-5-(trifluoromethyl)phenyl]propionic acid ethyl ester instead of (3*S*)-3-[[4-[3-[2-(2-(tert-butoxycarbonylamino)ethoxy)ethoxy]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid ethyl ester and stirring for 2 hours instead of 16 hours. White foam. MS *m/z*: 684.5 [M+H]⁺, ESI pos.

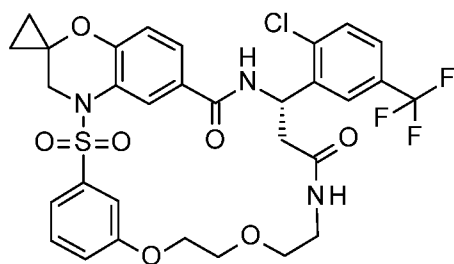
5 Step 3: (3*S*)-3-[[4-[3-[2-(2-Aminoethoxy)ethoxy]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-fluoro-5-(trifluoromethyl)phenyl]propionic acid

The title compound was prepared in analogy to **Example 1, step 3** starting from (3*S*)-3-[[4-[3-[2-(2-aminoethoxy)ethoxy]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-fluoro-5-(trifluoromethyl)phenyl]propionic acid ethyl ester instead of (3*S*)-3-[[4-[3-[2-(2-aminoethoxy)ethoxy]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-fluoro-5-(trifluoromethyl)phenyl]propionic acid ethyl ester. Off-white solid. MS *m/z*: 654.4 [M-H]⁻, ESI pos.

Step 4: (17*S*)-17-[2-Fluoro-5-(trifluoromethyl)phenyl]-2,2-diketo-8,11,24-trioxa-2λ⁶-thia-1,14,18-triazatetracyclo[18.6.2.13,7.0.23,27]nonacosa-3(29),4,6,20,22,27-hexaene-15,19-quinone

15 The title compound was prepared in analogy to **Example 1, step 4** starting from (3*S*)-3-[[4-[3-[2-(2-aminoethoxy)ethoxy]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-fluoro-5-(trifluoromethyl)phenyl]propionic acid instead (3*S*)-3-[[4-[3-[2-(2-aminoethoxy)ethoxy]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid and stirring for 6 hours instead of 18 hours. Off-white lyophilised powder. MS *m/z*: 638.2 [M+H]⁺, ESI pos.

Example 3: (17*S*)-17-[2-Chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-spiro[8,11,24-trioxa-2λ⁶-thia-1,14,18-triazatetracyclo[18.6.2.13,7.0.23,27]nonacosa-3,5,7(29),20(28),21,23(27)-hexaene-25,1'-cyclopropane]-15,19-dione



Step1: (3*S*)-3-[[4-[3-[2-[2-(tert-Butoxycarbonylamino)ethoxy]ethoxy]phenyl]sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid ethyl ester

The title compound was prepared in analogy to **Example 1** starting from of (3*S*)-3-[2-chloro-5-(trifluoromethyl)phenyl]-3-[[4-(3-hydroxyphenyl)sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]propionic acid ethyl ester (**Intermediate D2**) instead of ethyl (3*S*)-3-[2-chloro-5-(trifluoromethyl)phenyl]-3-[[4-(3-hydroxyphenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate (**Intermediate D1**). White foam, MS m/z: 700.3 [M+-BOC+H]⁺, ESI pos.

Step 2: (3*S*)-3-[[4-[3-[2-[2-(tert-Butoxycarbonylamino)ethoxy]ethoxy]phenyl]sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid

To a solution of (3*S*)-3-[[4-[3-[2-[2-(tert-butoxycarbonylamino)ethoxy]ethoxy]phenyl]sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid ethyl ester (188 mg, 228 μmol) in tetrahydrofuran (1 ml) and ethanol (500 μl) was added an aqueous solution of 1 M lithium hydroxide (700 μl, 700 μmol). The reaction mixture was stirred at room temperature for two hours and the solvents were evaporated. The solid was suspended in water and a solution of 1 M HCl (705 μl, 705 μmol) was added. After stirring for about 30 min, the white suspension was filtered and washed with water. The solid was dried under high vacuum to give (3*S*)-3-[[4-[3-[2-[2-(tert-butoxycarbonylamino)ethoxy]ethoxy]phenyl]sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid (168 mg, 93 % yield) as a white solid. MS m/z: 700.4 [M+H]⁺, ESI pos.

Step 3: (3*S*)-3-[[4-[3-[2-(2-Aminoethoxy)ethoxy]phenyl]sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid hydrogen chloride salt

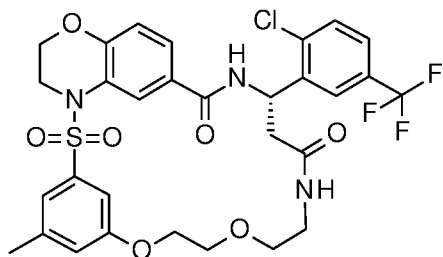
To a solution of (3*S*)-3-[[4-[3-[2-[2-(tert-butoxycarbonylamino)ethoxy]ethoxy]phenyl]sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid (168 mg, 0.210 mmol) in ethyl acetate (2 ml) was added a solution of 4 M HCl in dioxane (263 μl, 1.05 mmol). The reaction mixture was stirred at room temperature for 48 hours. At this time, the

reaction was completed as indicated by LC/MS. Diethylether was added and the white suspension obtained was filtered and dried to give (3*S*)-3-[[4-[3-[2-(2-aminoethoxy)ethoxy]phenyl]sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid hydrogen chloride salt (150 mg, 94 % yield) as a white solid. MS *m/z*: 672.2 [M+H]⁺, ESI pos.

Step 4: (17*S*)-17-[2-Chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-spiro[8,11,24-trioxa-2λ⁶-thia-1,14,18-triazatetracyclo[18.6.2.13,7.0.23,27]nonacosa-3,5,7(29),20(28),21,23(27)-hexaene-25,1'-cyclopropane]-15,19-dione

To a solution of (3*S*)-3-[[4-[3-[2-(2-aminoethoxy)ethoxy]phenyl]sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid hydrogen chloride salt (150 mg, 0.198 mmol) in dry N,N-dimethylformamide (5 ml) were added DIPEA (138 μl, 792 μmol) and HATU (301 mg, 792 μmol). The reaction mixture was stirred at room temperature for 4.5 hours. According to LC/MS, the reaction was finished. Water and ethyl acetate were added to the reaction mixture. The organic layer was separated and the aqueous phase was extracted twice with ethyl acetate. The combined organic layer was dried over Na₂SO₄, filtered and evaporated. The crude product was purified via flash column chromatography (Isolute-MN, SiO₂, methanol in dichloromethane, 0 % -10 %). The solid obtained was not pure enough, a second column was performed (SiO₂, ethyl acetate in heptane 30-100 %) to give (17*S*)-17-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-spiro[8,11,24-trioxa-2λ⁶-thia-1,14,18-triazatetracyclo[18.6.2.13,7.0.23,27]nonacosa-3,5,7(29),20(28),21,23(27)-hexaene-25,1'-cyclopropane]-15,19-dione (39.4 mg, 15 % yield) as a light yellow solid. MS *m/z*: 698.4 [M+H]⁺, ESI pos.

Example 4: (17*S*)-17-[2-Chloro-5-(trifluoromethyl)phenyl]-5-methyl-2,2-dioxo-8,11,24-trioxa-2λ⁶-thia-1,14,18-triazatetracyclo[18.6.2.13,7.0.23,27]nonacosa-3,5,7(29),20(28),21,23(27)-hexaene-15,19-dione



Step 1: Ethyl (3*S*)-3-[[4-[3-[2-[2-(tert-butoxycarbonylamino)ethoxy]ethoxy]-5-methyl-phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate

To a solution of ethyl (3*S*)-3-[2-chloro-5-(trifluoromethyl)phenyl]-3-[[4-(3-hydroxy-5-methyl-phenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate (**Intermediate D3**, 5 50.0 mg, 0.08 mmol) in N,N-dimethylformamide (1 ml) was added K₂CO₃ (22 mg, 0.16 mmol), KI (2.65 mg, 0.02 mmol) and tert-butyl N-[2-(2-bromoethoxy)ethyl]carbamate (32 mg, 0.12 mmol). The reaction mixture was stirred at 60 °C for 16 hours. After cooling, the reaction mixture was poured into water and extracted three times with ethyl acetate. The combined 10 organic layer was dried with Na₂SO₄, filtered and evaporated under reduced pressure. The residue was purified by reverse phase flash chromatography (Flash Spherical C18, 20-35 µm; 100 Å; 20 g, gradient water-acetonitrile). The combined purified fraction was concentrated and extracted again three times with ethyl acetate. The combined organic layer was dried with Na₂SO₄, filtered and evaporated under reduced pressure to give ethyl (3*S*)-3-[[4-[3-[2-[2-(tert- 15 butoxycarbonylamino)ethoxy]ethoxy]-5-methyl-phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate (70.0 mg, > 100 % yield) as a colorless oil. MS m/z: 714.2 [M-BOC+H]⁺, ESI pos.

Step 2: (3*S*)-3-[[4-[3-[2-[2-(tert-Butoxycarbonylamino)ethoxy]ethoxy]-5-methyl-phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid

To a solution of ethyl (3*S*)-3-[[4-[3-[2-[2-(tert-butoxycarbonylamino)ethoxy]ethoxy]-5-methyl-phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate (70.0 mg, 0.09 mmol, 1.0 eq) in methanol (1.5 ml) was added dropwise a solution of 1M NaOH (0.09 ml, 0.34 mmol). The reaction mixture was stirred 25 at 20 °C for 3 hours and was poured into water and acidified to pH = 3-5 with 1 N HCl at 0 °C. The solution was extracted three times with ethyl acetate. The combined organic layer was washed with brine, dried with Na₂SO₄, filtered and evaporated under reduced pressure to give (3*S*)-3-[[4-[3-[2-[2-(tert-butoxycarbonylamino)ethoxy]ethoxy]-5-methyl-phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic 30 acid (68.0 mg, 100 % yield) as a colorless oil. MS m/z: 686.2 [M-100+H]⁺, ESI pos.

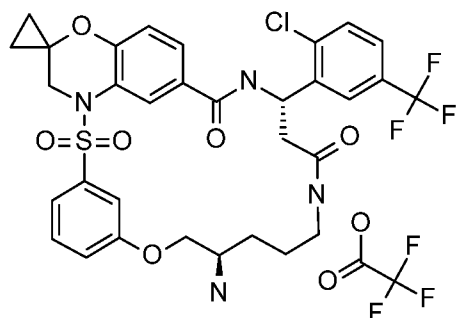
Step 3: (3S)-3-[[4-[3-[2-[2-(tert-Butoxycarbonylamino)ethoxy]ethoxy]-5-methyl-phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid hydrochloride salt.

To a solution of (3S)-3-[[4-[3-[2-[2-(tert-butoxycarbonylamino)ethoxy]ethoxy]-5-methyl-phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid (68.0 mg, 0.09 mmol) in dichloromethane (0.6 ml) was added a solution of 4 M HCl in 1,4-dioxane (0.6 ml) at 0 °C. After stirring at 20 °C for 1 hour, the reaction mixture was concentrated to give crude (3S)-3-[[4-[3-[2-(2-aminoethoxy)ethoxy]-5-methyl-phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid hydrochloride salt (62.0 mg, 94 % yield) as a colorless oil. MS m/z: 686.2 [M+H]⁺, ESI pos.

Step 4: (17S)-17-[2-Chloro-5-(trifluoromethyl)phenyl]-5-methyl-2,2-dioxo-8,11,24-trioxa-2λ⁶-thia-1,14,18-triazatetracyclo[18.6.2.13,7.0^{23,27}]nonacosa-3,5,7(29),20(28),21,23(27)-hexaene-15,19-dione

To a solution of (3S)-3-[[4-[3-[2-(2-aminoethoxy)ethoxy]-5-methyl-phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid hydrochloride salt (62 mg, 0.09 mmol) in N,N-dimethylformamide (2 ml) were added DIPEA (44.3 mg, 0.34 mmol) and HATU (39.1 mg, 0.1 mmol) in batches. The reaction mixture was stirred at 20 °C for 2 hours. The reaction mixture was filtered, concentrated and was purified by prep-HPLC (Phenomenex Luna C18 150 x 25 mm x 10 μm, gradient water with 2.25 % formic acid and acetonitrile). The combined purified phase was concentrated and lyophilized to give (17S)-17-[2-chloro-5-(trifluoromethyl)phenyl]-5-methyl-2,2-dioxo-8,11,24-trioxa-2λ⁶-thia-1,14,18-triazatetracyclo[18.6.2.13,7.0^{23,27}]nonacosa-3,5,7(29),20(28),21,23(27)-hexaene-15,19-dione as a white solid (17.5 mg, 28 % yield). MS m/z: 668.2 [M+H]⁺, ESI pos.

Example 5: (10R,17S)-10-Amino-17-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-spiro[8,24-dioxa-2λ⁶-thia-1,14,18-triazatetracyclo[18.6.2.13,7.0^{23,27}]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-25,1'-cyclopropane]-15,19-dione trifluoroacetic acid salt



Step 1: tert-Butyl (4R)-4-[(E)-3-ethoxy-3-oxo-prop-1-enyl]-2,2-dimethyl-oxazolidine-3-carboxylate

To a solution of triethyl phosphonoacetate (CAS 30492-56-9, 1.47 g, 6.54 mmol) in tetrahydrofuran (15 ml) was added sodium hydride 60 % (261.7 mg, 6.54 mmol). After stirring at 0 °C for 30 minutes, (S)-(-)-3-Boc-2,2-dimethyloxazolidine-4-carboxaldehyde (CAS 102308-32-7, 1.0 g, 4.36 mmol) was added at 0 °C. The mixture was stirred at 20 °C for 1 hour. The reaction mixture was poured into water and extracted three times with ethyl acetate. The combined organic layer was dried with Na₂SO₄, filtered and concentrated under reduced pressure. The crude product was purified by reverse phase flash chromatography (Flash Spherical C18, 20-35 µm; 100 Å; 40 g, gradient water with 0.05% TFA and acetonitrile). The combined fractions were concentrated and extracted three times with ethyl acetate. The combined organic layer was dried with Na₂SO₄, filtered and concentrated under reduced pressure to give *tert*-butyl (4R)-4-[(E)-3-ethoxy-3-oxo-prop-1-enyl]-2,2-dimethyl-oxazolidine-3-carboxylate (1.4 g > 100 % yield) as a colorless oil. MS m/z: 200.0 [M-100+H]⁺, ESI pos.

Step 2: tert-Butyl (4R)-4-(3-ethoxy-3-oxo-propyl)-2,2-dimethyl-oxazolidine-3-carboxylate

To a solution of *tert*-butyl (4R)-4-[(E)-3-ethoxy-3-oxo-prop-1-enyl]-2,2-dimethyl-oxazolidine-3-carboxylate (1.4 g, 4.68 mmol) in ethanol (20 ml) was added Pd/C (500 mg) and the reaction mixture was stirred at 20 °C for 2 hours under hydrogen atmosphere (1 atm, balloon). The mixture was filtered, the filtrate was concentrated to *tert*-butyl (4R)-4-(3-ethoxy-3-oxo-propyl)-2,2-dimethyl-oxazolidine-3-carboxylate (1.4 g, 85 % yield) as a colorless oil. MS m/z: 202.1 [M-100+H]⁺, ESI pos.

Step 3: tert-Butyl (4R)-4-(3-hydroxypropyl)-2,2-dimethyl-oxazolidine-3-carboxylate

To a mixture of LiAlH₄ (264 mg, 6.97 mmol) in tetrahydrofuran (25 ml) was added *tert*-butyl (4R)-4-(3-ethoxy-3-oxo-propyl)-2,2-dimethyl-oxazolidine-3-carboxylate at 0 °C (1.4 g,

4.65 mmol) and the reaction mixture was stirred at 0 °C for 1 hour. The reaction was quenched by addition of $\text{Na}_2\text{SO}_4 \cdot 10\text{H}_2\text{O}$ (1 g) at 0 °C. After stirring for 30 minutes, the filtrate was concentrated to give tert-butyl (4*R*)-4-(3-hydroxypropyl)-2,2-dimethyl-oxazolidine-3-carboxylate (1.1 g, 24 % yield) as a colorless oil. MS *m/z*: 204.1 [$\text{M}-\text{C}_4\text{H}_8+\text{H}$]⁺, ESI pos.

5 Step 4: tert-Butyl (4*R*)-4-[3-(1,3-dioxoisindolin-2-yl)propyl]-2,2-dimethyl-oxazolidine-3-carboxylate

To a solution of tert-butyl (4*R*)-4-(3-hydroxypropyl)-2,2-dimethyl-oxazolidine-3-carboxylate (1.1 g, 4.24 mmol) and phthalimide (CAS 85-41-6, 1.25 g, 8.5 mmol) in toluene (15 ml) was added (tributylphosphoranylidene)acetonitrile (CAS 157141-27-0, 3.07 g, 12.7 mmol). The mixture was stirred at 20 °C for 16 hours. The reaction mixture was poured into water and the mixture was extracted three times with ethyl acetate. The combined organic layer was dried with Na_2SO_4 , filtered and concentrated under reduced pressure. The crude product was purified by flash chromatography (silica gel, ethyl acetate in petroleum ether 0-20 %) followed by reverse phase flash chromatography (Flash Spherical C18, 20-35 μm ; 100 Å; gradient water with 0.05 % TFA and acetonitrile) to give tert-butyl (4*R*)-4-[3-(1,3-dioxoisindolin-2-yl)propyl]-2,2-dimethyl-oxazolidine-3-carboxylate (880 mg, 53 % yield) as a yellow solid. MS *m/z*: 289.1 [$\text{M}-\text{BOC}+\text{H}$]⁺, ESI pos.

Step 5: 2-[(4*R*)-4-Amino-5-hydroxy-pentyl]isoindoline-1,3-dione hydrochloride salt

To a solution of tert-butyl (4*R*)-4-[3-(1,3-dioxoisindolin-2-yl)propyl]-2,2-dimethyl-oxazolidine-3-carboxylate (800 mg, 2.06 mmol) in 1,4-Dioxane (5 ml) was added an aqueous solution of 5 M HCl (5.0 ml, 25.0 mmol) and stirred at 20 °C for 1 hour. Acetonitrile and water were added, and the mixture was directly lyophilized to give 2-[(4*R*)-4-amino-5-hydroxy-pentyl]isoindoline-1,3-dione hydrochloride salt (550 mg, 94 % yield) as a white solid. MS *m/z*: 249.1 [$\text{M}+\text{H}$]⁺, ESI pos.

Step 6: tert-Butyl *N*-[(1*R*)-4-(1,3-dioxoisindolin-2-yl)-1-(hydroxymethyl)butyl]carbamate

To a solution of 2-[(4*R*)-4-amino-5-hydroxy-pentyl]isoindoline-1,3-dione hydrochloride salt (550 mg, 1.9 mmol) in dichloromethane (10 ml) were added DIPEA (996.7 mg, 7.73 mmol) and di-tert-butyl dicarbonate (CAS 24424-99-5, 632.4 mg, 2.9 mmol). The reaction was stirred at 20 °C for 2 hours. The reaction mixture was then poured into water and was extracted three times with ethyl acetate. The combined organic layer was dried with Na_2SO_4 , filtered and evaporated. The crude product was purified by flash chromatography (silica gel, ethyl acetate in petroleum ether 20-50 %) to afford tert-butyl *N*-[(1*R*)-4-(1,3-dioxoisindolin-2-yl)-1-

(hydroxymethyl)butyl]carbamate (820 mg, 2.35 mmol, > 100 % yield) as a white oil. MS m/z: 249.1 [M-100+H]⁺, ESI pos.

Step 7: Ethyl (3*S*)-3-[[4-[3-[(2*R*)-2-(tert-butoxycarbonylamino)-5-(1,3-dioxoisindolin-2-yl)pentoxyl]phenyl]sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate

To a solution of ethyl (3*S*)-3-[2-chloro-5-(trifluoromethyl)phenyl]-3-[[4-(3-hydroxyphenyl)sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]propanoate (**Intermediate D4**, 300 mg, 0.47 mmol) in toluene (6 ml) was added (tributylphosphoranylidene)acetonitrile (CAS 157141-27-0, 339.9 mg). The mixture was stirred at 100 °C for 16 hours. The reaction mixture was poured into water and the mixture was extracted three times with ethyl acetate. The combined organic layer was dried with Na₂SO₄, filtered and concentrated under reduced pressure. The crude product was purified by reverse phase flash chromatography (Flash Spherical C18, 20-35 μm; 100 Å; gradient water with 0.05 % TFA and acetonitrile) to give ethyl (3*S*)-3-[[4-[3-[(2*R*)-2-(tert-butoxycarbonylamino)-5-(1,3-dioxoisindolin-2-yl)pentoxyl]phenyl]sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate (180 mg, 37 % yield) as a brown solid. MS m/z: 869.3 [M-100+H]⁺, ESI pos.

Step 8: 2-[[[(4*R*)-4-(tert-Butoxycarbonylamino)-5-[3-[6-[[1*S*]-2-carboxy-1-[2-chloro-5-(trifluoromethyl)phenyl]ethyl]carbamoyl]spiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-4-yl]sulfonylphenoxy]pentyl]carbamoyl]benzoic acid

To a solution of ethyl (3*S*)-3-[[4-[3-[(2*R*)-2-(tert-butoxycarbonylamino)-5-(1,3-dioxoisindolin-2-yl)pentoxyl]phenyl]sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate (160 mg, 0.17 mmol) in a mixture of tetrahydrofuran (1 ml), methanol (1 ml) and water (1 ml) was added LiOH · H₂O (20.8 mg, 0.5 mmol). The reaction mixture was stirred at 20 °C for 1 hour. The mixture was diluted with water and the pH was adjusted to 4 with 1 M aqueous HCl. The mixture was extracted three times with ethyl acetate. The combined organic layer was dried over Na₂SO₄, filtered and concentrated under reduced pressure to give 2-[[[(4*R*)-4-(tert-butoxycarbonylamino)-5-[3-[6-[[1*S*]-2-carboxy-1-[2-chloro-5-(trifluoromethyl)phenyl]ethyl]carbamoyl]spiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-4-yl]sulfonylphenoxy]pentyl]carbamoyl]benzoic acid (160 mg, 83 % yield) as a yellow solid. MS m/z: 959.4 [M+H]⁺, ESI pos.

Step 9: (3S)-3-[[4-[3-[(2R)-5-Amino-2-(tert-butoxycarbonylamino)pentoxy]phenyl]sulfonylspiro[3H-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid

To a solution of 2-[[[(4R)-4-(tert-butoxycarbonylamino)-5-[3-[6-[[[(1S)-2-carboxy-1-[2-chloro-5-(trifluoromethyl)phenyl]ethyl]carbamoyl]spiro[3H-1,4-benzoxazine-2,1'-cyclopropane]-4-yl]sulfonylphenoxy]pentyl]carbamoyl]benzoic acid (160.0 mg, 0.17 mmol) in ethanol (3 ml) was added N₂H₄.H₂O (0.08 ml, 1.67 mmol) and the reaction mixture was stirred at 80 °C for 16 hours. The reaction mixture was purified directly by reverse phase flash chromatography (Flash Spherical C18, 20-35 μm; 100 Å; 20 g, water-acetonitrile) to give (3S)-3-[[4-[3-[(2R)-5-amino-2-(tert-butoxycarbonylamino)pentoxy]phenyl]sulfonylspiro[3H-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid (100 mg, 74 % yield) as a white solid. MS m/z: 811.3 [M+H]⁺, ESI pos.

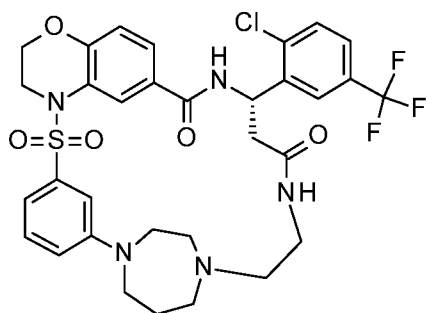
Step 10: tert-Butyl N-[(10R,17S)-17-[2-chloro-5-(trifluoromethyl)phenyl]-2,2,15,19-tetraoxo-spiro[8,24-dioxa-2λ⁶-thia-1,14,18-triazatetracyclo[18.6.2.13,7.023,27]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-25,1'-cyclopropane]-10-yl]carbamate

To a solution of (3S)-3-[[4-[3-[(2R)-5-amino-2-(tert-butoxycarbonylamino)pentoxy]phenyl]sulfonylspiro[3H-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid (90.0 mg, 0.11 mmol) in acetonitrile (10 ml) was added N-methylimidazole (CAS 616-47-7, 27.3 mg, 0.33 mmol). After stirring for 5 minutes at room temperature, N,N,N',N'-tetramethylchloroformamidinium hexafluorophosphate (CAS 207915-99-9, 37.35 mg) was added. The mixture was stirred at 20 °C for 1 hour. The solvent was evaporated and the residue was by reverse phase flash chromatography (Flash Spherical C18, 20-35 μm; 100 Å; 20 g, water-acetonitrile) to give tert-butyl N-[(10R,17S)-17-[2-chloro-5-(trifluoromethyl)phenyl]-2,2,15,19-tetraoxo-spiro[8,24-dioxa-2λ⁶-thia-1,14,18-triazatetracyclo[18.6.2.13,7.023,27]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-25,1'-cyclopropane]-10-yl]carbamate (30 mg, 32 % yield) as a white solid. MS m/z: 693.2 [M-100+H]⁺, ESI pos.

Step 11: (10R,17S)-10-Amino-17-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-spiro[8,24-dioxa-2λ⁶-thia-1,14,18-triazatetracyclo[18.6.2.13,7.023,27]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-25,1'-cyclopropane]-15,19-dione 2,2,2-trifluoroacetic acid salt.

To a solution of *tert*-butyl *N*-[(10*R*,17*S*)-17-[2-chloro-5-(trifluoromethyl)phenyl]-2,2,15,19-tetraoxo-spiro[8,24-dioxa-2λ⁶-thia-1,14,18-triazatetracyclo[18.6.2.13,7.0.23,27]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-25,1'-cyclopropane]-10-yl]carbamate (20.0 mg, 0.03 mmol) in dichloromethane (0.500 ml) was added 2,2,2-trifluoroacetic (0.5 ml, 0.03 mmol) and the
 5 reaction mixture was stirred at 20 °C for 1 hour. The mixture reaction was concentrated and was purified by prep-HPLC (Phenomenex Luna C18 150 x 25 mm x 10 μm, gradient water with 1% TFA and acetonitrile) to give (10*R*,17*S*)-10-amino-17-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-spiro[8,24-dioxa-2λ⁶-thia-1,14,18-triazatetracyclo[18.6.2.13,7.0.23,27]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-25,1'-cyclopropane]-15,19-dione 2,2,2-trifluoroacetic acid
 10 salt (5.76 mg, 33 % yield) as a white solid. MS *m/z*: 693.2 [M+H]⁺, ESI pos.

Example 6: (18*S*)-18-[2-Chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-25-oxa-2λ⁶-thia-1,8,12,15,19-pentazapentacyclo[19.6.2.28,12.13,7.0.24,28]dotriaconta-3,5,7(32),21(29),22,24(28)-hexaene-16,20-dione



Step1: Methyl 4-[3-(4-*tert*-butoxycarbonyl-1,4-diazepan-1-yl)phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate

To a solution of methyl 4-(3-bromophenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate (**Intermediate C4**, 1.0 g, 2.43 mmol) in 1,4-dioxane (10 ml) were added *tert*-butyl 1,4-diazepane-1-carboxylate (CAS 112275-50-0, 0.57 ml, 2.91 mmol), Cs₂CO₃ (1185.49 mg, 3.64 mmol), xantphos (140 mg, 0.24 mmol) and Pd₂(dba)₃ · CHCl₃ (125 mg, 0.12 mmol) were added while degassing with argon. The vial was sealed and the mixture was stirred at 100 °C for 16 hours. The reaction mixture was poured into water and was extracted three times with ethyl acetate. The combined organic layer was washed with brine, dried over Na₂SO₄, filtered, and
 25 concentrated in *vacuo*. The crude product was purified by flash chromatography (SiO₂, petroleum ether in ethyl acetate, 0-100 %) to yield methyl 4-[3-(4-*tert*-butoxycarbonyl-1,4-

diazepan-1-yl)phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate (1.3 g, 99 % yield) as a light yellow solid. m/z: 476.3 [M+H-C₄H₉]⁺, ESI pos.

Step 2: Methyl 4-[3-(1,4-diazepan-1-yl)phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate hydrochloride salt

- 5 To solution of methyl 4-[3-(4-*tert*-butoxycarbonyl-1,4-diazepan-1-yl)phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate (1.3 g, 2.45 mmol) in dichloromethane (12 ml) was added a solution of 4 M HCl in 1,4-dioxane (12.0 ml, 48.0 mmol) at 0 °C. The mixture was stirred at 20 °C for 1 hour. The reaction mixture was concentrated to give crude methyl 4-[3-(1,4-diazepan-1-yl)phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate hydrochloride
10 salt (1.1 g, 81 % yield) as a light yellow solid. MS m/z: 432.2 [M+H]⁺, ESI pos.

Step 3: Methyl 4-[3-[4-[2-(*tert*-butoxycarbonylamino)ethyl]-1,4-diazepan-1-yl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate

- To a solution of methyl 4-[3-(1,4-diazepan-1-yl)phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate hydrochloride salt (1.1 g, 2.35 mmol) and sodium iodide (35.23 mg, 0.24 mmol) in
15 N,N-dimethylformamide (20 ml) were added K₂CO₃ (972 mg, 7.05 mmol) and *tert*-butyl N-(2-bromoethyl)carbamate (CAS 39684-80-5, 1054 mg, 4.7 mmol). The mixture was stirred at 20 °C for 2 hours. The reaction mixture was poured into water and was extracted three times with ethyl acetate. The combined organic layer was washed with brine, dried over Na₂SO₄, filtered, and concentrated under reduced pressure. The crude product was purified by reverse phase flash
20 chromatography (Flash Spherical C18, 20-35µm; 100Å; gradient water with 0.05 % TFA and acetonitrile) to give methyl 4-[3-[4-[2-(*tert*-butoxycarbonylamino)ethyl]-1,4-diazepan-1-yl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate (700 mg, 51 % yield) as a light yellow lyophilized powder. MS m/z: 575.4 [M+H]⁺, ESI pos.

- Step 4: 4-[3-[4-[2-(*tert*-Butoxycarbonylamino)ethyl]-1,4-diazepan-1-yl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid
25

- The title compound was prepared in analogy to **Intermediate D3, step 1** starting from methyl 4-[3-[4-[2-(*tert*-butoxycarbonylamino)ethyl]-1,4-diazepan-1-yl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate instead of methyl 4-(3-methoxy-5-methyl-phenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate (**Intermediate C2**). Light yellow oil, MS m/z: 561.4
30 [M+H]⁺, ESI pos.

Step 5: Ethyl (3*S*)-3-[[4-[3-[4-[2-(*tert*-butoxycarbonylamino)ethyl]-1,4-diazepan-1-yl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate

The title compound was prepared in analogy to **Intermediate D3, step 3** starting from 4-[3-[4-[2-(*tert*-butoxycarbonylamino)ethyl]-1,4-diazepan-1-yl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid instead of 4-(3-hydroxy-5-methyl-phenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid and stirring at room temperature for 1 hour instead of 16 hours. White solid. MS *m/z*: 838.5 [M+H]⁺, ESI pos.

Step 6: (3*S*)-3-[[4-[3-[4-[2-(*tert*-Butoxycarbonylamino)ethyl]-1,4-diazepan-1-yl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid

To a solution of ethyl (3*S*)-3-[[4-[3-[4-[2-(*tert*-butoxycarbonylamino)ethyl]-1,4-diazepan-1-yl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate (150 mg, 0.18 mmol) in methanol (3 ml) was added a solution of 2 M NaOH (0.18 ml, 0.72 mmol). The reaction mixture was stirred at 20 °C for 16 hours. The pH of the mixture was adjusted to 4 with a solution of 1M HCl. The mixture was extracted three times with ethyl acetate. The combined organic layer was dried with Na₂SO₄, filtered and evaporated. The residue was purified by reverse phase flash chromatography (Flash Spherical C18, 20-35 μm; 100 Å; gradient water with 0.05 % HCl and acetonitrile) to give (3*S*)-3-[[4-[3-[4-[2-(*tert*-butoxycarbonylamino)ethyl]-1,4-diazepan-1-yl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid (120 mg, 77 % yield) as a white lyophilized solid. MS *m/z*: 808.3 [M+H]⁺, ESI pos.

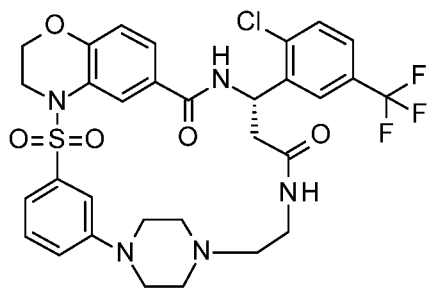
Step 7: (3*S*)-3-[[4-[3-[4-(2-Aminoethyl)-1,4-diazepan-1-yl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid hydrochloride salt

The title compound was prepared in analogy to **Example 4, step 3** starting from (3*S*)-3-[[4-[3-[4-[2-(*tert*-butoxycarbonylamino)ethyl]-1,4-diazepan-1-yl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid instead of (3*S*)-3-[[4-[3-[2-[2-(*tert*-butoxycarbonylamino)ethoxy]ethoxy]-5-methyl-phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid. Light yellow oil, MS *m/z*: 708.3 [M+H]⁺, ESI pos.

Step 8: (18*S*)-18-[2-Chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-25-oxa-2λ⁶-thia-1,8,12,15,19-pentazapentacyclo[19.6.2.28,12.13,7.024,28]dotriaconta-3,5,7(32),21(29),22,24(28)-hexaene-16,20-dione

The title compound was prepared in analogy to **Example 4, step 4** starting (3*S*)-3-[[4-[3-[4-(2-aminoethyl)-1,4-diazepan-1-yl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid hydrochloride salt instead of (3*S*)-3-[[4-[3-[2-(2-aminoethoxy)ethoxy]-5-methyl-phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid hydrochloride salt. White solid, MS *m/z*: 692.4 [M+H]⁺, ESI pos.

Example 7: (17*S*)-17-[2-Chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-24-oxa-2λ⁶-thia-1,8,11,14,18-pentazapentacyclo[18.6.2.28,11.13,7.023,27]hentriaconta-3,5,7(31),20(28),21,23(27)-hexaene-15,19-dione



Step 1: 4-[3-[4-[2-(*tert*-Butoxycarbonylamino)ethyl]piperazino]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid methyl ester

To a solution of 4-(3-bromophenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid methyl ester (**Intermediate C4**, 200 mg, 485 μmol) in dry 1,4-dioxane (1.4 ml) were added while degassing with argon *tert*-butyl *N*-(2-piperazin-1-ylethyl)carbamate (CAS 140447-78-5, 133 mg, 582 μmol), cesium carbonate (237 mg, 727 μmol), xantphos (11.2 mg, 19.4 μmol), and tris(dibenzylideneacetone)dipalladium-chloroform adduct (10 mg, 9.7 μmol). The vial was sealed and the mixture was stirred at 100 °C for 16 hours. LC/MS showed full conversion of the starting material. The mixture was extracted with three times with ethyl acetate. The combined organic layer was washed with brine, dried over Na₂SO₄, filtered, and concentrated in vacuo. The crude product was purified by flash chromatography (SiO₂, methanol in dichloromethane, 0-5 %) to yield 4-[3-[4-[2-(*tert*-butoxycarbonylamino)ethyl]piperazino]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid methyl (229 mg, 84 % yield) as a yellow viscous oil. MS *m/z*: 561.5[M+H]⁺, ESI pos.

Step 2: 4-[3-[4-[2-(*tert*-Butoxycarbonylamino)ethyl]piperazino]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid

To a solution of 4-[3-[4-[2-(*tert*-butoxycarbonylamino)ethyl]piperazino]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid methyl ester (225 mg, 401 μ mol) in methanol (803 μ l) and tetrahydrofuran (1.54 ml), a solution of 4 M sodium hydroxide (301 μ l, 1.2 mmol) was added dropwise at room temperature. The reaction was stirred overnight and was acidified with aqueous citric acid and extracted twice with ethyl acetate. The combined organic layer was washed with brine, dried over Na₂SO₄, filtered, and concentrated in vacuo to give 4-[3-[4-[2-(*tert*-butoxycarbonylamino)ethyl]piperazino]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid (224 mg, > 100 % yield) as a colorless waxy solid. MS m/z: 547.3 [M+H]⁺, ESI pos.

Step 3: (3*S*)-3-[[4-[3-[4-[2-(*tert*-Butoxycarbonylamino)ethyl]piperazino]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid ethyl ester

To a solution of 4-[3-[4-[2-(*tert*-butoxycarbonylamino)ethyl]piperazino]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid (215 mg, 393 μ mol) in dry N,N-dimethylformamide (1.24 ml), was added HATU (164 mg, 432 μ mol). The mixture was left to stir for 10 minutes, after which (3*S*)-3-amino-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid ethyl ester (**Intermediate B1**, 128 mg, 432 μ mol) and DIPEA (76.3 mg, 103 μ l, 590 μ mol) were added. The reaction was stirred at room temperature for 16 hours. LC/MS showed incomplete conversion of the starting material. HATU (82.3 mg, 216 μ mol) and DIPEA (76.3 mg, 103 μ l, 590 μ mol) were added. The reaction mixture was stirred for five hours and was extracted twice with ethyl acetate. The combined organic layer was washed with brine, dried over Na₂SO₄, filtered, and concentrated in vacuo. The crude product was purified by flash chromatography (silica gel, ethyl acetate in heptane 0-50%) to yield (3*S*)-3-[[4-[3-[4-[2-(*tert*-butoxycarbonylamino)ethyl]piperazino]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid ethyl ester (300 mg, 75 % yield) as an off-white foam. MS m/z: 824.2 [M+H]⁺, ESI pos.

Step 4: (3*S*)-3-[[4-[3-[4-(2-Aminoethyl)piperazino]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid ethyl ester hydrochloric salt

To a solution of (3*S*)-3-[[4-[3-[4-[2-(*tert*-butoxycarbonylamino)ethyl]piperazino]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid ethyl ester (297 mg, 292 μ mol) in ethyl acetate (2.45 ml), a solution of 4 M HCl in dioxane (450 μ l, 1.8 mmol) was added dropwise at 0 °C. The mixture was warmed to room temperature and stirred for 3 hours. The solid was filtered and dried under high *vacuo* to give (3*S*)-3-[[4-[3-[4-(2-aminoethyl)piperazino]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid ethyl ester hydrochloric salt (300 mg, 75 % yield) as an off-white foam. MS *m/z*: 722.2 [M-H]⁻, ESI neg.

10 Step 5: (3*S*)-3-[[4-[3-[4-(2-Aminoethyl)piperazino]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid hydrogen chloride salt

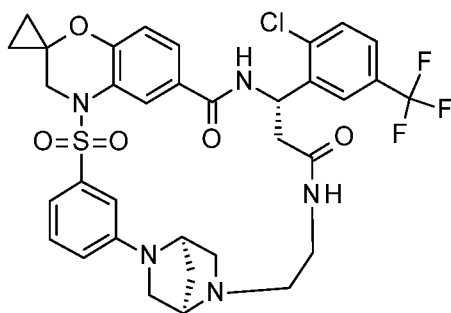
To a solution of (3*S*)-3-[[4-[3-[2-(2-aminoethoxy)ethoxy]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-fluoro-5-(trifluoromethyl)phenyl]propionic acid ethyl ester (300 mg, 439 μ mol) in methanol (878 μ l), a solution of 4 M NaOH (439 μ l, 1.76 mmol) was added at room temperature. The reaction was slightly exothermic. LC/MS after 16 hours showed incomplete conversion. The reaction mixture was diluted with tetrahydrofuran (963 μ l) and a solution of 4 M NaOH (138 μ l, 552 μ mol) was added. LC/MS after 3 hours showed improved but still poor conversion. A solution of 4 M NaOH (138 μ l, 552 μ mol) was added and the reaction was left to stir for another 16 hours. The mixture was concentrated in *vacuo*. The resulting solid was dissolved in water and acidified with a solution of 4M HCl aq. (1 ml). The solid was filtered to give (3*S*)-3-[[4-[3-[4-(2-aminoethyl)piperazino]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid hydrogen chloride salt (156 mg, 88 % yield) as an off-white solid. MS *m/z*: 694.1 [M-H]⁻, ESI neg.

Step 6: (17*S*)-17-[2-Chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-24-oxa-2 λ ⁶-thia-1,8,11,14,18-pentazapentacyclo[18.6.2.28,11.13,7.0^{23,27}]hentriaconta-3,5,7(31),20(28),21,23(27)-hexaene-15,19-dione

To a solution of (3*S*)-3-[[4-[3-[4-(2-aminoethyl)piperazino]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid hydrogen chloride (100 mg, 131 μ mol) in dry N,N-dimethylformamide (2.7 ml) were added HATU (199 mg, 524 μ mol) and DIPEA (92 μ l, 524 μ mol). The reaction was stirred at room temperature for 3

hours and was directly purified by prep-HPLC (Gemini NX, 12 nm, 5 μ m, 100 x 30 mm, gradient water with 0.1 % triethylamine). The sample was lyophilised to yield (17*S*)-17-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-24-oxa-2 λ ⁶-thia-1,8,11,14,18-pentazapentacyclo[18.6.2.28,11.13,7.023,27]hentriaconta-3,5,7(31),20(28),21,23(27)-hexaene-15,19-dione (11.2 mg, 13 % yield) as a white powder. MS *m/z*: 678.2 [M+H]⁺, ESI pos.

Example 8: (10*R*,12*R*,20*S*)-20-[2-Chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-spiro[27-oxa-2 λ ⁶-thia-1,8,14,17,21-pentazahexacyclo[21.6.2.13,7.08,12.010,14.026,30]dotriaconta-3(32),4,6,23(31),24,26(30)-hexaene-28,1'-cyclopropane]-18,22-dione



10 Step 1: Methyl 4-[3-[(1*R*,4*R*)-5-*tert*-butoxycarbonyl]-2,5-diazabicyclo[2.2.1]heptan-2-yl]phenyl]sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carboxylate

The title compound was prepared in analogy to **Example 6, step 1** starting with methyl 4-(3-bromophenyl)sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carboxylate (**Intermediate C5**) and *tert*-butyl (1*R*,4*R*)-2,5-diazabicyclo[2.2.1]heptane-2-carboxylate (CAS 134003-84-2) instead of methyl 4-(3-bromophenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate (**Intermediate C4**) and *tert*-butyl 1,4-diazepane-1-carboxylate. Black solid, MS *m/z*: 556.2 [M+H]⁺, ESI pos.

Step 2: Methyl 4-[3-[(1*R*,4*R*)-2,5-diazabicyclo[2.2.1]heptan-2-yl]phenyl]sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carboxylate

20 To a solution of methyl 4-[3-[(1*R*,4*R*)-5-*tert*-butoxycarbonyl]-2,5-diazabicyclo[2.2.1]heptan-2-yl]phenyl]sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carboxylate (640 mg, 1.15 mmol) in dichloromethane (5 ml) was added TFA (5.0 ml, 65 mmol) and stirred at 20 °C for 1 hour. The reaction mixture was poured into a saturated aqueous solution of NaHCO₃ at 20 °C and was extracted twice with ethyl acetate. The combined organic layer was washed with brine, dried
25 over Na₂SO₄, filtered, and concentrated in vacuo to yield crude methyl 4-[3-[(1*R*,4*R*)-2,5-

diazabicyclo[2.2.1]heptan-2-yl]phenyl]sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carboxylate (720 mg, > 100 % yield) as black solid. MS *m/z*: 456.1 [M+H]⁺, ESI pos.

Step 3: Methyl 4-[3-[(1*R*,4*R*)-5-[2-(*tert*-butoxycarbonylamino)ethyl]-2,5-

diazabicyclo[2.2.1]heptan-2-yl]phenyl]sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-
5 carboxylate

To solution of methyl 4-[3-[(1*R*,4*R*)-2,5-diazabicyclo[2.2.1]heptan-2-yl]phenyl]sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carboxylate (700 mg, 1.54 mmol) in methanol (10 ml) was added *N*-BOC-2-aminoacetaldehyde (CAS 89711-08-0, 734 mg, 4.61 mmol), ZnCl₂ (1.05 g, 7.68 mmol) and NaBH₃CN (290 mg, 4.61 mmol). The mixture was
10 stirred at 20 °C for 2 hours. The reaction mixture was concentrated and was purified by reverse phase flash chromatography (Flash Spherical C18, 20-35μm; 100Å; gradient water with 0.05% TFA and acetonitrile) to give methyl 4-[3-[(1*R*,4*R*)-5-[2-(*tert*-butoxycarbonylamino)ethyl]-2,5-diazabicyclo[2.2.1]heptan-2-yl]phenyl]sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carboxylate (720 mg, 67 % yield) as a brown oil. MS *m/z*: 599.2 [M+H]⁺, ESI pos.

15 Step 4: 4-[3-[(1*R*,4*R*)-5-[2-(*tert*-Butoxycarbonylamino)ethyl]-2,5-diazabicyclo[2.2.1]heptan-2-
yl]phenyl]sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carboxylic acid

To a solution of methyl 4-[3-[(1*R*,4*R*)-5-[2-(*tert*-butoxycarbonylamino)ethyl]-2,5-diazabicyclo[2.2.1]heptan-2-yl]phenyl]sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carboxylate (670 mg, 1.12 mmol) in methanol (3 ml), tetrahydrofuran (3 ml) and water (3 ml)
20 was added lithium hydroxide monohydrate (141 mg, 3.36 mmol) and stirred at 50 °C for 1 hour. The pH of the mixture was adjusted to 4 by addition of a solution of aqueous 1M HCl. The mixture was extracted three times with ethyl acetate. The combined organic layer was dried over Na₂SO₄, filtered and concentrated under reduced pressure to give 4-[3-[(1*R*,4*R*)-5-[2-(*tert*-butoxycarbonylamino)ethyl]-2,5-diazabicyclo[2.2.1]heptan-2-yl]phenyl]sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carboxylic acid (670 mg, 84 % yield) as a brown oil. MS *m/z*:
25 585.3 [M+H]⁺, ESI pos.

Step 5: Ethyl (3*S*)-3-[[4-[3-[(1*R*,4*R*)-5-[2-(*tert*-butoxycarbonylamino)ethyl]-2,5-

diazabicyclo[2.2.1]heptan-2-yl]phenyl]sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-
carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate

30 To a solution of 4-[3-[(1*R*,4*R*)-5-[2-(*tert*-butoxycarbonylamino)ethyl]-2,5-diazabicyclo[2.2.1]heptan-2-yl]phenyl]sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-

carboxylic acid (620 mg, 1.06 mmol) in N,N-dimethylformamide (10 ml) were added ethyl (3*S*)-3-amino-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate (**Intermediate B1**, 314 mg, 1.06 mmol), DIPEA (410 mg, 3.18 mmol) and HATU (300 mg, 1.3 mmol) and stirred at 20 °C for 1 hour. The reaction mixture was poured into water and extracted three times with ethyl acetate.

- 5 The combined organic layer was dried over Na₂SO₄, filtered and concentrated under reduced pressure. The crude product was purified by reverse phase flash chromatography (Flash Spherical C18, 20-35 µm; 100 Å; gradient water with 0.05 % TFA and acetonitrile) to ethyl (3*S*)-3-[[4-[3-[(1*R*,4*R*)-5-[2-(*tert*-butoxycarbonylamino)ethyl]-2,5-diazabicyclo[2.2.1]heptan-2-yl]phenyl]sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate (560 mg, 54 % yield) as a brown solid. MS m/z: 862.3 [M+H]⁺, ESI pos.

Step 6: (3*S*)-3-[[4-[3-[(1*R*,4*R*)-5-[2-(*tert*-Butoxycarbonylamino)ethyl]-2,5-diazabicyclo[2.2.1]heptan-2-yl]phenyl]sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid

- 15 The title compound was prepared in analogy to **Example 6, step 6** starting from ethyl (3*S*)-3-[[4-[3-[(1*R*,4*R*)-5-[2-(*tert*-butoxycarbonylamino)ethyl]-2,5-diazabicyclo[2.2.1]heptan-2-yl]phenyl]sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate instead of ethyl (3*S*)-3-[[4-[3-[4-[2-(*tert*-butoxycarbonylamino)ethyl]-1,4-diazepan-1-yl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate. White solid. MS m/z: 843.3 [M+H]⁺, ESI pos.

Step 7: (3*S*)-3-[2-Chloro-5-(trifluoromethyl)phenyl]-3-[[4-[3-[(1*R*,4*R*)-5-(2-aminoethyl)-2,5-diazabicyclo[2.2.1]heptan-2-yl]phenyl]sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]propanoic acid hydrochloride salt

- 25 To a solution of (3*S*)-3-[[4-[3-[(1*R*,4*R*)-5-[2-(*tert*-butoxycarbonylamino)ethyl]-2,5-diazabicyclo[2.2.1]heptan-2-yl]phenyl]sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid (100 mg, 0.12 mmol) in dichloromethane (1 ml) was added a solution of 4 M HCl in 1,4-dioxane (1.0 ml, 4.0 mmol) at 0 °C. The reaction mixture was stirred at 20 °C for 1 hour then was diluted with acetonitrile and
30 water and directly lyophilized to give (3*S*)-3-[2-chloro-5-(trifluoromethyl)phenyl]-3-[[4-[3-[(1*R*,4*R*)-5-(2-aminoethyl)-2,5-diazabicyclo[2.2.1]heptan-2-yl]phenyl]sulfonylspiro[3*H*-1,4-

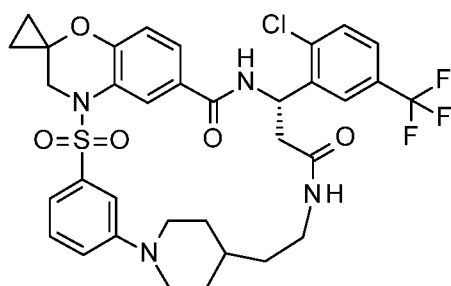
benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]propanoic acid hydrochloride salt (60 mg, 65 % yield) as a brown solid. MS m/z: 734.2 [M+H]⁺, ESI pos.

Step 8: (10*R*,12*R*,20*S*)-20-[2-Chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-spiro[27-oxa-2λ⁶-thia-1,8,14,17,21-pentazahexacyclo[21.6.2.13,7.08,12.010,14.026,30]dotriaconta-

5 3(32),4,6,23(31),24,26(30)-hexaene-28,1'-cyclopropane]-18,22-dione

To a solution of (3*S*)-3-[[4-[3-[(1*R*,4*R*)-5-(2-aminoethyl)-2,5-diazabicyclo[2.2.1]heptan-2-yl]phenyl]sulfonyl]spiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid hydrochloride salt (40 mg, 0.05 mmol) in N,N-dimethylformamide (4 ml) were added DIPEA (20.09 mg, 0.16 mmol) and HATU (14.65 mg, 10 0.06 mmol) and stirred at 20 °C for 16 hours. The reaction mixture was concentrated and purified prep-HPLC (Phenomenex Luna C18 150 x 25mm x 10 μm, gradient water with 2.25 % formic acid and acetonitrile) to give (10*R*,12*R*,20*S*)-20-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-spiro[27-oxa-2λ⁶-thia-1,8,14,17,21-pentazahexacyclo[21.6.2.13,7.08,12.010,14.026,30]dotriaconta-3(32),4,6,23(31),24,26(30)- 15 hexaene-28,1'-cyclopropane]-18,22-dione (9.15 mg, 24 % yield) as a yellow solid. MS m/z: 716.2 [M+H]⁺, ESI pos.

Example 9: (17*S*)-17-[2-Chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-spiro[24-oxa-2λ⁶-thia-1,8,14,18-tetrazapentacyclo[18.6.2.28,11.13,7.023,27]hentriaconta-3,5,7(31),20(28),21,23(27)-hexaene-25,1'-cyclopropane]-15,19-dione



20

Step 1: (3*S*)-3-[[4-[3-[4-[2-(tert-butoxycarbonylamino)ethyl]piperidino]phenyl]sulfonyl]spiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid ethyl ester

25 The title compound was prepared in analogy to **Example 7, step 1** starting from ethyl (3*S*)-3-[[4-(3-bromophenyl)sulfonyl]spiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-

chloro-5-(trifluoromethyl)phenyl]propanoate (**Intermediate D6**) and *tert*-butyl *N*-[2-(4-piperidyl)ethyl]carbamate

(CAS 165528-81-4) instead of of 4-(3-bromophenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid methyl ester (**Intermediate C4**) and *tert*-butyl *N*-(2-piperazin-1-

5 ylethyl)carbamate. White lyophilized powder, MS *m/z*: 849.4 [M+H]⁺, ESI pos.

Step 2: (3S)-3-[[4-[3-[4-[2-(*tert*-butoxycarbonylamino)ethyl]piperidino]phenyl]sulfonylspiro[3H-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid

The title compound was prepared in analogy to **Example 7, step 2** starting from (3S)-3-[[4-[3-
10 [4-[2-(*tert*-butoxycarbonylamino)ethyl]piperidino]phenyl]sulfonylspiro[3H-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid ethyl ester instead of 4-[3-[4-[2-(*tert*-butoxycarbonylamino)ethyl]piperazino]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid methyl ester. Off-white solid, MS *m/z*: 821.4 [M+H]⁺, ESI pos.

15 Step 3: (3S)-3-[[4-[3-[4-(2-Aminoethyl)piperidino]phenyl]sulfonylspiro[3H-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid hydrogen chloride

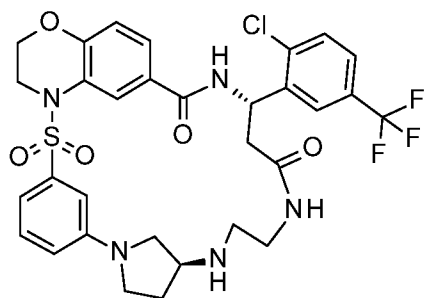
The title compound was prepared in analogy to **Example 7, step 4** starting from (3S)-3-[[4-[3-
20 [4-[2-(*tert*-butoxycarbonylamino)ethyl]piperidino]phenyl]sulfonylspiro[3H-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid instead of (3S)-3-[[4-[3-[4-[2-(*tert*-butoxycarbonylamino)ethyl]piperazino]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid ethyl ester. Off-white solid, MS *m/z*: 719.5 [M-H]⁻, ESI pos.

Step 4: (17S)-17-[2-Chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-spiro[24-oxa-2λ⁶-thia-
25 1,8,14,18-tetrazapentacyclo[18.6.2.28,11.13,7.023,27]hentriaconta-3,5,7(31),20(28),21,23(27)-hexaene-25,1'-cyclopropane]-15,19-dione

The title compound was prepared in analogy to **Example 7, step 6** starting from (3S)-3-[[4-[3-
[4-(2-Aminoethyl)piperidino]phenyl]sulfonylspiro[3H-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid hydrogen chloride
30 instead of (3S)-3-[[4-[3-[4-(2-aminoethyl)piperazino]phenyl]sulfonyl-2,3-dihydro-1,4-

benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid hydrogen chloride. White lyophilized powder, MS m/z: 703.1 [M+H]⁺, ESI pos.

Example 10: (11*S*,18*S*)-18-[2-Chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-25-oxa-2λ⁶-thia-1,8,12,15,19-pentazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-
5 3,5,7(31),21(29),22,24(28)-hexaene-16,20-dione



Step 1: Methyl 4-[3-[(3*S*)-3-(*tert*-butoxycarbonylamino)pyrrolidin-1-yl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate

The title compound was prepared in analogy to **Example 6, step 1** starting with (*S*)-3-(BOC-amino)pyrrolidine (CAS 122536-76-9) instead of *tert*-butyl 1,4-diazepane-1-carboxylate. Yellow
10 solid. MS m/z: 518.1 [M+H]⁺, ESI pos.

Step 2: Methyl 4-[3-[(3*S*)-3-aminopyrrolidin-1-yl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate

The title compound was prepared in analogy to **Example 8, step 2** starting with methyl 4-[3-[(3*S*)-3-(*tert*-butoxycarbonylamino)pyrrolidin-1-yl]phenyl]sulfonyl-2,3-dihydro-1,4-
15 benzoxazine-6-carboxylate instead of methyl 4-[3-[(1*R*,4*R*)-2,5-diazabicyclo[2.2.1]heptan-2-yl]phenyl]sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carboxylate. Yellow oil. MS m/z: 418.1 [M+H]⁺, ESI pos.

Step 3: Methyl 4-[3-[(3*S*)-3-[2-(*tert*-butoxycarbonylamino)ethylamino]pyrrolidin-1-yl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate
20

To solution of methyl 4-[3-[(3*S*)-3-aminopyrrolidin-1-yl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate (1.8 g, 4.31 mmol) in methanol (20 ml) was added acetic acid to adjust the pH of the mixture to 3-4 at 20 °C, then *N*-BOC-2-aminoacetaldehyde (1029 mg, 6.47 mmol) and NaBH₃CN (2718 mg, 43.1 mmol) were added and the reaction mixture was allowed
25 to stir at 20 °C for 16 hours. The reaction mixture was filtered, the filtrate was poured into water

and extracted three times with ethyl acetate. The combined organic layer was dried over Na₂SO₄, filtered and concentrated under reduced pressure. The crude product was purified by reverse phase flash chromatography (Flash Spherical C18, 20-35 µm; 100 Å; gradient water with 0.0 5% TFA and acetonitrile) to give methyl 4-[3-[(3*S*)-3-[2-(*tert*-

5 butoxycarbonylamino)ethylamino]pyrrolidin-1-yl]phenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate (930 mg, 39 % yield) as a white solid. MS m/z: 561.2 [M+H]⁺, ESI pos.

Step 4: Methyl 4-[3-[(3*S*)-3-[allyloxycarbonyl]-2-(*tert*-
butoxycarbonylamino)ethyl]amino]pyrrolidin-1-yl]phenyl)sulfonyl-2,3-dihydro-1,4-
benzoxazine-6-carboxylate

10 To a solution of methyl 4-[3-[(3*S*)-3-[2-(*tert*-butoxycarbonylamino)ethylamino]pyrrolidin-1-yl]phenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate (900.0 mg, 1.61 mmol) in tetrahydrofuran (5 ml) and water (5 ml) were added NaHCO₃ (202 mg, 2.41 mmol) and allyl chloroformate (CAS 2937-50-0, 232 mg, 1.93 mmol) at 0 °C. The mixture was stirred at 20 °C for 2 hours. The reaction mixture was poured into water and extracted three times with ethyl
15 acetate. The combined organic layer was washed with brine, dried over Na₂SO₄, filtered and concentrated under reduced pressure. The crude product was purified by flash chromatography (silica gel, ethyl acetate/petroleum ether 25-50%) to give methyl 4-[3-[(3*S*)-3-[allyloxycarbonyl]-2-(*tert*-butoxycarbonylamino)ethyl]amino]pyrrolidin-1-yl]phenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate (730 mg, 70% yield) as a colorless oil. MS m/z: 645.3 [M+H]⁺, ESI
20 pos.

Step 5: 4-[3-[(3*S*)-3-[Allyloxycarbonyl]-2-(*tert*-butoxycarbonylamino)ethyl]amino]pyrrolidin-1-yl]phenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid

The title compound was prepared in analogy to **Intermediate D3, step 1** starting from 4-[3-[(3*S*)-3-[allyloxycarbonyl]-2-(*tert*-butoxycarbonylamino)ethyl]amino]pyrrolidin-1-yl]phenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate instead of methyl 4-(3-methoxy-5-methyl-phenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate (**Intermediate C2**). White
25 oil. MS m/z: 631.3 [M+H]⁺, ESI pos.

Step 6: Ethyl (3*S*)-3-[[4-[3-[(3*S*)-3-[allyloxycarbonyl]-2-(*tert*-
butoxycarbonylamino)ethyl]amino]pyrrolidin-1-yl]phenyl)sulfonyl-2,3-dihydro-1,4-
30 benzoxazine-6-carboxyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate

To a solution of 4-[3-[(3*S*)-3-[allyloxycarbonyl]-2-(*tert*-butoxycarbonylamino)ethyl]amino]pyrrolidin-1-yl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid (680 mg, 1.08 mmol) in *N,N*-dimethylformamide (15 ml) were added ethyl (3*S*)-3-amino-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate (**Intermediate B1**, 5 478 mg, 1.62 mmol) and DIPEA (417 mg, 3.23 mmol) and HATU (304 mg, 1.29 mmol). The mixture was stirred at 20 °C for 16 hours. The reaction mixture was poured into water and extracted three times with ethyl acetate. The combined organic layer was washed with brine, dried over Na₂SO₄, filtered and concentrated under reduced pressure. The crude product was purified by flash chromatography (silica gel, ethyl acetate/ petroleum ether 25-50 %) to give 10 ethyl (3*S*)-3-[[4-[3-[(3*S*)-3-[allyloxycarbonyl]-2-(*tert*-butoxycarbonylamino)ethyl]amino]pyrrolidin-1-yl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate (730 mg, 70 % yield) as a white oil, MS *m/z*: 908.4 [M+H]⁺, ESI pos.

Step 7: (3*S*)-3-[[4-[3-[(3*S*)-3-[Allyloxycarbonyl]-2-(*tert*-butoxycarbonylamino)ethyl]amino]pyrrolidin-1-yl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid

The title compound was prepared in analogy to **Example 4, step 2** starting with ethyl (3*S*)-3-[[4-[3-[(3*S*)-3-[allyloxycarbonyl]-2-(*tert*-butoxycarbonylamino)ethyl]amino]pyrrolidin-1-yl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate instead of ethyl (3*S*)-3-[[4-[3-[2-[2-(*tert*-butoxycarbonylamino)ethoxy]ethoxy]-5-methyl-phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate. White solid. MS *m/z*: 880.4 [M+H]⁺, ESI pos.

Step 8: (3*S*)-3-[[4-[3-[(3*S*)-3-[Allyloxycarbonyl(2-aminoethyl)amino]pyrrolidin-1-yl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid hydrochloride salt

The title compound was prepared in analogy to **Example 4, step 3** starting from (3*S*)-3-[[4-[3-[(3*S*)-3-[allyloxycarbonyl]-2-(*tert*-butoxycarbonylamino)ethyl]amino]pyrrolidin-1-yl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid instead of (3*S*)-3-[[4-[3-[2-[2-(*tert*-butoxycarbonylamino)ethoxy]ethoxy]-5-methyl-phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-

carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid. White solid, MS m/z: 780.3 [M+H]⁺, ESI pos.

Step 9: Allyl (11S,18S)-18-[2-chloro-5-(trifluoromethyl)phenyl]-2,2,16,20-tetraoxo-25-oxa-2λ⁶-thia-1,8,12,15,19-pentazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-
5 3,5,7(31),21(29),22,24(28)-hexaene-12-carboxylate

To a solution of (3S)-3-[[4-[3-[(3S)-3-[allyloxycarbonyl(2-aminoethyl)amino]pyrrolidin-1-yl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid hydrochloride salt (230 mg, 0.28 mmol) in acetonitrile (20 ml) was added N-methyl imidazole (69.4 mg, 0.84 mmol). After stirring for 5 minutes at
10 room temperature, *N,N,N',N'*-tetramethylchloroformamidinium hexafluorophosphate (CAS 207915-99-9, 94.8 mg, 0.34 mmol) was added. The mixture was stirred at 20 °C for 2 hours. The reaction mixture was poured into water and extracted with ethyl acetate. The organic layer was washed with brine, dried over Na₂SO₄, filtered and concentrated under reduced pressure. The crude product was purified by reverse phase flash chromatography (Flash Spherical C18, 20-
15 35 μm; 100 Å; gradient water with 0.05 % TFA and acetonitrile) to give (11S,18S)-18-[2-chloro-5-(trifluoromethyl)phenyl]-2,2,16,20-tetraoxo-25-oxa-2λ⁶-thia-1,8,12,15,19-pentazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-3,5,7(31),21(29),22,24(28)-hexaene-12-carboxylate (250 mg, 78 % yield) as a white solid. MS m/z: 780.3 [M+H]⁺, ESI pos.

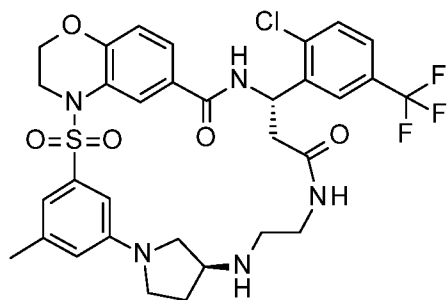
Step 10: (11S,18S)-18-[2-Chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-25-oxa-2λ⁶-thia-
20 1,8,12,15,19-pentazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-
3,5,7(31),21(29),22,24(28)-hexaene-16,20-dione

To a solution of allyl (11S,18S)-18-[2-chloro-5-(trifluoromethyl)phenyl]-2,2,16,20-tetraoxo-25-oxa-2λ⁶-thia-1,8,12,15,19-pentazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-3,5,7(31),21(29),22,24(28)-hexaene-12-carboxylate (140.0 mg, 0.18 mmol) in tetrahydrofuran (5
25 ml) was added 1,3-dimethylbarbituric acid (CAS 769-42-6, 201 mg, 1.29 mmol) and Pd(PPh₃)₄ (42.5 mg, 0.04 mmol) under nitrogen. The mixture was stirred at 20 °C for 2 hours. The reaction mixture was concentrated to give a residue and was then purified by prep-TLC (chromatography (silica gel, ethyl acetate/petroleum ether) followed by prep-HPLC (Phenomenex Luna C18 150 x 25 mm x 10 μm, gradient water with 2.25 % formic acid and acetonitrile and re-purified by
30 prep-HPLC (Waters Xbridge 150 x 25 mm x 5 μm, water with 0.5% ammonium hydroxide and acetonitrile). The purified solution was concentrated and lyophilized to give (11S,18S)-18-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-25-oxa-2λ⁶-thia-1,8,12,15,19-

pentazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-3,5,7(31),21(29),22,24(28)-hexaene-16,20-dione (36 mg, 25 % yield) as a white solid. MS m/z: 678.2 [M+H]⁺, ESI pos.

Example 11: (11*S*,18*S*)-18-[2-Chloro-5-(trifluoromethyl)phenyl]-5-methyl-2,2-dioxo-25-oxa-2λ⁶-thia-1,8,12,15,19-pentazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-

5 3,5,7(31),21(29),22,24(28)-hexaene-16,20-dione



Step 1: Methyl 4-[3-[(3*S*)-3-(*tert*-butoxycarbonylamino)pyrrolidin-1-yl]-5-methyl-phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate

The title compound was prepared in analogy to **Example 6, step 1** starting with methyl 4-(3-bromo-5-methyl-phenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate (**Intermediate C6**) and (S)-3-(BOC-amino)pyrrolidine (CAS 122536-76-9) instead of methyl 4-(3-bromophenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate (**Intermediate C4**) and *tert*-butyl 1,4-diazepane-1-carboxylate. Yellow solid. MS m/z: 532.3 [M+H]⁺, ESI pos.

15 Step 2: Methyl 4-[3-[(3*S*)-3-aminopyrrolidin-1-yl]-5-methyl-phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate

The title compound was prepared in analogy to **Example 8, step 2** starting with methyl 4-[3-[(3*S*)-3-(*tert*-butoxycarbonylamino)pyrrolidin-1-yl]-5-methyl-phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate instead of methyl 4-[3-[(1*R*,4*R*)-5-*tert*-butoxycarbonyl-2,5-diazabicyclo[2.2.1]heptan-2-yl]phenyl]sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carboxylate. Colorless oil. MS m/z: 432.2 [M+H]⁺, ESI pos.

Step 3: Methyl 4-[3-[(3*S*)-3-[2-(*tert*-butoxycarbonylamino)ethylamino]pyrrolidin-1-yl]-5-methyl-phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate

To a solution of methyl 4-[3-[(3*S*)-3-aminopyrrolidin-1-yl]-5-methyl-phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate (1.4 g, 3.24 mmol) in methanol (30 ml) was added N-25 BOC-2-aminoacetaldehyde (775 mg, 4.87 mmol), ZnCl₂ (2.21 g, 16.2 mmol) and NaBH₃CN

(612 mg, 9.73 mmol). The mixture was stirred at 20 °C for 2 hours. The reaction mixture was poured into water and extracted three times with ethyl acetate. The organic layer was washed with brine, dried over Na₂SO₄, filtered and concentrated under reduced pressure. The crude product was purified by prep-HPLC (Phenomenex Luna C18 250 x 25 mm x 10 µm, gradient water with 1 % TFA and acetonitrile) to give methyl 4-[3-[(3*S*)-3-[2-(*tert*-butoxycarbonylamino)ethylamino]pyrrolidin-1-yl]-5-methyl-phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate (1.5 g, 79 % yield) as a white solid. MS m/z: 575.4 [M+H]⁺, ESI pos.

Step 4: Methyl 4-[3-[(3*S*)-3-[allyloxycarbonyl]-2-(*tert*-butoxycarbonylamino)ethyl]amino]pyrrolidin-1-yl]-5-methyl-phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate

To a solution of methyl 4-[3-[(3*S*)-3-[2-(*tert*-butoxycarbonylamino)ethylamino]pyrrolidin-1-yl]-5-methyl-phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate (650 mg, 1.13 mmol) in tetrahydrofuran (13 ml) and water (13 ml) were added NaHCO₃ (143 mg, 1.7 mmol) and allyl chloroformate (164 mg, 1.36 mmol) at 0 °C. The mixture was stirred at 20 °C for 2 hours. The reaction mixture was poured into water and extracted three times with ethyl acetate. The combined organic layer was washed with brine, dried over Na₂SO₄, filtered and concentrated under reduced pressure to give methyl 4-[3-[(3*S*)-3-[allyloxycarbonyl]-2-(*tert*-butoxycarbonylamino)ethyl]amino]pyrrolidin-1-yl]-5-methyl-phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate (800 mg, < 100 % yield) as a light yellow oil. MS m/z: 659.5 [M+H]⁺, ESI pos.

Step 5: 4-[3-[(3*S*)-3-[Allyloxycarbonyl]-2-(*tert*-butoxycarbonylamino)ethyl]amino]pyrrolidin-1-yl]-5-methyl-phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid

The title compound was prepared in analogy to **Intermediate D3, step 1** starting from 4-[3-[(3*S*)-3-[allyloxycarbonyl]-2-(*tert*-butoxycarbonylamino)ethyl]amino]pyrrolidin-1-yl]-5-methyl-phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate instead of methyl 4-(3-methoxy-5-methyl-phenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate (**Intermediate C2**) and stirring at 20 °C instead of 50 °C. Yellow solid, MS m/z: 645.4 [M+H]⁺, ESI pos.

Step 6: Ethyl (3*S*)-3-[[4-[3-[(3*S*)-3-[allyloxycarbonyl]-2-(*tert*-butoxycarbonylamino)ethyl]amino]pyrrolidin-1-yl]-5-methyl-phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate

To a solution of 4-[3-[(3*S*)-3-[allyloxycarbonyl]-2-(*tert*-butoxycarbonylamino)ethyl]amino]pyrrolidin-1-yl]-5-methyl-phenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid (700 mg, 1.09 mmol) in N,N-dimethylformamide (10 ml) were added DIPEA (0.57 ml, 3.26 mmol), ethyl (3*S*)-3-amino-3-[2-chloro-5-

- 5 (trifluoromethyl)phenyl]propanoate (482 mg) and HATU (495 mg, 1.3 mmol). The reaction mixture was stirred at 20 °C for 1 hour. The reaction mixture was poured into water and extracted with ethyl acetate. The organic layer was washed with brine, dried over Na₂SO₄, filtered and concentrated under reduced pressure. The crude product was purified by reverse phase flash chromatography (Flash Spherical C18, 20-35 µm; 100 Å; 20 g, water-acetonitrile) to
10 give ethyl (3*S*)-3-[[4-[3-[(3*S*)-3-[allyloxycarbonyl]-2-(*tert*-butoxycarbonylamino)ethyl]amino]pyrrolidin-1-yl]-5-methyl-phenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate (800 mg, 69 % yield) as a white solid MS m/z: 922.6 [M+H]⁺, ESI pos.

Step 7: (3*S*)-3-[[4-[3-[(3*S*)-3-[Allyloxycarbonyl]-2-(*tert*-

- 15 butoxycarbonylamino)ethyl]amino]pyrrolidin-1-yl]-5-methyl-phenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid

The title compound was prepared in analogy to **Example 4, step 2** starting with ethyl (3*S*)-3-[[4-[3-[(3*S*)-3-[allyloxycarbonyl]-2-(*tert*-butoxycarbonylamino)ethyl]amino]pyrrolidin-1-yl]-5-methyl-phenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-

20 (trifluoromethyl)phenyl]propanoate instead of ethyl (3*S*)-3-[[4-[3-[2-(2-(*tert*-butoxycarbonylamino)ethoxy]ethoxy)-5-methyl-phenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate. Light yellow solid. MS m/z: 894.5 [M+H]⁺, ESI pos.

Step 8: (3*S*)-3-[[4-[3-[(3*S*)-3-[Allyloxycarbonyl(2-aminoethyl)amino]pyrrolidin-1-yl]-5-methyl-

25 phenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid hydrochloride salt

The title compound was prepared in analogy to **Example 8, step 7** starting with (3*S*)-3-[[4-[3-[(3*S*)-3-[allyloxycarbonyl]-2-(*tert*-butoxycarbonylamino)ethyl]amino]pyrrolidin-1-yl]-5-methyl-phenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-

30 (trifluoromethyl)phenyl]propanoic acid instead of (3*S*)-3-[[4-[3-[(1*R*,4*R*)-5-[2-(*tert*-butoxycarbonylamino)ethyl]-2,5-diazabicyclo[2.2.1]heptan-2-yl]phenyl)sulfonyl]spiro[3*H*-1,4-

benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid. White solid. MS m/z: 794.4 [M+H]⁺, ESI pos.

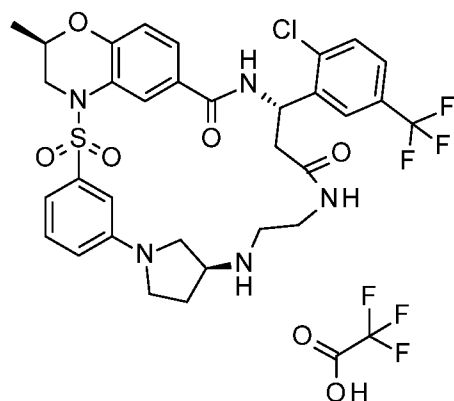
Step 9: Allyl (11S,18S)-18-[2-chloro-5-(trifluoromethyl)phenyl]-5-methyl-2,2,16,20-tetraoxo-25-oxa-2λ⁶-thia-1,8,12,15,19-pentazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-3,5,7(31),21(29),22,24(28)-hexaene-12-carboxylate

To a solution of (3S)-3-[[4-[3-[(3S)-3-[allyloxycarbonyl(2-aminoethyl)amino]pyrrolidin-1-yl]-5-methyl-phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid hydrochloride salt (140 mg, 0.17 mmol) in acetonitrile (14 ml) was added N-methyl imidazole (55.3 mg, 0.67 mmol) and stirred at 20 °C for 5 minutes, then *N,N,N',N'*-tetramethylchloroformamidinium hexafluorophosphate (56.7 mg, 0.2 mmol) was added. The reaction mixture was stirred at 20 °C for 1 hour. The mixture was evaporated and purified by reverse phase flash chromatography (Flash Spherical C18, 20-35 μm; 100 Å; water-acetonitrile) to give allyl (11S,18S)-18-[2-chloro-5-(trifluoromethyl)phenyl]-5-methyl-2,2,16,20-tetraoxo-25-oxa-2λ⁶-thia-1,8,12,15,19-pentazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-3,5,7(31),21(29),22,24(28)-hexaene-12-carboxylate (80 mg, 60 % yield) as a white solid. MS m/z: 776.5 [M+H]⁺, ESI pos.

Step 10: (11S,18S)-18-[2-Chloro-5-(trifluoromethyl)phenyl]-5-methyl-2,2-dioxo-25-oxa-2λ⁶-thia-1,8,12,15,19-pentazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-3,5,7(31),21(29),22,24(28)-hexaene-16,20-dione

To a solution of allyl (11S,18S)-18-[2-chloro-5-(trifluoromethyl)phenyl]-5-methyl-2,2,16,20-tetraoxo-25-oxa-2λ⁶-thia-1,8,12,15,19-pentazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-3,5,7(31),21(29),22,24(28)-hexaene-12-carboxylate (70 mg, 0.09 mmol) in tetrahydrofuran (4 ml) were added 1,3-dimethylbarbituric acid (98.6 mg, 0.63 mmol) and Pd(PPh₃)₄ (20.8 mg, 0.02 mmol) under nitrogen. The mixture was stirred at 20 °C for 2 hours. The solvent was evaporated and the residue was purified via prep-TLC chromatography (silica gel, ethyl acetate/petroleum ether 0-100 %) by prep-HPLC (Phenomenex Luna C18 150 x 25 mm x 10 μm, gradient water with 0.5 % ammonium hydroxide and acetonitrile) to afford (11S,18S)-18-[2-chloro-5-(trifluoromethyl)phenyl]-5-methyl-2,2-dioxo-25-oxa-2λ⁶-thia-1,8,12,15,19-pentazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-3,5,7(31),21(29),22,24(28)-hexaene-16,20-dione (22.2 mg, 36 % yield) as a white solid. MS m/z: 692.4 [M+H]⁺, ESI pos.

Example 12: (11*S*,18*S*,26*R*)-18-[2-Chloro-5-(trifluoromethyl)phenyl]-26-methyl-2,2-dioxo-25-oxa-2 λ^6 -thia-1,8,12,15,19-pentazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-3(31),4,6,21(29),22,24(28)-hexaene-16,20-dione 2,2,2-trifluoroacetic acid salt



5 Step 1: Ethyl (3*S*)-3-[[*(2R)*-4-[3-[(3*S*)-3-(*tert*-butoxycarbonylamino)pyrrolidin-1-yl]phenyl]sulfonyl-2-methyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate

To a solution of ethyl (3*S*)-3-[[*(2R)*-4-(3-bromophenyl)sulfonyl-2-methyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate

10 **(Intermediate D8**, 1.55 g, 2.25 mmol) in 1,4-dioxane (20 ml), were added (*S*)-3-(BOC-amino)pyrrolidine (837 mg, 4.49 mmol), Pd₂(dba)₃ · CHCl₃ (116 mg, 0.11 mmol), xantphos (130 mg, 0.22 mmol) and Cs₂CO₃ (1098 mg, 3.37 mmol) while degassing with argon. The vial was sealed and the mixture was stirred at 100 °C for 16 hours. The reaction mixture was filtered, the filtrate was poured into water and was extracted three times with ethyl acetate. The combined
15 organic layer was washed with brine, dried with Na₂SO₄ filtered and evaporated. The crude product was purified by flash chromatography (silica gel, ethyl acetate/petroleum ether 50 %) to give ethyl (3*S*)-3-[[*(2R)*-4-[3-[(3*S*)-3-(*tert*-butoxycarbonylamino)pyrrolidin-1-yl]phenyl]sulfonyl-2-methyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate (1 g, 52 % yield) as a white solid. MS m/z: 795.2 [M+H]⁺,
20 ESI pos.

Step 2: Ethyl (3*S*)-3-[[*(2R)*-4-[3-[(3*S*)-3-aminopyrrolidin-1-yl]phenyl]sulfonyl-2-methyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate

To solution of ethyl (3*S*)-3-[[*(2R)*-4-[3-[(3*S*)-3-(*tert*-butoxycarbonylamino)pyrrolidin-1-yl]phenyl]sulfonyl-2-methyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-

(trifluoromethyl)phenyl]propanoate (950 mg, 1.19 mmol) in dichloromethane (5 ml) was added TFA (5 ml) at 0 °C. The mixture was stirred at 20 °C for 1 hour. The reaction mixture was poured into saturated aqueous NaHCO₃ and was extracted three times with ethyl acetate. The combined organic layer was washed with brine, dried with Na₂SO₄, filtered and concentrated to give crude ethyl (3*S*)-3-[[*(2R)*-4-[3-[(3*S*)-3-aminopyrrolidin-1-yl]phenyl]sulfonyl-2-methyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate (970 mg, > 100 % yield) as a yellow solid. MS *m/z*: 695.2 [M+H]⁺, ESI pos.

Step 3: Ethyl (3*S*)-3-[2-chloro-5-(trifluoromethyl)phenyl]-3-[[*(2R)*-4-[3-[(3*S*)-3-[2-(1,3-dioxoisindolin-2-yl)ethylamino]pyrrolidin-1-yl]phenyl]sulfonyl-2-methyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate

To solution of ethyl (3*S*)-3-[[*(2R)*-4-[3-[(3*S*)-3-aminopyrrolidin-1-yl]phenyl]sulfonyl-2-methyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate (910 mg, 1.31 mmol) in methanol (10 ml) were added *N*-(2-oxoethyl)phthalimide (CAS 2913-97-5, 297 mg, 1.57 mmol), NaBH₃CN (247 mg, 3.93 mmol) and ZnCl₂ (268 mg, 1.96 mmol). The mixture was stirred at 20 °C for 2 hours. The reaction mixture was filtered, the filtrate was poured into water and was extracted three times with ethyl acetate. The combined organic layer was washed with brine, dried with Na₂SO₄ filtered and evaporated. The crude product was purified by prep-HPLC (Phenomenex Luna C18 150 x 25 mm x 10 μm, gradient water with 0.5 % TFA and acetonitrile) to give ethyl (3*S*)-3-[2-chloro-5-(trifluoromethyl)phenyl]-3-[[*(2R)*-4-[3-[(3*S*)-3-[2-(1,3-dioxoisindolin-2-yl)ethylamino]pyrrolidin-1-yl]phenyl]sulfonyl-2-methyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate (730 mg, 62 % yield) as a white solid. MS *m/z*: 868.4 [M+H]⁺, ESI pos.

Step 4: Ethyl (3*S*)-3-[[*(2R)*-4-[3-[3-*tert*-butoxycarbonyl]-2-(1,3-dioxoisindolin-2-yl)ethyl]amino]pyrrolidin-1-yl]phenyl]sulfonyl-2-methyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate

To a solution of ethyl (3*S*)-3-[2-chloro-5-(trifluoromethyl)phenyl]-3-[[*(2R)*-4-[3-[(3*S*)-3-[2-(1,3-dioxoisindolin-2-yl)ethylamino]pyrrolidin-1-yl]phenyl]sulfonyl-2-methyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate (700 mg, 0.81 mmol) in dichloromethane (10 ml) were added DIPEA (420 μl, 2.4 mmol) and di-*t*-butyldicarbonate (528 mg, 2.4 mmol). The reaction mixture was stirred at 20 °C for 16 hours. The reaction mixture was filtered, the filtrate was poured into water and was extracted three times with ethyl acetate. The combined organic

layer was washed with brine, dried with Na₂SO₄ filtered and evaporated. The crude product was purified by prep-HPLC (Phenomenex Luna C18 150 x 25 mm x 10 µm, gradient water and acetonitrile) to give ethyl (3*S*)-3-[[*(2R)*-4-[3-[3-*tert*-butoxycarbonyl]-2-(1,3-dioxoisindolin-2-yl)ethyl]amino]pyrrolidin-1-yl]phenyl]sulfonyl-2-methyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate (600 mg, 75 % yield) as a white solid. MS m/z: 968.4 [M+H]⁺, ESI pos.

Step 5: 2-[2-[*tert*-Butoxycarbonyl]-[(3*S*)-1-[3-[[*(2R)*-6-[[*(1S)*-2-carboxy-1-[2-chloro-5-(trifluoromethyl)phenyl]ethyl]carbamoyl]-2-methyl-2,3-dihydro-1,4-benzoxazin-4-yl]sulfonyl]phenyl]pyrrolidin-3-yl]amino]ethylcarbamoyl]benzoic acid

To a solution of ethyl (3*S*)-3-[[*(2R)*-4-[3-[(3*S*)-3-*tert*-butoxycarbonyl]-2-(1,3-dioxoisindolin-2-yl)ethyl]amino]pyrrolidin-1-yl]phenyl]sulfonyl-2-methyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate (300.0 mg, 0.31 mmol) in tetrahydrofuran (2 ml) was added LiOH · H₂O (39.0 mg, 0.93 mmol). The mixture was stirred at 20 °C for 2 hours. The mixture was poured into water and treated with an aqueous solution of 1M HCl at 0 °C. The mixture was extracted three times with ethyl acetate. The combined organic layer was washed with brine, dried with Na₂SO₄, filtered and evaporated to give 2-[2-[*tert*-butoxycarbonyl]-[(3*S*)-1-[3-[[*(2R)*-6-[[*(1S)*-2-carboxy-1-[2-chloro-5-(trifluoromethyl)phenyl]ethyl]carbamoyl]-2-methyl-2,3-dihydro-1,4-benzoxazin-4-yl]sulfonyl]phenyl]pyrrolidin-3-yl]amino]ethylcarbamoyl]benzoic acid (300 mg, 97 % yield) as a white solid. MS m/z: 958.4 [M+H]⁺, ESI pos.

Step 6: (3*S*)-3-[[*(2R)*-4-[3-[(3*S*)-3-[2-Aminoethyl(*tert*-butoxycarbonyl)amino]pyrrolidin-1-yl]phenyl]sulfonyl-2-methyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid

To a solution of 2-[2-[*tert*-butoxycarbonyl]-1-[3-[[*(2R)*-6-[[*(1S)*-2-carboxy-1-[2-chloro-5-(trifluoromethyl)phenyl]ethyl]carbamoyl]-2-methyl-2,3-dihydro-1,4-benzoxazin-4-yl]sulfonyl]phenyl]pyrrolidin-3-yl]amino]ethylcarbamoyl]benzoic acid (300 mg, 0.31 mmol) in ethanol (5 ml) was added N₂H₄ · H₂O (157 mg, 3.13 mmol). The reaction mixture was stirred at 80 °C for 3 hours. The reaction mixture was hot-filtered, the filtrate was evaporated and purified by prep-HPLC (Phenomenex Luna C18, 150 x 25mm x 10 µm, gradient water and acetonitrile) to give (3*S*)-3-[[*(2R)*-4-[3-[(3*S*)-3-[2-aminoethyl (*tert*-butoxycarbonyl)amino]pyrrolidin-1-yl]phenyl]sulfonyl-2-methyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-

(trifluoromethyl)phenyl]propanoic acid (170 mg, 63 % yield) as a white solid. MS m/z: 810.3 [M+H]⁺, ESI pos.

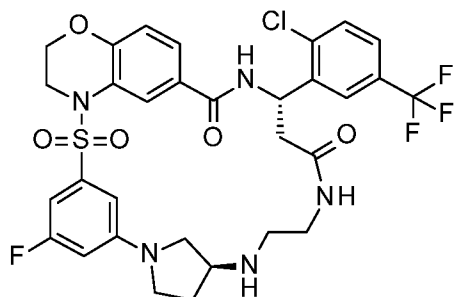
Step 7: *tert*-Butyl (11*S*,18*S*,26*R*)-18-[2-chloro-5-(trifluoromethyl)phenyl]-26-methyl-2,2,16,20-tetraoxo-25-oxa-2λ⁶-thia-1,8,12,15,19-pentazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-3,5,7(31),21(29),22,24(28)-hexaene-12-carboxylate

To a solution of (3*S*)-3-[[*(2R)*-4-[3-[(3*S*)-3-[2-aminoethyl(*tert*-butoxycarbonyl)amino]pyrrolidin-1-yl]phenyl]sulfonyl-2-methyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid (170 mg, 0.21 mmol) 5.0 eq) in acetonitrile (25 ml) was added *N*-methylimidazole (86 mg, 1.05 mmol) and stirred at 20 °C for 5 min , then *N,N,N',N'*-tetramethylchloroformamidinium hexafluorophosphate (56.7 mg, 0.2 mmol) was added. The reaction mixture was stirred at 20 °C for 1 hour. The reaction mixture was filtered, the filtrate was poured into water and was extracted three times with ethyl acetate. The combined organic layer was washed with brine, dried with Na₂SO₄ filtered and evaporated. The crude product was purified by prep-HPLC (Phenomenex Luna C18 150 x 25mm x 10 μm, gradient water with 1
15 1% TFA and acetonitrile) to give *tert*-butyl (11*S*,18*S*,26*R*)-18-[2-chloro-5-(trifluoromethyl)phenyl]-26-methyl-2,2,16,20-tetraoxo-25-oxa-2λ⁶-thia-1,8,12,15,19-pentazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-3,5,7(31),21(29),22,24(28)-hexaene-12-carboxylate (90 mg, 54 % yield) as a white solid. MS m/z: 792.3 [M+H]⁺, ESI pos.

Step 8: (11*S*,18*S*,26*R*)-18-[2-Chloro-5-(trifluoromethyl)phenyl]-26-methyl-2,2-dioxo-25-oxa-2λ⁶-thia-1,8,12,15,19-pentazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-3,5,7(31),21(29),22,24(28)-hexaene-16,20-dione 2,2,2-trifluoroacetic acid salt.

To solution of *tert*-butyl (11*S*,18*S*,26*R*)-18-[2-chloro-5-(trifluoromethyl)phenyl]-26-methyl-2,2,16,20-tetraoxo-25-oxa-2λ⁶-thia-1,8,12,15,19-pentazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-3,5,7(31),21(29),22,24(28)-hexaene-12-carboxylate (80 mg, 0.1 mmol) in dichloromethane (1 ml) was added 2,2,2-trifluoroacetic acid_ (1.0 ml) at 0 °C. The mixture was stirred at 20 °C for 1 hour. The solvent and excess of TFA were evaporated and the residue was purified by prep-HPLC (Phenomenex Luna C18, 150 x 25 mm x10 μm, gradient water with 1% TFA and acetonitrile) to give (11*S*,18*S*,26*R*)-18-[2-chloro-5-(trifluoromethyl)phenyl]-26-methyl-2,2-dioxo-25-oxa-2λ⁶-thia-1,8,12,15,19-pentazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-3,5,7(31),21(29),22,24(28)-hexaene-16,20-dione 2,2,2-trifluoroacetic acid salt (70.2 mg, 85 % yield) as a white solid. MS m/z: 692.2 [M+H]⁺, ESI pos.

Example 13: (11*S*,18*S*)-18-[2-Chloro-5-(trifluoromethyl)phenyl]-5-fluoro-2,2-dioxo-25-oxa-2λ⁶-thia-1,8,12,15,19-pentazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-3,5,7(31),21(29),22,24(28)-hexaene-16,20-dione



5 Step 1: Methyl 4-[3-[(3*S*)-3-(*tert*-butoxycarbonylamino)pyrrolidin-1-yl]-5-fluoro-phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate

The title compound was prepared in analogy to **Example 6, step 1** starting with methyl 4-(3-bromo-5-fluoro-phenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate (**Intermediate C7**) and (*S*)-3-(BOC-amino)pyrrolidine (CAS 122536-76-9) instead of methyl 4-(3-

10 bromophenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate (**Intermediate C4**) and *tert*-butyl 1,4-diazepane-1-carboxylate. Brown solid. MS *m/z*: 536.3 [M+H]⁺, ESI pos.

Step 2: Methyl 4-[3-[(3*S*)-3-aminopyrrolidin-1-yl]-5-fluoro-phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate

The title compound was prepared in analogy to **Example 8, step 2** starting with methyl 4-[3-
15 [(3*S*)-3-(*tert*-butoxycarbonylamino)pyrrolidin-1-yl]-5-fluoro-phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate instead of methyl 4-[3-[(1*R*,4*R*)-2,5-diazabicyclo[2.2.1]heptan-2-yl]phenyl]sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carboxylate. Yellow solid. MS *m/z*: 436.1 [M+H]⁺, ESI pos.

20 Step 3: Methyl 4-[3-[(3*S*)-3-[2-(*tert*-butoxycarbonylamino)ethylamino]pyrrolidin-1-yl]-5-fluoro-phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate

The title compound was prepared in analogy to **Example 10, step 3** starting with methyl 4-[3-[(3*S*)-3-aminopyrrolidin-1-yl]-5-fluoro-phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate instead of methyl 4-[3-[(3*S*)-3-aminopyrrolidin-1-yl]-5-fluoro-phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate. Yellow solid. MS *m/z*: 579.5 [M+H]⁺, ESI pos.

Step 4: Methyl 4-[3-[(3*S*)-3-[allyloxycarbonyl]-2-(*tert*-butoxycarbonylamino)ethyl]amino]pyrrolidin-1-yl]-5-fluoro-phenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate

The title compound was prepared in analogy to **Example 10, step 4** starting with methyl 4-[3-[(3*S*)-3-[2-(*tert*-butoxycarbonylamino)ethylamino]pyrrolidin-1-yl]-5-fluoro-phenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate instead of methyl 4-[3-[(3*S*)-3-[2-(*tert*-butoxycarbonylamino)ethylamino]pyrrolidin-1-yl]phenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate. Yellow solid. MS *m/z*: 663.1 [M+H]⁺, ESI pos.

Step 5: 4-[3-[(3*S*)-3-[Allyloxycarbonyl]-2-(*tert*-butoxycarbonylamino)ethyl]amino]pyrrolidin-1-yl]phenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid

The title compound was prepared in analogy to **Intermediate D3, step 1** starting from methyl 4-[3-[(3*S*)-3-[allyloxycarbonyl]-2-(*tert*-butoxycarbonylamino)ethyl]amino]pyrrolidin-1-yl]-5-fluoro-phenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate instead of methyl 4-(3-methoxy-5-methyl-phenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate (**Intermediate C2**). White solid, MS *m/z*: 649.4 [M+H]⁺, ESI pos.

Step 6: Ethyl (3*S*)-3-[[4-[3-[(3*S*)-3-[allyloxycarbonyl]-2-(*tert*-butoxycarbonylamino)ethyl]amino]pyrrolidin-1-yl]-5-fluoro-phenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate

The title compound was prepared in analogy to **Example 11, step 6** starting with 4-[3-[(3*S*)-3-[allyloxycarbonyl]-2-(*tert*-butoxycarbonylamino)ethyl]amino]pyrrolidin-1-yl]phenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid instead of 4-[3-[(3*S*)-3-[allyloxycarbonyl]-2-(*tert*-butoxycarbonylamino)ethyl]amino]pyrrolidin-1-yl]-5-methyl-phenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid. White solid. MS *m/z*: 926.1 [M+H]⁺, ESI pos.

Step 7: (3*S*)-3-[[4-[3-[(3*S*)-3-[Allyloxycarbonyl]-2-(*tert*-butoxycarbonylamino)ethyl]amino]pyrrolidin-1-yl]-5-fluoro-phenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid

The title compound was prepared in analogy to **Example 4, step 2** starting with ethyl (3*S*)-3-[[4-[3-[(3*S*)-3-[allyloxycarbonyl]-2-(*tert*-butoxycarbonylamino)ethyl]amino]pyrrolidin-1-yl]-5-fluoro-phenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate instead of ethyl (3*S*)-3-[[4-[3-[2-(*tert*-butoxycarbonylamino)ethoxy]ethoxy]-5-methyl-phenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-

carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate. White solid. MS m/z: 898.1 [M+H]⁺, ESI pos.

Step 8: (3S)-3-[[4-[3-[(3S)-3-[Allyloxycarbonyl(2-aminoethyl)amino]pyrrolidin-1-yl]-5-fluoro-phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid hydrochloride salt

The title compound was prepared in analogy to **Example 8, step 7** starting with (3S)-3-[[4-[3-[(3S)-3-[allyloxycarbonyl-[2-(*tert*-butoxycarbonylamino)ethyl]amino]pyrrolidin-1-yl]-5-fluoro-phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid instead of (3S)-3-[[4-[3-[(1R,4R)-5-[2-(*tert*-butoxycarbonylamino)ethyl]-2,5-diazabicyclo[2.2.1]heptan-2-yl]phenyl]sulfonyl]spiro[3H-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid. White solid. MS m/z: 798.4 [M+H]⁺, ESI pos.

Step 9: Allyl (11S,18S)-18-[2-chloro-5-(trifluoromethyl)phenyl]-5-fluoro-2,2,16,20-tetraoxo-25-oxa-2 λ⁶-thia-1,8,12,15,19-pentazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-3,5,7(31),21(29),22,24(28)-hexaene-12-carboxylate

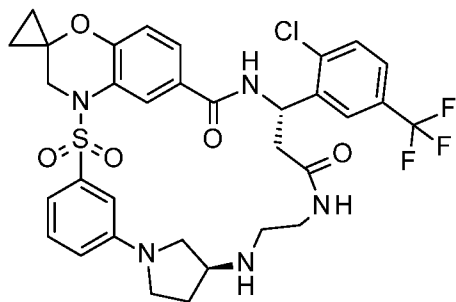
The title compound was prepared in analogy to **Example 11, step 9** starting with (3S)-3-[[4-[3-[(3S)-3-[allyloxycarbonyl(2-aminoethyl)amino]pyrrolidin-1-yl]-5-fluoro-phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid hydrochloride salt instead of (3S)-3-[[4-[3-[(3S)-3-[allyloxycarbonyl(2-aminoethyl)amino]pyrrolidin-1-yl]-5-fluoro-phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid hydrochloride salt. White solid. MS m/z: 780.2 [M+H]⁺, ESI pos.

Step 10: (11S,18S)-18-[2-Chloro-5-(trifluoromethyl)phenyl]-5-fluoro-2,2-dioxo-25-oxa-2 λ⁶-thia-1,8,12,15,19-pentazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-3,5,7(31),21(29),22,24(28)-hexaene-16,20-dione

The title compound was prepared in analogy to **Example 11, step 10** starting with allyl (11S,18S)-18-[2-chloro-5-(trifluoromethyl)phenyl]-5-fluoro-2,2,16,20-tetraoxo-25-oxa-2 λ⁶-thia-1,8,12,15,19-pentazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-3,5,7(31),21(29),22,24(28)-hexaene-12-carboxylate instead of allyl (11S,18S)-18-[2-chloro-5-(trifluoromethyl)phenyl]-5-methyl-2,2,16,20-tetraoxo-25-oxa-2 λ⁶-thia-1,8,12,15,19-

pentazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-3,5,7(31),21(29),22,24(28)-hexaene-12-carboxylate. Off-white solid. MS m/z: 696.1 [M+H]⁺, ESI pos

Example 14: (11*S*,18*S*)-18-[2-Chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-spiro[25-oxa-2λ⁶-thia-1,8,12,15,19-pentazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-3(31),4,6,21(29),22,24(28)-hexaene-26,1'-cyclopropane]-16,20-dione



Step 1: Methyl 4-[3-[(3*S*)-3-(*tert*-butoxycarbonylamino)pyrrolidin-1-yl]phenyl]sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carboxylate

The title compound was prepared in analogy to **Example 6, step 1** starting with methyl 4-(3-bromophenyl)sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carboxylate (**Intermediate C5**) and (*S*)-3-(BOC-amino)pyrrolidine (CAS 122536-76-9) instead of methyl 4-(3-bromophenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate (**Intermediate C4**) and *tert*-butyl 1,4-diazepane-1-carboxylate. Yellow solid. MS m/z: 544.2 [M+H]⁺, ESI pos.

Step 2: Methyl 4-[3-[(3*S*)-3-aminopyrrolidin-1-yl]phenyl]sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carboxylate

The title compound was prepared in analogy to **Example 8, step 2** starting with methyl 4-[3-[(3*S*)-3-(*tert*-butoxycarbonylamino)pyrrolidin-1-yl]phenyl]sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carboxylate instead of methyl 4-[3-[(1*R*,4*R*)-2,5-diazabicyclo[2.2.1]heptan-2-yl]phenyl]sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carboxylate. Yellow oil. MS m/z: 444.2 [M+H]⁺, ESI pos.

Step 3: Methyl 4-[3-[(3*S*)-3-[2-(*tert*-butoxycarbonylamino)ethylamino]pyrrolidin-1-yl]phenyl]sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carboxylate

The title compound was prepared in analogy to **Example 10, step 3** starting with methyl 4-[3-[(3*S*)-3-aminopyrrolidin-1-yl]phenyl]sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-

carboxylate instead of methyl 4-[3-[(3*S*)-3-aminopyrrolidin-1-yl]-5-fluoro-phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate. Yellow oil. MS *m/z*: 587.3 [M+H]⁺, ESI pos.

Step 4: Methyl 4-[3-[(3*S*)-3-[allyloxycarbonyl]-2-(*tert*-butoxycarbonylamino)ethyl]amino]pyrrolidin-1-yl]phenyl]sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carboxylate

The title compound was prepared in analogy to **Example 10, step 4** starting with methyl 4-[3-[(3*S*)-3-[2-(*tert*-butoxycarbonylamino)ethylamino]pyrrolidin-1-yl]phenyl]sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carboxylate instead of methyl 4-[3-[(3*S*)-3-[2-(*tert*-butoxycarbonylamino)ethylamino]pyrrolidin-1-yl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate. Colorless oil. MS *m/z*: 671.3 [M+H]⁺, ESI pos.

Step 5: 4-[3-[(3*S*)-3-[Allyloxycarbonyl]-2-(*tert*-butoxycarbonylamino)ethyl]amino]pyrrolidin-1-yl]phenyl]sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carboxylic acid

The title compound was prepared in analogy to **Intermediate D3, step 1** starting from methyl 4-[3-[(3*S*)-3-[allyloxycarbonyl]-2-(*tert*-butoxycarbonylamino)ethyl]amino]pyrrolidin-1-yl]phenyl]sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carboxylate instead of methyl 4-(3-methoxy-5-methyl-phenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate (**Intermediate C2**). Yellow oil. MS *m/z*: 657.3 [M+H]⁺, ESI pos.

Step 6: Ethyl (3*S*)-3-[[4-[3-[(3*S*)-3-[allyloxycarbonyl]-2-(*tert*-butoxycarbonylamino)ethyl]amino]pyrrolidin-1-yl]phenyl]sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carboxyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate

The title compound was prepared in analogy to **Example 10, step 6** starting with 4-[3-[(3*S*)-3-[allyloxycarbonyl]-2-(*tert*-butoxycarbonylamino)ethyl]amino]pyrrolidin-1-yl]phenyl]sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carboxylic acid instead of 4-[3-[(3*S*)-3-[allyloxycarbonyl]-2-(*tert*-butoxycarbonylamino)ethyl]amino]pyrrolidin-1-yl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid. Yellow oil. MS *m/z*: 934.4 [M+H]⁺, ESI pos.

Step 7: (3*S*)-3-[[4-[3-[(3*S*)-3-[Allyloxycarbonyl]-2-(*tert*-butoxycarbonylamino)ethyl]amino]pyrrolidin-1-yl]phenyl]sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carboxyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid

The title compound was prepared in analogy to **Example 4, step 2** starting with ethyl (3*S*)-3-[[4-[3-[(3*S*)-3-[allyloxycarbonyl]-[2-(*tert*-butoxycarbonylamino)ethyl]amino]pyrrolidin-1-yl]phenyl]sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate instead of ethyl (3*S*)-3-[[4-[3-[2-(*tert*-butoxycarbonylamino)ethoxy]ethoxy]-5-methyl-phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate. Yellow solid. MS *m/z*: 906.3 [M+H]⁺, ESI pos.

Step 8: (3*S*)-3-[[4-[3-[(3*S*)-3-[Allyloxycarbonyl(2-aminoethyl)amino]pyrrolidin-1-yl]phenyl]sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid hydrochloride salt

The title compound was prepared in analogy to **Example 8, step 7** starting with (3*S*)-3-[[4-[3-[(3*S*)-3-[allyloxycarbonyl]-[2-(*tert*-butoxycarbonylamino)ethyl]amino]pyrrolidin-1-yl]phenyl]sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid instead of (3*S*)-3-[[4-[3-[(1*R*,4*R*)-5-[2-(*tert*-butoxycarbonylamino)ethyl]-2,5-diazabicyclo[2.2.1]heptan-2-yl]phenyl]sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid. Red solid. MS *m/z*: 806.3 [M+H]⁺, ESI pos.

Step 9: Allyl (11*S*,18*S*)-18-[2-chloro-5-(trifluoromethyl)phenyl]-2,2,16,20-tetraoxo-spiro[25-oxa-2λ⁶-thia-1,8,12,15,19-pentazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-3,5,7(31),21(29),22,24(28)-hexaene-26,1'-cyclopropane]-12-carboxylate

To a solution of (3*S*)-3-[[4-[3-[(3*S*)-3-[allyloxycarbonyl(2-aminoethyl)amino]pyrrolidin-1-yl]phenyl]sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid hydrochloride salt (210 mg, 0.25 mmol) in acetonitrile (6 ml) was added *N*-methylimidazole (61.4 mg, 0.75 mmol) and stirred at 20 °C for 5 min, then *N,N,N',N'*-tetramethylchloroformamidinium hexafluorophosphate (83.9 mg, 0.3 mmol) was added. The reaction mixture was stirred at 20 °C for 16 hours and was poured into water. The aqueous phase was extracted three times with ethyl acetate. The combined organic layer was washed with brine, dried over Na₂SO₄, filtered and concentrated under reduced pressure. The crude product was purified by reverse phase by reverse phase flash chromatography (Flash Spherical C18, 20-35 μm; 100 Å; gradient water with 0.05 % NH₄OH and acetonitrile) to give allyl (11*S*,18*S*)-18-[2-chloro-5-(trifluoromethyl)phenyl]-2,2,16,20-tetraoxo-spiro[25-oxa-2λ⁶-thia-1,8,12,15,19-pentazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-

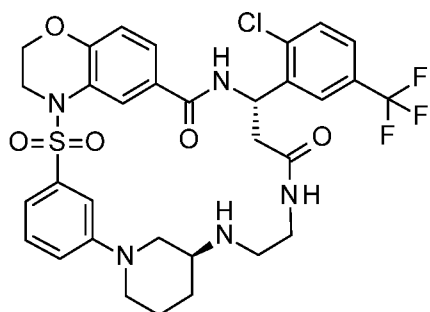
3,5,7(31),21(29),22,24(28)-hexaene-26,1'-cyclopropane]-12-carboxylate (85 mg, 40 % yield) as an orange solid. MS m/z: 788.3 [M+H]⁺, ESI pos.

Step 10: (11*S*,18*S*)-18-[2-Chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-spiro[25-oxa-2λ₆-thia-1,8,12,15,19-pentazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-

5 3,5,7(31),21(29),22,24(28)-hexaene-26,1'-cyclopropane]-16,20-dione

To a solution of allyl (11*S*,18*S*)-18-[2-chloro-5-(trifluoromethyl)phenyl]-2,2,16,20-tetraoxo-spiro[25-oxa-2λ₆-thia-1,8,12,15,19-pentazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-3,5,7(31),21(29),22,24(28)-hexaene-26,1'-cyclopropane]-12-carboxylate (50 mg, 0.06 mmol) in tetrahydrofuran (1 ml) were added 1,3-dimethylbarbituric acid (69.3 mg, 0.44 mmol) and
10 Pd(PPh₃)₄ (14.7 mg, 0.01 mmol) in tetrahydrofuran (1 ml) under nitrogen. The mixture was stirred at 20 °C for 2 hours. The solvent was evaporated and the residue was purified via prep-TLC chromatography (silica gel, ethyl acetate in petroleum ether) followed by prep-HPLC (Phenomenex Luna C18, 150 x 25 mm x 10 μm, gradient water with 0.5 % ammonium hydroxide and acetonitrile) and finally by SFC (DAICEL CHIRALPAK IC (250 mm x 30
15 mm,10 μm), ethanol, 0.1% NH₄OH) to give compound (11*S*,18*S*)-18-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-spiro[25-oxa-2λ₆-thia-1,8,12,15,19-pentazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-3,5,7(31),21(29),22,24(28)-hexaene-26,1'-cyclopropane]-16,20-dione (15.8 mg, 35 % yield) as a yellow solid. MS m/z: 704.2 [M+H]⁺, ESI pos.

20 **Example 15:** (12*S*,19*S*)-19-[2-Chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-26-oxa-2λ₆-thia-1,8,13,16,20-pentazapentacyclo[20.6.2.13,7.18,12.025,29]dotriaconta-3,5,7(32),22(30),23,25(29)-hexaene-17,21-dione



Step 1: Methyl 4-[3-[(3*S*)-3-(*tert*-butoxycarbonylamino)-1-piperidyl]phenyl]sulfonyl-2,3-
25 dihydro-1,4-benzoxazine-6-carboxylate

The title compound was prepared in analogy to **Example 6, step 1** starting with *tert*-butyl-*N*-3-*S*-3-piperidyl]carbamate (CAS 216854-23-8) instead of *tert*-butyl 1,4-diazepane-1-carboxylate. Yellow solid. MS *m/z*: 532.2 [M+H]⁺, ESI pos.

Step 2: Methyl 4-[3-[(3*S*)-3-amino-1-piperidyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate

The title compound was prepared in analogy to **Example 8, step 2** starting with methyl 4-[3-[(3*S*)-3-(*tert*-butoxycarbonylamino)-1-piperidyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate instead of methyl 4-[3-[(1*R*,4*R*)-2,5-diazabicyclo[2.2.1]heptan-2-yl]phenyl]sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carboxylate. Yellow solid. MS *m/z*: 432.1 [M+H]⁺, ESI pos.

Step 3: Methyl 4-[3-[(3*S*)-3-[2-(*tert*-butoxycarbonylamino)ethylamino]-1-piperidyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate

The title compound was prepared in analogy to **Example 10, step 3** starting with methyl 4-[3-[(3*S*)-3-amino-1-piperidyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate instead of methyl 4-[3-[(3*S*)-3-aminopyrrolidin-1-yl]-5-fluoro-phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate. Yellow solid. MS *m/z*: 575.3 [M+H]⁺, ESI pos.

Step 4: Methyl 4-[3-[(3*S*)-3-[2-(*tert*-butoxycarbonylamino)ethyl-vinyloxycarbonyl-amino]-1-piperidyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate

To a solution of methyl 4-[3-[(3*S*)-3-[2-(*tert*-butoxycarbonylamino)ethylamino]-1-piperidyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate (990 mg, 1.72 mmol) in a mixture of tetrahydrofuran (10 ml) and water (10 ml) were added NaHCO₃ (217 mg, 2.58 mmol) and allyl chloroformate (282 mg, 2.07 mmol) at 0 °C. The mixture was stirred at 20 °C for 2 hours. The mixture was poured into water and extracted three times with ethyl acetate. The combined organic layer was dried with Na₂SO₄, filtered and evaporated. The crude product was purified by flash chromatography (silica gel, ethyl acetate/petroleum ether 30-50%) to give methyl 4-[3-[(3*S*)-3-[2-(*tert*-butoxycarbonylamino)ethyl-vinyloxycarbonyl-amino]-1-piperidyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate (830 mg, 70 % yield) as a white solid. MS *m/z*: 659.2 [M+H]⁺, ESI pos.

Step 5: 4-[3-[(3*S*)-3-[Allyloxycarbonyl]-[2-(*tert*-butoxycarbonylamino)ethyl]amino]-1-piperidyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid

The title compound was prepared in analogy to **Intermediate D3, step 1** starting from methyl 4-[3-[(3*S*)-3-[2-(*tert*-butoxycarbonylamino)ethyl-vinyloxycarbonyl-amino]-1-piperidyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate instead of methyl 4-(3-methoxy-5-methyl-phenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate (**Intermediate C2**). White solid, MS m/z: 645.2 [M+H]⁺, ESI pos.

Step 6: Ethyl (3*S*)-3-[[4-[3-[(3*R*)-3-[allyloxycarbonyl]-2-(*tert*-butoxycarbonylamino)ethyl]amino]-1-piperidyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate

The title compound was prepared in analogy to **Example 8, step 5** starting with 4-[3-[(3*S*)-3-[allyloxycarbonyl]-2-(*tert*-butoxycarbonylamino)ethyl]amino]-1-piperidyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid instead of 4-[3-[(1*R*,4*R*)-5-[2-(*tert*-butoxycarbonylamino)ethyl]-2,5-diazabicyclo[2.2.1]heptan-2-yl]phenyl]sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carboxylic acid. Yellow solid. MS m/z: 922.3 [M+H]⁺, ESI pos.

Step 7: (3*S*)-3-[[4-[3-[(3*S*)-3-[Allyloxycarbonyl]-2-(*tert*-butoxycarbonylamino)ethyl]amino]-1-piperidyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid

The title compound was prepared in analogy to **Example 4, step 2** starting with ethyl (3*S*)-3-[[4-[3-[(3*R*)-3-[allyloxycarbonyl]-2-(*tert*-butoxycarbonylamino)ethyl]amino]-1-piperidyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate instead of ethyl (3*S*)-3-[[4-[3-[2-(*tert*-butoxycarbonylamino)ethoxy]ethoxy]-5-methyl-phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate. White solid. MS m/z: 894.0 [M+H]⁺, ESI pos.

Step 8: (3*S*)-3-[[4-[3-[(3*S*)-3-[Allyloxycarbonyl(2-aminoethyl)amino]-1-piperidyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid hydrochloride salt

The title compound was prepared in analogy to **Example 4, step 3** starting from (3*S*)-3-[[4-[3-[(3*S*)-3-[allyloxycarbonyl]-2-(*tert*-butoxycarbonylamino)ethyl]amino]-1-piperidyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid instead of (3*S*)-3-[[4-[3-[2-(*tert*-

butoxycarbonylamino)ethoxy]ethoxy]-5-methyl-phenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid. White solid, MS m/z: 794.2 [M+H]⁺, ESI pos.

Step 9: Allyl (12*S*,19*S*)-19-[2-chloro-5-(trifluoromethyl)phenyl]-2,2,17,21-tetraoxo-26-oxa-2λ⁶-thia-1,8,13,16,20-pentazapentacyclo[20.6.2.13,7.18,12.025,29]dotriaconta-3,5,7(32),22(30),23,25(29)-hexaene-13-carboxylate

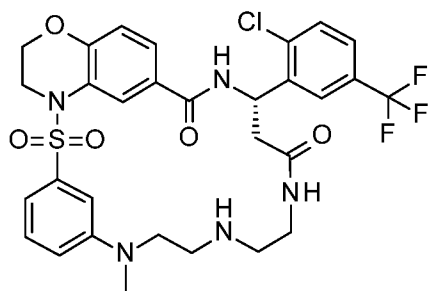
To a solution of (3*S*)-3-[[4-[3-[(3*S*)-3-allyloxycarbonyl(2-aminoethyl)amino]-1-piperidyl]phenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid hydrochloride salt (430 mg, 0.52 mmol) in *N,N*-dimethylformamide (10 ml) were added DIPEA (267 mg, 2.07 mmol) and HATU (236 mg, 0.62 mmol). The reaction mixture was stirred at 20 °C for 1 hour. The mixture was poured into water and extracted three times with ethyl acetate. The combined organic layer was dried with Na₂SO₄, filtered and evaporated. The crude product was purified by reverse phase flash chromatography (Flash Spherical C18, 20-35 μm; 100 Å; gradient water with 0.05 % TFA and acetonitrile) to give allyl (12*S*,19*S*)-19-[2-chloro-5-(trifluoromethyl)phenyl]-2,2,17,21-tetraoxo-26-oxa-2λ⁶-thia-1,8,13,16,20-pentazapentacyclo[20.6.2.13,7.18,12.025,29]dotriaconta-3,5,7(32),22(30),23,25(29)-hexaene-13-carboxylate (180 mg, 41 % yield) as a white solid. MS m/z: 776.1 [M+H]⁺, ESI pos.

Step 10: (12*S*,19*S*)-19-[2-Chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-26-oxa-2λ⁶-thia-1,8,13,16,20-pentazapentacyclo[20.6.2.13,7.18,12.025,29]dotriaconta-3,5,7(32),22(30),23,25(29)-hexaene-17,21-dione

To a solution of allyl (12*S*,19*S*)-19-[2-chloro-5-(trifluoromethyl)phenyl]-2,2,17,21-tetraoxo-26-oxa-2λ⁶-thia-1,8,13,16,20-pentazapentacyclo[20.6.2.13,7.18,12.025,29]dotriaconta-3,5,7(32),22(30),23,25(29)-hexaene-13-carboxylate (160.0 mg, 0.21 mmol) in tetrahydrofuran (3 ml), were added 1,3-dimethylbarbituric acid (225 mg, 1.44 mmol) and Pd(PPh₃)₄ (48 mg, 0.04 mmol) under a nitrogen atmosphere. The mixture was stirred at 20 °C for 2 hours. The mixture was poured into water and extracted three times with ethyl acetate. The combined organic layer was concentrated. The crude product was purified by prep-TLC (silica gel, methanol in dichloromethane 10 %). The combined fractions were evaporated and the residue was purified by prep-HPLC (Phenomenex Luna C18 150 x 25 mm x 10 μm, gradient water with 2.25 % formic acid and acetonitrile) to give (12*S*,19*S*)-19-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-26-oxa-2λ⁶-thia-1,8,13,16,20-pentazapentacyclo[20.6.2.13,7.18,12.025,29]dotriaconta-

3,5,7(32),22(30),23,25(29)-hexaene-17,21-dione (50 mg, 26 % yield) as a white solid. MS m/z: 692.0 [M+H]⁺, ESI pos.

Example 16: (17*S*)-17-[2-Chloro-5-(trifluoromethyl)phenyl]-8-methyl-2,2-dioxo-24-oxa-2λ⁶-thia-1,8,11,14,18-pentazatetracyclo[18.6.2.13,7.0.23,27]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-15,19-dione



Step 1: Methyl 4-[3-[2-(*tert*-butoxycarbonylamino)ethyl-methyl-amino]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate

The title compound was prepared in analogy to **Example 6, step 1** starting with *tert*-butyl 2-(methylamino)ethylcarbamate (CAS 122734-32-1) instead of *tert*-butyl 1,4-diazepane-1-carboxylate. Yellow solid. MS m/z: 506.1 [M+H]⁺, ESI pos.

Step 2: 4-[3-[2-(*tert*-Butoxycarbonylamino)ethyl-methyl-amino]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid

To solution of methyl 4-[3-[2-(*tert*-butoxycarbonylamino)ethyl-methyl-amino]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate (1.47 g, 2.91 mmol) dissolved in a mixture of tetrahydrofuran (7 ml), methanol (7 ml) and water (7 ml) was added LiOH · H₂O (366 mg, 8.72 mmol). The mixture was stirred at 20 °C for 1 hour. The mixture was poured into water and extracted three times with ethyl acetate. The aqueous layer was collected and the pH was adjusted to 3-4 by addition of 1 N HCl before extracting three times with ethyl acetate. The combined organic layer was dried with Na₂SO₄, filtered and evaporated to give 4-[3-[2-(*tert*-butoxycarbonylamino)ethyl-methyl-amino]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid (1.5 g, > 100 % yield). Yellow oil. MS m/z: 492.2 [M+H]⁺, ESI pos.

Step 3: Ethyl (3*S*)-3-[[4-[3-[2-(*tert*-butoxycarbonylamino)ethyl-methyl-amino]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate

To a solution of 4-[3-[2-(*tert*-butoxycarbonylamino)ethyl-methyl-amino]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid (1.45 g, 2.95 mmol) in *N,N*-dimethylformamide (15 ml) were added, ethyl (3*S*)-3-amino-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate (1.31 g, 4.42 mmol), DIPEA (1.56 ml, 8.8 mmol) and HATU (0.83 g, 3.54 mmol). The mixture was stirred at 20 °C for 16 hours. The mixture was poured into water and extracted three times with ethyl acetate. The combined organic layer was dried with Na₂SO₄, filtered and evaporated. The crude product was purified via flash chromatography (silica gel, ethyl acetate/ petroleum ether 25-50 %) to give ethyl (3*S*)-3-[[4-[3-[2-(*tert*-butoxycarbonylamino)ethyl-methyl-amino]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate (1.5 g, 48 % yield). White oil. MS *m/z*: 769.3 [M+H]⁺, ESI pos.

Step 4: Ethyl (3*S*)-3-[[4-[3-[2-aminoethyl(methyl)amino]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate

The title compound was prepared in analogy to **Example 8, step 2** starting with ethyl (3*S*)-3-[[4-[3-[2-(*tert*-butoxycarbonylamino)ethyl-methyl-amino]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate instead of methyl 4-[3-[(1*R*,4*R*)-2,5-diazabicyclo[2.2.1]heptan-2-yl]phenyl]sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carboxylate. Yellow oil. MS *m/z*: 669.2 [M+H]⁺, ESI pos.

Step 5: Ethyl (3*S*)-3-[[4-[3-[2-[2-(*tert*-butoxycarbonylamino)ethylamino]ethyl-methyl-amino]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate

To solution of ethyl (3*S*)-3-[[4-[3-[2-aminoethyl(methyl)amino]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate (1.1 g, 1.64 mmol) in methanol (10 ml) were added acetic acid until pH 3-4, *N*-BOC-2-aminoacetaldehyde (393 mg, 2.47 mmol) and NaBH₃CN (1.04 g, 16.4 mmol). The mixture was stirred at 20 °C for 2 hours. The mixture was poured into water and extracted three times with ethyl acetate. The combined organic layer was dried with Na₂SO₄, filtered and evaporated. The crude product was purified by prep-HPLC (Phenomenex Luna C18 150 x 25mm x 10 μm, gradient water with 0.5% TFA and acetonitrile) to give ethyl (3*S*)-3-[[4-[3-[2-[2-(*tert*-butoxycarbonylamino)ethylamino]ethyl-methyl-amino]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate (520 mg, 37 % yield). White solid. MS *m/z*: 812.3 [M+H]⁺, ESI pos.

Step 6: Ethyl (3*S*)-3-[[4-[3-[2-[allyloxycarbonyl]-2-(*tert*-butoxycarbonylamino)ethyl]amino]ethyl-methyl-amino]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate

The title compound was prepared in analogy to **Example 10, step 4** starting with ethyl (3*S*)-3-[[4-[3-[2-[2-(*tert*-butoxycarbonylamino)ethylamino]ethyl-methyl-amino]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate instead of methyl 4-[3-[(3*S*)-3-[2-(*tert*-butoxycarbonylamino)ethylamino]pyrrolidin-1-yl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate. Yellow solid. MS *m/z*: 896.4 [M+H]⁺, ESI pos.

Step 7: (3*S*)-3-[[4-[3-[2-[Allyloxycarbonyl]-2-(*tert*-butoxycarbonylamino)ethyl]amino]ethyl-methyl-amino]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid

The title compound was prepared in analogy to **Example 4, step 2** starting with ethyl (3*S*)-3-[[4-[3-[2-[allyloxycarbonyl]-2-(*tert*-butoxycarbonylamino)ethyl]amino]ethyl-methyl-amino]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate instead of ethyl (3*S*)-3-[[4-[3-[2-[2-(*tert*-butoxycarbonylamino)ethoxy]ethoxy]-5-methyl-phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate. White solid. MS *m/z*: 868.4 [M+H]⁺, ESI pos.

Step 8: (3*S*)-3-[[4-[3-[2-[Allyloxycarbonyl(2-aminoethyl)amino]ethyl-methyl-amino]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid hydrochloride acid

To a solution of (3*S*)-3-[[4-[3-[2-[allyloxycarbonyl]-2-(*tert*-butoxycarbonylamino)ethyl]amino]ethyl-methyl-amino]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid (370 mg, 0.43 mmol) in dichloromethane (3 ml) was added HCl/1,4-dioxane (3.0 ml, 12 mmol) at 0 °C and the mixture was stirred at 20 °C for 1 hour. Water was added and the aqueous solution was lyophilized to give (3*S*)-3-[[4-[3-[2-[allyloxycarbonyl(2-aminoethyl)amino]ethyl-methyl-amino]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid hydrochloride salt (300 mg, 81 % yield) as a white solid. MS *m/z*: 768.3 [M+H]⁺, ESI pos.

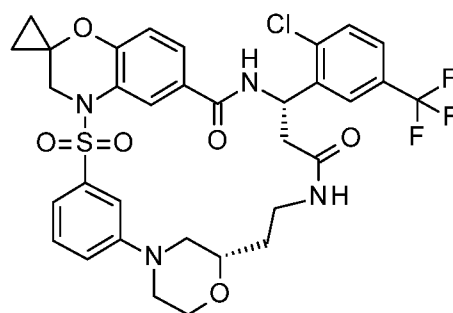
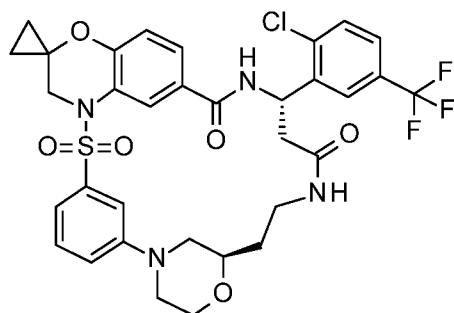
Step 9: Allyl (17S)-17-[2-chloro-5-(trifluoromethyl)phenyl]-8-methyl-2,2,15,19-tetraoxo-24-oxa-2 λ^6 -thia-1,8,11,14,18-pentazatetracyclo[18.6.2.13,7.023,27]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-11-carboxylate

The title compound was prepared in analogy to **Example 14, step 9** starting with (3S)-3-[[4-[3-[2-[allyloxycarbonyl(2-aminoethyl)amino]ethyl-methyl-amino]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid hydrochloride salt instead of (3S)-3-[[4-[3-[(3S)-3-[allyloxycarbonyl(2-aminoethyl)amino]pyrrolidin-1-yl]phenyl]sulfonylspiro[3H-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid hydrochloride salt and stirring for 20 hours instead of 1 hour. White solid. MS m/z: 750.3 [M+H]⁺, ESI pos.

Step 10: (17S)-17-[2-Chloro-5-(trifluoromethyl)phenyl]-8-methyl-2,2-dioxo-24-oxa-2 λ^6 -thia-1,8,11,14,18-pentazatetracyclo[18.6.2.13,7.023,27]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-15,19-dione

To a solution of allyl (17S)-17-[2-chloro-5-(trifluoromethyl)phenyl]-8-methyl-2,2,15,19-tetraoxo-24-oxa-2 λ^6 -thia-1,8,11,14,18-pentazatetracyclo[18.6.2.13,7.023,27]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-11-carboxylate (140 mg, 0.19 mmol) in tetrahydrofuran (5 ml) was added 1,3-dimethylbarbituric acid (204 mg, 1.31 mmol) and Pd(PPh₃)₄ (43.1 mg, 0.04 mmol) under nitrogen. The mixture was stirred at 20 °C for 1 hour. The reaction mixture was filtered and the filtrate was evaporated. The residue was purified firstly by prep-TLC (SiO₂, petroleum ether/ethyl acetate) followed by by prep-HPLC (Phenomenex Luna C18 150 x 25 mm x 10 μ m, gradient water with 0.5 % ammonium hydroxide and acetonitrile) and with SFC (REGIS(S,S)WHELK-O1 (250 mm x 25 mm, 10 μ m), methanol with 0.1 % NH₄OH) to give (17S)-17-[2-chloro-5-(trifluoromethyl)phenyl]-8-methyl-2,2-dioxo-24-oxa-2 λ^6 -thia-1,8,11,14,18-pentazatetracyclo[18.6.2.13,7.023,27]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-15,19-dione (68 mg, 55 % yield) as a white solid. MS m/z: 666.2 [M+H]⁺, ESI pos.

Example 17 and Example 18: (12R,18S)-18-[2-Chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-spiro[11,25-dioxa-2 λ^6 -thia-1,8,15,19-tetrazapentacyclo[19.6.2.13,7.18,12.024,28]hentriaconta-3(31),4,6,21(29),22,24(28)-hexaene-26,1'-cyclopropane]-16,20-dione and (12S,18S)-18-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-spiro[11,25-dioxa-2 λ^6 -thia-1,8,15,19-tetrazapentacyclo[19.6.2.13,7.18,12.024,28]hentriaconta-3(31),4,6,21(29),22,24(28)-hexaene-26,1'-cyclopropane]-16,20-dione (12R or 12S arbitrarily assigned)



Step 1: Methyl 4-[3-[2-[2-(*tert*-butoxycarbonylamino)ethyl]morpholin-4-yl]phenyl]sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carboxylate

To a solution of methyl 4-(3-bromophenyl)sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carboxylate (**Intermediate C5**, 500 mg, 1.14 mmol) in 1,4-dioxane (10 ml) were added *tert*-butyl N-(2-morpholin-2-ylethyl)carbamate (394 mg, 1.71 mmol), Cs₂CO₃ (558 mg, 1.71 mmol), xantphos (66 mg, 0.11 mmol) and Pd₂(dba)₃ · CHCl₃ (118 mg, 0.11 mmol). The reaction mixture was stirred at 100 °C for 16 hours under nitrogen. The reaction mixture was filtered, concentrated and purified by flash chromatography (silica gel, ethyl acetate/petroleum ether 30-60 %) to give methyl 4-[3-[2-[2-(*tert*-butoxycarbonylamino)ethyl]morpholin-4-yl]phenyl]sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carboxylate (520 mg, 72 % yield) as a brown solid MS *m/z*: 588.2 [M+H]⁺, ESI pos.

Step 2: 4-[3-[2-[2-(*tert*-Butoxycarbonylamino)ethyl]morpholin-4-yl]phenyl]sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carboxylic acid

The title compound was prepared in analogy to **Example 16, step 2** starting with methyl 4-[3-[2-[2-(*tert*-butoxycarbonylamino)ethyl]morpholin-4-yl]phenyl]sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carboxylate instead of methyl 4-[3-[2-(*tert*-butoxycarbonylamino)ethyl-methyl-amino]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate and stirring at 50 °C for 2 hours instead of 20 °C for 1 hour. White solid. MS *m/z*: 574.3 [M+H]⁺, ESI pos.

Step 3: Ethyl (3*S*)-3-[[4-[3-[2-[2-(*tert*-butoxycarbonylamino)ethyl]morpholin-4-yl]phenyl]sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate

The title compound was prepared in analogy to **Example 16, step 3** starting with 4-[3-[2-[2-(*tert*-butoxycarbonylamino)ethyl]morpholin-4-yl]phenyl]sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carboxylic acid instead of 4-[3-[2-(*tert*-butoxycarbonylamino)ethyl-methyl-

amino]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid and stirring for 2 hours instead of 16 hours. Yellow oil. MS m/z: 851.1 [M+H]⁺, ESI pos.

Step 4: (3S)-3-[[4-[3-[2-[2-(tert-Butoxycarbonylamino)ethyl]morpholin-4-yl]phenyl]sulfonyl]spiro[3H-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid

The title compound was prepared in analogy to **Intermediate D3, step 1** starting from ethyl (3S)-3-[[4-[3-[2-[2-(tert-butoxycarbonylamino)ethyl]morpholin-4-yl]phenyl]sulfonyl]spiro[3H-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate instead of methyl 4-(3-methoxy-5-methyl-phenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate (**Intermediate C2**). Yellow oil. MS m/z: 823.2 [M+H]⁺, ESI pos.

Step 5: (3S)-3-[[4-[3-[2-(2-Aminoethyl)morpholin-4-yl]phenyl]sulfonyl]spiro[3H-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid 2,2,2-trifluoroacetic acid salt

To solution of (3S)-3-[[4-[3-[2-[2-(tert-butoxycarbonylamino)ethyl]morpholin-4-yl]phenyl]sulfonyl]spiro[3H-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid (190 mg, 0.23 mmol) in dichloromethane (1.5 ml) was added TFA (1.5 ml) at 0 °C. The mixture was allowed to stirred at 20 °C for 1 hour. The reaction mixture was evaporated and lyophilised to give (3S)-3-[[4-[3-[2-(2-aminoethyl)morpholin-4-yl]phenyl]sulfonyl]spiro[3H-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid 2,2,2-trifluoroacetic acid salt (250 mg, > 100 % yield) as a white solid. MS m/z: 723.1 [M+H]⁺, ESI pos.

Step 6: (12R,18S)-18-[2-Chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-spiro[11,25-dioxa-2λ⁶-thia-1,8,15,19-tetrazapentacyclo[19.6.2.13,7.18,12.024,28]hentriaconta-3(31),4,6,21(29),22,24(28)-hexaene-26,1'-cyclopropane]-16,20-dione and (12S,18S)-18-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-spiro[11,25-dioxa-2λ⁶-thia-1,8,15,19-tetrazapentacyclo[19.6.2.13,7.18,12.024,28]hentriaconta-3(31),4,6,21(29),22,24(28)-hexaene-26,1'-cyclopropane]-16,20-dione

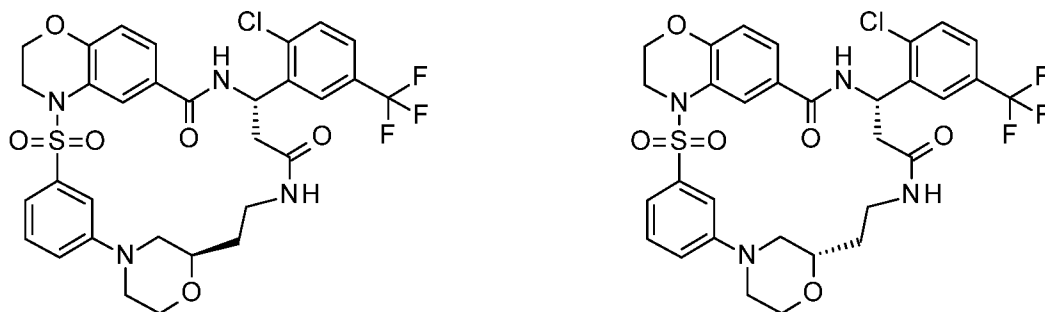
To a solution of (3S)-3-[[4-[3-[2-(2-aminoethyl)morpholin-4-yl]phenyl]sulfonyl]spiro[3H-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid 2,2,2-trifluoroacetic acid salt (230 mg, 0.27 mmol) in

acetonitrile (23 ml) was added N-methyl imidazole (67mg, 0.82 mmol). After stirring for 5 minutes at 20 ° C, *N,N,N',N'*-tetramethylchloroformamidinium hexafluorophosphate (56.8 mg, 0.2 mmol) was added. The solution was stirred at 20 ° C for 1 hour. The reaction mixture was concentrated and purified by prep-HPLC (Waters Xbridge 150 x 25 mm x 5 µm, water with 0.5 % ammonium hydroxide and acetonitrile) to give the two diastereomers.

First eluting diastereomer (absolute stereochemistry arbitrarily assigned): (12*R*,18*S*)-18-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-spiro[11,25-dioxa-2λ⁶-thia-1,8,15,19-tetrazapentacyclo[19.6.2.13,7.18,12.024,28]hentriaconta-3(31),4,6,21(29),22,24(28)-hexaene-26,1'-cyclopropane]-16,20-dione (18.8 mg, 9 % yield). White solid. MS *m/z*: 705.1 [M+H]⁺, ESI pos.

Second eluting diastereomer (absolute stereochemistry arbitrarily assigned): (12*S*,18*S*)-18-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-spiro[11,25-dioxa-2λ⁶-thia-1,8,15,19-tetrazapentacyclo[19.6.2.13,7.18,12.024,28]hentriaconta-3(31),4,6,21(29),22,24(28)-hexaene-26,1'-cyclopropane]-16,20-dione (16.4 mg, 8 % yield). White solid. MS *m/z*: 705.1 [M+H]⁺, ESI pos.

Example 19 and Example 20: (12*R*,18*S*)-18-[2-Chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-11,25-dioxa-2λ⁶-thia-1,8,15,19-tetrazapentacyclo[19.6.2.13,7.18,12.024,28]hentriaconta-3(31),4,6,21(29),22,24(28)-hexaene-16,20-dione and (12*S*,18*S*)-18-[2-Chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-11,25-dioxa-2λ⁶-thia-1,8,15,19-tetrazapentacyclo[19.6.2.13,7.18,12.024,28]hentriaconta-3(31),4,6,21(29),22,24(28)-hexaene-16,20-dione (12*R* or 12*S* arbitrarily assigned)



Step 1: Methyl 4-[3-[2-[2-(*tert*-butoxycarbonylamino)ethyl]morpholin-4-yl]phenyl]sulfonylspro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carboxylate

The title compound was prepared in analogy to **Example 17 and 18, step 1** starting with methyl 4-(3-bromophenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate (**Intermediate C4**) instead of methyl 4-(3-bromophenyl)sulfonylspro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carboxylate (**Intermediate C5**). Brown solid. MS *m/z*: 506.2 [M+H]⁺, ESI pos.

5 Step 2: 4-[3-[2-[2-(*tert*-Butoxycarbonylamino)ethyl]morpholin-4-yl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid

The title compound was prepared in analogy to **Example 16, step 2** starting with methyl 4-[3-[2-[2-(*tert*-butoxycarbonylamino)ethyl]morpholin-4-yl]phenyl]sulfonylspro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carboxylate instead of methyl 4-[3-[2-(*tert*-butoxycarbonylamino)ethyl-methyl-amino]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate and stirring at 50 ° C
10 for 2 hours instead at 20 ° C for 1 hour. White solid. MS *m/z*: 548.2 [M+H]⁺, ESI pos.

Step 3: Ethyl (3*S*)-3-[[4-[3-[2-[2-(*tert*-butoxycarbonylamino)ethyl]morpholin-4-yl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate

15 The title compound was prepared in analogy to **Example 16, step 3** starting with 4-[3-[2-[2-(*tert*-butoxycarbonylamino)ethyl]morpholin-4-yl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid instead of 4-[3-[2-(*tert*-butoxycarbonylamino)ethyl-methyl-amino]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid and stirring for 2 hours instead of 16 hours. White solid. MS *m/z*: 825.2 [M+H]⁺, ESI pos.

20 Step 4: (3*S*)-3-[[4-[3-[2-[2-(*tert*-Butoxycarbonylamino)ethyl]morpholin-4-yl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid

To a solution of ethyl (3*S*)-3-[[4-[3-[2-[2-(*tert*-butoxycarbonylamino)ethyl]morpholin-4-yl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate (200 mg, 0.24 mmol) in methanol (4 ml) was added NaOH
25 (0.24 ml, 0.97 mmol). The reaction mixture was stirred at 20 °C for 3 hours. The reaction mixture was adjusted to pH to 4 by addition of an aqueous solution of 1 M HCl and was then extracted twice with ethyl acetate. The combined organic layer was dried with Na₂SO₄, filtered and evaporated to afford (3*S*)-3-[[4-[3-[2-[2-(*tert*-butoxycarbonylamino)ethyl]morpholin-4-yl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-
30

(trifluoromethyl)phenyl]propanoic acid (200 mg, 97 % yield) as a white solid. MS m/z: 797.2 [M+H]⁺, ESI pos.

Step 5: (3*S*)-3-[[4-[3-[2-(2-Aminoethyl)morpholin-4-yl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid hydrochloride salt

To a solution of (3*S*)-3-[[4-[3-[2-(2-*tert*-butoxycarbonylamino)ethyl]morpholin-4-yl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid (190 mg, 0.24 mmol) in dichloromethane (1.5 ml) was added a solution of 4 M HCl in dioxane (1.5 ml, 6.0 mmol). The reaction mixture was stirred at 20 °C for 2 hours. The reaction mixture was concentrated to give crude (3*S*)-3-[[4-[3-[2-(2-aminoethyl)morpholin-4-yl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid hydrochloride salt (170 mg, 97 % yield) as a grey solid. MS m/z: 697.3 [M+H]⁺, ESI pos.

Step 6: : (12*R*,18*S*)-18-[2-Chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-11,25-dioxo-2λ⁶-thia-1,8,15,19-tetrazapentacyclo[19.6.2.13,7.18,12.024,28]hentriaconta-3(31),4,6,21(29),22,24(28)-hexaene-16,20-dione and (12*S*,18*S*)-18-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-11,25-dioxo-2λ⁶-thia-1,8,15,19-tetrazapentacyclo[19.6.2.13,7.18,12.024,28]hentriaconta-3(31),4,6,21(29),22,24(28)-hexaene-16,20-dione

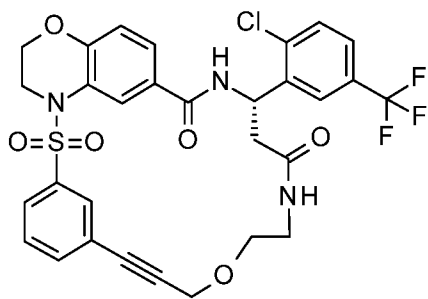
The title compounds were prepared in analogy to **Example 17 and 18, step 6** starting with (3*S*)-3-[[4-[3-[2-(2-aminoethyl)morpholin-4-yl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid hydrochloride salt instead of (3*S*)-3-[[4-[3-[2-(2-aminoethyl)morpholin-4-yl]phenyl]sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid 2,2,2-trifluoroacetic acid salt.

The mixture of diastereomers was separated using prep-HPLC (Phenomenex Luna C18 150 x 25 mm x 10 μm, gradient water with 2.25 % formic acid and acetonitrile).

First eluting diastereomer (absolute stereochemistry arbitrarily assigned): (12*R*,18*S*)-18-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-11,25-dioxo-2λ⁶-thia-1,8,15,19-tetrazapentacyclo[19.6.2.13,7.18,12.024,28]hentriaconta-3(31),4,6,21(29),22,24(28)-hexaene-16,20-dione (25.8 mg, 16 % yield). White solid. MS m/z: 679.1 [M+H]⁺, ESI pos.

Second eluting diastereomer (absolute stereochemistry arbitrarily assigned): (12*S*,18*S*)-18-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-11,25-dioxa-2 λ^6 -thia-1,8,15,19-tetrazapentacyclo[19.6.2.13,7.18,12.024,28]hentriaconta-3(31),4,6,21(29),22,24(28)-hexaene-16,20-dione (30.1 mg, 18 % yield). White solid. MS *m/z*: 679.1 [M+H]⁺, ESI pos.

- 5 **Example 21:** (17*S*)-17-[2-Chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-11,24-dioxa-2 λ^6 -thia-1,14,18-triazatetracyclo[18.6.2.13,7.023,27]nonacosa-3,5,7(29),20(28),21,23(27)-hexaen-8-yne-15,19-dione



- 10 Step 1: (3*S*)-3-[[4-[3-[3-[2-(*tert*-Butoxycarbonylamino)ethoxy]prop-1-ynyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid ethyl ester

To a solution of (3*S*)-3-[[4-(3-bromophenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid ethyl ester (**Intermediate**

- 15 **D5**, 200 mg, 295.9 μ mol) in dry 1,4-dioxane (1.5 ml), were added while bubbling argon, cuprous iodide (11.3 mg, 59 μ mol), DIPEA (210 μ l, 1.18 mmol), bis(triphenylphosphine)palladium(II) dichloride (20.8 mg, 29.6 μ mol) and *N*-(2-propargyloxyethyl)carbamic acid *tert*-butyl ester (CAS 634926-63-9, 177 mg, 888 μ mol). The reaction mixture was stirred at 80 °C for 3 hours. The reaction was partitioned between water and ethyl acetate. The combined organic layer was washed with brine, dried over Na₂SO₄, vacuum-filtered, and concentrated in vacuo. The crude product was purified by flash chromatography (SiO₂, ethyl acetate/heptane, 0-50 %) to give (3*S*)-3-[[4-[3-[3-[2-(*tert*-butoxycarbonylamino)ethoxy]prop-1-ynyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid ethyl ester (179 mg, 25 73 %) as a light brown waxy solid. MS *m/z*: 794.2 [M+H]⁺, ESI pos.

Step 2: (3S)-3-[[4-[3-[3-[2-(*tert*-Butoxycarbonylamino)ethoxy]prop-1-ynyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid

- 5 To a solution of (3S)-3-[[4-[3-[3-[2-(*tert*-butoxycarbonylamino)ethoxy]prop-1-ynyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid ethyl ester (178 mg, 224 μ mol) in methanol (448 μ l) and tetrahydrofuran (896 μ l), was added dropwise a solution of 4 M sodium hydroxide (168 μ l, 672 μ mol). The reaction was stirred at room temperature for 3 hours. The mixture was acidified
10 to precipitate the free acid. The solid was filtered and dried under high vacuo to give (3S)-3-[[4-[3-[3-[2-(*tert*-butoxycarbonylamino)ethoxy]prop-1-ynyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid (143 mg, 81 % yield) as a white powder. MS m/z: 710.1 [M+H-Buten]⁺, ESI pos.

- Step 3: (3S)-3-[[4-[3-[3-(2-Aminoethoxy)prop-1-ynyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid hydrogen chloride salt
- 15

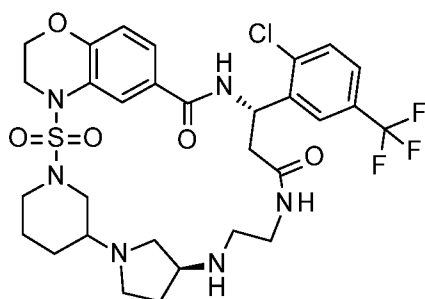
- To a solution of (3S)-3-[[4-[3-[3-[2-(*tert*-butoxycarbonylamino)ethoxy]prop-1-ynyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid (143 mg, 186 μ mol) in ethyl acetate (1.27 ml), was added
20 dropwise at 0 °C a solution of 4 M HCl in 1,4-dioxane (232 μ l, 930 μ mol). The mixture was warmed to room temperature and stirred for 24 hours. The reaction mixture was concentrated in *vacuo* to give (3S)-3-[[4-[3-[3-(2-aminoethoxy)prop-1-ynyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid hydrogen chloride salt (150 mg, 100 %) as a dark brown solid. MS m/z: 666.1 [M+H]⁺, ESI pos.

- 25 Step 4: (17S)-17-[2-Chloro-5-(trifluoromethyl)phenyl]-2,2-diketo-11,24-dioxa-2 λ ⁶-thia-1,14,18-triazatetracyclo[18.6.2.13,7.0^{23,27}]nonacosa-3(29),4,6,20,22,27-hexaen-8-yne-15,19-quinone

- To a solution of (3S)-3-[[4-[3-(5-aminopentoxo)phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid hydrochloride salt (21 mg, 28.2 μ mol) in dichloromethane (565 μ l), were added 1-(3-dimethylaminopropyl)-3-
30 ethylcarbodiimide hydrochloride (6.5 mg, 33.9 μ mol), 1-hydroxybenzotriazole hydrate (4.32 mg, 28.2 μ mol) and DIPEA (15 μ l, 84.7 μ mol). The mixture was stirred at room temperature for 17 hours and was concentrated in *vacuo*. The crude product was purified by reverse phase flash

chromatography (Flash Spherical C18, 20-35 μm ; 100 $\text{\AA}$; gradient water with 0.05 % TFA and acetonitrile) to give (17*S*)-17-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-diketo-11,24-dioxa-2 λ^6 -thia-1,14,18-triazatetracyclo[18.6.2.13,7.0^{23,27}]nonacosa-3(29),4,6,20,22,27-hexaen-8-yne-15,19-quinone (5.3 mg, 29 % yield) as a white lyophilised powder. MS *m/z*: 666.1 [M+H]⁺, ESI pos.

Example 22: (11*S*,18*S*)-18-[2-Chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-25-oxa-2 λ^6 -thia-1,3,8,12,15,19-hexazapentacyclo[19.6.2.13,7.18,11.0^{24,28}]hentriaconta-21(29),22,24(28)-triene-16,20-dione, mixture of two diastereomers.



Step 1: Methyl 4-fluorosulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate

To a solution of methyl 3,4-dihydro-2*H*-1,4-benzoxazine-6-carboxylate (**Intermediate A1**, 2.0 g, 10.4 mmol) in acetonitrile (40 ml) were added *N,N*-diisopropylethylamine (7.21 ml, 41.4 mmol) and 4-dimethylaminopyridine (3.79 g, 3 mmol) and (4-

acetamidophenyl)(fluorosulfonyl)sulfamoyl fluoride (9.76 g, 30 mmol) at room temperature. The reaction mixture was stirred at 80 °C for 16 hours. The reaction mixture was filtered and the filter cake was washed three times with ethyl acetate. The filtrate was concentrated and purified via flash chromatography (silica gel, ethyl acetate/petroleum ether 0-20 %) to give methyl 4-fluorosulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate (2.8 g, 98 % yield) as a brown solid.

MS *m/z*: 276.1 [M+H]⁺, ESI pos.

Step 2: tert-Butyl *N*-[(3*S*)-1-(1-benzyl-3-piperidyl)pyrrolidin-3-yl]carbamate

To solution of (*S*)-3-(BOC-amino)pyrrolidine (CAS 122536-76-9, 8.0 g, 42.9 mmol) and 1-benzyl-3-piperidone (CAS 50606-58-1, 16.3 g, 86 mmol) in dichloroethane (80 ml) was added NaBH(OAc)₃ (45.5 g, 215 mmol). The mixture was stirred at 20 °C for 2 hours.

The mixture was poured into water and extracted three times with dichloromethane. The crude product was purified via flash chromatography (silica gel, ethyl acetate 100 % then

methanol/dichloromethane 0-10 %) to give *tert*-butyl *N*-[(3*S*)-1-(1-benzyl-3-piperidyl)pyrrolidin-3-yl]carbamate (22 g, 61.2 mmol, > 100 % yield) as an orange oil. MS *m/z*: 360.3 [M+H]⁺, ESI pos.

Step 3: *tert*-Butyl *N*-[(3*S*)-1-(3-piperidyl)pyrrolidin-3-yl]carbamate

- 5 To a solution of *tert*-butyl *N*-[(3*S*)-1-(1-benzyl-3-piperidyl)pyrrolidin-3-yl]carbamate (1.8 g, 5.01 mmol) in methanol (30 ml) was added Pd/C (0.9 g, 0.85 mmol). The mixture was stirred at 30 °C for 2 hours under an atmosphere of hydrogen (50 psi). The reaction mixture was filtered and the filtrate was evaporated to give *tert*-butyl *N*-[(3*S*)-1-(3-piperidyl)pyrrolidin-3-yl]carbamate (1.5 g, 90 % yield) as a white solid. MS *m/z*: 270.2 [M+H]⁺, ESI pos.

10 Step 4: Methyl 4-[[3-[(3*S*)-3-(*tert*-butoxycarbonylamino)pyrrolidin-1-yl]-1-piperidyl]sulfonyl]-2,3-dihydro-1,4-benzoxazine-6-carboxylate

- To a solution of methyl 4-fluorosulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate (750 mg, 2.72 mmol) in 2-methylbutan-2-ol (15.0 ml) were added calcium(II) bis(trifluoromethanesulfonyl)imide (2.46 g, 4.09 mmol), DABCO (CAS 280-57-9, 611 mg, 5.45 mmol) and *tert*-butyl *N*-[(3*S*)-1-(3-piperidyl)pyrrolidin-3-yl]carbamate (1.47 g, 5.45 mmol). The reaction mixture was stirred at 90 °C for 16 hours. The mixture was poured into water and extracted three times with ethyl acetate. The combined organic layer was washed with brine, dried with Na₂SO₄, filtered and evaporated. The crude product was purified by prep-HPLC (Phenomenex Luna C18 150 x 25 mm x 10 μm, gradient water with 2.25 % formic acid and acetonitrile to give methyl 4-[[3-[(3*S*)-3-(*tert*-butoxycarbonylamino)pyrrolidin-1-yl]-1-piperidyl]sulfonyl]-2,3-dihydro-1,4-benzoxazine-6-carboxylate (500 mg, 17 % yield) as a yellow solid. MS *m/z*: 525.2 [M+H]⁺, ESI pos.

Step 4: Methyl 4-[[3-[(3*S*)-3-aminopyrrolidin-1-yl]-1-piperidyl]sulfonyl]-2,3-dihydro-1,4-benzoxazine-6-carboxylate

- 25 To a solution of methyl 4-[[3-[(3*S*)-3-(*tert*-butoxycarbonylamino)pyrrolidin-1-yl]-1-piperidyl]sulfonyl]-2,3-dihydro-1,4-benzoxazine-6-carboxylate (450 mg, 0.86 mmol) in dichloromethane (3 ml) was added TFA (3.0 ml) at 20 °C. The reaction mixture was stirred at 20 °C for 1 hour and was carefully poured into a solution of aqueous NaHCO₃. The pH was adjusted to pH > 7 and the mixture was extracted three times with ethyl acetate. The combined organic layer was dried with Na₂SO₄, filtered and concentrated under reduced pressure to give
- 30

methyl 4-[[3-[(3*S*)-3-aminopyrrolidin-1-yl]-1-piperidyl]sulfonyl]-2,3-dihydro-1,4-benzoxazine-6-carboxylate (500 mg, >100 % yield) as a yellow oil. MS *m/z*: 425.3 [M+H]⁺, ESI pos.

Step 5: Methyl 4-[[3-[(3*S*)-3-[2-(*tert*-butoxycarbonylamino)ethylamino]pyrrolidin-1-yl]-1-piperidyl]sulfonyl]-2,3-dihydro-1,4-benzoxazine-6-carboxylate

- 5 The title compound was prepared in analogy to **Example 11, step 3** starting with methyl 4-[[3-[(3*S*)-3-aminopyrrolidin-1-yl]-1-piperidyl]sulfonyl]-2,3-dihydro-1,4-benzoxazine-6-carboxylate instead of methyl 4-[3-[(3*S*)-3-aminopyrrolidin-1-yl]-5-methyl-phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate. Yellow solid. MS *m/z*: 568.3 [M+H]⁺, ESI pos.

Step 6: Methyl 4-[[3-[(3*S*)-3-[allyloxycarbonyl]-2-(*tert*-

- 10 butoxycarbonylamino)ethyl]amino]pyrrolidin-1-yl]-1-piperidyl]sulfonyl]-2,3-dihydro-1,4-benzoxazine-6-carboxylate

The title compound was prepared in analogy to **Example 10, step 4** starting with methyl 4-[[3-[(3*S*)-3-[2-(*tert*-butoxycarbonylamino)ethylamino]pyrrolidin-1-yl]-1-piperidyl]sulfonyl]-2,3-dihydro-1,4-benzoxazine-6-carboxylate instead of methyl 4-[3-[(3*S*)-3-[2-(*tert*-

- 15 butoxycarbonylamino)ethylamino]pyrrolidin-1-yl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate and using ethyl acetate/ petroleum ether 80-100% instead of 25-50%. Yellow solid. MS *m/z*: 652.3 [M+H]⁺, ESI pos.

Step 7: 4-[[3-[(3*S*)-3-[Allyloxycarbonyl]-2-(*tert*-butoxycarbonylamino)ethyl]amino]pyrrolidin-1-yl]-1-piperidyl]sulfonyl]-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid

- 20 The title compound was prepared in analogy to **Intermediate D3, step 1** starting from methyl 4-[[3-[(3*S*)-3-[allyloxycarbonyl]-2-(*tert*-butoxycarbonylamino)ethyl]amino]pyrrolidin-1-yl]-1-piperidyl]sulfonyl]-2,3-dihydro-1,4-benzoxazine-6-carboxylate instead of methyl 4-(3-methoxy-5-methyl-phenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate (**Intermediate C2**) and stirring at 25 °C for 16 hours instead of 50 °C for 2 hours. Yellow solid. MS *m/z*: 638.2 [M+H]⁺,
25 ESI pos.

Step 8: Ethyl (3*S*)-3-[[4-[[3-[(3*S*)-3-[allyloxycarbonyl]-2-(*tert*-butoxycarbonylamino)ethyl]amino]pyrrolidin-1-yl]-1-piperidyl]sulfonyl]-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate

The title compound was prepared in analogy to **Example 4, step 4** starting with 4-[[3-[(3*S*)-3-[allyloxycarbonyl]-2-(*tert*-butoxycarbonylamino)ethyl]amino]pyrrolidin-1-yl]-1-

- 30

piperidyl)sulfonyl]-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid instead of (3*S*)-3-[[4-[3-[2-(2-aminoethoxy)ethoxy]-5-methyl-phenyl]sulfonyl]-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid hydrochloride salt. White solid, MS *m/z*: 915.2 [M+H]⁺, ESI pos.

5 Step 9: (3*S*)-3-[[4-[3-[(3*S*)-3-[Allyloxycarbonyl]-2-(*tert*-butoxycarbonylamino)ethyl]amino]pyrrolidin-1-yl]-1-piperidyl)sulfonyl]-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid

The title compound was prepared in analogy to **Example 19 and Example 20, step 4** starting with ethyl (3*S*)-3-[[4-[3-[(3*S*)-3-[allyloxycarbonyl]-2-(*tert*-

10 butoxycarbonylamino)ethyl]amino]pyrrolidin-1-yl]-1-piperidyl)sulfonyl]-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate instead of ethyl (3*S*)-3-[[4-[3-[2-[2-(*tert*-butoxycarbonylamino)ethyl]morpholin-4-yl]phenyl]sulfonyl]-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate. White solid. MS *m/z*: 887.4 [M+H]⁺, ESI pos.

15 Step 10: (3*S*)-3-[[4-[3-[(3*S*)-3-[Allyloxycarbonyl(2-aminoethyl)amino]pyrrolidin-1-yl]-1-piperidyl)sulfonyl]-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid hydrochloride salt

To a solution of (3*S*)-3-[[4-[3-[(3*S*)-3-[allyloxycarbonyl]-2-(*tert*-butoxycarbonylamino)ethyl]amino]pyrrolidin-1-yl]-1-piperidyl)sulfonyl]-2,3-dihydro-1,4-

20 benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid (180 mg, 0.2 mmol) in dichloromethane (2 ml) was added a solution of 4 M HCl in dioxane (2.0 ml, 8.0 mmol). The mixture was stirred at 20 °C for 1 hour. The reaction mixture was diluted with water (20ml) and acetonitrile (5 ml) and was lyophilized to give (3*S*)-3-[[4-[3-[(3*S*)-3-[allyloxycarbonyl(2-aminoethyl)amino]pyrrolidin-1-yl]-1-piperidyl)sulfonyl]-2,3-dihydro-1,4-

25 benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid hydrochloride salt (150 mg, 86 % yield) as a white solid. MS *m/z*: 787.3 [M+H]⁺, ESI pos.

Step 11: Allyl (11*S*,18*S*)-18-[2-chloro-5-(trifluoromethyl)phenyl]-2,2,16,20-tetraoxo-25-oxa-2λ⁶-thia-1,3,8,12,15,19-hexazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-21(29),22,24(28)-triene-12-carboxylate

30 The title compound was prepared in analogy to **Example 5, step 10** starting with (3*S*)-3-[[4-[3-[(3*S*)-3-[allyloxycarbonyl(2-aminoethyl)amino]pyrrolidin-1-yl]-1-piperidyl)sulfonyl]-2,3-

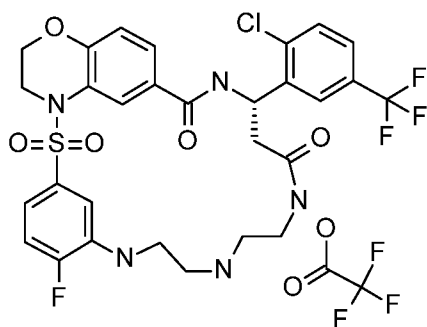
dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid hydrochloride salt instead of (3*S*)-3-[[4-[3-[(2*R*)-5-amino-2-(*tert*-butoxycarbonylamino)pentoxy]phenyl]sulfonyl]spiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid. White solid. MS *m/z*:

5 769.3 [M+H]⁺, ESI pos.

Step 12: (11*S*,18*S*)-18-[2-Chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-25-oxa-2λ⁶-thia-1,3,8,12,15,19-hexazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-21(29),22,24(28)-triene-16,20-dione

To a solution of allyl (11*S*,18*S*)-18-[2-chloro-5-(trifluoromethyl)phenyl]-2,2,16,20-tetraoxo-25-oxa-2λ⁶-thia-1,3,8,12,15,19-hexazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-21(29),22,24(28)-triene-12-carboxylate (20 mg, 0.03 mmol) in tetrahydrofuran (0.5 ml) were added 1,3-dimethylbarbituric acid (28.4 mg, 0.18 mmol) and Pd(PPh₃)₄ (6.01 mg, 0.01 mmol) under nitrogen. The reaction mixture was stirred at 20 °C for 1 hour. The reaction mixture was concentrated was then purified by prep-TLC (chromatography (silica gel, ethyl acetate/petroleum ether) followed by prep-HPLC (Waters Xbridge 150 x 25 mm x 10 μm, water with 0.5% ammonia hydroxide and acetonitrile) to give (11*S*,18*S*)-18-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-25-oxa-2λ⁶-thia-1,3,8,12,15,19-hexazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-21(29),22,24(28)-triene-16,20-dione (4 mg, 19 % yield) as an off-white solid. MS *m/z*: 685.4 [M+H]⁺, ESI pos.

20 **Example 23:** (17*S*)-17-[2-Chloro-5-(trifluoromethyl)phenyl]-6-fluoro-2,2-dioxo-24-oxa-2λ⁶-thia-1,8,11,14,18-pentazatetracyclo[18.6.2.13,7.023,27]nonacosa-3,5,7(29),20(28),21,23(27)-hexaene-15,19-dione 2,2,2-trifluoroacetic acid salt



Step 1: *tert*-Butyl *N*-[2-(benzyloxycarbonylamino)ethyl]-*N*-(2-hydroxyethyl)carbamate

To a solution of *tert*-butyl *N*-(2-aminoethyl)-*N*-(2-hydroxyethyl)carbamate (1 g, 4.9 mmol) in a mixture of 1,4-dioxane (10 ml) and water (10 ml) were added NaHCO₃ (1.23 g, 14.7 mmol) and benzyl chloroformate (1.25 g, 7.34 mmol) at 0 °C. The mixture was stirred at 20 °C for 2 hours. The mixture was poured into water and extracted three times with ethyl acetate. The combined organic layer was washed three times with brine, dried over Na₂SO₄, filtered and evaporated. The crude product was purified via flash chromatography (silica gel, ethyl acetate/petroleum ether 0-100 %) to give *tert*-butyl *N*-[2-(benzyloxycarbonylamino)ethyl]-*N*-(2-hydroxyethyl)carbamate (1 g, 56 % yield) as a colorless oil. MS *m/z*: 361.2 [M+Na]⁺, ESI pos.

Step 2: *tert*-Butyl *N*-[2-(benzyloxycarbonylamino)ethyl]-*N*-[2-(1,3-dioxoisindolin-2-yl)ethyl]carbamate

To a solution of *tert*-butyl *N*-[2-(benzyloxycarbonylamino)ethyl]-*N*-(2-hydroxyethyl)carbamate (1.06 g, 3.13 mmol) and phthalimide (461 mg, 3.13 mmol) in tetrahydrofuran (20 ml) was added (tributylphosphoranylidene)acetonitrile (CAS 157141-27-0, 3.02 g, 12.5 mmol). The mixture was stirred at 60 °C for 16 hours. The reaction mixture was poured into water and the mixture was extracted three times with ethyl acetate. The combined organic layer was dried with Na₂SO₄, filtered and concentrated under reduced pressure. The crude product was purified by reverse phase flash chromatography (Flash Spherical C18, 20-35 µm; 100 Å; 20 g, water-acetonitrile) to give crude *tert*-butyl *N*-[2-(benzyloxycarbonylamino)ethyl]-*N*-[2-(1,3-dioxoisindolin-2-yl)ethyl]carbamate (700 mg, 48 % yield) as a light yellow oil. MS *m/z*: 368.2 [M-100+Na]⁺, ESI pos.

Step 3: *tert*-Butyl *N*-(2-aminoethyl)-*N*-[2-(1,3-dioxoisindolin-2-yl)ethyl]carbamate

To a solution of *tert*-butyl *N*-[2-(benzyloxycarbonylamino)ethyl]-*N*-[2-(1,3-dioxoisindolin-2-yl)ethyl]carbamate (600 mg, 1.28 mmol) in methanol (12 ml) were added acetic acid (7.3 µl, 0.13 mmol) and Pd/C(wet) (300.0 mg, 0.28 mmol) at 20 °C. The reaction mixture was stirred at 30 °C for 2 hours under a hydrogen pressure (50 psi). The reaction mixture was filtered over diatomite. According to LC/MS, starting material was still present. The reaction was run a second time with acetic acid (73 µl 1.28 mmol) and Pd/C(wet) (300 mg, 0.28 mmol) at 30 °C for 3 hours under H₂ (50 psi). The reaction mixture was filtered and the filtrate was evaporated and was purified by reverse phase flash chromatography (Flash Spherical C18, 20-35 µm; 100 Å; 20 g, water-acetonitrile) to give *tert*-butyl *N*-(2-aminoethyl)-*N*-[2-(1,3-dioxoisindolin-2-yl)ethyl]carbamate (120 mg, 17 % yield) as a colorless oil. MS *m/z*: 334.2[M+H]⁺, ESI pos.

Step 4: Ethyl (3*S*)-3-[[4-[3-[2-[*tert*-butoxycarbonyl]-2-(1,3-dioxoisindolin-2-yl)ethyl]amino]ethylamino]-4-fluoro-phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate

The title compound was prepared in analogy to **Example 6, step 1** starting with ethyl (3*S*)-3-[[4-(3-bromo-4-fluoro-phenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate (**Intermediate C7**) and *tert*-butyl *N*-(2-aminoethyl)-*N*-[2-(1,3-dioxoisindolin-2-yl)ethyl]carbamate instead of methyl 4-(3-bromophenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate (**Intermediate C4**) and *tert*-butyl 1,4-diazepane-1-carboxylate. Yellow solid. MS *m/z*: 946.5 [M+H]⁺, ESI pos.

Step 5: Ethyl (3*S*)-3-[[4-[3-[2-[2-aminoethyl(*tert*-butoxycarbonyl)amino]ethylamino]-4-fluoro-phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate

To a solution of ethyl (3*S*)-3-[[4-[3-[2-[*tert*-butoxycarbonyl]-2-(1,3-dioxoisindolin-2-yl)ethyl]amino]ethylamino]-4-fluoro-phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate (60.0 mg, 0.06 mmol) in ethanol (1 ml) was added N₂H₄ · H₂O (0.03 ml, 0.63 mmol). The reaction mixture was stirred at 25 °C for 16 hours. The reaction mixture was evaporated to give crude ethyl (3*S*)-3-[[4-[3-[2-[2-aminoethyl(*tert*-butoxycarbonyl)amino]ethylamino]-4-fluoro-phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate (50 mg, 97 % yield). MS *m/z*: 816.4 [M+H]⁺, ESI pos.

Step 6: (3*S*)-3-[[4-[3-[2-[2-Aminoethyl(*tert*-butoxycarbonyl)amino]ethylamino]-4-fluoro-phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid

To solution of ethyl (3*S*)-3-[[4-[3-[2-[2-aminoethyl(*tert*-butoxycarbonyl)amino]ethylamino]-4-fluoro-phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate (20 mg, 0.02 mmol) in a mixture of water (1 ml) and tetrahydrofuran (1 ml) was added LiOH·H₂O (3.09 mg, 0.07 mmol). The reaction mixture was stirred at 20 °C for 2 hours. A solution of 1 M HCl was added until pH = 4 was obtained and the solution was extracted three times with ethyl acetate. The combined organic layer was dried with Na₂SO₄, filtered and evaporated to give (3*S*)-3-[[4-[3-[2-[2-aminoethyl(*tert*-butoxycarbonyl)amino]ethylamino]-4-fluoro-phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-

carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid (25 mg, > 100 % yield) as a light yellow solid. MS m/z: 788.4 [M+H]⁺, ESI pos.

Step 7: *tert*-Butyl (17*S*)-17-[2-chloro-5-(trifluoromethyl)phenyl]-6-fluoro-2,2,15,19-tetraoxo-24-oxa-2λ⁶-thia-1,8,11,14,18-pentazatetracyclo[18.6.2.13,7.0^{23,27}]nonacosa-

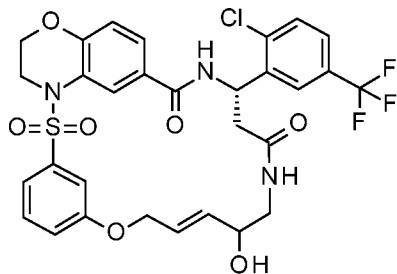
5 3,5,7(29),20(28),21,23(27)-hexaene-11-carboxylate

The title compound was prepared in analogy to **Example 5, step 10** starting with (3*S*)-3-[[4-[3-[2-[2-aminoethyl(*tert*-butoxycarbonyl)amino]ethylamino]-4-fluoro-phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid instead of (3*S*)-3-[[4-[3-[(2*R*)-5-amino-2-(*tert*-

10 butoxycarbonylamino)pentoxy]phenyl]sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid. Off-white solid. MS m/z: 670.2 [M+H]⁺, ESI pos.

Example 24: (10*E*,17*S*)-17-[2-Chloro-5-(trifluoromethyl)phenyl]-12-hydroxy-2,2-dioxo-8,24-dioxa-2λ⁶-thia-1,14,18-triazatetracyclo[18.6.2.13,7.0^{23,27}]nonacosa-

15 3(29),4,6,10,20(28),21,23(27)-heptaene-15,19-dione, mixture of two diastereomers.



Step 1: (3*S*)-3-[[4-(3-Allyloxyphenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid ethyl ester

The title compound was prepared in analogy to **Example 1, step 1** using allyl bromide (CAS 106-95-6) instead of *N*-[2-(2-bromoethoxy)ethyl]carbamic acid *tert*-butyl ester (CAS 165963-71-3). Colorless viscous oil. MS m/z: 653.1[M+H]⁺, ESI pos.

Step 2: (3*S*)-3-[[4-(3-Allyloxyphenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid

The title compound was prepared in analogy to **Example 1, step 3**. The free acid was precipitated by addition of HCl and filtered instead of extracting. White powder. MS m/z: 625.1[M+H]⁺, ESI pos.

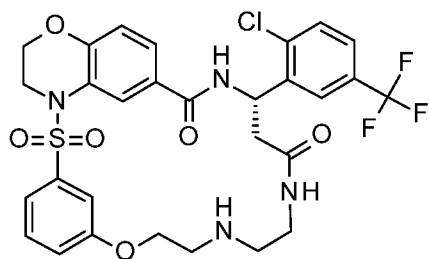
Step 3: 4-(3-Allyloxyphenyl)sulfonyl-N-[(1S)-1-[2-chloro-5-(trifluoromethyl)phenyl]-3-(2-hydroxybut-3-enylamino)-3-keto-propyl]-2,3-dihydro-1,4-benzoxazine-6-carboxamide

To a solution of (3S)-3-[[4-(3-allyloxyphenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid (150 mg, 240 µmol) in dry *N,N*-dimethylformamide (2.33 ml), were added 1-amino-3-buten-2-ol (CAS 15426-17-2, 27.2 mg, 312 µmol), DIPEA (167 µl, 960 µmol), and HATU (137 mg, 360 µmol). The reaction mixture was stirred at room temperature for 16 hours, after which LC/MS showed complete conversion of the starting material. Upon addition of water, the amide precipitated and was subsequently isolated by filtration and dried to give 4-(3-allyloxyphenyl)sulfonyl-N-[(1S)-1-[2-chloro-5-(trifluoromethyl)phenyl]-3-(2-hydroxybut-3-enylamino)-3-keto-propyl]-2,3-dihydro-1,4-benzoxazine-6-carboxamide as a pink powder. (300 mg, 75 % yield). m/z: 694.1 [M+H]⁺, ESI pos.

Step 4: (10E, 17S)-17-[2-Chloro-5-(trifluoromethyl)phenyl]-12-hydroxy-2,2-diketo-8,24-dioxo-2λ⁶-thia-1,14,18-triazatetracyclo[18.6.2.13,7.0.23,27]nonacosa-3,5,7(29),10,20,22,27-heptaene-15,19-quinone

To a solution of *N*-[(1S)-3-(allylamino)-1-[2-chloro-5-(trifluoromethyl)phenyl]-3-keto-propyl]-4-(3-allyloxyphenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxamide (165 mg, 223 µmol) in dry dichloromethane (13.1 ml) was added Grubbs II catalyst (CAS 246047-72-3, 38 mg, 44.7 µmol) while degassing the reaction mixture with argon. The vial was sealed and the mixture was heated at 70°C. LC/MS after 16 hours showed full conversion of the starting material. The mixture was filtered, concentrated in vacuo and purified by prep-HPLC (YMC-Triart C18, 12 nm, 5 µm, 100 x 30 mm, water with 0.1% formic acid and acetonitrile) to yield (10E,17S)-17-[2-chloro-5-(trifluoromethyl)phenyl]-12-hydroxy-2,2-diketo-8,24-dioxo-2λ⁶-thia-1,14,18-triazatetracyclo[18.6.2.13,7.0.23,27]nonacosa-3,5,7(29),10,20,22,27-heptaene-15,19-quinone as a white lyophilized powder (18.6 mg, 12 % yield). m/z: 666.2 [M+H]⁺, ESI pos.

Example 25: (17S)-17-[2-Chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-8,24-dioxo-2λ⁶-thia-1,11,14,18-tetrazatetracyclo[18.6.2.13,7.0.23,27]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-15,19-dione



Step 1: *N*-(2-Aminoethyl)-*N*-(2-hydroxyethyl)carbamic acid *tert*-butyl ester hydrochloride salt

To a solution of *N*-[2-(benzyloxycarbonylamino)ethyl]-*N*-(2-hydroxyethyl)carbamic acid *tert*-butyl ester (CAS 169773-08-4, 1.5 g, 4.43 mmol) in methanol (30 ml) was added a solution of
 5 4 M HCl in dioxane (1.33 g, 1.11 ml, 4.43 mmol). The reaction mixture was stirred under a hydrogen atmosphere (balloon) overnight. The catalyst was filtered over dicalite and the filtrate was evaporated to *N*-(2-aminoethyl)-*N*-(2-hydroxyethyl)carbamic acid *tert*-butyl ester hydrochloride salt (1.03 g, 96 % yield) as a colorless oil. MS *m/z*: 205.5 [M+H]⁺, ESI pos.

Step 2: *N*-[2-(9*H*-Fluoren-9-ylmethoxycarbonylamino)ethyl]-*N*-(2-hydroxyethyl)carbamic acid *tert*-butyl ester

To a solution of *N*-(2-aminoethyl)-*N*-(2-hydroxyethyl)carbamic acid *tert*-butyl ester hydrochloride salt (1.03 g, 4.44 mmol) in acetone (30 ml) were added a saturated sodium bicarbonate solution (14.8 ml, 17.6 mol) and *N*-(9*H*-fluoren-9-ylmethoxycarbonyloxy)succinimide (CAS 82911-69-1, 1.5 g, 4.44 mmol). The reaction mixture
 15 was stirred at room temperature for 4 hours. The acetone was removed under reduced pressure and a solution of 1 M aq HCl (15 ml) and ethyl acetate (15 ml) were added. The layers were separated, and the organic layer was washed twice with 1 M HCl, then with a solution of saturated aqueous NaHCO₃ and finally with brine. The organic layer was dried over magnesium sulfate, filtered and evaporated to afford *N*-[2-(9*H*-fluoren-9-ylmethoxycarbonylamino)ethyl]-*N*-(2-hydroxyethyl)carbamic acid *tert*-butyl ester (1.96 g, 99 % yield) as a light green semisolid.
 20 MS *m/z*: 427.2 [M+H]⁺, ESI pos.

Step 3: (3*S*)-3-[[4-[3-[2-[*tert*-Butoxycarbonyl]-[2-(9*H*-fluoren-9-ylmethoxycarbonylamino)ethyl]amino]ethoxy]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid ethyl ester

To a solution of (3*S*)-3-[2-chloro-5-(trifluoromethyl)phenyl]-3-[[4-(3-hydroxyphenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propionic acid ethyl ester (**Intermediate D1**,
 25 250 mg, 408 μmol) in tetrahydrofuran (2.0 ml) were added *N*-[2-(9*H*-fluoren-9-

ylmethoxycarbonylamino)ethyl]-*N*-(2-hydroxyethyl)carbamic acid *tert*-butyl ester (193 mg, 408 μ mol) followed by Ph_3P (214 mg, 816 μ mol) and DIAD (159 μ l, 816 μ mol). The solution was stirred overnight at room temperature. The reaction was not finished. Ph_3P (214 mg, 816 μ mol) and DIAD (159 μ l, 816 μ mol) were added and stirring was continued until completion. The
5 reaction mixture was quenched with water and extracted twice with ethyl acetate. The combined organic layer was washed with brine, dried over magnesium sulfate, filtered and concentrated. The crude product was purified via flash chromatography (silica gel, TBME in heptane, 4-90%) to give (3*S*)-3-[[4-[3-[2-[*tert*-butoxycarbonyl]-[2-(9*H*-fluoren-9-ylmethoxycarbonylamino)ethyl]amino]ethoxy]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-
10 carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid ethyl ester (143 mg, 34 %) as a colorless oil. MS *m/z*: 921.2 [*M*-100+*H*]⁺, ESI pos.

Step 4: (3*S*)-3-[[4-[3-[2-[2-Aminoethyl(*tert*-butoxycarbonyl)amino]ethoxy]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic
15 acid ethyl ester

To a solution of (3*S*)-3-[[4-[3-[2-[*tert*-butoxycarbonyl]-[2-(9*H*-fluoren-9-ylmethoxycarbonylamino)ethyl]amino]ethoxy]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid ethyl ester (140 mg, 137 μ mol) in *N,N*-dimethylformamide (3.7 ml) was added pyrrolidine (CAS 123-75-1, 768 μ l, 9.28 mmol). The reaction mixture was stirred at room temperature overnight. The reaction mixture was directly purified via preparative HPLC (Phenomenex Gemini C18, 110 mm A Axia, 5 μ m, acetonitrile/water + 0.1 % formic acid) to afford (3*S*)-3-[[4-[3-[2-[2-aminoethyl(*tert*-butoxycarbonyl)amino]ethoxy]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid ethyl ester (76.7 mg, 70 %) as a colorless
25 viscous oil. MS *m/z* = 779.3 [*M*+*H*]⁺, ESI pos

Step 6: (17*S*)-17-[2-Chloro-5-(trifluoromethyl)phenyl]-2,2,15,19-tetraketo-8,24-dioxa-2 λ ⁶-thia-1,11,14,18-tetrazatetracyclo[18.6.2.13,7.0.23,27]nonacosa-3,5,7(29),20,22,27-hexaene-11-carboxylic acid *tert*-butyl ester

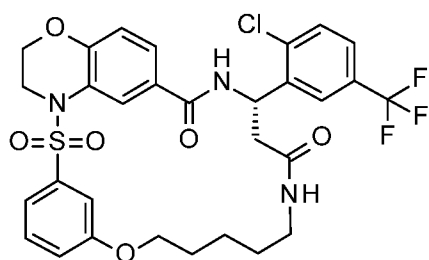
30 To a solution of (3*S*)-3-[[4-[3-[2-[2-aminoethyl(*tert*-butoxycarbonyl)amino]ethoxy]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid ethyl ester (109 mg, 136 μ mol), in 1,4-dioxane (1 ml) were added LiOH (6.53 mg, 273 μ mol) and water (59 μ l, 3.27 mmol). The

suspension was stirred at 50 °C over the weekend and was then cooled to room temperature. The reaction mixture was diluted with *N,N*-dimethylformamide (5 ml) and dichloromethane (5 ml). This solution was added dropwise to a solution of HATU (63.3 mg, 166 μmol) and DIPEA (71 μl, 409 μmol) dissolved in *N,N*-dimethylformamide, extra dry (2 ml) and was fitted with a dropping funnel. The reaction mixture was concentrated under high vacuum and purified by preparative HPLC (column Phenomenex Gemini C18, 110 mm A Axia, 5 μm, acetonitrile/ water + 0.1% formic acid) to afford (17*S*)-17-[2-chloro-5-(trifluoromethyl)phenyl]-2,2,15,19-tetraketo-8,24-dioxa-2λ⁶-thia-1,11,14,18-tetrazatetracyclo[18.6.2.13,7.0^{23,27}]nonacosa-3,5,7(29),20,22,27-hexaene-11-carboxylic acid *tert*-butyl ester (56.8 mg, 46 %) as a white powder. MS *m/z* = 753.1 [M+H]⁺, ESI pos

Step 7: (17*S*)-17-[2-Chloro-5-(trifluoromethyl)phenyl]-2,2-diketo-8,24-dioxa-2λ⁶-thia-1,11,14,18-tetrazatetracyclo[18.6.2.13,7.0^{23,27}]nonacosa-3(29),4,6,20,22,27-hexaene-15,19-quinone

To a solution of (17*S*)-17-[2-chloro-5-(trifluoromethyl)phenyl]-2,2,15,19-tetraketo-8,24-dioxa-2λ⁶-thia-1,11,14,18-tetrazatetracyclo[18.6.2.13,7.0^{23,27}]nonacosa-3,5,7(29),20(28),21,23(27)-hexaene-11-carboxylic acid *tert*-butyl ester (55 mg, 60.6 μmol) in ethyl acetate (788 μl) were added *p*-toluenesulfonic acid monohydrate (12.1 mg, 63.6 μmol) and ethanol (78 μl). The mixture was stirred at 50 °C until completion was shown in LC/MS. The solution was concentrated and purified by preparative HPLC (Phenomenex Gemini C18, 110 mm A Axia, 5 μm, acetonitrile/ water + 0.1% formic acid) to afford (17*S*)-17-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-diketo-8,24-dioxa-2λ⁶-thia-1,11,14,18-tetrazatetracyclo[18.6.2.13,7.0^{23,27}]nonacosa-3(29),4,6,20,22,27-hexaene-15,19-quinone (4.6 mg, 12 %) as an orange solid. MS *m/z* = 653.1 [M+H]⁺, ESI pos

Example 26: (17*S*)-17-[2-Chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-8,24-dioxa-2λ⁶-thia-1,14,18-triazatetracyclo[18.6.2.13,7.0^{23,27}]nonacosa-3,5,7(29),20(28),21,23(27)-hexaene-15,19-dione



Step 1: (3*S*)-3-[[4-[3-[5-(*tert*-Butoxycarbonylamino)pentoxy]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid ethyl ester

To a solution of (3*S*)-3-[2-chloro-5-(trifluoromethyl)phenyl]-3-[[4-(3-hydroxyphenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propionic acid ethyl ester (Intermediate **D1**, 200 mg, 326 μ mol) in dry tetrahydrofuran (1.1 ml) were added under argon, N-(5-hydroxypentyl)carbamic acid *tert*-butyl ester (CAS 75178-90-4, 166 mg, 816 μ mol) and Ph₃P (214 mg, 816 μ mol). The solution was cooled to 0 °C and diisopropyl azodicarboxylate (159 μ l, 816 μ mol) was added dropwise afterwards the reaction mixture was allowed to reach room temperature and stirred for 24 hours. LC/MS showed incomplete conversion of the starting material, therefore additional N-(5-hydroxypentyl)carbamic acid *tert*-butyl ester (66.3 mg, 326 μ mol) and diisopropyl azodicarboxylate (63 μ l, 326 μ mol) were added and the reaction was stirred for another 16 hours. The reaction mixture was quenched with water and extracted twice with ethyl acetate. The combined organic layer was washed with brine, dried over Na₂SO₄, filtered and concentrated in vacuo. The crude product was purified via flash chromatography (silica gel, 25 g, ethyl acetate in heptane 0-50 %) to yield (3*S*)-3-[[4-[3-[5-(*tert*-butoxycarbonylamino)pentoxy]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid ethyl ester (209 mg, 77 % yield) as a colorless waxy solid. MS m/z: 698.1 [M-BOC+H]⁺, ESI pos.

Step 2: (3*S*)-3-[[4-[3-[5-(*tert*-Butoxycarbonylamino)pentoxy]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid

The title compound was prepared in analogy to **Example 3, step 2** starting from (3*S*)-3-[[4-[3-[5-(*tert*-butoxycarbonylamino)pentoxy]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid ethyl ester instead of (3*S*)-3-[[4-[3-[2-[2-(*tert*-butoxycarbonylamino)ethoxy]ethoxy]phenyl]sulfonyl]spiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid ethyl ester. The free acid was precipitated by addition of HCl and filtered instead of extracting. White powder. MS m/z: 670.1[M-BOC+H]⁺, ESI pos.

Step 3: (3*S*)-3-[[4-[3-(5-Aminopentoxy)phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid hydrogen chloride salt

The title compound was prepared in analogy to **Example 3, step 3** starting from (3*S*)-3-[[4-[3-[5-(*tert*-butoxycarbonylamino)pentoxy]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid instead of (3*S*)-3-[[4-[3-

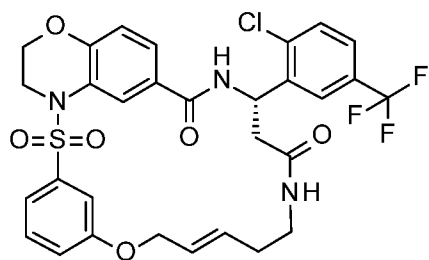
[2-[2-(*tert*-butoxycarbonylamino)ethoxy]ethoxy]phenyl]sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid. White solid. MS *m/z*: 670.1[M+H]⁺, ESI pos.

Step 4: (17*S*)-17-[2-Chloro-5-(trifluoromethyl)phenyl]-2,2-diketo-8,24-dioxa-2λ⁶-thia-1,14,18-

5 triazatetracyclo[18.6.2.13,7.0^{23,27}]nonacosa-3(29),4,6,20,22,27-hexaene-15,19-quinone

To a solution of (3*S*)-3-[[4-[3-(5-aminopentoxo)phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid hydrochloride salt (21 mg, 28.2 μmol) in dichloromethane (565 μl), 1-(3-dimethylaminopropyl)-3-ethylcarbodiimide hydrochloride (6.5 mg, 33.9 μmol), 1-hydroxybenzotriazole hydrate (4.32 mg, 28.2 μmol)
 10 and DIPEA (15 μl, 84.7 μmol) were added. The mixture was stirred at room temperature for 17 hours, after which it was concentrated under reduced pressure and purified by prep-HPLC (Gemini NX, 12 nm, 5 μm, 100 x 30 mm, water with 0.1 % formic acid and acetonitrile) to yield (17*S*)-17-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-diketo-8,24-dioxa-2λ⁶-thia-1,14,18-triazatetracyclo[18.6.2.13,7.0^{23,27}]nonacosa-3(29),4,6,20,22,27-hexaene-15,19-quinone as a
 15 white lyophilized powder (5.3 mg, 29 %).

Example 27: (17*S*)-17-[2-Chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-8,24-dioxa-2λ⁶-thia-1,14,18-triazatetracyclo[18.6.2.13,7.0^{23,27}]nonacosa-3,5,7(29),10,20(28),21,23(27)-heptaene-15,19-dione



Step 1: (3*S*)-3-[[4-[3-[(*E*)-5-(*tert*-Butoxycarbonylamino)pent-2-enoxo]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid ethyl ester

The title compound was prepared in analogy to **Example 25, step 3** starting from *N*-[(*E*)-5-hydroxypent-3-enyl]carbamic acid *tert*-butyl ester (CAS 333438-48-5) instead of *N*-[2-(9*H*-fluoren-9-ylmethoxycarbonylamino)ethyl]-*N*-(2-hydroxyethyl)carbamic acid *tert*-butyl ester.
 25 Colorless waxy solid. MS *m/z*: 796.2 [M+H]⁺, ESI pos.

Step 2: (3*S*)-3-[[4-[3-[(*E*)-5-Aminopent-2-enoxy]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid ethyl ester hydrochloride salt

To a solution of (3*S*)-3-[[4-[3-[(*E*)-5-(*tert*-butoxycarbonylamino)pent-2-enoxy]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid ethyl ester (220 mg, 276 μ mol) in ethyl acetate (1.9 ml), was added dropwise at 0 °C a solution of 4 M HCl in 1,4-dioxane (207 μ l, 829 μ mol). The mixture was allowed to reach room temperature and to stir for 48 hours. The mixture was concentrated in vacuo to give (3*S*)-3-[[4-[3-[(*E*)-5-aminopent-2-enoxy]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid ethyl ester hydrochloride salt (228 mg, > 100 % yield) as a light brown viscous oil. MS *m/z*: 696.2 [M+H]⁺, ESI pos.

Step 3: (3*S*)-3-[[4-[3-[(*E*)-5-Aminopent-2-enoxy]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid sodium salt

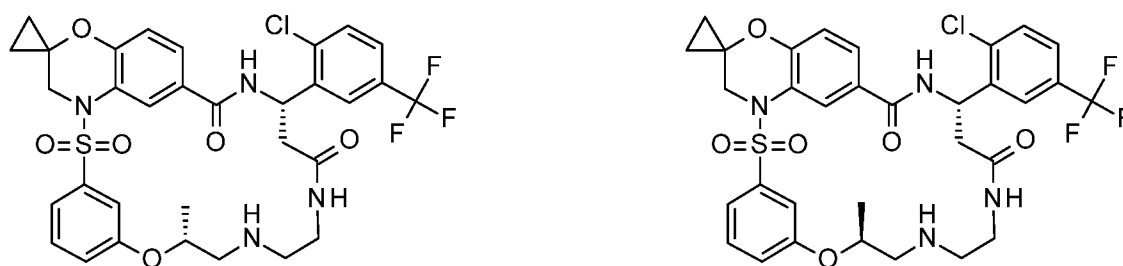
To a solution of (3*S*)-3-[[4-[3-[2-(2-aminoethoxy)ethoxy]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-fluoro-5-(trifluoromethyl)phenyl]propionic acid ethyl ester (300 mg, 439 μ mol) in a mixture of methanol (877 μ l) and tetrahydrofuran (1.61 ml), was added a solution of aqueous 4 M NaOH (439 μ l, 1.76 mmol) at room temperature over night. The reaction mixture was concentrated in vacuo and the residue was co-evaporated with toluene twice to give (3*S*)-3-[[4-[3-[(*E*)-5-aminopent-2-enoxy]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid sodium salt (261 mg, > 100 % yield) as an orange waxy solid. MS *m/z*: 668.3 [M+H]⁺, ESI pos.

Step 4: (10*E*,17*S*)-17-[2-Chloro-5-(trifluoromethyl)phenyl]-2,2-diketo-8,24-dioxa-2 λ ⁶-thia-1,14,18-triazatetracyclo[18.6.2.13,7.0.23,27]nonacosa-3(29),4,6,10,20,22,27-heptaene-15,19-quinone

A suspension of (3*S*)-3-[[4-[3-[(*E*)-5-aminopent-2-enoxy]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid sodium salt (180 mg, 269 μ mol) in *N,N*-dimethylformamide (1 ml) was added dropwise to a solution of HATU (410 mg, 1.08 mmol) dissolved in dry *N,N*-dimethylformamide (6.74 ml) followed by addition of DIPEA (188 μ l, 1.08 mmol). The reaction was left to stir for 72 hours and was concentrated in vacuo. The crude product was purified via prep HPLC (YMC-Triart C18, 12 nm, 5 μ m, 100 x 30 mm, water with 0.1% formic acid and acetonitrile) to yield (10*E*, 17*S*)-17-[2-

chloro-5-(trifluoromethyl)phenyl]-2,2-diketo-8,24-dioxa-2 λ^6 -thia-1,14,18-triazatetracyclo[18.6.2.13,7.023,27]nonacosa-3(29),4,6,10,20,22,27-heptaene-15,19-quinone (27 mg, 17 %) as a white lyophilised powder. MS m/z: 650.1 [M+H]⁺, ESI pos.

Example 28 and Example 29: (9*R*,17*S*)-17-[2-Chloro-5-(trifluoromethyl)phenyl]-9-methyl-2,2-dioxo-spiro[8,24-dioxa-2 λ^6 -thia-1,11,14,18-tetrazatetracyclo[18.6.2.13,7.023,27]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-25,1'-cyclopropane]-15,19-dione, mixture of two diastereomers and (9*S*, 17*S*)-17-[2-Chloro-5-(trifluoromethyl)phenyl]-9-methyl-2,2-dioxo-spiro[8,24-dioxa-2 λ^6 -thia-1,11,14,18-tetrazatetracyclo[18.6.2.13,7.023,27]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-25,1'-cyclopropane]-15,19-dione, mixture of two diastereomers



Step 1: Ethyl (3*S*)-3-[[4-[3-[2-(*tert*-butoxycarbonylamino)-1-methyl-ethoxy]phenyl]sulfonyl]spiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate

To a solution of ethyl (3*S*)-3-[2-chloro-5-(trifluoromethyl)phenyl]-3-[[4-(3-hydroxyphenyl)sulfonyl]spiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]propanoate (**Intermediate D4**, 500 mg, 0.78 mmol) and N-BOC-1-amino-2-propanol (411 mg, 2.35 mmol) in toluene (10 ml) was added (tributylphosphoranylidene)acetonitrile (CAS 157141-27-0, 566 mg, 2.35 mmol). The reaction mixture was stirred at 100 °C for 2 hours. After cooling, the reaction mixture was poured into water and was extracted three times with ethyl acetate. The combined organic layer was dried with Na₂SO₄, filtered and concentrated under reduced pressure. The crude product was purified by flash chromatography (silica gel, ethyl acetate in petroleum ether 20-50 %) followed by reverse phase flash chromatography (Flash Spherical C18, 20-35 μ m; 100 Å; gradient water with 0.05 % TFA and acetonitrile). The combined fractions were concentrated under reduced pressure and the aqueous layer was extracted three times with ethyl acetate. The combined organic layer was dried with Na₂SO₄, filtered and concentrated under reduced pressure to give ethyl (3*S*)-3-[[4-[3-[2-(*tert*-butoxycarbonylamino)-1-methyl-ethoxy]phenyl]sulfonyl]spiro[3*H*-

1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate (350 mg, 56 % yield) as a yellow oil. MS m/z: 696.2 [M-100+H]⁺, ESI pos.

Step 2: Ethyl (3*S*)-3-[[4-[3-(2-amino-1-methyl-ethoxy)phenyl]sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate

The title compound was prepared in analogy to **Example 8, step 2** starting from ethyl (3*S*)-3-[[4-[3-[2-(*tert*-butoxycarbonylamino)-1-methyl-ethoxy]phenyl]sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate instead of methyl 4-[3-[(1*R*,4*R*)-5-*tert*-butoxycarbonyl-2,5-diazabicyclo[2.2.1]heptan-2-yl]phenyl]sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carboxylate. Brown oil. MS m/z: 696.2 [M+H]⁺, ESI pos.

Step 3: Ethyl (3*S*)-3-[2-chloro-5-(trifluoromethyl)phenyl]-3-[[4-[3-[2-[2-(1,3-dioxoisindolin-2-yl)ethylamino]-1-methyl-ethoxy]phenyl]sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]propanoate

To solution of ethyl (3*S*)-3-[[4-[3-(2-amino-1-methyl-ethoxy)phenyl]sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate (350 mg, 0.5 mmol) in methanol (6 ml) was added *N*-(2-oxoethyl)phthalimide (CAS 2913-97-5, 104.62 mg, 0.55 mmol) and NaBH₃CN (317 mg, 5.03 mmol). The mixture was stirred at 20 °C for 2 hours. The mixture was poured into water and extracted three times with ethyl acetate. The combined organic layer was dried with Na₂SO₄, filtered and evaporated. The mixture was purified by reverse phase flash chromatography (Flash Spherical C18, 20-35 μm; 100 Å; gradient water with 0.05 % TFA and acetonitrile) to give ethyl (3*S*)-3-[2-chloro-5-(trifluoromethyl)phenyl]-3-[[4-[3-[2-[2-(1,3-dioxoisindolin-2-yl)ethylamino]-1-methyl-ethoxy]phenyl]sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]propanoate (260 mg, 55 % yield) as a yellow solid. MS m/z: 869.3 [M+H]⁺, ESI pos.

Step 4: Ethyl (3*S*)-3-[[4-[3-[2-[*tert*-butoxycarbonyl]-2-(1,3-dioxoisindolin-2-yl)ethyl]amino]-1-methyl-ethoxy]phenyl]sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate

To a solution of ethyl (3*S*)-3-[[4-[3-[2-[2-(1,3-dioxoisindolin-2-yl)ethylamino]-1-methyl-ethoxy]phenyl]sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]propanoate (260 mg, 0.3 mmol) in dichloromethane (6 ml) was added DIPEA (116 mg, 0.9 mmol) followed by di-*tert*-butyldicarbonate (196 mg, 0.9 mmol) and stirred at 20 °C for 2 hours. The mixture was poured into water and extracted three times with ethyl acetate. The combined organic layer was dried with Na₂SO₄, filtered and evaporated. The crude product was purified by flash chromatography (silica gel, ethyl acetate in petroleum ether 20-50 %) to give ethyl (3*S*)-3-[[4-[3-[2-[*tert*-butoxycarbonyl]-2-(1,3-dioxoisindolin-2-yl)ethyl]amino]-1-methyl-ethoxy]phenyl]sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate (330 mg, > 100 % yield) as a white solid. MS m/z: 869.3 [M-100+H]⁺, ESI pos.

Step 5: Ethyl (3*S*)-3-[[4-[3-[2-[2-aminoethyl(*tert*-butoxycarbonyl)amino]-1-methyl-ethoxy]phenyl]sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate

To a solution of ethyl (3*S*)-3-[[4-[3-[2-[*tert*-butoxycarbonyl]-2-(1,3-dioxoisindolin-2-yl)ethyl]amino]-1-methyl-ethoxy]phenyl]sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate (330 mg, 0.34 mmol) in ethanol (3 ml) was added N₂H₄ · H₂O (0.02 ml, 0.51 mmol). The reaction mixture was stirred at 20 °C for 2 hours, the solvents were evaporated to give crude ethyl (3*S*)-3-[[4-[3-[2-[2-aminoethyl(*tert*-butoxycarbonyl)amino]-1-methyl-ethoxy]phenyl]sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate (87 mg, 100 % yield), colorless liquid. MS m/z: 839.3 [M-100+H]⁺, ESI pos.

Step 6: (3*S*)-3-[[4-[3-[2-[2-Aminoethyl(*tert*-butoxycarbonyl)amino]-1-methyl-ethoxy]phenyl]sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid

To a solution of ethyl (3*S*)-3-[[4-[3-[2-[2-aminoethyl(*tert*-butoxycarbonyl)amino]-1-methyl-ethoxy]phenyl]sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate (285.0 mg, 0.34 mmol) in a mixture of tetrahydrofuran (3 ml) and water (3 ml) was added LiOH·H₂O (42.74 mg, 1.02 mmol). The reaction mixture was stirred at 20 °C for 1 hour. The reaction mixture was directly purified by

reverse phase flash chromatography (Flash Spherical C18, 20-35 μm ; 100 $\text{\AA}$; 20 g, water-acetonitrile) to give (3*S*)-3-[[4-[3-[2-[2-aminoethyl(*tert*-butoxycarbonyl)amino]-1-methyl-ethoxy]phenyl]sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid (140.0 mg, 45 % yield) as a white solid. MS
5 m/z: 811.3 [M+H]⁺, ESI pos.

Step 7: *tert*-Butyl-(17*S*)-17-[2-chloro-5-(trifluoromethyl)phenyl]-9-methyl-2,2,15,19-tetraoxo-spiro[8,24-dioxa-2 λ^6 -thia-1,11,14,18-tetrazatetracyclo[18.6.2.13,7.023,27]nonacosa-3,5,7(29),20(28),21,23(27)-hexaene-25,1'-cyclopropane]-11-carboxylate

The title compound was prepared in analogy to **Example 5, step 10** starting with (3*S*)-3-[[4-[3-[2-[2-aminoethyl(*tert*-butoxycarbonyl)amino]-1-methyl-ethoxy]phenyl]sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid instead of (3*S*)-3-[[4-[3-[(2*R*)-5-amino-2-(*tert*-butoxycarbonylamino)pentoxy]phenyl]sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid. White solid. MS m/z:
15 693.2 [M-100+H]⁺, ESI pos.

Step 8: (9*R*,17*S*)-17-[2-Chloro-5-(trifluoromethyl)phenyl]-9-methyl-2,2-dioxo-spiro[8,24-dioxa-2 λ^6 -thia-1,11,14,18-tetrazatetracyclo[18.6.2.13,7.023,27]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-25,1'-cyclopropane]-15,19-dione and (9*S*,17*S*)-17-[2-chloro-5-(trifluoromethyl)phenyl]-9-methyl-2,2-dioxo-spiro[8,24-dioxa-2 λ^6 -thia-1,11,14,18-tetrazatetracyclo[18.6.2.13,7.023,27]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-25,1'-cyclopropane]-15,19-dione

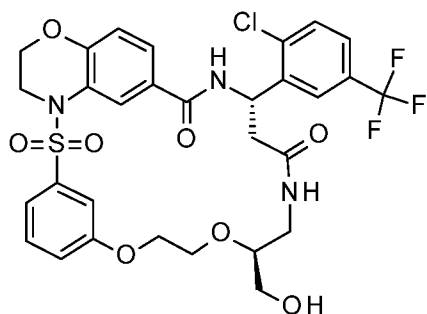
To a solution of *tert*-butyl (17*S*)-17-[2-chloro-5-(trifluoromethyl)phenyl]-9-methyl-2,2,15,19-tetraoxo-spiro[8,24-dioxa-2 λ^6 -thia-1,11,14,18-tetrazatetracyclo[18.6.2.13,7.023,27]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-25,1'-cyclopropane]-11-carboxylate (100 mg, 0.13 mmol) in
25 dichloromethane (2 ml) was added 2,2,2 trifluoroacetic acid (2 ml). The reaction mixture was stirred at 20 °C for 1 hour. The solvent and excess of 2,2,2 trifluoroacetic acid were evaporated and the residue was purified by prep-HPLC (Phenomenex Luna C18 150 x 25 mm x 10 μm , gradient water with 1% TFA and acetonitrile) to give the mixture of diastereomers. The racemic mixture was separated by SFC (DAICEL CHIRALPAK IC 250 mm x 30 mm, 5 μm ,
30 acetonitrile/*i*-PrOH with 0.1 % NH₄OH) to give the two diastereomers.

First eluting diastereomer (R_t = 0.879 min, absolute stereochemistry arbitrarily assigned): (9*S*, 17*S*)-17-[2-chloro-5-(trifluoromethyl)phenyl]-9-methyl-2,2-dioxo-spiro[8,24-dioxa-2 λ^6 -thia-

1,11,14,18-tetrazatetracyclo[18.6.2.13,7.023,27]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-25,1'-cyclopropane]-15,19-dione (30.2 mg, 34 % yield). White solid. MS m/z : 693.2 $[M+H]^+$, ESI pos.

Second eluting diastereomer (R_t = 0.884 min, absolute stereochemistry arbitrarily assigned): (9*R*, 17*S*)-17-[2-chloro-5-(trifluoromethyl)phenyl]-9-methyl-2,2-dioxo-spiro[8,24-dioxa-2 λ^6 -thia-1,11,14,18-tetrazatetracyclo[18.6.2.13,7.023,27]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-25,1'-cyclopropane]-15,19-dione (24.75 mg, 28 % yield). White solid. MS m/z : 693.2 $[M+H]^+$, ESI pos.

Example 30: (12*S*, 17*S*)-17-[2-Chloro-5-(trifluoromethyl)phenyl]-12-(hydroxymethyl)-2,2-dioxo-8,11,24-trioxa-2 λ^6 -thia-1,14,18-triazatetracyclo[18.6.2.13,7.023,27]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-15,19-dione



Step 1: *tert*-Butyl *N*-[3-[*tert*-butyl(diphenyl)silyl]oxy-2-hydroxy-propyl]carbamate

To solution of *tert*-butyl (2,3-dihydroxypropyl)carbamate (8 g, 41.8 mmol) and imidazole (8.54 g, 125 mmol) in dichloromethane (100 ml) was added *tert*-butylchlorodiphenylsilane (16.3 ml, 62.8 mmol) at 0 °C. The mixture was stirred at 20 °C for 16 hours. The reaction mixture was filtered, the filtrate was poured into water and extracted three times with ethyl acetate. The combined organic layer was washed three times with brine, dried with Na₂SO₄, filtered and evaporated. The crude product was purified via flash column chromatography (silica gel, ethyl acetate/ petroleum ether, 50 %) to give *tert*-butyl *N*-[3-[*tert*-butyl(diphenyl)silyl]oxy-2-hydroxy-propyl]carbamate (16 g, 83 % yield) as a colorless oil. MS m/z : 452.3 $[M+Na]^+$, ESI pos.

Step 2: *tert*-Butyl 2-[1-[(*tert*-butoxycarbonylamino)methyl]-2-[*tert*-butyl(diphenyl)silyl]oxy-ethoxy]acetate

To a solution of *tert*-butyl *N*-[3-[*tert*-butyl(diphenyl)silyl]oxy-2-hydroxy-propyl]carbamate (7 g, 16.3 mmol) in dichloromethane (80 ml), were added at 0 °C tetrabutylammonium chloride (1.36

g, 4.89 mmol), NaOH (6.11 g, 48.9 mmol) and *tert*-butyl bromoacetate (9.53 g, 48.9 mmol). The reaction mixture was stirred at 20 °C for 12 hours. The mixture was poured into water and extracted three times with ethyl acetate. The combined organic layer was dried with Na₂SO₄, filtered and evaporated. The crude product was purified via flash column chromatography (silica gel, ethyl acetate/ petroleum ether, 0-50 %) to give *tert*-butyl 2-[1-[(*tert*-butoxycarbonylamino)methyl]-2-[*tert*-butyl(diphenyl)silyl]oxy-ethoxy]acetate (7.2 g, 78 % yield) as a colorless oil. MS m/z: 566.5 [M+Na]⁺, ESI pos.

Step 3: *tert*-Butyl N-[3-[*tert*-butyl(diphenyl)silyl]oxy-2-(2-hydroxyethoxy)propyl]carbamate

To a solution of *tert*-butyl 2-[1-[(*tert*-butoxycarbonylamino)methyl]-2-[*tert*-butyl(diphenyl)silyl]oxy-ethoxy]acetate (7 g, 12.9 mmol) in tetrahydrofuran (70 ml) at 0 °C was added lithium aluminium hydride (979 mg, 25.8 mmol). The reaction mixture was stirred at 20 °C for 2 hours. The reaction was quenched by addition of Na₂SO₄·H₂O, then the mixture was filtered and concentrated under reduced pressure. The crude product was purified via flash column chromatography (silica gel, ethyl acetate/ petroleum ether, 0-100 %) to give *tert*-butyl N-[3-[*tert*-butyl(diphenyl)silyl]oxy-2-(2-hydroxyethoxy)propyl]carbamate (3.6 g, 58 % yield) as a colorless oil. MS m/z: 496.4 [M+Na]⁺, ESI pos.

Step 4: Ethyl (3*S*)-3-[[4-[3-[2-[1-[(*tert*-butoxycarbonylamino)methyl]-2-[*tert*-butyl(diphenyl)silyl]oxy-ethoxy]ethoxy]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate

To a solution of ethyl (3*S*)-3-[2-chloro-5-(trifluoromethyl)phenyl]-3-[[4-(3-hydroxyphenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate (**Intermediate D1**, 647 mg, 1.06 mmol) and *tert*-butyl N-[3-[*tert*-butyl(diphenyl)silyl]oxy-2-(2-hydroxyethoxy)propyl]carbamate (500 mg, 1.06 mmol) in tetrahydrofuran (10 ml) was added (tributylphosphoranylidene)acetonitrile (CAS 157141-27-0, 1019 mg, 4.22 mmol) at 0 °C. The reaction mixture was stirred at 60 °C for 16 hours. The reaction mixture was stirred at 100 °C for 2 hours. After cooling, the reaction mixture was poured into water and was extracted three times with ethyl acetate. The combined organic layer was dried with Na₂SO₄, filtered and concentrated under reduced pressure. The crude product was purified by flash chromatography (silica gel, ethyl acetate in petroleum ether 30-50 %) to give ethyl (3*S*)-3-[[4-[3-[2-[1-[(*tert*-butoxycarbonylamino)methyl]-2-[*tert*-butyl(diphenyl)silyl]oxy-ethoxy]ethoxy]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-

(trifluoromethyl)phenyl]propanoate (870 mg, 72 % yield) as a white solid. MS m/z: 1068.7 [M+H]⁺, ESI pos.

Step 5: (3S)-3-[[4-[3-[2-[1-[(*tert*-Butoxycarbonylamino)methyl]-2-[*tert*-butyl(diphenyl)silyl]oxy-ethoxy]ethoxy]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid

To a solution of ethyl (3S)-3-[[4-[3-[2-[1-[(*tert*-butoxycarbonylamino)methyl]-2-[*tert*-butyl(diphenyl)silyl]oxy-ethoxy]ethoxy]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate (870 mg, 0.81 mmol) in methanol (7 ml) was added a solution of aqueous 4 M NaOH (0.81 ml, 3.26 mmol) at 20 °C.

The reaction mixture was stirred at room temperature for 16 hours. A solution of 1 M HCl was added until pH = 4 and the mixture was three times with ethyl acetate. The combined organic layer was dried with Na₂SO₄, filtered and concentrated under reduced pressure to give (3S)-3-[[4-[3-[2-[1-[(*tert*-butoxycarbonylamino)methyl]-2-[*tert*-butyl(diphenyl)silyl]oxy-ethoxy]ethoxy]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid (738 mg, containing 18 % of (3S)-3-[[4-[3-[2-[1-[(*tert*-butoxycarbonylamino)methyl]-2-hydroxy-ethoxy]ethoxy]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid) as a yellow oil. MS m/z: 1068.7 [M+H]⁺, ESI pos.

Step 6: (3S)-3-[[4-[3-[2-[1-(Aminomethyl)-2-hydroxy-ethoxy]ethoxy]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic hydrochloride salt

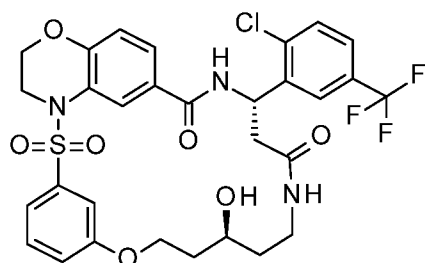
To solution of (3S)-3-[[4-[3-[2-[1-[(*tert*-butoxycarbonylamino)methyl]-2-[*tert*-butyl(diphenyl)silyl]oxy-ethoxy]ethoxy]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid (738 mg, 0.74 mmol, 82 % purity) in dichloromethane (4 ml) was added a solution of 4 M HCl/1,4-dioxane (4.0 ml, 16.0 mmol) at 0 °C. The mixture was stirred at 20 °C for 1 hour and was evaporated to give (3S)-3-[[4-[3-[2-[1-(aminomethyl)-2-hydroxy-ethoxy]ethoxy]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic hydrochloride salt (670 mg, light yellow solid, containing 19 % of (3S)-3-[[4-[3-[2-[1-(aminomethyl)-2-hydroxy-ethoxy]ethoxy]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid hydrochloride salt).

Step 7: (12*S*,17*S*)-17-[2-Chloro-5-(trifluoromethyl)phenyl]-12-(hydroxymethyl)-2,2-dioxo-8,11,24-trioxa-2 λ^6 -thia-1,14,18-triazatetracyclo[18.6.2.13,7.0^{23,27}]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-15,19-dione

To a solution of (3*S*)-3-[[4-[3-[2-[1-(aminomethyl)-2-*tert*-butyl(diphenyl)silyl]oxyethoxy]ethoxy]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid hydrochloride salt (645 mg, 0.54 mmol, 81 % purity) in N,N-dimethylformamide (35 ml) were added DIPEA (694 mg, 4.99 mmol), HATU (352 mg, 1.5 mmol). The reaction mixture was stirred at 20 °C for 2 hours. The mixture was poured into water and extracted three times with ethyl acetate. The combined organic layer was dried with Na₂SO₄,
 10 filtered and evaporated. The residue was purified by prep-HPLC (Phenomenex Luna C18 150 x 25 mm x 10 μ m, gradient water with 2.25 % formic acid and acetonitrile) to give the mixture of two diastereomers (17*S*)-12-[[*tert*-butyl(diphenyl)silyl]oxymethyl]-17-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-8,11,24-trioxa-2 λ^6 -thia-1,14,18-triazatetracyclo[18.6.2.13,7.0^{23,27}]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-15,19-dione.
 15 The mixture was separated by SFC (Column REGIS(S,S)WHELK-O1(250 mm x 25 mm, 10 μ m), EtOH with 0.1 % NH₄OH).

Second eluting diastereomer (absolute stereochemistry arbitrarily assigned) (12*S*, 17*S*)-17-[2-chloro-5-(trifluoromethyl)phenyl]-12-(hydroxymethyl)-2,2-dioxo-8,11,24-trioxa-2 λ^6 -thia-1,14,18-triazatetracyclo[18.6.2.13,7.0^{23,27}]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-
 20 15,19-dione (22 mg, 20 % yield). White solid. MS *m/z*: 684.4 [M+H]⁺, ESI pos.

Example 31: (11*S*, 17*S*)-17-[2-Chloro-5-(trifluoromethyl)phenyl]-11-hydroxy-2,2-dioxo-8,24-dioxa-2 λ^6 -thia-1,14,18-triazatetracyclo[18.6.2.13,7.0^{23,27}]nonacosa-3,5,7(29),20(28),21,23(27)-hexaene-15,19-dione



25 Step 1: Ethyl (3*S*)-3-[*tert*-butyl(diphenyl)silyl]oxy-4-cyano-butanoate

To solution of ethyl (3*S*)-4-cyano-3-hydroxy-butanoate (1 g, 6.36 mmol) and imidazole (1.3 g, 19.1 mmol) in dichloromethane (20 ml) was added *tert*-butylchlorodiphenylsilane (2.62 g, 9.54

mmol) at 0 °C. The mixture was stirred at 20 °C for 16 hours. The mixture was poured into water and extracted three times with ethyl acetate. The combined organic layer was dried with Na₂SO₄, filtered and evaporated. The crude product was purified via flash chromatography (silica gel, ethyl acetate/ petroleum ether, 0-30 %) to give ethyl (3*S*)-3-[*tert*-butyl(diphenyl)silyl]oxy-4-cyano-butanoate (2.5 g, 88 % yield) as a colorless oil. MS m/z: 418.1 [M+H]⁺, ESI pos.

Step 2: (3*S*)-3-[*tert*-Butyl(diphenyl)silyl]oxy-5-hydroxy-pentanenitrile

To a solution of ethyl (3*S*)-3-[*tert*-butyl(diphenyl)silyl]oxy-4-cyano-butanoate (2.1g, 5.31 mmol) in tetrahydrofuran (40 ml) was added dropwise at 0°C a solution of 2 M LiBH₄ in tetrahydrofuran (13.3 ml, 26.54 mmol) under nitrogen. The reaction mixture was allowed to warm to 25 °C and to stir for 16 h under nitrogen. The reaction mixture was slowly added to a cooled saturated aqueous solution of NH₄Cl (150 ml) and was stirred at 0 °C for 10 min. The mixture was extracted three times with ethyl acetate. The combined organic layer was dried with Na₂SO₄, filtered and evaporated. The crude product was purified via flash chromatography (silica gel, ethyl acetate/ petroleum ether, 0-30 %) to give (3*S*)-3-[*tert*-butyl(diphenyl)silyl]oxy-5-hydroxy-pentanenitrile (1.5 g, 79 % yield) as a colorless oil. MS m/z: 354.2 [M+H]⁺, ESI pos.

Step 3: Ethyl (3*S*)-3-[[4-[3-[(3*R*)-3-[*tert*-butyl(diphenyl)silyl]oxy-4-cyano-butoxy]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate

The title compound was prepared in analogy to **Example 30, step 4** starting from (3*S*)-3-[*tert*-butyl(diphenyl)silyl]oxy-5-hydroxy-pentanenitrile instead of *tert*-butyl *N*-[3-[*tert*-butyl(diphenyl)silyl]oxy-2-(2-hydroxyethoxy)propyl]carbamate and stirring at 100 °C for 16 hours instead of 100 °C for 2 hours. Light yellow oil. MS m/z: 948.5 [M+H]⁺, ESI pos.

Step 4: Methyl (3*S*)-3-[[4-[3-[(3*S*)-5-amino-3-[*tert*-butyl(diphenyl)silyl]oxy-pentoxo]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate

To a solution of ethyl (3*S*)-3-[[4-[3-[(3*S*)-3-[*tert*-butyl(diphenyl)silyl]oxy-4-cyano-butoxy]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate (500 mg, 0.53 mmol) in methanol (12.5 ml) and aqueous ammonia was added Raney Ni (250 mg, 2.92 mmol). The reaction mixture was stirred at 30 °C for 4 hours under a hydrogen atmosphere (50 psi). The reaction mixture was filtered, and the filtrate was evaporated to give crude methyl (3*S*)-3-[[4-[3-[(3*S*)-5-amino-3-[*tert*-

butyl(diphenyl)silyl]oxy-pentoxylphenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate (430 mg, 64 % yield). MS m/z: 938.6 [M+H]⁺, ESI pos.

Step 5: (3*S*)-3-[[4-[3-[(3*S*)-5-Amino-3-[*tert*-butyl(diphenyl)silyl]oxy-pentoxylphenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid

To a solution of methyl (3*S*)-3-[[4-[3-[(3*S*)-5-amino-3-[*tert*-butyl(diphenyl)silyl]oxy-pentoxylphenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate (430 mg, 0.46 mmol) in methanol (10 ml) was added an aqueous solution of 4 M NaOH (0.86 ml, 3.43 mmol). The mixture was stirred at 20 °C for 16 hours. The mixture was poured into water and extracted three times with ethyl acetate. The combined organic layer was dried with Na₂SO₄, filtered and evaporated. The crude product was purified via flash chromatography (silica gel, methanol in ethyl acetate, 0-10 %) to give (3*S*)-3-[[4-[3-[(3*S*)-5-amino-3-[*tert*-butyl(diphenyl)silyl]oxy-pentoxylphenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid (230 mg, 48 % yield) as a white solid. MS m/z: 924.6 [M+H]⁺, ESI pos.

Step 6: (11*S*,17*S*)-11-[*tert*-Butyl(diphenyl)silyl]oxy-17-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-8,24-dioxa-2λ⁶-thia-1,14,18-triazatetracyclo[18.6.2.13,7.0^{23,27}]nonacosa-3,5,7(29),20(28),21,23(27)-hexaene-15,19-dione

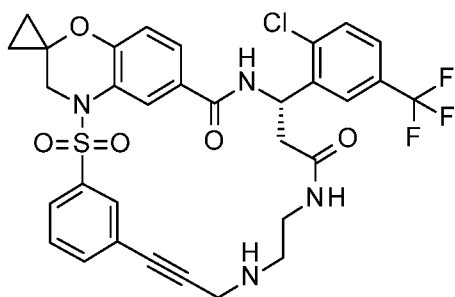
The title compound was prepared in analogy to **Example 11, step 9** starting with (3*S*)-3-[[4-[3-[(3*S*)-5-amino-3-[*tert*-butyl(diphenyl)silyl]oxy-pentoxylphenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid instead of (3*S*)-3-[[4-[3-[(3*S*)-3-[allyloxycarbonyl(2-aminoethyl)amino]pyrrolidin-1-yl]-5-methylphenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid hydrochloride salt. White solid. MS m/z: 906.5 [M+H]⁺, ESI pos.

Step 7: (11*S*, 17*S*)-17-[2-Chloro-5-(trifluoromethyl)phenyl]-11-hydroxy-2,2-dioxo-8,24-dioxa-2λ⁶-thia-1,14,18-triazatetracyclo[18.6.2.13,7.0^{23,27}]nonacosa-3,5,7(29),20(28),21,23(27)-hexaene-15,19-dione

To a solution of (11*S*, 17*S*)-11-[*tert*-butyl(diphenyl)silyl]oxy-17-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-8,24-dioxa-2λ⁶-thia-1,14,18-

triazatetracyclo[18.6.2.13,7.0²³,27]nonacosa-3,5,7(29),20(28),21,23(27)-hexaene-15,19-dione (50.0 mg, 0.06 mmol, 1.0 eq) in *N,N*-dimethylformamide (2.5 ml) was added CsF (19.18 mg, 0.17 mmol). The reaction mixture was stirred at 55 °C for 2 hours. The mixture was purified by prep-HPLC (Phenomenex Luna C18 150 x 25 mm x 10 µm, gradient water with 2.25 % formic acid and acetonitrile) to give (11*S*, 17*S*)-17-[2-chloro-5-(trifluoromethyl)phenyl]-11-hydroxy-2,2-dioxo-8,24-dioxa-2λ⁶-thia-1,14,18-triazatetracyclo[18.6.2.13,7.0²³,27]nonacosa-3,5,7(29),20(28),21,23(27)-hexaene-15,19-dione (33.5 mg, 85 % yield) as a white solid. MS *m/z*: 668.3 [M+H]⁺, ESI pos.

Example 32: (17*S*)-17-[2-Chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-spiro[24-oxa-2λ⁶-thia-1,11,14,18-tetrazatetracyclo[18.6.2.13,7.0²³,27]nonacosa-3,5,7(29),20(28),21,23(27)-hexaen-8-yne-25,1'-cyclopropane]-15,19-dione



Step 1: *N*-[2-(Propargylamino)ethyl]carbamic acid *tert*-butyl ester

To a solution of propargylamine (CAS 2450-71-7, 1.71 ml, 26.8 mmol) in *N,N*-dimethylformamide (4 ml), were added sodium bicarbonate (412 mg, 4.91 mmol) and *N*-(2-bromoethyl)carbamic acid *tert*-butyl ester (CAS 39684-80-5), 1 g, 4.46 mmol). The vial was sealed and the mixture was stirred at 60 °C for 4 hours. The reaction mixture was extracted twice with ethyl acetate, and the combined organic layer was dried over Na₂SO₄, filtered, and concentrated in vacuo. The crude product was purified by flash chromatography (silica gel, ethyl acetate/heptane, 0-50 %) to yield *N*-[2-(propargylamino)ethyl]carbamic acid *tert*-butyl ester (705 mg, 73 % yield) as a yellow oil.

¹H NMR (300 MHz, CHLOROFORM-*d*) δ = 4.95 (br s, 1H), 4.13 - 4.00 (m, 1H), 3.43 (d, *J* = 2.4 Hz, 2H), 3.34 - 3.19 (m, 2H), 2.89 - 2.78 (m, 2H), 2.29 - 2.19 (m, 1H), 1.45 (s, 9H)

Step 2: (3*S*)-3-[[4-[3-[3-[2-(*tert*-Butoxycarbonylamino)ethylamino]prop-1-

ynyl]phenyl]sulfonyl]spiro[3H-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid ethyl ester

The title compound was prepared in analogy to **Example 21, step 1** starting with ethyl (3*S*)-3-[[4-(3-bromophenyl)sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate (**Intermediate D6**) and N-(2-(propargylamino)ethyl)carbamic acid *tert*-butyl ester instead of (3*S*)-3-[[4-(3-bromophenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid ethyl ester (**Intermediate D5**) and N-(2-propargyloxyethyl)carbamic acid *tert*-butyl ester. Yellow waxy solid. MS *m/z*: 819.2 [M+H]⁺, ESI pos.

Step 3: (3*S*)-3-[[4-[3-[3-[2-(*tert*-Butoxycarbonylamino)ethylamino]prop-1-

ynyl]phenyl]sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid

The title compound was prepared in analogy to **Example 21, step 3** starting with (3*S*)-3-[[4-[3-[3-[2-(*tert*-butoxycarbonylamino)ethylamino]prop-1-ynyl]phenyl]sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid ethyl ester instead of (3*S*)-3-[[4-[3-[3-[2-(*tert*-butoxycarbonylamino)ethylamino]prop-1-ynyl]phenyl]sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid ethyl ester. Brown solid. MS *m/z*: 791.5 [M+H]⁺, ESI pos.

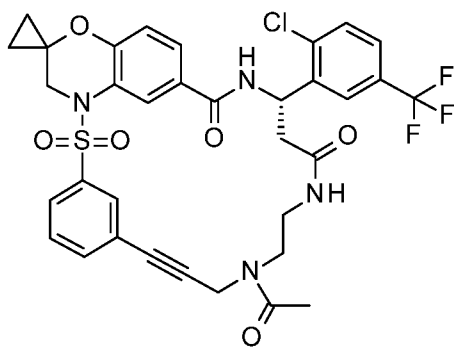
Step 4: (3*S*)-3-[[4-[3-[3-(2-Aminoethylamino)prop-1-ynyl]phenyl]sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid

To a solution of (3*S*)-3-[[4-[3-[3-[2-(*tert*-butoxycarbonylamino)ethylamino]prop-1-ynyl]phenyl]sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid (118 mg, 115 μmol) in 1,4-dioxane (1.22 ml), was added dropwise at 0 °C a solution of 4 M HCl in dioxane (201 μl, 804 μmol). The mixture was allowed to stir at room temperature for 24 hours. The reaction mixture was concentrated in vacuo and purified by preparative HPLC (column YMC-Triart C18, 12 nm, 5 μm, 100 x 30 mm, acetonitrile/water + 0.1 % formic acid) to give (3*S*)-3-[[4-[3-[3-(2-aminoethylamino)prop-1-ynyl]phenyl]sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid (30 mg, 38 %) as a white lyophilised powder. MS *m/z*: 689.4 [M-H]⁻, ESI neg.

Step 5: (17*S*)-17-[2-Chloro-5-(trifluoromethyl)phenyl]-2,2-diketo-spiro[24-oxa-2 λ ⁶-thia-1,11,14,18-tetrazatetracyclo[18.6.2.13,7.0^{23,27}]nonacosa-3(29),4,6,20,22,27-hexaen-8-yne-25,1'-cyclopropane]-15,19-quinone

To a solution of (3*S*)-3-[[4-[3-[4-(2-aminoethyl)piperazino]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid (30 mg, 43.4 μ mol) in dry *N,N*-dimethylformamide (0.89 ml) were added HATU (66 mg, 174 μ mol) and DIPEA (91.3 μ l, 524 μ mol). The reaction was stirred at room temperature for 4 hours. The reaction mixture was directly separated by preparative HPLC (column YMC-Triart C18, 12 nm, 5 μ m, 100 x 30 mm, ACN /water + 0.1 % formic acid) to give (17*S*)-17-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-diketo-spiro[24-oxa-2 λ ⁶-thia-1,11,14,18-tetrazatetracyclo[18.6.2.13,7.0^{23,27}]nonacosa-3(29),4,6,20,22,27-hexaen-8-yne-25,1'-cyclopropane]-15,19-quinone (2 mg, 6 % yield) as a white lyophilised powder. MS *m/z*: 673.1 [M+H]⁺, ESI pos.

Example 33: (17*S*)-11-Acetyl-17-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-spiro[24-oxa-2 λ ⁶-thia-1,11,14,18-tetrazatetracyclo[18.6.2.13,7.0^{23,27}]nonacosa-3,5,7(29),20(28),21,23(27)-hexaen-8-yne-25,1'-cyclopropane]-15,19-dione



Step 1: (3*S*)-3-[[4-[3-[3-(2-aminoethylamino)prop-1-ynyl]phenyl]sulfonyl]spiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid hydrochloride salt

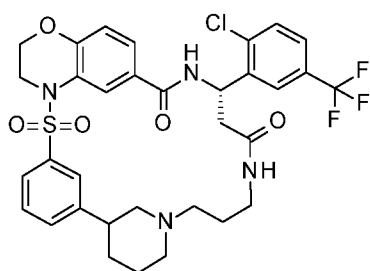
To a solution of (3*S*)-3-[[4-[3-[3-(2-*tert*-butoxycarbonylamino)ethylamino]prop-1-ynyl]phenyl]sulfonyl]spiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid (36 mg, 45.5 μ mol) in ethyl acetate (482 μ l), was added dropwise at 0 °C a solution of 4 M HCl in dioxane (79.6 μ l, 319 μ mol). The mixture was allowed to stir at room temperature for 24 hours. The reaction mixture was concentrated under reduced pressure to give (3*S*)-3-[[4-[3-[3-(2-aminoethylamino)prop-1-

ynyl]phenyl]sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid (30 mg, 38 %, contains traces of acetic acid, saponification of ethyl acetate) as a white lyophilised powder. MS *m/z*: 689.5 [M-H]⁻, ESI neg.

Step 2: (17*S*)-11-Acetyl-17-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-diketo-spiro[24-oxa-2λ⁶-thia-1,11,14,18-tetrazatetracyclo[18.6.2.13,7.0^{23,27}]nonacosa-3,5,7(29),20,22,27-hexaen-8-yne-25,1'-cyclopropane]-15,19-quinone

To a solution of (3*S*)-3-[[4-[3-[4-(2-aminoethyl)piperazino]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid hydrochloride salt (30 mg, 43.4 μmol) in dry *N,N*-dimethylformamide (0.89 ml) were added HATU (66 mg, 173.6 μmol) and DIPEA (91.3 μl, 524 μmol). The reaction was stirred at room temperature for 4 hours. The reaction mixture was directly separated by preparative HPLC (column YMC-Triart C18, 12 nm, 5 μm, 100 x 30 mm, ACN /water + 0.1 % formic acid) to give (17*S*)-17-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-diketo-spiro[24-oxa-2λ⁶-thia-1,11,14,18-tetrazatetracyclo[18.6.2.13,7.0^{23,27}]nonacosa-3(29),4,6,20,22,27-hexaen-8-yne-25,1'-cyclopropane]-15,19-quinone (2 mg, 6 % yield) as a white lyophilised powder. MS *m/z*: 673.1 [M+H]⁺, ESI pos.

Example 34: (19*S*)-19-[2-Chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-26-oxa-2λ⁶-thia-1,12,16,20-tetrazapentacyclo[20.6.2.13,7.18,12.0^{25,29}]dotriaconta-3(32),4,6,22(30),23,25(29)-hexaene-17,21-dione, mixture of two diastereomers.



Step 1: 4-[3-(1-*tert*-Butoxycarbonyl-3,6-dihydro-2*H*-pyridin-5-yl)phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid methyl ester

To a solution of 4-(3-bromophenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid methyl ester (**Intermediate C4**, 200 mg, 485 μmol) in a mixture of 1,4-dioxane (12.4 ml) and water (1.4 ml), were added while degasing with nitrogen, 5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-3,6-dihydro-2*H*-pyridine-1-carboxylic acid *tert*-butyl ester (CAS 286961-14-6, 180 mg, 582 μmol), potassium carbonate (134 mg, 970 μmol), and Pd(dppf)Cl₂ · CH₂Cl₂ (39.6

mg, 48.5 μ mol). The vial was sealed and the mixture stirred at 100 °C for 4 hours. After cooling, the reaction mixture was partitioned between water and dichloromethane. The aqueous phase was extracted with dichloromethane. The combined organics were washed with brine, dried over Na₂SO₄, vacuum-filtered, and concentrated in vacuo. The crude product was purified via flash chromatography (silica gel, ethyl acetate/heptane, 0-100 %) to give 4-[3-(1-*tert*-butoxycarbonyl-3,6-dihydro-2*H*-pyridin-5-yl)phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid methyl ester (235.8 mg, 93 %) as a light yellow waxy solid. MS *m/z*: 415.2 [M+H]⁺, ESI pos.

Step 2: 4-[3-(1-*tert*-Butoxycarbonyl-3-piperidyl)phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid methyl ester

To a solution of 4-[3-(1-*tert*-butoxycarbonyl-3,6-dihydro-2*H*-pyridin-5-yl)phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid methyl ester (570 mg, 1.11 mmol) in methanol (12.3 ml) was added Pd/C 10 % (61.6 mg, 57.9 μ mol). The reaction mixture was stirred under an hydrogen atmosphere at room temperature and 3 bars for 1 hour. The reaction mixture was filtered and the filtrate was evaporated to give 4-[3-(1-*tert*-butoxycarbonyl-3-piperidyl)phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid methyl ester (543 mg, 94 %) as an off-white foam. MS *m/z*: 417.1 [M+H-BOC]⁺, ESI pos.

Step 3: 4-[3-(3-Piperidyl)phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid methyl ester hydrochloride salt

To a solution of 4-[3-(1-*tert*-butoxycarbonyl-3-piperidyl)phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid methyl ester (542 mg, 1.05 mmol) in ethyl acetate (7.14 ml) was added dropwise at 0 °C a solution of 4 M HCl in 1,4-dioxane (1.31 ml, 5.25 mmol). The mixture was stirred at room temperature for 24 hours. Diethylether was added to the reaction mixture. The suspension was filtered. The sticky filter cake was dissolved in ethyl acetate and concentrated in vacuo to afford 4-[3-(3-piperidyl)phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid methyl ester hydrogen salt (501 mg, > 100 % yield) as a white foam. MS *m/z*: 417.1 [M+H]⁺, ESI pos.

Step 4: 4-[3-[1-[3-(*tert*-Butoxycarbonylamino)propyl]-3-piperidyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid methyl ester

To a solution of 4-[3-(3-piperidyl)phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid methyl ester hydrochloride salt (270 mg, 596 μ mol) in acetonitrile (6 ml) were added potassium carbonate (412 mg, 2.98 mmol) and *N*-(3-bromopropyl)carbamic acid *tert*-butyl ester

(CAS 83948-53-2, 284 mg, 1.19 mmol). The reaction mixture was stirred at 80 °C for 16 hours. The mixture was extracted twice with ethyl acetate, and the combined organic layer was dried over Na₂SO₄, filtered, and concentrated in vacuo. The crude product was purified via flash chromatography (silica gel, ethyl acetate/heptane, 0-100%) to give 4-[3-[1-[3-(*tert*-butoxycarbonylamino)propyl]-3-piperidyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid methyl ester (220 mg, 64 % yield) as a colorless waxy solid. MS m/z: 574.2 [M+H]⁺, ESI pos.

Step 5: 4-[3-[1-[3-(*tert*-Butoxycarbonylamino)propyl]-3-piperidyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid

To a solution of 4-[3-[1-[3-(*tert*-butoxycarbonylamino)propyl]-3-piperidyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid methyl ester (220 mg, 383 μmol) in methanol (767 μl) and tetrahydrofuran (1.53 ml), 4 M sodium hydroxide (288 μl, 1.15 mmol) was added dropwise at room temperature. The reaction was stirred at room temperature for 3 hours. The mixture was acidified with citric acid 1M and was extracted with ethyl acetate. The organic layer was dried over Na₂SO₄, filtered and concentrated in vacuo to give 4-[3-[1-[3-(*tert*-butoxycarbonylamino)propyl]-3-piperidyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid (206 mg, 91 %) as a white solid. MS m/z: 560.2 [M+H]⁺, ESI pos.

Step 6: (3*S*)-3-[[4-[3-[1-[3-(*tert*-Butoxycarbonylamino)propyl]-3-piperidyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid ethyl ester

To a solution of 4-[3-[1-[3-(*tert*-butoxycarbonylamino)propyl]-3-piperidyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid (200 mg, 357 μmol) in dry *N,N*-dimethylformamide (3.5 ml) were added (3*S*)-3-amino-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid ethyl ester (**Intermediate B1**, 137 mg, 465 μmol), DIPEA (185 mg, 249 μl) and HATU (204 mg, 536 μmol). The reaction mixture was stirred at room temperature for 3 hours. Water was added to the reaction mixture and the suspension obtained was filtered, the solid was dried under high vacuum to give (3*S*)-3-[[4-[3-[1-[3-(*tert*-butoxycarbonylamino)propyl]-3-piperidyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid ethyl ester (183 mg, 56 % yield) as a light yellow solid. MS m/z: 837.2 [M+H]⁺, ESI pos.

Step 7: (3*S*)-3-[[4-[3-[1-[3-(*tert*-Butoxycarbonylamino)propyl]-3-piperidyl]phenyl]sulfonyl-2,3-

5 dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid

The title compound was prepared in analogy to **step 5** starting with (3*S*)-3-[[4-[3-[1-[3-(*tert*-butoxycarbonylamino)propyl]-3-piperidyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid instead of 4-[3-[1-[3-(*tert*-butoxycarbonylamino)propyl]-3-piperidyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid methyl ester (99.4 mg, 51 % yield). White waxy solid. MS *m/z*: 809.4 [M+H]⁺, ESI pos.

10 Step 8: (3*S*)-3-[[4-[3-[1-(3-Aminopropyl)-3-piperidyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid hydrogenschloride salt

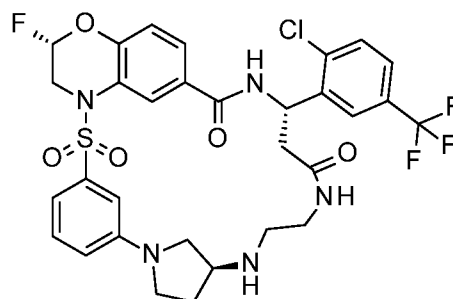
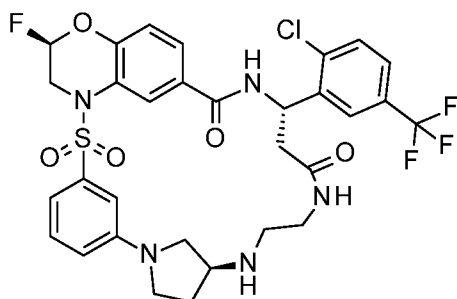
The title compound was prepared in analogy to **Example 21, step 3** starting with (3*S*)-3-[[4-[3-[1-[3-(*tert*-butoxycarbonylamino)propyl]-3-piperidyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid instead of (3*S*)-3-[[4-[3-[3-[2-(*tert*-butoxycarbonylamino)ethoxy]prop-1-ynyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid (40 mg, 44%). White waxy solid. MS *m/z*: 707.4 [M-H]⁻, ESI neg.

20 Step 9: (1*S*)-19-[2-Chloro-5-(trifluoromethyl)phenyl]-2,2-diketo-26-oxa-2λ⁶-thia-1,12,16,20-tetrazapentacyclo[20.6.2.13,7.18,12.025,29]dotriaconta-3(32),4,6,22,24,29-hexaene-17,21-quinone

The title compound was prepared in analogy to **Example 1, step 4** starting from (3*S*)-3-[[4-[3-[1-(3-aminopropyl)-3-piperidyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid hydrogenschloride salt instead of (3*S*)-3-[[4-[3-[2-(2-aminoethoxy)ethoxy]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid. White lyophilised powder. MS *m/z*: 691.3 [M+H]⁺, ESI pos.

Example 35 and Example 36: (1*S*,18*S*,26*R*)-18-[2-Chloro-5-(trifluoromethyl)phenyl]-26-fluoro-2,2-dioxo-25-oxa-2λ⁶-thia-1,8,12,15,19-pentazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-3(31),4,6,21(29),22,24(28)-hexaene-16,20-dione and (1*S*,18*S*,26*S*)-18-[2-chloro-5-(trifluoromethyl)phenyl]-26-fluoro-2,2-dioxo-25-

oxa-2λ⁶-thia-1,8,12,15,19-pentazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-3(31),4,6,21(29),22,24(28)-hexaene-16,20-dione



5 Step 1: Benzyl (3*S*)-3-[2-(1,3-dioxisoindolin-2-yl)ethylamino]pyrrolidine-1-carboxylate

The title compound was prepared in analogy to **Example 28 and Example 29, step 3** starting from benzyl (3*S*)-3-aminopyrrolidine-1-carboxylate (CAS 122536-72-5) instead of ethyl (3*S*)-3-[[4-[3-(2-amino-1-methyl-ethoxy)phenyl]sulfonyl]spiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate. Colorless oil. MS m/z:

10 394.2 [M+H]⁺, ESI pos.

Step 2: Benzyl (3*S*)-3-[*tert*-butoxycarbonyl]-[2-(1,3-dioxisoindolin-2-yl)ethyl]amino]pyrrolidine-1-carboxylate

The title compound was prepared in analogy to **Example 28 and Example 29, step 4** starting from benzyl (3*S*)-3-[2-(1,3-dioxisoindolin-2-yl)ethylamino]pyrrolidine-1-carboxylate instead of
15 ethyl (3*S*)-3-[2-chloro-5-(trifluoromethyl)phenyl]-3-[[4-[3-[2-[2-(1,3-dioxisoindolin-2-yl)ethylamino]-1-methyl-ethoxy]phenyl]sulfonyl]spiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]propanoate. Colorless oil. MS m/z: 394.2 [M-100+H]⁺, ESI pos.

Step 3: *tert*-Butyl *N*-[2-(1,3-dioxisoindolin-2-yl)ethyl]-*N*-[(3*S*)-pyrrolidin-3-yl]carbamate

To a solution of benzyl (3*S*)-3-[*tert*-butoxycarbonyl]-[2-(1,3-dioxisoindolin-2-yl)ethyl]amino]pyrrolidine-1-carboxylate (4.9 g, 9.93 mmol) in methanol (80 ml) was added
20 Pd/C (wet). The reaction mixture was stirred at 30 °C for 16 hours under an atmosphere of hydrogen (50 psi). The reaction mixture was filtered over diatomite and the filtrate was evaporated. The crude product was purified by reverse phase flash chromatography (Flash Spherical C18, 20-35 μm; 100 Å; gradient water with 0.05 % TFA and acetonitrile) to give (*tert*-

butyl *N*-[2-(1,3-dioxoisindolin-2-yl)ethyl]-*N*-[(3*S*)-pyrrolidin-3-yl]carbamate (840 mg, 23 % yield) as an orange oil. MS *m/z*: 360.2 [M-100+H]⁺, ESI pos.

Step 4: Ethyl (3*S*)-3-[[4-[3-[(3*S*)-3-[*tert*-butoxycarbonyl]-2-(1,3-dioxoisindolin-2-yl)ethyl]amino]pyrrolidin-1-yl]phenyl]sulfonyl-2-fluoro-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate

The title compound was prepared in analogy to **Example 17 and Example 18, step 1** starting from ethyl (3*S*)-3-[[4-(3-bromophenyl)sulfonyl-2-fluoro-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate (**Intermediate D10**) and (*tert*-butyl *N*-[2-(1,3-dioxoisindolin-2-yl)ethyl]-*N*-[(3*S*)-pyrrolidin-3-yl]carbamate instead of methyl 4-(3-bromophenyl)sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carboxylate (**Intermediate C5**) and *tert*-butyl *N*-(2-morpholin-2-ylethyl)carbamate. Colorless oil. MS *m/z*: 972.1 [M+H]⁺, ESI pos.

Step 5: Ethyl (3*S*)-3-[[4-[3-[(3*S*)-3-[2-aminoethyl(*tert*-butoxycarbonyl)amino]pyrrolidin-1-yl]phenyl]sulfonyl-2-fluoro-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate

The title compound was prepared in analogy to **Example 23, step 4** starting from ethyl (3*S*)-3-[[4-[3-[(3*S*)-3-[*tert*-butoxycarbonyl]-2-(1,3-dioxoisindolin-2-yl)ethyl]amino]pyrrolidin-1-yl]phenyl]sulfonyl-2-fluoro-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate instead of ethyl (3*S*)-3-[[4-[3-[2-[*tert*-butoxycarbonyl]-2-(1,3-dioxoisindolin-2-yl)ethyl]amino]ethylamino]-4-fluoro-phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate. White oil. MS *m/z*: 697.3 [M+H]⁺, ESI pos.

Step 6: (3*S*)-3-[[4-[3-[(3*S*)-3-[2-Aminoethyl(*tert*-butoxycarbonyl)amino]pyrrolidin-1-yl]phenyl]sulfonyl-2-fluoro-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid

The title compound was prepared in analogy to **Example 28 and Example 29, step 6** starting from ethyl (3*S*)-3-[[4-[3-[(3*S*)-3-[2-aminoethyl(*tert*-butoxycarbonyl)amino]pyrrolidin-1-yl]phenyl]sulfonyl-2-fluoro-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate instead of ethyl (3*S*)-3-[[4-[3-[2-[2-aminoethyl(*tert*-butoxycarbonyl)amino]-1-methyl-ethoxy]phenyl]sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-

cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate and stirring at 25 °C for 16 hours instead of 20 °C for 1 hour. Orange oil. MS m/z: 814.2 [M-100+H]⁺, ESI pos.

Step 7: *tert*-Butyl (11*S*,18*S*)-18-[2-chloro-5-(trifluoromethyl)phenyl]-26-fluoro-2,2,16,20-

5 tetraoxo-25-oxa-2λ⁶-thia-1,8,12,15,19-pentazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-3,5,7(31),21(29),22,24(28)-hexaene-12-carboxylate

The title compound was prepared in analogy to **Example 5, step 10** starting with (3*S*)-3-[[4-[3-[(3*S*)-3-[2-aminoethyl(*tert*-butoxycarbonyl)amino]pyrrolidin-1-yl]phenyl]sulfonyl-2-fluoro-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic
10 acid instead of (3*S*)-3-[[4-[3-[(2*R*)-5-amino-2-(*tert*-butoxycarbonylamino)pentoxylphenyl]sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid. White solid. MS m/z: 796.2 [M+H]⁺, ESI pos

Step 8: (11*S*,18*S*,26*R*)-18-[2-Chloro-5-(trifluoromethyl)phenyl]-26-fluoro-2,2-dioxo-25-oxa-2λ⁶-
15 thia-1,8,12,15,19-pentazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-3,5,7(31),21(29),22,24(28)-hexaene-16,20-dione and (11*S*,18*S*,26*S*)-18-[2-chloro-5-(trifluoromethyl)phenyl]-26-fluoro-2,2-dioxo-25-oxa-2λ⁶-thia-1,8,12,15,19-
pentazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-3,5,7(31),21(29),22,24(28)-hexaene-16,20-dione

20 To a solution of *tert*-butyl (11*S*,18*S*)-18-[2-chloro-5-(trifluoromethyl)phenyl]-26-fluoro-2,2,16,20-tetraoxo-25-oxa-2λ⁶-thia-1,8,12,15,19-pentazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-3,5,7(31),21(29),22,24(28)-hexaene-12-carboxylate (55 mg, 0.07 mmol) in dichloromethane (1 ml) was added 2,2,2 trifluoroacetic acid (1.0 ml, 0.07 mmol). The reaction mixture was stirred at 25 °C for 1 hour. The reaction
25 mixture was concentrated and lyophilized to give the crude mixture of diastereomers.

The diastereomers were separated using SFC (REGIS(S,S)WHELK-O1, 250 mm x 25 mm, 10 μm, methanol with 0.1 % ammonium hydroxide and acetonitrile)

First eluting diastereomer (R_t = 0.523 min, absolute stereochemistry arbitrarily assigned):
(11*S*,18*S*,26*R*)-18-[2-chloro-5-(trifluoromethyl)phenyl]-26-fluoro-2,2-dioxo-25-oxa-2λ⁶-thia-
30 1,8,12,15,19-pentazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-

3,5,7(31),21(29),22,24(28)-hexaene-16,20-dione (17.3 mg, 36 % yield). Yellow solid. MS m/z: 696.1 [M+H]⁺, ESI pos.

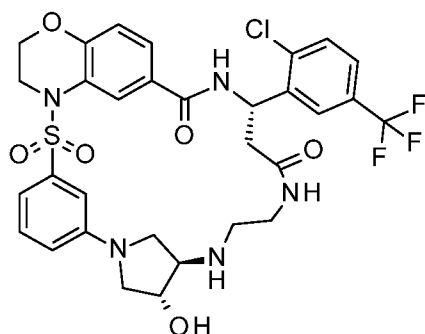
Second eluting diastereomer (R_t = 0.533 min, absolute stereochemistry arbitrarily assigned): (11*S*,18*S*,26*S*)-18-[2-chloro-5-(trifluoromethyl)phenyl]-26-fluoro-2,2-dioxo-25-oxa-2λ⁶-thia-

5 1,8,12,15,19-pentazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-

3,5,7(31),21(29),22,24(28)-hexaene-16,20-dione (17.9 mg, 36 % yield). Yellow solid. MS m/z: 696.1 [M+H]⁺, ESI pos.

Example 37: (10*R*,11*R*,18*S*)-18-[2-Chloro-5-(trifluoromethyl)phenyl]-10-hydroxy-2,2-dioxo-25-oxa-2λ⁶-thia-1,8,12,15,19-pentazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-

10 3(31),4,6,21(29),22,24(28)-hexaene-16,20-dione



Step 1: Ethyl (3*S*)-3-[[4-[3-[(3*R*,4*R*)-3-(*tert*-butoxycarbonylamino)-4-hydroxy-pyrrolidin-1-yl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate

15 The title compound was prepared in analogy to **Example 17 and Example 18, step 1** starting from of ethyl (3*S*)-3-[[4-(3-bromophenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate (**Intermediate D5**) and *tert*-butyl *N*-[(3*R*,4*R*)-4-hydroxypyrrolidin-3-yl]carbamate (CAS 1203566-77-1) instead of methyl 4-(3-bromophenyl)sulfonylspro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carboxylate (**Intermediate C5**) and *tert*-butyl *N*-(2-morpholin-2-ylethyl)carbamate. Orange oil. MS m/z: 797.2 [M+H]⁺, ESI pos.

Step 2: Ethyl -(3*S*)-3-[2-chloro-5-(trifluoromethyl)phenyl]-3-[[4-[3-[(3*R*,4*R*)-3-amino-4-hydroxy-pyrrolidin-1-yl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate 2,2,2-trifluoroacetic acid salt

The title compound was prepared in analogy to **Example 17 and 18, step 5** starting with ethyl (3*S*)-3-[[4-[3-[(3*R*,4*R*)-3-(*tert*-butoxycarbonylamino)-4-hydroxy-pyrrolidin-1-yl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate instead of (3*S*)-3-[[4-[3-[2-(*tert*-

- 5 butoxycarbonylamino)ethyl]morpholin-4-yl]phenyl]sulfonylspro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid. Orange solid. MS *m/z*: 697.1 [M+H]⁺, ESI pos.

Step 3: Ethyl (3*S*)-3-[2-chloro-5-(trifluoromethyl)phenyl]-3-[[4-[3-[(3*R*,4*R*)-3-[2-(1,3-dioxoisindolin-2-yl)ethylamino]-4-hydroxy-pyrrolidin-1-yl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate

- 10 The title compound was prepared in analogy to **Example 12, step 3** starting with ethyl (3*S*)-3-[2-chloro-5-(trifluoromethyl)phenyl]-3-[[4-[3-[(3*R*,4*R*)-3-amino-4-hydroxy-pyrrolidin-1-yl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate 2,2,2-trifluoroacetic acid salt instead of ethyl (3*S*)-3-[[4-[3-[(3*S*)-3-aminopyrrolidin-1-yl]phenyl]sulfonyl-2-methyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate. White solid. MS *m/z*: 870.1 [M+H]⁺, ESI pos.

Step 4: 2-[2-[[4-[(3*R*,4*R*)-1-[3-[[6-[(1*S*)-2-Carboxy-1-[2-chloro-5-(trifluoromethyl)phenyl]ethyl]carbonyl]-2,3-dihydro-1,4-benzoxazin-4-yl]sulfonyl]phenyl]-4-hydroxy-pyrrolidin-3-yl]amino]ethyl]carbonyl]benzoic acid

- 20 The title compound was prepared in analogy to **Example 5, step 8** starting with ethyl (3*S*)-3-[2-chloro-5-(trifluoromethyl)phenyl]-3-[[4-[3-[(3*R*,4*R*)-3-[2-(1,3-dioxoisindolin-2-yl)ethylamino]-4-hydroxy-pyrrolidin-1-yl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate instead of ethyl (3*S*)-3-[[4-[3-[(2*R*)-2-(*tert*-butoxycarbonylamino)-5-(1,3-dioxoisindolin-2-yl)pentoxyl]phenyl]sulfonylspro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate. White solid. MS *m/z*: 860.1 [M+H]⁺, ESI pos.

Step 5: (3*S*)-3-[[4-[3-[(3*R*,4*R*)-3-(2-Aminoethylamino)-4-hydroxy-pyrrolidin-1-yl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid

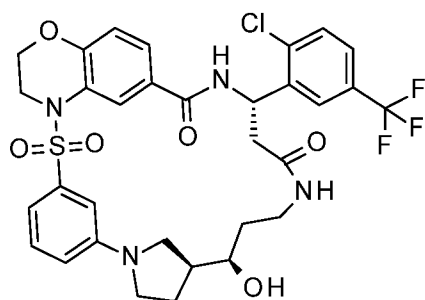
- 30 To a solution of 2-[2-[[4-[(3*R*,4*R*)-1-[3-[[6-[(1*S*)-2-carboxy-1-[2-chloro-5-(trifluoromethyl)phenyl]ethyl]carbonyl]-2,3-dihydro-1,4-benzoxazin-4-yl]sulfonyl]phenyl]-4-

hydroxy-pyrrolidin-3-yl]amino]ethylcarbamoyl]benzoic acid (160 mg, 0.19 mmol) in ethanol (3 ml) was added $\text{N}_2\text{H}_4 \cdot \text{H}_2\text{O}$ (0.09 ml, 1.86 mmol). The reaction mixture was stirred at 80 °C for 2 hours. The reaction mixture was directly purified by reverse phase flash chromatography (Flash Spherical C18, 20-35 μm ; 100 Å; gradient water with 0.05 % TFA and acetonitrile) to give ((3*S*)-3-[[4-[3-[(3*R*,4*R*)-3-(2-aminoethylamino)-4-hydroxy-pyrrolidin-1-yl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid (100 mg, 76 % yield) as a yellow solid. MS m/z : 712.1 $[\text{M}+\text{H}]^+$, ESI pos.

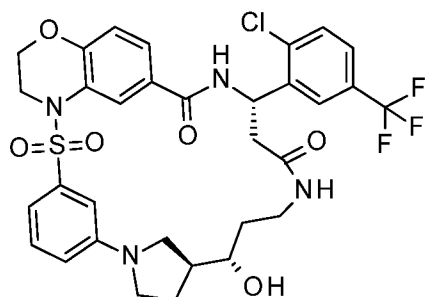
Step 6: (10*R*,11*R*,18*S*)-18-[2-Chloro-5-(trifluoromethyl)phenyl]-10-hydroxy-2,2-dioxo-25-oxa-2 λ^6 -thia-1,8,12,15,19-pentazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-3(31),4,6,21(29),22,24(28)-hexaene-16,20-dione

The title compound was prepared in analogy to **Example 4, step 4** starting with ((3*S*)-3-[[4-[3-[(3*R*,4*R*)-3-(2-aminoethylamino)-4-hydroxy-pyrrolidin-1-yl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid instead of (3*S*)-3-[[4-[3-[2-(2-aminoethoxy)ethoxy]-5-methyl-phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid hydrochloride salt. White solid, MS m/z : 694.2 $[\text{M}+\text{H}]^+$, ESI pos.

Example 38 to 41: (11*S*,12*R*,18*S*)-18-[2-Chloro-5-(trifluoromethyl)phenyl]-12-hydroxy-2,2-dioxo-25-oxa-2 λ^6 -thia-1,8,15,19-tetrazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-3(31),4,6,21(29),22,24(28)-hexaene-16,20-dione (Example 38, Diastereomer A, absolute stereochemistry arbitrarily assigned) and

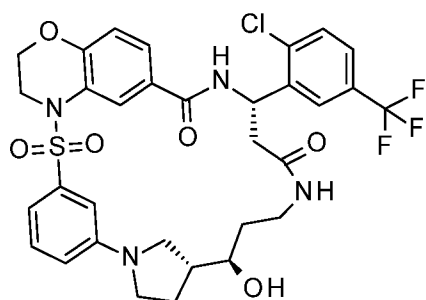


(11*S*,12*S*,18*S*)-18-[2-chloro-5-(trifluoromethyl)phenyl]-12-hydroxy-2,2-dioxo-25-oxa-2 λ^6 -thia-1,8,15,19-tetrazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-3(31),4,6,21(29),22,24(28)-hexaene-16,20-dione (Example 39, Diastereomer B, absolute stereochemistry arbitrarily assigned) and



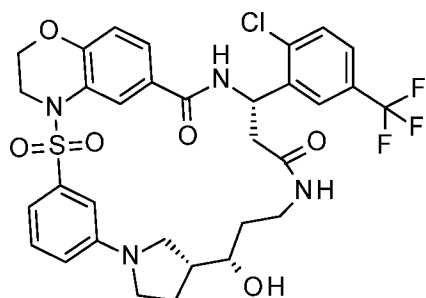
(11*R*,12*R*,18*S*)-18-[2-chloro-5-(trifluoromethyl)phenyl]-12-hydroxy-2,2-dioxo-25-oxa-2 λ^6 -thia-1,8,15,19-tetrazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-3(31),4,6,21(29),22,24(28)-hexaene-16,20-dione (Example 40, Diastereomer C, absolute stereochemistry arbitrarily

5 assigned) and



(11*R*,12*S*,18*S*)-18-[2-chloro-5-(trifluoromethyl)phenyl]-12-hydroxy-2,2-dioxo-25-oxa-2 λ^6 -thia-1,8,15,19-tetrazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-3(31),4,6,21(29),22,24(28)-hexaene-16,20-dione (Example 41, Diastereomer D, absolute stereochemistry arbitrarily

10 assigned).



Step1: *tert*-Butyl 3-(2-cyano-1-hydroxy-ethyl)pyrrolidine-1-carboxylate

To a solution of acetonitrile (315 μ l, 6.02 mmol) in tetrahydrofuran (12 ml) was added dropwise
 15 a solution of 2.5 M *n*-BuLi in tetrahydrofuran (2.65 ml, 6.62 mmol) under nitrogen at -70 °C.

The reaction mixture was stirred at -70 °C for 1 hour then a solution of 1-BOC-3-pyrrolidinecarbaldehyde (CAS 59379-02-1, 600 mg, 3.01 mmol) in tetrahydrofuran (4 ml) was added at the same temperature. The reaction mixture was stirred at -70 °C for 2.5 hours. The reaction mixture was slowly poured into saturated NH₄Cl aqueous solution (50 ml) at 0 °C, then the mixture was extracted three times with ethyl acetate. The combined organic layer was dried by Na₂SO₄, filtered and concentrated. The crude product was purified via flash chromatography (silica gel, ethyl acetate/ petroleum ether 0-75 %) to give *tert*-butyl 3-(2-cyano-1-hydroxy-ethyl)pyrrolidine-1-carboxylate (400 mg, 56 % yield) as a colorless oil.

¹H NMR (400 MHz, CHLOROFORM-*d*) δ ppm 1.44 - 1.48 (m, 9 H) 1.56 - 1.70 (m, 1 H) 2.06 (br d, J=3.2 Hz, 1 H) 2.31 - 2.45 (m, 1 H) 2.48 - 2.69 (m, 2 H) 2.97 - 3.73 (m, 6 H) 3.78 - 3.98 (m, 1 H)

Step 2: 3-Hydroxy-3-pyrrolidin-3-yl-propanenitrile 2,2,2-trifluoroacetic acid salt

To a solution of *tert*-butyl 3-(2-cyano-1-hydroxy-ethyl)pyrrolidine-1-carboxylate (400 mg, 1.66 mmol) in dichloromethane (4 ml) was added dropwise 2,2,2 trifluoroacetic acid (4.0 ml). The reaction mixture was stirred at 25 °C for 2 hours then water (20 ml) was carefully added to the mixture. The solution was lyophilized to give 3-hydroxy-3-pyrrolidin-3-yl-propanenitrile 2,2,2-trifluoroacetic acid salt (350 mg, 83 % yield) as a light yellow oil.

¹H NMR (400 MHz, DMSO-*d*₆) δ ppm 1.55 - 1.86 (m, 1 H) 1.89 - 2.02 (m, 1 H) 2.27 - 2.37 (m, 1 H) 2.56 - 2.77 (m, 2 H) 2.81 - 3.06 (m, 1 H) 3.07 - 3.33 (m, 3 H) 3.60 - 3.74 (m, 2 H) 8.53 - 9.01 (m, 2 H)

Step 3: Ethyl (3*S*)-3-[2-chloro-5-(trifluoromethyl)phenyl]-3-[[4-[3-[3-(2-cyano-1-hydroxy-ethyl)pyrrolidin-1-yl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate

To a solution of ethyl (3*S*)-3-[[4-(3-bromophenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate (**Intermediate D5**, 904 mg, 1.3 mmol) 3-hydroxy-3-pyrrolidin-3-yl-propanenitrile 2,2,2-trifluoroacetic acid salt (340 mg, 1.34 mmol) in 1,4-dioxane (5 ml) were added Cs₂CO₃ (1.31 g, 4.01 mmol), xantphos (77.4 mg, 0.13 mmol) and Pd₂(dba)₃ · CHCl₃ (69.2 mg, 0.07 mmol). The reaction mixture was stirred at 100 °C for 2 hours under nitrogen. The mixture was filtered and the filtrate was concentrated under reduced pressure. The crude product was first purified via flash chromatography (silica gel, ethyl acetate/petroleum ether 0-100 %) then by reverse phase flash chromatography (Flash

Spherical C18, 20-35 μm ; 100 $\text{\AA}$; gradient water with 0.1 % TFA and acetonitrile) to give ethyl (3*S*)-3-[2-chloro-5-(trifluoromethyl)phenyl]-3-[[4-[3-[3-(2-cyano-1-hydroxy-ethyl)pyrrolidin-1-yl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate (350 mg, 34 % yield) as a white solid. MS m/z : 735.4 $[\text{M}+\text{H}]^+$, ESI

5 Step 4: Ethyl (3*S*)-3-[[4-[3-[3-(3-amino-1-hydroxy-propyl)pyrrolidin-1-yl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate

To a solution of ethyl (3*S*)-3-[2-chloro-5-(trifluoromethyl)phenyl]-3-[[4-[3-[3-(2-cyano-1-hydroxy-ethyl)pyrrolidin-1-yl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate (350 mg, 0.48 mmol) in methanol (10 ml) were added aqueous
10 ammonia (2.0 ml, 0.48 mmol) and Raney Ni (175 mg, 2.04 mmol). The reaction was stirred at 30 $^{\circ}\text{C}$ for 2 hours under a hydrogen atmosphere (15 psi). The reaction mixture was filtered and the filtrate was evaporated. The crude product was purified by reverse phase flash chromatography (Flash Spherical C18, 20-35 μm ; 100 $\text{\AA}$; gradient water with 0.1 % TFA and acetonitrile) to give ethyl (3*S*)-3-[[4-[3-[3-(3-amino-1-hydroxy-propyl)pyrrolidin-1-yl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-
15 (trifluoromethyl)phenyl]propanoate (250 mg, 71 % yield). White solid. MS m/z : 739.4 $[\text{M}+\text{H}]^+$, ESI

Step 5: (3*S*)-3-[[4-[3-[3-(3-Amino-1-hydroxy-propyl)pyrrolidin-1-yl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid 2,2,2-trifluoroacetic acid salt

20

To a solution of ethyl (3*S*)-3-[[4-[3-[3-(3-amino-1-hydroxy-propyl)pyrrolidin-1-yl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate (200 mg, 0.27 mmol) in a mixture of methanol (2 ml) , tetrahydrofuran (2 ml) and water (2 ml) was added LiOH.H₂O (45.51 mg, 1.08 mmol). The
25 mixture was stirred for 2 hours at 25 $^{\circ}\text{C}$. A solution of 1 M HCl was added until pH = 4 was obtained and the solution was extracted three times with ethyl acetate. The combined organic layer was dried with Na₂SO₄, filtered and evaporated. The crude product was purified by reverse phase flash chromatography (Flash Spherical C18, 20-35 μm ; 100 $\text{\AA}$; gradient water with 0.05 % TFA and acetonitrile) to give (3*S*)-3-[[4-[3-[3-(3-amino-1-hydroxy-propyl)pyrrolidin-1-yl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-
30 (trifluoromethyl)phenyl]propanoic acid 2,2,2-trifluoroacetic acid salt (170 mg, 87 % yield) as a white solid. MS m/z : 711.4 $[\text{M}+\text{H}]^+$, ESI

Step 6: (18*S*)-18-[2-Chloro-5-(trifluoromethyl)phenyl]-12-hydroxy-2,2-dioxo-25-oxa-2λ⁶-thia-1,8,15,19-tetrazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-3,5,7(31),21(29),22,24(28)-hexaene-16,20-dione (mixture of 4 diastereomers)

To a solution of (3*S*)-3-[[4-[3-[3-(3-amino-1-hydroxy-propyl)pyrrolidin-1-yl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid 2,2,2-trifluoroacetic acid salt (150 mg, 0.18 mmol) in *N,N*-dimethylformamide (15 ml) were added DIPEA (70.5 mg, 0.55 mmol) and HATU (51.3 mg, 0.22 mmol). The reaction mixture was stirred at 25 °C for 2 hours. The mixture was purified by reverse phase flash chromatography (Flash Spherical C18, 20-35 μm; 100 Å; 20 g, water-acetonitrile) and concentrated to give (18*S*)-18-[2-chloro-5-(trifluoromethyl)phenyl]-12-hydroxy-2,2-dioxo-25-oxa-2λ⁶-thia-1,8,15,19-tetrazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-3,5,7(31),21(29),22,24(28)-hexaene-16,20-dione (60 mg, 40 % yield) as a white solid. MS *m/z*: 693.4 [M+H]⁺, ESI

The four diastereomers were separated using two chiral SFC chromatography.

Column 1: REGIS(S,S)WHELK-O1(250 mm x 25 mm, 10 μm), EtOH with 0.1 % NH₄OH, 60 %

First eluting diastereomer (R_t = 1.607 min, absolute stereochemistry arbitrarily assigned): (11*S*,12*R*,18*S*)-18-[2-chloro-5-(trifluoromethyl)phenyl]-12-hydroxy-2,2-dioxo-25-oxa-2λ⁶-thia-1,8,15,19-tetrazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-3(31),4,6,21(29),22,24(28)-hexaene-16,20-dione (Example 38, Diastereomer A) (14.5 mg, 23 % yield). White solid. MS *m/z*: 693.4 [M+H]⁺, ESI pos.

Diastereomers B, C and D were not separated.

Column 2: DAICEL CHIRALPAK IG (250 mm x 30 mm, 10 μm), EtOH with 0.1% NH₄OH, 45%

First eluting diastereomer (R_t = 1.657 min, absolute stereochemistry arbitrarily assigned): (11*S*,12*S*,18*S*)-18-[2-chloro-5-(trifluoromethyl)phenyl]-12-hydroxy-2,2-dioxo-25-oxa-2λ⁶-thia-1,8,15,19-tetrazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-3(31),4,6,21(29),22,24(28)-hexaene-16,20-dione (Example 39, Diastereomer B) (5.27 mg, 8 % yield). White solid. MS *m/z*: 693.4 [M+H]⁺, ESI pos.

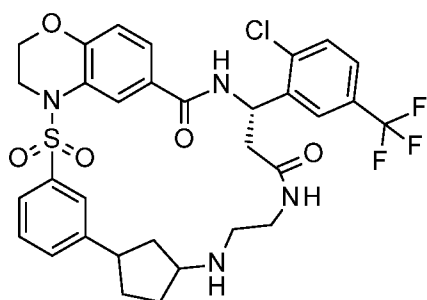
Second eluting diastereomer (R_t = 1.811 min, absolute stereochemistry arbitrarily assigned): (11*R*,12*R*,18*S*)-18-[2-chloro-5-(trifluoromethyl)phenyl]-12-hydroxy-2,2-dioxo-25-oxa-2λ⁶-thia-

1,8,15,19-tetrazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-3(31),4,6,21(29),22,24(28)-hexaene-16,20-dione (Example 40, Diastereomer C) (17.9 mg, 36 % yield). White solid. MS m/z : 693.4 $[M+H]^+$, ESI pos.

Third eluting diastereomer (R_t = 1.929 min, absolute stereochemistry arbitrarily assigned):

5 (11*R*,12*R*,18*S*)-18-[2-chloro-5-(trifluoromethyl)phenyl]-12-hydroxy-2,2-dioxo-25-oxa-2 λ^6 -thia-1,8,15,19-tetrazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-3(31),4,6,21(29),22,24(28)-hexaene-16,20-dione (Example 41, Diastereomer D) (7.91 mg, 13 % yield). White solid. MS m/z : 693.4 $[M+H]^+$, ESI pos.

Example 42: (18*S*)-18-[2-Chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-25-oxa-2 λ^6 -thia-1,12,15,19-tetrazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-3(31),4,6,21(29),22,24(28)-hexaene-16,20-dione, mixture of four diastereomers



Step 1: 4-[3-(4,4,5,5-Tetramethyl-1,3,2-dioxaborolan-2-yl)phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid methyl ester

To a solution of 4-(3-bromophenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid methyl ester (**Intermediate C4**, 100 mg, 243 μ mol) in 1,4-dioxane (3 ml) were added bis(pinacolato)diboron (73.9 mg, 291 μ mol), potassium acetate (71.4 mg, 728 μ mol) and 1,1'-bis(diphenylphosphino)ferrocene-palladium(II)dichloride dichloromethane complex (20.1 mg, 24.3 μ mol) at room temperature under argon. The reaction mixture was stirred at 90 °C for 2 hours. The reaction mixture was cooled down to room temperature, diluted with ethyl acetate, filtered over celite, and concentrated. The crude product was adsorbed on ISOLUTE HM-N and purified by flash chromatography (silica gel, ethyl acetate/heptane, 0-100 %) to give 4-[3-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid methyl ester (110 mg, 89 % yield) as a light brown oil.

¹H NMR (300 MHz, CHLOROFORM-d) δ = 8.51 (d, J = 2.0 Hz, 1H), 8.14 (s, 1H), 7.98 (d, J = 7.3 Hz, 1H), 7.76 (dd, J = 2.0, 8.7 Hz, 1H), 7.68 (td, J = 1.7, 7.9 Hz, 1H), 7.44 (t, J = 7.7 Hz, 1H), 6.83 (d, J = 8.7 Hz, 1H), 3.92 (s, 3H), 3.91 - 3.81 (m, 4H), 1.34 - 1.31 (m, 12H)

Step 2: 4-[3-(3-Ketocyclopentyl)phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid methyl ester

To a solution of 4-[3-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid methyl ester (217 mg, 425 μ mol) in a mixture of 1,4-dioxane (4 ml) and water (1 ml) were added DIPEA (110 μ l, 638 μ mol), cyclopent-2-en-1-one (105 mg, 107 μ l, 1.28 mmol) and chloro(1,5-cyclooctadiene)rhodium (I) dimer (10.5 mg, 21.3 μ mol). The reaction mixture was stirred at 80 °C for 1.5 hour. After cooling to room temperature, the reaction mixture was poured into water and extracted twice with ethyl acetate. The combined organic layer was washed with brine, dried over Na₂SO₄, filtered and concentrated in vacuo. The crude material was absorbed on ISOLUTE HM-N and purified by flash chromatography (silica gel, ethyl acetate/heptane, 0-80 %) to afford 4-[3-(3-ketocyclopentyl)phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid methyl ester (102 mg, 49 % yield) as a colorless oil. MS m/z: 416.2 [M+H]⁺, ESI pos.

Step 3: 4-[3-[3-[2-(tert-Butoxycarbonylamino)ethylamino]cyclopentyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid methyl ester

To a solution of 4-[3-(3-ketocyclopentyl)phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid methyl ester (130 mg, 313 μ mol) in dry dichloromethane (4 ml) was added N-(2-aminoethyl)carbamic acid *tert*-butyl ester (CAS 57260-73-8, 65.2 mg, 64.4 μ l, 407 μ mol). After stirring for 10 min, sodium triacetoxymethylborohydride (106 mg, 500 μ mol) was added and the reaction mixture was stirred at room temperature for 4 hours. The reaction mixture was poured into a saturated solution of NaHCO₃ and extracted twice with ethyl acetate. The combined organic layer was washed with brine, dried over Na₂SO₄, filtered and concentrated in vacuo. The residue was adsorbed on ISOLUTE HM-N and purified by flash chromatography (Si-amine silicagel, ethyl acetate/heptane, 0-100 %) to afford 4-[3-[3-[2-(*tert*-butoxycarbonylamino)ethylamino]cyclopentyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid methyl ester (150 mg, 77 % yield) as a white foam. MS m/z: 560.4 [M+H]⁺, ESI pos.

Step 4: 4-[3-[3-[2-(tert-Butoxycarbonylamino)ethylamino]cyclopentyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid

To a solution of 4-[3-[3-[2-(*tert*-butoxycarbonylamino)ethylamino]cyclopentyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid methyl ester (147 mg, 236 μ mol) in a mixture of tetrahydrofuran (1.5 ml) and methanol (1.5 ml) was added a solution of aqueous 4 M NaOH (142 μ l, 567 μ mol). The reaction mixture was stirred at room temperature overnight. A solution of
5 aqueous 4 M NaOH (59.1 μ l, 236 μ mol) was added and the reaction mixture was further stirred at room temperature for 1 day. The solvents were evaporated in vacuo, the residue was dissolved in water (2 ml) and acetic acid (46 μ l, 804 μ mol) was added. The solid was filtered, washed with water and dried on high vacuo to afford 4-[3-[3-[2-(*tert*-
butoxycarbonylamino)ethylamino]cyclopentyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-
10 carboxylic acid (129 mg, 90 % yield) as an off-white solid. MS *m/z*: 546.4 [M+H]⁺, ESI pos.

Step 5: (3*S*)-3-[[4-[3-[3-[2-(*tert*-Butoxycarbonylamino)ethylamino]cyclopentyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid ethyl ester

To a solution of 4-[3-[3-[2-(*tert*-butoxycarbonylamino)ethylamino]cyclopentyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid (287 mg, 526 μ mol) in N,N-dimethylformamide
15 (7.5 ml) were added (3*S*)-3-amino-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid ethyl ester (**Intermediate B1**, 202 mg, 685 μ mol), DIPEA (276 μ l, 1.58 mmol) and HATU (260 mg, 684 μ mol). The reaction mixture was stirred at room temperature for 1 hour. The reaction mixture was poured into water and extracted twice with ethyl acetate. The combined organic
20 layer was washed with brine, dried over Na₂SO₄, filtered and concentrated in vacuo. The residue was adsorbed on ISOLUTE HM-N and purified by flash chromatography (Si-amine silicagel, ethyl acetate/heptane, 0-100 %) to afford (3*S*)-3-[[4-[3-[3-[2-(*tert*-
butoxycarbonylamino)ethylamino]cyclopentyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid ethyl ester (399 mg,
25 76 %) as a white foam. MS *m/z*: 823.6 [M+H]⁺, ESI pos.

Step 6: (3*S*)-3-[[4-[3-[3-[2-(*tert*-Butoxycarbonylamino)ethylamino]cyclopentyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid

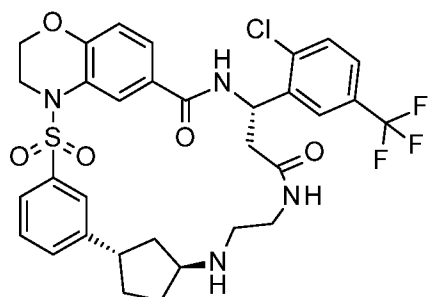
To a solution of (3*S*)-3-[[4-[3-[3-[2-(*tert*-
30 butoxycarbonylamino)ethylamino]cyclopentyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid ethyl ester (397 mg, 482 mmol) in a mixture of tetrahydrofuran (2 ml) and ethanol (1 ml) was added a solution of 1 M

LiOH (1.45 ml, 1.45 mmol). The reaction mixture was stirred at room temperature for 90 minutes. The mixture was poured into an aqueous solution of KHSO₄ and was extracted with ethyl acetate. The organic layer was washed with brine, dried over Na₂SO₄ and concentrated in *vacuo*. The residue was suspended in dichloromethane/heptane (1:1), concentrated, filtered and washed with heptane to afford (3*S*)-3-[[4-[3-[3-[2-(*tert*-butoxycarbonylamino)ethylamino]cyclopentyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid (441 mg, 92 %) as a white solid. MS *m/z*: 795.5 [M+H]⁺, ESI pos.

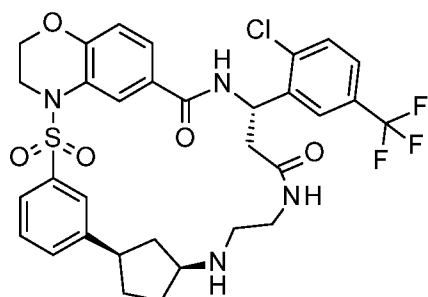
Step 7: (18*S*)-18-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-25-oxa-2λ⁶-thia-1,12,15,19-tetrazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-3(31),4,6,21(29),22,24(28)-hexaene-16,20-dione

To a solution of (3*S*)-3-[[4-[3-[3-[2-(*tert*-butoxycarbonylamino)ethylamino]cyclopentyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propionic acid (437 mg, 550 μmol) in a mixture of dry 1,4-dioxane (5 ml) and dry dichloromethane (2.5 ml) was added at 0 °C a solution of 4 M HCl in 1,4-dioxane (1.24 ml, 4.95 mmol). The reaction mixture was allowed to stir at room temperature for 3 hours and was concentrated in *vacuo*. The crude material was dissolved in *N,N*-dimethylformamide (12 ml) then DIPEA (480 μl, 2.75 mmol) and HATU (251 mg, 659 μmol) were added. The reaction mixture was stirred at room temperature for 45 min. The reaction mixture was poured into water and extracted twice with ethyl acetate. The combined organic layer was washed with brine, dried over Na₂SO₄, filtered and concentrated in *vacuo*. The crude material was purified by reverse phase chromatography (Redisep Gold, HP C18, 30 g, 0 % to 70% acetonitrile in water) to afford (18*S*)-18-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-25-oxa-2λ⁶-thia-1,12,15,19-tetrazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-3(31),4,6,21(29),22,24(28)-hexaene-16,20-dione (185 mg, 47 %) as an off-white solid. MS *m/z*: 677.4 [M+H]⁺, ESI pos.

Example 43 to 46: (8*S*, 11*S*, 18*S*)-18-[2-Chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-25-oxa-2λ⁶-thia-1,12,15,19-tetrazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-3(31),4,6,21(29),22,24(28)-hexaene-16,20-dione (Example 43, Diastereomer A, absolute stereochemistry arbitrarily assigned) and

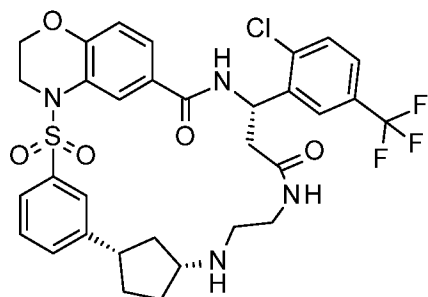


(8*R*, 11*S*, 18*S*)-18-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-25-oxa-2 λ^6 -thia-1,12,15,19-tetrazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-3(31),4,6,21(29),22,24(28)-hexaene-16,20-dione (Example 44, Diastereomer B, absolute stereochemistry arbitrarily assigned) and

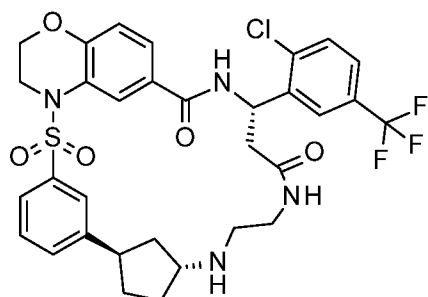


5

(8*S*, 11*R*, 18*S*)-18-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-25-oxa-2 λ^6 -thia-1,12,15,19-tetrazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-3(31),4,6,21(29),22,24(28)-hexaene-16,20-dione (Example 45, Diastereomer C, absolute stereochemistry arbitrarily assigned) and



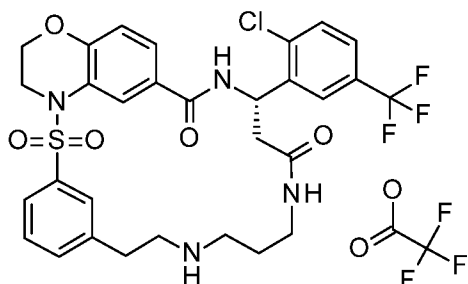
10 (8*R*, 11*R*, 18*S*)-18-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-25-oxa-2 λ^6 -thia-1,12,15,19-tetrazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-3(31),4,6,21(29),22,24(28)-hexaene-16,20-dione (Example 46, Diastereomer D, absolute stereochemistry arbitrarily assigned)



The four diastereomers were separated using chiral SFC chromatography (Column Chiral SZ, 5 μ m, 250 x 20 mm, 35 % MeOH + 0.2 % diethylamine). The absolute stereochemistry was arbitrarily assigned, the CIS/TRANS assignment was confirmed by NMR analysis.

- 5 First eluting diastereomer (TRANS, R_t = 3.804 min): (11*S*,12*R*,18*S*)-18-[2-chloro-5-(trifluoromethyl)phenyl]-12-hydroxy-2,2-dioxo-25-oxa-2 λ^6 -thia-1,8,15,19-tetrazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-3(31),4,6,21(29),22,24(28)-hexaene-16,20-dione (Example 43, Diastereomer A) (24 mg, 12 % yield). Off-white solid. MS m/z : 677.2 $[M+H]^+$, ESI pos.
- 10 Second eluting diastereomer (CIS, R_t = 3.864 min): (8*R*, 11*S*,18*S*)-18-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-25-oxa-2 λ^6 -thia-1,12,15,19-tetrazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-3(31),4,6,21(29),22,24(28)-hexaene-16,20-dione (Example 44, Diastereomer B) (38 mg, 18 % yield). Off-white solid. MS m/z : 677.2 $[M+H]^+$, ESI pos.
- 15 Third eluting diastereomer (CIS, R_t = 4.573 min): (8*S*,11*R*,18*S*)-18-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-25-oxa-2 λ^6 -thia-1,12,15,19-tetrazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-3(31),4,6,21(29),22,24(28)-hexaene-16,20-dione (Example 45, Diastereomer C) (40 mg, 19 % yield). Off-white solid. MS m/z : 677.2 $[M+H]^+$, ESI pos.
- 20 Fourth eluting diastereomer (probably TRANS, R_t = 5.366 min): (8*R*,11*R*,18*S*)-18-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-25-oxa-2 λ^6 -thia-1,12,15,19-tetrazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-3(31),4,6,21(29),22,24(28)-hexaene-16,20-dione (Example 46, Diastereomer D) (21 mg, 10 % yield). Light brown solid. MS m/z : 677.2 $[M+H]^+$, ESI pos.

Example 47: (17*S*)-17-[2-Chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-24-oxa-2 λ ⁶-thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.0^{23,27}]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-15,19-dione 2,2,2-trifluoroacetic acid salt



5

Step 1: Ethyl (3*S*)-3-[2-chloro-5-(trifluoromethyl)phenyl]-3-[[4-[3-[2-[3-(1,3-dioxoisindolin-2-yl)propylamino]ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate

The title compound was prepared in analogy to **Example 16, step 5** starting with ethyl (3*S*)-3-[[4-[3-(2-aminoethyl)phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate (**Intermediate D9**) and 3-(1,3-dioxoisindolin-2-yl)propanal (CAS 2436-29-5) instead of ethyl (3*S*)-3-[[4-[3-[2-aminoethyl(methyl)amino]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate and N-BOC-2-aminoacetaldehyde. Yellow solid. MS *m/z*: 827.1 [M+H]⁺, ESI pos.

Step 2: Ethyl (3*S*)-3-[[4-[3-[2-[*tert*-butoxycarbonyl]-3-(1,3-dioxoisindolin-2-yl)propyl]amino]ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate

To a solution of ethyl (3*S*)-3-[2-chloro-5-(trifluoromethyl)phenyl]-3-[[4-[3-[2-[3-(1,3-dioxoisindolin-2-yl)propylamino]ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate (300 mg, 0.32 mmol) in dichloromethane (6 ml) were added DIPEA (0.16 ml, 0.96 mmol) and di-*t*-butyldicarbonate (209 mg, 0.96 mmol). The reaction mixture was stirred at 20 °C for 2 hours. The mixture was poured into water and extracted three times with ethyl acetate. The combined organic layer was dried with Na₂SO₄, filtered and evaporated to afford (ethyl (3*S*)-3-[[4-[3-[2-[*tert*-butoxycarbonyl]-3-(1,3-dioxoisindolin-2-yl)propyl]amino]ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-

chloro-5-(trifluoromethyl)phenyl]propanoate (250 mg, 72 % yield) as a white solid. MS m/z: 827.1 [M-100+H]⁺, ESI pos.

Step 3: Ethyl (3*S*)-3-[[4-[3-[2-[3-aminopropyl(*tert*-butoxycarbonyl)amino]ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate

To a solution of ethyl (3*S*)-3-[[4-[3-[2-[*tert*-butoxycarbonyl]-3-(1,3-dioxoisindolin-2-yl)propyl]amino]ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate (230 mg, 0.25 mmol) in ethanol (3 ml) was added N₂H₄·H₂O (0.02 ml, 0.37 mmol). The reaction mixture was stirred at 20 °C for 16 hours. The reaction mixture was evaporated to give crude ethyl (3*S*)-3-[[4-[3-[2-[3-aminopropyl(*tert*-butoxycarbonyl)amino]ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate (195 mg, 99 % yield). Colorless liquid. MS m/z: 797.2 [M+H]⁺, ESI pos.

Step 4: (3*S*)-3-[[4-[3-[2-[3-Aminopropyl(*tert*-butoxycarbonyl)amino]ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid

To a solution of ethyl (3*S*)-3-[[4-[3-[2-[3-aminopropyl(*tert*-butoxycarbonyl)amino]ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate (195 mg, 0.1 mmol) in tetrahydrofuran (3 ml) and water (3 ml) was added LiOH · H₂O (12.6 mg, 0.3 mmol). The reaction mixture was stirred at 20 °C for 2 hours. A solution of 1 M HCl was added until pH = 4 was obtained and the solution was extracted three times with ethyl acetate. The combined organic layer was dried with Na₂SO₄, filtered and evaporated. The crude product was purified by reverse phase flash chromatography (Flash Spherical C18, 20-35 μm; 100 Å; gradient water with 0.05 % TFA and acetonitrile) to afford ((3*S*)-3-[[4-[3-[2-[3-aminopropyl(*tert*-butoxycarbonyl)amino]ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid (130 mg, 67 % yield) as a white solid. MS m/z: 769.1 [M+H]⁺, ESI pos.

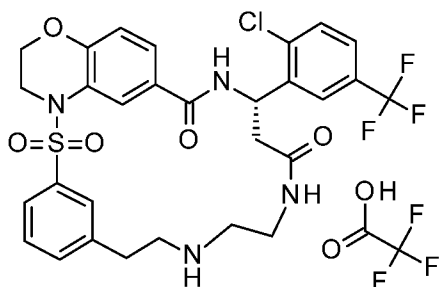
Step 5: *tert*-Butyl -(17*S*)-17-[2-chloro-5-(trifluoromethyl)phenyl]-2,2,15,19-tetraoxo-24-oxa-2λ⁶-thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.0.23,27]nonacos-3,5,7(29),20(28),21,23(27)-hexaene-10-carboxylate

The title compound was prepared in analogy to **Example 5, step 10** starting with ((3*S*)-3-[[4-[3-[2-[3-aminopropyl(*tert*-butoxycarbonyl)amino]ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid instead of (3*S*)-3-[[4-[3-[(2*R*)-5-amino-2-(*tert*-butoxycarbonylamino)pentoxy]phenyl]sulfonylspiro[3H-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid. White solid. MS *m/z*: 651.1 [M-100+H]⁺, ESI pos.

Step 6: (17*S*)-17-[2-Chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-24-oxa-2λ⁶-thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.0^{23,27}]nonacosa-3,5,7(29),20(28),21,23(27)-hexaene-15,19-dione 2,2,2-trifluoroacetic acid salt

The title compound was prepared in analogy to **Example 5, step 11** starting from *tert*-butyl - (17*S*)-17-[2-chloro-5-(trifluoromethyl)phenyl]-2,2,15,19-tetraoxo-24-oxa-2λ⁶-thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.0^{23,27}]nonacosa-3,5,7(29),20(28),21,23(27)-hexaene-10-carboxylate instead of *tert*-butyl *N*-[(10*R*, 17*S*)-17-[2-chloro-5-(trifluoromethyl)phenyl]-2,2,15,19-tetraoxo-spiro[8,24-dioxa-2λ⁶-thia-1,14,18-triazatetracyclo[18.6.2.13,7.0^{23,27}]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-25,1'-cyclopropane]-10-yl]carbamate. White solid. MS *m/z*: 651.1 [M+H]⁺, ESI pos.

Example 48: (16*S*)-16-[2-Chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-23-oxa-2λ⁶-thia-1,10,13,17-tetrazatetracyclo[17.6.2.13,7.0^{22,26}]octacosa-3(28),4,6,19(27),20,22(26)-hexaene-14,18-dione 2,2,2-trifluoroacetic acid salt



Step 1: Ethyl (3*S*)-3-[2-chloro-5-(trifluoromethyl)phenyl]-3-[[4-[3-[2-[2-(1,3-dioxoisindolin-2-yl)ethylamino]ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate

The title compound was prepared in analogy to **Example 16, step 5** starting with ethyl (3*S*)-3-[[4-[3-(2-aminoethyl)phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate (**Intermediate D9**) and *N*-(2-oxoethyl)phthalimide

instead of ethyl (3*S*)-3-[[4-[3-[2-aminoethyl(methyl)amino]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate and *N*-BOC-2-aminoacetaldehyde. Yellow solid. MS *m/z*: 813.1 [M+H]⁺, ESI pos.

Step 2: Ethyl (3*S*)-3-[[4-[3-[2-[*tert*-butoxycarbonyl]-2-(1,3-dioxoisindolin-2-

5 yl)ethyl]amino]ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate

The title compound was prepared in analogy to **Example 47, step 2** starting with ethyl (3*S*)-3-[2-chloro-5-(trifluoromethyl)phenyl]-3-[[4-[3-[2-[2-(1,3-dioxoisindolin-2-yl)ethylamino]ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate
10 instead of ethyl (3*S*)-3-[2-chloro-5-(trifluoromethyl)phenyl]-3-[[4-[3-[2-[3-(1,3-dioxoisindolin-2-yl)propylamino]ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate. Yellow solid. MS *m/z*: 813.1 [M-100+H]⁺, ESI pos.

Step 3: Ethyl (3*S*)-3-[[4-[3-[2-[2-aminoethyl(*tert*-butoxycarbonyl)amino]ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-

15 (trifluoromethyl)phenyl]propanoate

The title compound was prepared in analogy to **Example 47, step 3** starting with ethyl (3*S*)-3-[[4-[3-[2-[*tert*-butoxycarbonyl]-2-(1,3-dioxoisindolin-2-yl)ethyl]amino]ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate instead of ethyl (3*S*)-3-[[4-[3-[2-[*tert*-butoxycarbonyl]-3-(1,3-dioxoisindolin-2-yl)propyl]amino]ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate. Colorless liquid. MS *m/z*:
20 783.2 [M+H]⁺, ESI pos.

Step 4: (3*S*)-3-[[4-[3-[2-[2-Aminoethyl(*tert*-butoxycarbonyl)amino]ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic

25 acid

The title compound was prepared in analogy to **Example 47, step 4** starting with ethyl (3*S*)-3-[[4-[3-[2-[2-aminoethyl(*tert*-butoxycarbonyl)amino]ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate instead of ethyl (3*S*)-3-[[4-[3-[2-[3-aminopropyl(*tert*-butoxycarbonyl)amino]ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate.
30 White solid. MS *m/z*: 755.2 [M+H]⁺, ESI pos.

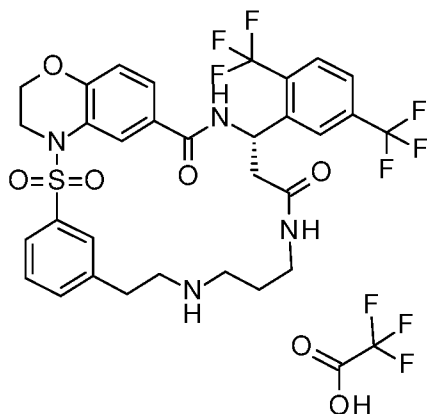
Step 5: *tert*-Butyl (16*S*)-16-[2-chloro-5-(trifluoromethyl)phenyl]-2,2,14,18-tetraoxo-23-oxa-2 λ^6 -thia-1,10,13,17-tetrazatetracyclo[17.6.2.13,7.022,26]octacos-3(28),4,6,19(27),20,22(26)-hexaene-10-carboxylate

The title compound was prepared in analogy to **Example 5, step 10** starting with (3*S*)-3-[[4-[3-[2-[2-aminoethyl(*tert*-butoxycarbonyl)amino]ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid instead of rac-(3*S*)-3-[[4-[3-[(2*R*)-5-amino-2-(*tert*-butoxycarbonylamino)pentoxy]phenyl]sulfonyl]spiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid. Yellow solid. MS *m/z*: 637.1 [M-100+H]⁺, ESI pos.

Step 6: (16*S*)-16-[2-Chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-23-oxa-2 λ^6 -thia-1,10,13,17-tetrazatetracyclo[17.6.2.13,7.022,26]octacos-3(28),4,6,19(27),20,22(26)-hexaene-14,18-dione 2,2,2-trifluoroacetic acid salt

The title compound was prepared in analogy to **Example 5, step 11** starting with *tert*-butyl (16*S*)-16-[2-chloro-5-(trifluoromethyl)phenyl]-2,2,14,18-tetraoxo-23-oxa-2 λ^6 -thia-1,10,13,17-tetrazatetracyclo[17.6.2.13,7.022,26]octacos-3(28),4,6,19(27),20,22(26)-hexaene-10-carboxylate instead of *tert*-butyl *N*-[(10*R*, 17*S*)-17-[2-chloro-5-(trifluoromethyl)phenyl]-2,2,15,19-tetraoxo-spiro[8,24-dioxo-2 λ^6 -thia-1,14,18-triazatetracyclo[18.6.2.13,7.023,27]nonacos-3(29),4,6,20(28),21,23(27)-hexaene-25,1'-cyclopropane]-10-yl]carbamate. White solid. MS *m/z*: 637.1 [M+H]⁺, ESI pos.

Example 49: (17*S*)-17-[2,5-Bis(trifluoromethyl)phenyl]-2,2-dioxo-24-oxa-2 λ^6 -thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.023,27]nonacos-3(29),4,6,20(28),21,23(27)-hexaene-15,19-dione 2,2,2-trifluoroacetic acid salt



Step 1: Ethyl (3*S*)-3-[2,5-bis(trifluoromethyl)phenyl]-3-[[4-[3-[2-[3-(1,3-dioxoisindolin-2-yl)propylamino]ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate 2,2,2-trifluoroacetic acid salt

The title compound was prepared in analogy to **Example 11, step 3** starting with ethyl (3*S*)-3-[2,5-bis(trifluoromethyl)phenyl]-3-[[4-[3-[2-(*tert*-butoxycarbonylamino)ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate (**Intermediate D11**) and 3-(1,3-dioxoisindolin-2-yl)propanal (CAS 2436-29-5) instead of methyl 4-[3-[(3*S*)-3-aminopyrrolidin-1-yl]-5-methyl-phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate and N-BOC-2-aminoacetaldehyde. Yellow solid. MS *m/z*: 861.5 [M+H]⁺, ESI pos.

Step 2: Ethyl (3*S*)-3-[2,5-bis(trifluoromethyl)phenyl]-3-[[4-[3-[2-[*tert*-butoxycarbonyl]-3-(1,3-dioxoisindolin-2-yl)propyl]amino]ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate

The title compound was prepared in analogy to **Example 47, step 2** starting with ethyl (3*S*)-3-[2,5-bis(trifluoromethyl)phenyl]-3-[[4-[3-[2-[3-(1,3-dioxoisindolin-2-yl)propylamino]ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate 2,2,2-trifluoroacetic acid salt instead of ethyl (3*S*)-3-[2-chloro-5-(trifluoromethyl)phenyl]-3-[[4-[3-[2-[3-(1,3-dioxoisindolin-2-yl)propylamino]ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate. Colorless oil. MS *m/z*: 961.5 [M+H]⁺, ESI pos.

Step 3: 2-[3-[2-[3-[[6-[(1*S*)-1-[2,5-Bis(trifluoromethyl)phenyl]-2-carboxy-ethyl]carbamoyl]-2,3-dihydro-1,4-benzoxazin-4-yl]sulfonyl]phenyl]ethyl-*tert*-butoxycarbonyl]-amino]propylcarbamoyl]benzoic acid

To a solution of ethyl (3*S*)-3-[2,5-bis(trifluoromethyl)phenyl]-3-[[4-[3-[2-[*tert*-butoxycarbonyl]-3-(1,3-dioxoisindolin-2-yl)propyl]amino]ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate (230 mg, 0.24 mmol) in a mixture of tetrahydrofuran (3 ml), methanol (3 ml) and water (3 ml) was added lithium hydroxide monohydrate (30.1 mg, 0.72 mmol). The reaction mixture was stirred at 25 °C for 2 hours. A solution of 1 M HCl was added until pH = 4 was obtained and the solution was extracted three times with ethyl acetate. The combined organic layer was dried with Na₂SO₄, filtered and evaporated to give 2-[3-[2-[3-[[6-[(1*S*)-1-[2,5-bis(trifluoromethyl)phenyl]-2-carboxy-ethyl]carbamoyl]-2,3-dihydro-1,4-benzoxazin-4-yl]sulfonyl]phenyl]ethyl-*tert*-butoxycarbonyl]-amino]propylcarbamoyl]benzoic acid (220 mg, 95 % yield) as a white solid. MS *m/z*: 951.5 [M+H]⁺, ESI pos.

Step 4: (3*S*)-3-[[4-[3-[2-[3-Aminopropyl(*tert*-butoxycarbonyl)amino]ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2,5-bis(trifluoromethyl)phenyl]propanoic acid 2,2,2-trifluoroacetic acid salt

The title compound was prepared in analogy to **Example 5, step 9** starting with 2-[3-[2-[3-[[6-
5 [[(1*S*)-1-[2,5-bis(trifluoromethyl)phenyl]-2-carboxy-ethyl]carbamoyl]-2,3-dihydro-1,4-benzoxazin-4-yl]sulfonyl]phenyl]ethyl-*tert*-butoxycarbonyl-amino]propylcarbamoyl]benzoic acid instead of 2-[[4(*R*)-4-(*tert*-butoxycarbonylamino)-5-[3-[6-[[1(*S*)-2-carboxy-1-[2-chloro-5-(trifluoromethyl)phenyl]ethyl]carbamoyl]spiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-4-yl]sulfonylphenoxy]pentyl]carbamoyl]benzoic acid. White solid. MS *m/z*: 803.5 [M+H]⁺, ESI
10 pos.

Step 5: *tert*-Butyl (17*S*)-17-[2,5-bis(trifluoromethyl)phenyl]-2,2,15,19-tetraoxo-24-oxa-2λ⁶-thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.0^{23,27}]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-10-carboxylate

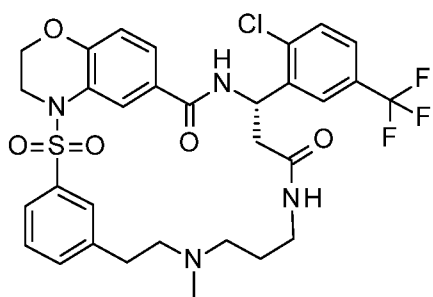
To a solution of (3*S*)-3-[[4-[3-[2-[3-aminopropyl(*tert*-
15 butoxycarbonyl)amino]ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2,5-bis(trifluoromethyl)phenyl]propanoic acid 2,2,2-trifluoroacetic acid salt (130 mg, 0.14 mmol) in *N,N*-dimethylformamide (12 ml) were added DIPEA (73.3 mg, 0.57 mmol) and HATU (40 mg, 0.17 mmol). The reaction mixture was stirred at 25 °C for 2 hours. The mixture was directly purified by reverse phase flash chromatography (Flash Spherical C18, 20-35 μm; 100 Å;
20 gradient water with 0.1 % TFA and acetonitrile) to give *tert*-butyl (17*S*)-17-[2,5-bis(trifluoromethyl)phenyl]-2,2,15,19-tetraoxo-24-oxa-2λ⁶-thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.0^{23,27}]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-10-carboxylate 2,2,2-trifluoroacetic acid salt (100 mg, 70 % yield) as a white solid. MS *m/z*: 785.5 [M+H]⁺, ESI pos.

25 Step 6: (17*S*)-17-[2,5-Bis(trifluoromethyl)phenyl]-2,2-dioxo-24-oxa-2λ⁶-thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.0^{23,27}]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-15,19-dione;2,2,2-trifluoroacetic acid

To a solution of *tert*-butyl (17*S*)-17-[2,5-bis(trifluoromethyl)phenyl]-2,2,15,19-tetraoxo-24-oxa-2λ⁶-thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.0^{23,27}]nonacosa-3(29),4,6,20(28),21,23(27)-
30 hexaene-10-carboxylate 2,2,2-trifluoroacetic acid salt (80 mg, 0.09 mmol) in dichloromethane (4 ml) was added TFA (0.8 ml, 0.09 mmol). The reaction mixture was stirred at 25 °C for 1 hour. The mixture was concentrated and the residue was dissolved in a mixture of water (15 ml) and

acetonitrile (10 ml) and was lyophilized to give (17*S*)-17-[2,5-bis(trifluoromethyl)phenyl]-2,2-dioxo-24-oxa-2 λ ⁶-thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.0^{23,27}]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-15,19-dione 2,2,2-trifluoroacetic acid salt (79 mg, > 100 % yield) as a white solid. MS *m/z*: 685.4 [M+H]⁺, ESI pos.

- 5 **Example 50:** (17*S*)-17-[2-Chloro-5-(trifluoromethyl)phenyl]-10-methyl-2,2-dioxo-24-oxa-2 λ ⁶-thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.0^{23,27}]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-15,19-dione



- 10 Step 1: Ethyl (3*S*)-3-[2-chloro-5-(trifluoromethyl)phenyl]-3-[[4-[3-[2-[3-(1,3-dioxoisindolin-2-yl)propyl-methyl-amino]ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate 2,2,2-trifluoroacetic acid salt

- To a solution of ethyl (3*S*)-3-[2-chloro-5-(trifluoromethyl)phenyl]-3-[[4-[3-[2-[3-(1,3-dioxoisindolin-2-yl)propylamino]ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate (200 mg, 0.24 mmol) in methanol (4 ml) were added NaBH₃CN (45.6 mg, 0.73 mmol) and ZnCl₂ (165 mg, 1.21 mmol) and formaldehyde (98.1 mg, 1.21 mmol) .
15 The reaction mixture was stirred at 25 °C for 1 hour. The mixture was poured into water and extracted three times with ethyl acetate. The combined organic layer was dried with Na₂SO₄, filtered and evaporated. The crude product was purified by reverse phase flash chromatography (Flash Spherical C18, 20-35 μm; 100 Å; gradient water with 0.05 % TFA and acetonitrile)
20 to afford (ethyl (3*S*)-3-[2-chloro-5-(trifluoromethyl)phenyl]-3-[[4-[3-[2-[3-(1,3-dioxoisindolin-2-yl)propyl-methyl-amino]ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate 2,2,2-trifluoroacetic acid salt (180 mg, 87 % yield) as a yellow solid. MS *m/z*: 841.1 [M+H]⁺, ESI pos.

- 25 Step 2: Ethyl (3*S*)-3-[[4-[3-[2-[3-aminopropyl(methyl)amino]ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate

To a solution of ethyl (3*S*)-3-[[4-[3-[2-[3-(1,3-dioxoisindolin-2-yl)propyl-methyl-amino]ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate (180 mg, 0.21 mmol) in ethanol (1.5 ml) was added N₂H₄ · H₂O (0.1 ml, 2.14 mmol) and stirred at 25 °C for 3 hours. The solvent was evaporated to give ethyl (3*S*)-3-[[4-[3-[2-[3-aminopropyl(methyl)amino]ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate (150 mg, 40 % yield) as a colorless liquid. MS m/z: 711.2 [M+H]⁺, ESI pos.

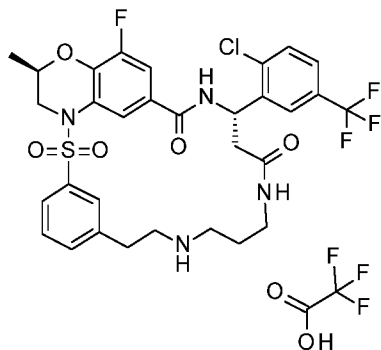
Step 3: (3*S*)-3-[[4-[3-[2-[3-Aminopropyl(methyl)amino]ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid

To a solution of ethyl (3*S*)-3-[[4-[3-[2-[3-aminopropyl(methyl)amino]ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate (150.0 mg, 0.21 mmol) in a mixture of tetrahydrofuran (0.5 ml) and water (0.5 ml) was added LiOH · H₂O (26.6 mg, 0.63 mmol). The reaction mixture was stirred at 25 °C for 16 hours. The solution was purified by prep-HPLC (Phenomenex Luna C18 150 x 25mm x 10 µm, gradient water with 1% TFA and acetonitrile) to give (3*S*)-3-[[4-[3-[2-[3-aminopropyl(methyl)amino]ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid (60 mg, 42 % yield) as a yellow solid. MS m/z: 683.2 [M+H]⁺, ESI pos.

Step 4: (17*S*)-17-[2-Chloro-5-(trifluoromethyl)phenyl]-10-methyl-2,2-dioxo-24-oxa-2λ⁶-thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.0^{23,27}]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-15,19-dione

To a solution of (3*S*)-3-[[4-[3-[2-[3-aminopropyl(methyl)amino]ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid (55 mg, 0.08 mmol) in acetonitrile (10 ml) was added *N*-methyl imidazole (19.8 mg, 0.24 mmol) and *N,N,N',N'*-tetramethylchloroformamidinium hexafluorophosphate (27.1 mg, 0.1 mmol). The reaction mixture was stirred at 25 °C for 2 hours. The mixture was concentrated and the residue was purified by prep-HPLC (Waters Xbridge 150 x 25 mm x 5 µm, water with 0.5 % ammonium hydroxide and acetonitrile) to give (17*S*)-17-[2-chloro-5-(trifluoromethyl)phenyl]-10-methyl-2,2-dioxo-24-oxa-2λ⁶-thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.0^{23,27}]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-15,19-dione (5.45 mg, 10 % yield) as a white solid. MS m/z: 665.2 [M+H]⁺, ESI pos.

Example 51: (17S,25R)-17-[2-chloro-5-(trifluoromethyl)phenyl]-22-fluoro-25-methyl-2,2-dioxo-24-oxa-2λ⁶-thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.0^{23,27}]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-15,19-dione 2,2,2-trifluoroacetic acid salt



5 Step 1: Ethyl (3S)-3-[2-chloro-5-(trifluoromethyl)phenyl]-3-[[2R)-4-[3-[2-[3-(1,3-dioxoisindolin-2-yl)propylamino]ethyl]phenyl]sulfonyl-8-fluoro-2-methyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate

The title compound was prepared in analogy to **Example 11, step 3** starting with ethyl (3S)-3-[[2R)-4-[3-(2-aminoethyl)phenyl]sulfonyl-8-fluoro-2-methyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate (**Intermediate D12**) and 3-(1,3-dioxoisindolin-2-yl)propanal (CAS 2436-29-5) instead of methyl 4-[3-[(3S)-3-aminopyrrolidin-1-yl]-5-methyl-phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate and N-BOC-2-aminoacetaldehyde. Yellow solid. MS m/z: 859.3 [M+H]⁺, ESI pos.

15 Step 2: Ethyl (3S)-3-[[2R)-4-[3-[2-[tert-butoxycarbonyl]-3-(1,3-dioxoisindolin-2-yl)propyl]amino]ethyl]phenyl]sulfonyl-8-fluoro-2-methyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate

To a solution of ethyl (3S)-3-[2-chloro-5-(trifluoromethyl)phenyl]-3-[[2R)-4-[3-[2-[3-(1,3-dioxoisindolin-2-yl)propylamino]ethyl]phenyl]sulfonyl-8-fluoro-2-methyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate (120 mg, 0.1 mmol) in dichloromethane (2 ml) was added DIPEA (52.7 mg, 0.41 mmol), then di-t-butylidicarbonate (33.4 mg, 0.15 mmol) was added to the mixture. The reaction mixture was stirred at 25°C for 2 hours. The mixture was poured into water and extracted three times with dichloromethane. The combined organic layer was dried with Na₂SO₄, filtered and evaporated. The crude product was purified via flash chromatography (silica gel, ethyl acetate/petroleum ether 10-40 %) to give ethyl (3S)-3-[[2R)-4-[3-[2-[tert-butoxycarbonyl]-3-(1,3-dioxoisindolin-2-yl)propyl]amino]ethyl]phenyl]sulfonyl-8-fluoro-2-methyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-

(trifluoromethyl)phenyl]propanoate (95 mg, 95 % yield) as a colorless oil. MS m/z: 859.2 [M-BOC+H]⁺, ESI pos.

Step 3: 2-[3-[2-[3-[[6-[(1S)-1-[2,5-Bis(trifluoromethyl)phenyl]-2-carboxy-ethyl]carbamoyl]-2,3-dihydro-1,4-benzoxazin-4-yl]sulfonyl]phenyl]ethyl-*tert*-butoxycarbonyl]-amino]propylcarbamoyl]benzoic acid

The title compound was prepared in analogy to **Example 49, step 3** starting with ethyl (3*S*)-3-[[*(2R)*]-4-[3-[2-[*tert*-butoxycarbonyl]-[3-(1,3-dioxoisindolin-2-yl)propyl]amino]ethyl]phenyl]sulfonyl-8-fluoro-2-methyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate instead of ethyl (3*S*)-3-[2,5-bis(trifluoromethyl)phenyl]-3-[[4-[3-[2-[*tert*-butoxycarbonyl]-[3-(1,3-dioxoisindolin-2-yl)propyl]amino]ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate. Yellow oil. MS m/z: 849.2 [M-BOC+H]⁺, ESI pos.

Step 4: (3*S*)-3-[2-chloro-5-(trifluoromethyl)phenyl]-3-[[*(2R)*]-4-[3-[2-[3-aminopropyl(*tert*-butoxycarbonyl)amino]ethyl]phenyl]sulfonyl-8-fluoro-2-methyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoic acid

The title compound was prepared in analogy to **Example 5, step 9** starting with 2-[3-[2-[3-[[6-[(1*S*)-1-[2,5-bis(trifluoromethyl)phenyl]-2-carboxy-ethyl]carbamoyl]-2,3-dihydro-1,4-benzoxazin-4-yl]sulfonyl]phenyl]ethyl-*tert*-butoxycarbonyl-amino]propylcarbamoyl]benzoic acid instead of 2-[[*(4R)*]-4-(*tert*-butoxycarbonylamino)-5-[3-[6-[[*(1S)*]-2-carboxy-1-[2-chloro-5-(trifluoromethyl)phenyl]ethyl]carbamoyl]spiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-4-yl]sulfonylphenoxy]pentyl]carbamoyl]benzoic acid and stirring at 80 °C for 8 hours instead of 16 hours. Yellow solid. MS m/z: 801.3 [M+H]⁺, ESI pos.

Step 5: *tert*-Butyl (17*S*,25*R*)-17-[2-chloro-5-(trifluoromethyl)phenyl]-22-fluoro-25-methyl-2,2,15,19-tetraoxo-24-oxa-2λ⁶-thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.0^{23,27}]nonacos-3,5,7(29),20(28),21,23(27)-hexaene-10-carboxylate

To a solution of (3*S*)-3-[2-chloro-5-(trifluoromethyl)phenyl]-3-[[*(2R)*]-4-[3-[2-[3-aminopropyl(*tert*-butoxycarbonyl)amino]ethyl]phenyl]sulfonyl-8-fluoro-2-methyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoic acid (55.0 mg, 0.06 mmol) in acetonitrile (6 ml) was added *N*-methyl imidazole (CAS 616-47-7, 24.7 mg, 0.3 mmol). After stirring at 20 °C for 5 min, *N,N,N',N'*-tetramethylchloroformamidinium hexafluorophosphate (CAS 207915-99-9, 20.2 mg, 0.07 mmol) was added. The reaction mixture was stirred at 20 °C for 2 hours. The

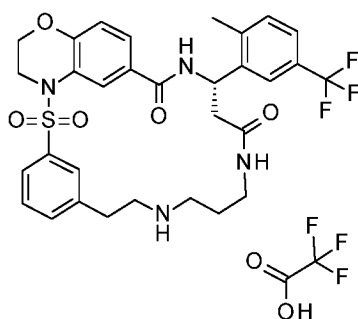
solvent was evaporated and the residue was purified by reverse phase flash chromatography (Flash Spherical C18, 20-35 μm ; 100 $\text{\AA}$; gradient water with 1% TFA and acetonitrile) to give *tert*-butyl (1*S*,25*R*)-17-[2-chloro-5-(trifluoromethyl)phenyl]-22-fluoro-25-methyl-2,2,15,19-tetraoxo-24-oxa-2 λ^6 -thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.0^{23,27}]nonacosa-

- 5 3,5,7(29),20(28),21,23(27)-hexaene-10-carboxylate (35.0 mg, 71 % yield) as an off-white solid. MS *m/z*: 682.2 [M-BOC+H]⁺, ESI pos.

Step 6: (1*S*,25*R*)-17-[2-chloro-5-(trifluoromethyl)phenyl]-22-fluoro-25-methyl-2,2-dioxo-24-oxa-2 λ^6 -thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.0^{23,27}]nonacosa-3,5,7(29),20(28),21,23(27)-hexaene-15,19-dione 2,2,2-trifluoroacetic acid salt

- 10 To a solution of *tert*-butyl (1*S*,25*R*)-17-[2-chloro-5-(trifluoromethyl)phenyl]-22-fluoro-25-methyl-2,2,15,19-tetraoxo-24-oxa-2 λ^6 -thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.0^{23,27}]nonacosa-3,5,7(29),20(28),21,23(27)-hexaene-10-carboxylate (30 mg, 0.04 mmol) in dichloromethane (2.5 ml) was added 2,2,2 trifluoroacetic acid (0.5 ml, 0.04 mmol). The reaction mixture was stirred at 25 °C for 1 hour. The mixture was
- 15 concentrated and the residue was dissolved in a mixture of water (15 ml) and acetonitrile (10 ml) and was lyophilized to give (1*S*,25*R*)-17-[2-chloro-5-(trifluoromethyl)phenyl]-22-fluoro-25-methyl-2,2-dioxo-24-oxa-2 λ^6 -thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.0^{23,27}]nonacosa-3,5,7(29),20(28),21,23(27)-hexaene-15,19-dione 2,2,2-trifluoroacetic acid salt (27.6 mg, 90 % yield) as a yellow solid. MS *m/z*: 683.2 [M+H]⁺, ESI pos.

- 20 **Example 52:** (1*S*)-17-[2-methyl-5-(trifluoromethyl)phenyl]-2,2-dioxo-24-oxa-2 λ^6 -thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.0^{23,27}]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-15,19-dione 2,2,2-trifluoroacetic acid salt



- Step 1: Ethyl (3*S*)-3-[[4-[3-[2-[3-(1,3-dioxoisindolin-2-yl)propylamino]ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-methyl-5-(trifluoromethyl)phenyl]propanoate
- 25

The title compound was prepared in analogy to **Example 28 and Example 29, step 3** starting from ethyl (3*S*)-3-[[4-[3-(2-aminoethyl)phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-methyl-5-(trifluoromethyl)phenyl]propanoate 2,2,2-trifluoroacetic acid salt (**Intermediate D14**) instead of ethyl (3*S*)-3-[[4-[3-(2-amino-1-methyl-ethoxy)phenyl]sulfonyl]spiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate. White solid. MS *m/z*: 807.4 [M+H]⁺, ESI pos.

Step 2: Ethyl (3*S*)-3-[[4-[3-[2-[*tert*-butoxycarbonyl]-3-(1,3-dioxoisindolin-2-yl)propyl]amino]ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-methyl-5-(trifluoromethyl)phenyl]propanoate

The title compound was prepared in analogy to **Example 51, step 2** starting with ethyl (3*S*)-3-[[4-[3-[2-[3-(1,3-dioxoisindolin-2-yl)propylamino]ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-methyl-5-(trifluoromethyl)phenyl]propanoate instead of ethyl (3*S*)-3-[2-chloro-5-(trifluoromethyl)phenyl]-3-[[2*R*]-4-[3-[2-[3-(1,3-dioxoisindolin-2-yl)propylamino]ethyl]phenyl]sulfonyl-8-fluoro-2-methyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate. Colorless oil. MS *m/z*: 807.4 [M-BOC+H]⁺, ESI pos.

Step 3: 2-[3-[*tert*-butoxycarbonyl]-2-[3-[[6-[(1*S*)-2-carboxy-1-[2-methyl-5-(trifluoromethyl)phenyl]ethyl]carbamoyl]-2,3-dihydro-1,4-benzoxazin-4-yl]sulfonyl]phenyl]ethyl]amino]propylcarbamoyl]benzoic acid

The title compound was prepared in analogy to **Example 49, step 3** starting with ethyl (3*S*)-3-[[4-[3-[2-[*tert*-butoxycarbonyl]-3-(1,3-dioxoisindolin-2-yl)propyl]amino]ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-methyl-5-(trifluoromethyl)phenyl]propanoate instead of ethyl (3*S*)-3-[2,5-bis(trifluoromethyl)phenyl]-3-[[4-[3-[2-[*tert*-butoxycarbonyl]-3-(1,3-dioxoisindolin-2-yl)propyl]amino]ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]. White solid. MS *m/z*: 797.3 [M-BOC+H]⁺, ESI pos.

Step 4: (3*S*)-3-[[4-[3-[2-[3-aminopropyl(*tert*-butoxycarbonyl)amino]ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-methyl-5-(trifluoromethyl)phenyl]propanoic acid 2,2,2-trifluoroacetic acid salt

The title compound was prepared in analogy to **Example 5, step 9** starting with 2-[3-[*tert*-butoxycarbonyl]-2-[3-[[6-[(1*S*)-2-carboxy-1-[2-methyl-5-(trifluoromethyl)phenyl]ethyl]carbamoyl]-2,3-dihydro-1,4-benzoxazin-4-

yl)sulfonyl]phenyl]ethyl]amino]propylcarbamoyl]benzoic acid instead of 2-[[[(4*R*)-4-(*tert*-butoxycarbonylamino)-5-[3-[6-[[[(1*S*)-2-carboxy-1-[2-chloro-5-(trifluoromethyl)phenyl]ethyl]carbamoyl]spiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-4-yl)sulfonyl]phenoxy]pentyl]carbamoyl]benzoic acid and stirring at 80°C for 8 hours instead of 16 hours. White solid. MS *m/z*: 749.3 [M+H]⁺, ESI pos.

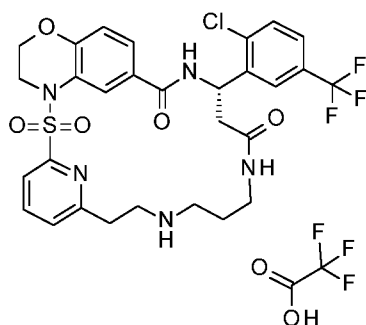
Step 5: *tert*-Butyl (17*S*)-17-[2-methyl-5-(trifluoromethyl)phenyl]-2,2,15,19-tetraoxo-24-oxa-2λ⁶-thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.0^{23,27}]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-10-carboxylate

The title compound was prepared in analogy to **Example 49, step 5** starting (3*S*)-3-[[4-[3-[2-[3-aminopropyl(*tert*-butoxycarbonyl)amino]ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-methyl-5-(trifluoromethyl)phenyl]propanoic acid 2,2,2-trifluoroacetic acid salt instead of *tert*-butyl (17*S*)-17-[2,5-bis(trifluoromethyl)phenyl]-2,2,15,19-tetraoxo-24-oxa-2λ⁶-thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.0^{23,27}]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-10-carboxylate. Yellow solid. MS *m/z*: 631.3 [M-BOC+H]⁺, ESI pos.

Step 6: (17*S*)-17-[2-Methyl-5-(trifluoromethyl)phenyl]-2,2-dioxo-24-oxa-2λ⁶-thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.0^{23,27}]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-15,19-dione 2,2,2-trifluoroacetic acid salt

The title compound was prepared in analogy to **Example 51, step 6** starting with *tert*-butyl (17*S*)-17-[2-methyl-5-(trifluoromethyl)phenyl]-2,2,15,19-tetraoxo-24-oxa-2λ⁶-thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.0^{23,27}]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-10-carboxylate instead of *tert*-butyl (17*S*,25*R*)-17-[2-chloro-5-(trifluoromethyl)phenyl]-22-fluoro-25-methyl-2,2,15,19-tetraoxo-24-oxa-2λ⁶-thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.0^{23,27}]nonacosa-3,5,7(29),20(28),21,23(27)-hexaene-10-carboxylate. White solid. MS *m/z*: 631.2 [M+H]⁺, ESI pos.

Example 53: (17*S*)-17-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-24-oxa-2λ⁶-thia-1,10,14,18,29-pentazatetracyclo[18.6.2.13,7.0^{23,27}]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-15,19-dione 2,2,2-trifluoroacetic acid salt



Step 1: Ethyl (3S)-3-[[4-[[6-(2-aminoethyl)-2-pyridyl]sulfonyl]-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate

- 5 To a solution of ethyl (3S)-3-[[4-[[6-(2-aminoethyl)-2-pyridyl]sulfonyl]-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate 2,2,2-trifluoroacetic acid salt (**Intermediate D13**, 180 mg, 0.24 mmol) in methanol (4 ml) was added NaBH₃CN (45 mg, 0.72 mmol), ZnCl₂ (162 mg, 1.19 mmol) and 3-(1,3-dioxoisindolin-2-yl)propanal (48.4 mg, 0.24 mmol). The reaction mixture was stirred at 25 °C for 4 hours. The
- 10 mixture was poured into water and extracted three times with ethyl acetate. The combined organic layer was dried with Na₂SO₄, filtered and evaporated. The crude product was purified by reverse phase flash chromatography (Flash Spherical C18, 20-35 µm; 100 Å; gradient water with 1% formic acid and acetonitrile), the combined fractions were concentrated and extracted to give
- 15 ethyl (3S)-3-[2-chloro-5-(trifluoromethyl)phenyl]-3-[[4-[[6-[2-[3-(1,3-dioxoisindolin-2-yl)propylamino]ethyl]-2-pyridyl]sulfonyl]-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate (180 mg, 87 % yield) as a white solid. MS m/z: 828.4 [M+H]⁺, ESI pos.

Step 2: Ethyl (3S)-3-[[4-[[6-[2-[tert-butoxycarbonyl]-3-(1,3-dioxoisindolin-2-yl)propyl]amino]ethyl]-2-pyridyl]sulfonyl]-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate

- 20 The title compound was prepared in analogy to **Example 51, step 2** starting with ethyl (3S)-3-[[4-[[6-(2-aminoethyl)-2-pyridyl]sulfonyl]-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate instead of ethyl (3S)-3-[2-chloro-5-(trifluoromethyl)phenyl]-3-[[4-[[6-(2-aminoethyl)-2-pyridyl]sulfonyl]-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate.

yl)propylamino]ethyl]phenyl]sulfonyl-8-fluoro-2-methyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate. White solid. MS m/z: 828.1 [M-100+H]⁺, ESI pos.

Step 3: 2-[3-[*tert*-butoxycarbonyl]-2-[6-[[6-[(1*S*)-2-carboxy-1-[2-chloro-5-(trifluoromethyl)phenyl]ethyl]carbamoyl]-2,3-dihydro-1,4-benzoxazin-4-yl]sulfonyl]-2-pyridyl]ethyl]amino]propylcarbamoyl]benzoic acid

The title compound was prepared in analogy to **Example 49, step 3** starting with ethyl (3*S*)-3-[[4-[[6-[2-[*tert*-butoxycarbonyl]-3-(1,3-dioxoisindolin-2-yl)propyl]amino]ethyl]-2-pyridyl]sulfonyl]-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate instead of ethyl (3*S*)-3-[2,5-bis(trifluoromethyl)phenyl]-3-[[4-[3-[2-[*tert*-butoxycarbonyl]-3-(1,3-dioxoisindolin-2-yl)propyl]amino]ethyl]phenyl]sulfonyl]-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate. White solid. MS m/z: 918.1 [M+H]⁺, ESI pos.

Step 4: (3*S*)-3-[[4-[(2*S*)-4-[3-[2-[3-aminopropyl(*tert*-butoxycarbonyl)amino]ethyl]phenyl]sulfonyl]-2-(hydroxymethyl)-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid 2,2,2-trifluoroacetic acid salt

The title compound was prepared in analogy to **Example 5, step 9** starting with 2-[3-[*tert*-butoxycarbonyl]-2-[3-[[6-[(1*S*)-2-carboxy-1-[2-chloro-5-(trifluoromethyl)phenyl]ethyl]carbamoyl]-2-(hydroxymethyl)-2,3-dihydro-1,4-benzoxazin-4-yl]sulfonyl]phenyl]ethyl]amino]propylcarbamoyl]benzoic acid instead of 2-[[4-[(4*R*)-4-(*tert*-butoxycarbonylamino)-5-[3-[6-[[6-[(1*S*)-2-carboxy-1-[2-chloro-5-(trifluoromethyl)phenyl]ethyl]carbamoyl]spiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-4-yl]sulfonylphenoxy]pentyl]carbamoyl]benzoic acid and stirring at 80°C for 6 hours instead of 16 hours. Yellow solid. MS m/z: 770.2 [M+H]⁺, ESI pos.

Step 5: *tert*-Butyl (17*S*)-17-[2-chloro-5-(trifluoromethyl)phenyl]-2,2,15,19-tetraoxo-24-oxa-2λ⁶-thia-1,10,14,18,29-pentazatetracyclo[18.6.2.13,7.0^{23,27}]nonacosa-3,5,7(29),20(28),21,23(27)-hexaene-10-carboxylate

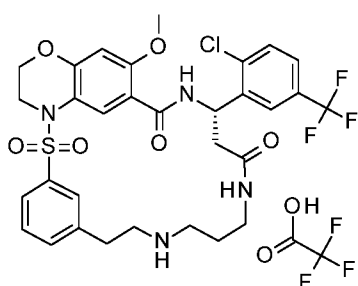
To a solution of (3*S*)-3-[[4-[[6-[2-[3-aminopropyl(*tert*-butoxycarbonyl)amino]ethyl]-2-pyridyl]sulfonyl]-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid 2,2,2-trifluoroacetic acid salt (80 mg, 0.09 mmol) in acetonitrile (8 ml) was added *N*-methylimidazole (37.1 mg, 0.45 mmol) and *N,N,N',N'*-tetramethylchloroformamidinium hexafluorophosphate (30.5 mg, 0.11 mmol). The reaction

mixture was stirred at 25 °C for 2 hours. The solvent was evaporated and the residue was purified by prep-HPLC (Phenomenex Luna C18 150 x 25 mm x 10 µm, gradient water with 1 % TFA and acetonitrile) to give *tert*-butyl (17*S*)-17-[2-chloro-5-(trifluoromethyl)phenyl]-2,2,15,19-tetraoxo-24-oxa-2λ⁶-thia-1,10,14,18,29-pentazatetracyclo[18.6.2.13,7.023,27]nonacosa-3,5,7(29),20(28),21,23(27)-hexaene-10-carboxylate (40.0 mg, 0.05 mmol, 56 % yield) as a white solid. MS *m/z*: 752.2 [M+H]⁺, ESI pos.

Step 6: (17*S*)-17-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-24-oxa-2λ⁶-thia-1,10,14,18,29-pentazatetracyclo[18.6.2.13,7.023,27]nonacosa-3,5,7(29),20(28),21,23(27)-hexaene-15,19-dione 2,2,2-trifluoroacetic acid salt

The title compound was prepared in analogy to **Example 51, step 6** starting with *tert*-butyl (17*S*)-17-[2-chloro-5-(trifluoromethyl)phenyl]-2,2,15,19-tetraoxo-24-oxa-2λ⁶-thia-1,10,14,18,29-pentazatetracyclo[18.6.2.13,7.023,27]nonacosa-3,5,7(29),20(28),21,23(27)-hexaene-10-carboxylate instead of *tert*-butyl (17*S*,25*R*)-17-[2-chloro-5-(trifluoromethyl)phenyl]-22-fluoro-25-methyl-2,2,15,19-tetraoxo-24-oxa-2λ⁶-thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.023,27]nonacosa-3,5,7(29),20(28),21,23(27)-hexaene-10-carboxylate. Yellow solid. MS *m/z*: 652.2 [M+H]⁺, ESI pos.

Example 54: (17*S*)-17-[2-chloro-5-(trifluoromethyl)phenyl]-21-methoxy-2,2-dioxo-24-oxa-2λ⁶-thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.023,27]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-15,19-dione 2,2,2-trifluoroacetic acid salt



Step 1: Ethyl (3*S*)-3-[2-chloro-5-(trifluoromethyl)phenyl]-3-[[4-[3-[2-[3-(1,3-dioxoisindolin-2-yl)propylamino]ethyl]phenyl]sulfonyl-8-fluoro-2-methyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate

The title compound was prepared in analogy to **Example 28 and Example 29, step 3** starting from ethyl (3*S*)-3-[2-chloro-5-(trifluoromethyl)phenyl]-3-[[4-[3-[2-[3-(1,3-dioxoisindolin-2-yl)propylamino]ethyl]phenyl]sulfonyl-7-methoxy-2,3-dihydro-1,4-benzoxazine-6-

carbonyl]amino]propanoate 2,2,2-trifluoroacetic acid (**Intermediate D14**) instead of ethyl (3*S*)-3-[[4-[3-(2-amino-1-methyl-ethoxy)phenyl]sulfonyl]spiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate. White solid. MS *m/z*: 857.4 [M+H]⁺, ESI pos.

5 Step 2: Ethyl (3*S*)-3-[[4-[3-[2-[*tert*-butoxycarbonyl]-3-(1,3-dioxoisindolin-2-yl)propyl]amino]ethyl]phenyl]sulfonyl-7-methoxy-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate

The title compound was prepared in analogy to **Example 51, step 2** starting with ethyl (3*S*)-3-[2-chloro-5-(trifluoromethyl)phenyl]-3-[[4-(2*R*)-4-[3-[2-[3-(1,3-dioxoisindolin-2-yl)propylamino]ethyl]phenyl]sulfonyl-8-fluoro-2-methyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate instead of ethyl (3*S*)-3-[2-chloro-5-(trifluoromethyl)phenyl]-3-[[4-(2*R*)-4-[3-[2-[3-(1,3-dioxoisindolin-2-yl)propylamino]ethyl]phenyl]sulfonyl-8-fluoro-2-methyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate. Colorless oil. MS *m/z*: 947.5 [M+H]⁺, ESI pos.

15 Step 3: 2-[3-[*tert*-butoxycarbonyl]-2-[3-[[6-[[1*S*]-2-carboxy-1-[2-chloro-5-(trifluoromethyl)phenyl]ethyl]carbamoyl]-7-methoxy-2,3-dihydro-1,4-benzoxazin-4-yl]sulfonyl]phenyl]ethyl]amino]propylcarbamoyl]benzoic acid

The title compound was prepared in analogy to **Example 49, step 3** starting with ethyl (3*S*)-3-[[4-[3-[2-[*tert*-butoxycarbonyl]-3-(1,3-dioxoisindolin-2-yl)propyl]amino]ethyl]phenyl]sulfonyl-7-methoxy-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate instead of ethyl (3*S*)-3-[2,5-bis(trifluoromethyl)phenyl]-3-[[4-[3-[2-[*tert*-butoxycarbonyl]-3-(1,3-dioxoisindolin-2-yl)propyl]amino]ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate stirring at 80 °C for 6 hours instead of 25 °C for 2 hours. White solid. MS *m/z*: 799.5 [M+H]⁺, ESI pos.

25 Step 4: (3*S*)-3-[2-chloro-5-(trifluoromethyl)phenyl]-3-[[4-(2*R*)-4-[3-[2-[3-aminopropyl(*tert*-butoxycarbonyl)amino]ethyl]phenyl]sulfonyl-8-fluoro-2-methyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoic acid

The title compound was prepared in analogy to **Example 5, step 9** starting with 2-[3-[2-[3-[[6-[[1*S*]-1-[2,5-Bis(trifluoromethyl)phenyl]-2-carboxy-ethyl]carbamoyl]-2,3-dihydro-1,4-benzoxazin-4-yl]sulfonyl]phenyl]ethyl-*tert*-butoxycarbonyl-amino]propylcarbamoyl]benzoic acid instead of 2-[[4-(2*R*)-4-(*tert*-butoxycarbonylamino)-5-[3-[6-[[1*S*]-2-carboxy-1-[2-chloro-5-

(trifluoromethyl)phenyl]ethyl]carbamoyl]spiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-4-yl]sulfonylphenoxy]pentyl]carbamoyl]benzoic acid and stirring at 80°C for 8 hours instead of 16 hours. Yellow solid. MS *m/z*: 801.3 [M+H]⁺, ESI pos.

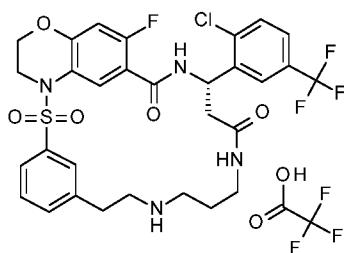
Step 5: *tert*-Butyl (17*S*)-17-[2-chloro-5-(trifluoromethyl)phenyl]-21-methoxy-2,2,15,19-tetraoxo-24-oxa-2λ⁶-thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.0^{23,27}]nonacosa-3,5,7(29),20(28),21,23(27)-hexaene-10-carboxylate

The title compound was prepared in analogy to **Example 49, step 5** starting with (3*S*)-3-[2-chloro-5-(trifluoromethyl)phenyl]-3-[[[(2*R*)-4-[3-[2-[3-aminopropyl(*tert*-butoxycarbonyl)amino]ethyl]phenyl]sulfonyl-8-fluoro-2-methyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoic acid instead of *tert*-butyl (17*S*)-17-[2,5-bis(trifluoromethyl)phenyl]-2,2,15,19-tetraoxo-24-oxa-2λ⁶-thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.0^{23,27}]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-10-carboxylate. White solid. MS *m/z*: 781.4 [M+H]⁺, ESI pos.

Step 6: (17*S*)-17-[2-chloro-5-(trifluoromethyl)phenyl]-21-methoxy-2,2-dioxo-24-oxa-2λ⁶-thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.0^{23,27}]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-15,19-dione 2,2,2-trifluoroacetic acid salt

The title compound was prepared in analogy to **Example 51, step 6** starting with *tert*-butyl (17*S*)-17-[2-chloro-5-(trifluoromethyl)phenyl]-21-methoxy-2,2,15,19-tetraoxo-24-oxa-2λ⁶-thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.0^{23,27}]nonacosa-3,5,7(29),20(28),21,23(27)-hexaene-10-carboxylate instead of *tert*-butyl (17*S*,25*R*)-17-[2-chloro-5-(trifluoromethyl)phenyl]-22-fluoro-25-methyl-2,2,15,19-tetraoxo-24-oxa-2λ⁶-thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.0^{23,27}]nonacosa-3,5,7(29),20(28),21,23(27)-hexaene-10-carboxylate. White solid. MS *m/z*: 681.4 [M+H]⁺, ESI pos.

Example 55: (17*S*)-17-[2-chloro-5-(trifluoromethyl)phenyl]-21-fluoro-2,2-dioxo-24-oxa-2λ⁶-thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.0^{23,27}]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-15,19-dione 2,2,2-trifluoroacetic acid salt



Step 1: Ethyl (3S)-3-[2-chloro-5-(trifluoromethyl)phenyl]-3-[[4-[3-[2-[3-(1,3-dioxoisindolin-2-yl)propylamino]ethyl]phenyl]sulfonyl-7-fluoro-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate

The title compound was prepared in analogy to **Example 8, step 3** starting with ethyl (3S)-3-[[4-[3-(2-aminoethyl)phenyl]sulfonyl-7-fluoro-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate 2,2,2-trifluoroacetic acid salt and 3-(1,3-dioxoisindolin-2-yl)propanal instead of methyl 4-[3-[(1R,4R)-2,5-diazabicyclo[2.2.1]heptan-2-yl]phenyl]sulfonylspro[3H-1,4-benzoxazine-2,1'-cyclopropane]-6-carboxylate and *N*-BOC-2-aminoacetaldehyde and stirring at 20 °C for 2 hours instead of 16 hours. White solid. MS *m/z*:
845.2 [M+H]⁺, ESI pos.

Step 2: Ethyl (3S)-3-[[4-[3-[2-[*tert*-butoxycarbonyl]-3-(1,3-dioxoisindolin-2-yl)propyl]amino]ethyl]phenyl]sulfonyl-7-fluoro-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate

The title compound was prepared in analogy to **Example 51, step 2** starting with ethyl (3S)-3-[2-chloro-5-(trifluoromethyl)phenyl]-3-[[4-[3-[2-[3-(1,3-dioxoisindolin-2-yl)propylamino]ethyl]phenyl]sulfonyl-7-fluoro-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate instead of ethyl (3S)-3-[2-chloro-5-(trifluoromethyl)phenyl]-3-[[[(2R)-4-[3-[2-[3-(1,3-dioxoisindolin-2-yl)propylamino]ethyl]phenyl]sulfonyl-8-fluoro-2-methyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate. Colorless oil. MS *m/z*:
845.3 [M-BOC+H]⁺, ESI pos.

Step 3: 2-[3-[*tert*-butoxycarbonyl]-2-[3-[[6-[(1S)-2-carboxy-1-[2-chloro-5-(trifluoromethyl)phenyl]ethyl]carbamoyl]-7-fluoro-2,3-dihydro-1,4-benzoxazin-4-yl]sulfonyl]phenyl]ethyl]amino]propylcarbonyl]benzoic acid

The title compound was prepared in analogy to **Example 49, step 3** starting with ethyl (3S)-3-[[4-[3-[2-[*tert*-butoxycarbonyl]-3-(1,3-dioxoisindolin-2-yl)propyl]amino]ethyl]phenyl]sulfonyl-7-fluoro-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate instead of ethyl (3S)-3-[2,5-bis(trifluoromethyl)phenyl]-3-[[4-[3-[2-[*tert*-butoxycarbonyl]-3-(1,3-dioxoisindolin-2-yl)propyl]amino]ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate. White crystalline solid. MS *m/z*:
835.3 [M-BOC+H]⁺, ESI pos.

Step 4: (3*S*)-3-[[4-[3-[2-[3-aminopropyl(*tert*-butoxycarbonyl)amino]ethyl]phenyl]sulfonyl-7-fluoro-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid 2,2,2-trifluoroacetic acid salt

The title compound was prepared in analogy to **Example 37, step 5** starting with 2-[3-[*tert*-butoxycarbonyl]-[2-[3-[[6-[[*(1S)*]-2-carboxy-1-[2-chloro-5-(trifluoromethyl)phenyl]ethyl]carbamoyl]-7-fluoro-2,3-dihydro-1,4-benzoxazin-4-yl]sulfonyl]phenyl]ethyl]amino]propylcarbamoyl]benzoic acid instead of 2-[2-[[*(3R,4R)*]-1-[3-[[6-[[*(1S)*]-2-carboxy-1-[2-chloro-5-(trifluoromethyl)phenyl]ethyl]carbamoyl]-2,3-dihydro-1,4-benzoxazin-4-yl]sulfonyl]phenyl]-4-hydroxy-pyrrolidin-3-yl]amino]ethylcarbamoyl]benzoic acid and stirring at 80 °C for 8 hours instead of 2 hours. White solid. MS *m/z*: 787.3 [*M*+*H*]⁺, ESI pos.

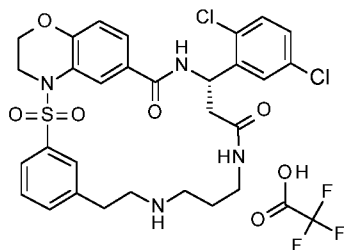
Step 5: *tert*-butyl (17*S*)-17-[2-chloro-5-(trifluoromethyl)phenyl]-21-fluoro-2,2,15,19-tetraoxo-24-oxa-2λ⁶-thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.0^{23,27}]nonacosa-3,5,7(29),20(28),21,23(27)-hexaene-10-carboxylate

The title compound was prepared in analogy to **Example 53, step 5** starting with (3*S*)-3-[[4-[3-[2-[3-aminopropyl(*tert*-butoxycarbonyl)amino]ethyl]phenyl]sulfonyl-7-fluoro-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid 2,2,2-trifluoroacetic acid salt instead of (3*S*)-3-[[4-[[6-[2-[3-aminopropyl(*tert*-butoxycarbonyl)amino]ethyl]-2-pyridyl]sulfonyl]-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid 2,2,2-trifluoroacetic acid salt. White solid. MS *m/z*: 669.2 [*M*-BOC+*H*]⁺, ESI pos.

Step 6: (17*S*)-17-[2-chloro-5-(trifluoromethyl)phenyl]-21-fluoro-2,2-dioxo-24-oxa-2λ⁶-thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.0^{23,27}]nonacosa-3,5,7(29),20(28),21,23(27)-hexaene-15,19-dione 2,2,2-trifluoroacetic acid salt

The title compound was prepared in analogy to **Example 51, step 6** starting with *tert*-butyl (17*S*)-17-[2-chloro-5-(trifluoromethyl)phenyl]-21-fluoro-2,2,15,19-tetraoxo-24-oxa-2λ⁶-thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.0^{23,27}]nonacosa-3,5,7(29),20(28),21,23(27)-hexaene-10-carboxylate instead of *tert*-butyl (17*S*,25*R*)-17-[2-chloro-5-(trifluoromethyl)phenyl]-22-fluoro-25-methyl-2,2,15,19-tetraoxo-24-oxa-2λ⁶-thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.0^{23,27}]nonacosa-3,5,7(29),20(28),21,23(27)-hexaene-10-carboxylate. White solid. MS *m/z*: 669.2 [*M*+*H*]⁺, ESI pos.

Example 56: (17*S*)-17-(2,5-dichlorophenyl)-2,2-dioxo-24-oxa-2 λ ⁶-thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.0^{23,27}]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-15,19-dione 2,2,2-trifluoroacetic acid salt



5 Step 1: Ethyl (3*S*)-3-(2,5-dichlorophenyl)-3-[[4-[3-[2-[3-(1,3-dioxoisindolin-2-yl)propylamino]ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate

The title compound was prepared in analogy to **Example 53, step 1** starting with ethyl-(3*S*)-3-[[4-[3-(2-aminoethyl)phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-(2,5-dichlorophenyl)propanoate 2,2,2-trifluoroacetic acid salt (**Intermediate D17**) instead of ethyl (3*S*)-3-[[4-[[6-(2-aminoethyl)-2-pyridyl]sulfonyl]-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate 2,2,2-trifluoroacetic acid salt (**Intermediate D13**). White solid. MS *m/z*: 793.3 [M+H]⁺, ESI pos.

15 Step 2: Ethyl (3*S*)-3-[[4-[3-[2-[*tert*-butoxycarbonyl]-3-(1,3-dioxoisindolin-2-yl)propyl]amino]ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-(2,5-dichlorophenyl)propanoate

The title compound was prepared in analogy to **Example 51, step 2** starting with ethyl (3*S*)-3-(2,5-dichlorophenyl)-3-[[4-[3-[2-[3-(1,3-dioxoisindolin-2-yl)propylamino]ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate instead of ethyl (3*S*)-3-[2-chloro-5-(trifluoromethyl)phenyl]-3-[[4-[[6-(2-aminoethyl)-2-pyridyl]sulfonyl]-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate. Colorless oil. MS *m/z*: 793.4 [M-100+H]⁺, ESI pos.

25 Step 3: 2-[3-[*tert*-Butoxycarbonyl]-2-[3-[[6-[(1*S*)-2-carboxy-1-(2,5-dichlorophenyl)ethyl]carbamoyl]-2,3-dihydro-1,4-benzoxazin-4-yl]sulfonyl]phenyl]ethyl]amino]propylcarbonyl]benzoic acid

The title compound was prepared in analogy to **Example 49, step 3** starting with ethyl (3*S*)-3-[[4-[3-[2-[*tert*-butoxycarbonyl-[3-(1,3-dioxoisindolin-2-yl)propyl]amino]ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-(2,5-dichlorophenyl)propanoate instead of ethyl (3*S*)-3-[2,5-bis(trifluoromethyl)phenyl]-3-[[4-[3-[2-[*tert*-butoxycarbonyl-[3-(1,3-dioxoisindolin-2-yl)propyl]amino]ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate. White solid. MS *m/z*: 883.3 [M+H]⁺, ESI pos.

Step 4: (3*S*)-3-[[4-[3-[2-[3-aminopropyl(*tert*-butoxycarbonyl)amino]ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-(2,5-dichlorophenyl)propanoic acid 2,2,2-trifluoroacetic acid salt

The title compound was prepared in analogy to **Example 5, step 9** starting with 2-[3-[*tert*-butoxycarbonyl-[2-[3-[[6-[[1*S*]-2-carboxy-1-(2,5-dichlorophenyl)ethyl]carbamoyl]-2,3-dihydro-1,4-benzoxazin-4-yl]sulfonyl]phenyl]ethyl]amino]propylcarbamoyl]benzoic acid instead of 2-[[4*R*]-4-(*tert*-butoxycarbonylamino)-5-[3-[6-[[1*S*]-2-carboxy-1-[2-chloro-5-(trifluoromethyl)phenyl]ethyl]carbamoyl]spiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-4-yl]sulfonylphenoxy]pentyl]carbamoyl]benzoic acid and stirring at 80°C for 4 hours instead of 16 hours. White solid. MS *m/z*: 735.4 [M+H]⁺, ESI pos.

Step 5: *tert*-Butyl (17*S*)-17-(2,5-dichlorophenyl)-2,2,15,19-tetraoxo-24-oxa-2λ⁶-thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.023,27]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-10-carboxylate

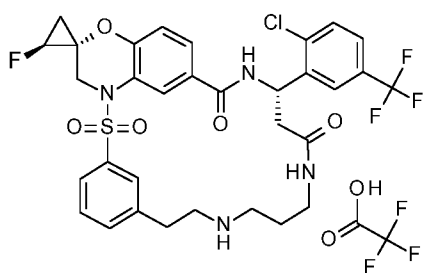
The title compound was prepared in analogy to **Example 49, step 5** starting with (3*S*)-3-[[4-[3-[2-[3-aminopropyl(*tert*-butoxycarbonyl)amino]ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-(2,5-dichlorophenyl)propanoic acid 2,2,2-trifluoroacetic acid salt instead of *tert*-butyl (17*S*)-17-[2,5-bis(trifluoromethyl)phenyl]-2,2,15,19-tetraoxo-24-oxa-2λ⁶-thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.023,27]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-10-carboxylate. White solid. MS *m/z*: 717.3 [M+H]⁺, ESI pos.

Step 6: (17*S*)-17-(2,5-dichlorophenyl)-2,2-dioxo-24-oxa-2λ⁶-thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.023,27]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-15,19-dione 2,2,2-trifluoroacetic acid salt

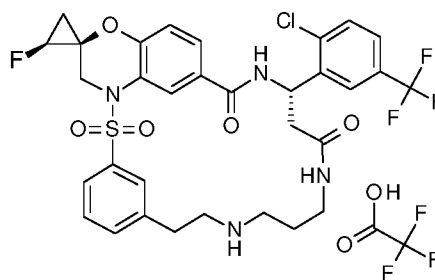
The title compound was prepared in analogy to **Example 51, step 6** starting with *tert*-butyl (17*S*)-17-(2,5-dichlorophenyl)-2,2,15,19-tetraoxo-24-oxa-2λ⁶-thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.023,27]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-10-

carboxylate instead of *tert*-butyl (17*S*,25*R*)-17-[2-chloro-5-(trifluoromethyl)phenyl]-22-fluoro-25-methyl-2,2,15,19-tetraoxo-24-oxa-2 λ ⁶-thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.0^{23,27}]nonacosa-3,5,7(29),20(28),21,23(27)-hexaene-10-carboxylate. Yellow solid. MS *m/z*: 617.1 [M+H]⁺, ESI pos.

- 5 **Example 57 and 58:** (17*S*)-17-[2-chloro-5-(trifluoromethyl)phenyl]-2'-fluoro-2,2-dioxo-spiro[24-oxa-2 λ ⁶-thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.0^{23,27}]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-25,1'-cyclopropane]-15,19-dione 2,2,2-trifluoroacetic acid salt (mixture of TRANS diastereomers arbitrarily assigned) and (17*S*)-17-[2-chloro-5-(trifluoromethyl)phenyl]-2'-fluoro-2,2-dioxo-spiro[24-oxa-2 λ ⁶-thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.0^{23,27}]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-25,1'-cyclopropane]-15,19-dione 2,2,2-trifluoroacetic acid salt (mixture of CIS diastereomers arbitrarily assigned)



2 TRANS diastereomers



2 CIS diastereomers

- 15 Step 1: Ethyl (3*S*)-3-[2-chloro-5-(trifluoromethyl)phenyl]-3-[[4-[3-[2-[3-(1,3-dioxoisindolin-2-yl)propylamino]ethyl]phenyl]sulfonyl-2'-fluoro-spiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]propanoate

The title compound was prepared in analogy to **Example 53, step 1** starting with ethyl-(3*S*)-3-[[4-[3-(2-aminoethyl)phenyl]sulfonyl-2'-fluoro-spiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate 2,2,2-trifluoroacetic acid salt (**Intermediate D18**) instead of ethyl (3*S*)-3-[[4-[[6-(2-aminoethyl)-2-pyridyl]sulfonyl]-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate 2,2,2-trifluoroacetic acid salt (**Intermediate D13**). White solid. MS *m/z*: 871.1 [M+H]⁺, ESI pos.

- 25 Step 2: Ethyl (3*S*)-3-[[4-[3-[2-[*tert*-butoxycarbonyl]-3-(1,3-dioxoisindolin-2-yl)propyl]amino]ethyl]phenyl]sulfonyl-2'-fluoro-spiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate

The title compound was prepared in analogy to **Example 51, step 2** starting with ethyl (3*S*)-3-[2-chloro-5-(trifluoromethyl)phenyl]-3-[[4-[3-[2-[3-(1,3-dioxoisindolin-2-yl)propylamino]ethyl]phenyl]sulfonyl-2'-fluoro-spiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]propanoate instead of ethyl (3*S*)-3-[2-chloro-5-(trifluoromethyl)phenyl]-3-
 5 [[(2*R*)-4-[3-[2-[3-(1,3-dioxoisindolin-2-yl)propylamino]ethyl]phenyl]sulfonyl-8-fluoro-2-methyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate and stirring at 20 °C for 16 hours instead of 25 °C for 2 hours. Colorless oil. MS *m/z*: 871.3 [M-BOC+H]⁺, ESI pos.

Step 3: 2-[3-[*tert*-Butoxycarbonyl]-2-[3-[6-[[(1*S*)-2-carboxy-1-[2-chloro-5-(trifluoromethyl)phenyl]ethyl]carbamoyl]-2'-fluoro-spiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-4-yl]sulfonylphenyl]ethyl]amino]propylcarbamoyl]benzoic acid

 10

The title compound was prepared in analogy to **Example 49, step 3** starting with ethyl (3*S*)-3-[[4-[3-[2-[*tert*-butoxycarbonyl]-3-(1,3-dioxoisindolin-2-yl)propyl]amino]ethyl]phenyl]sulfonyl-2'-fluoro-spiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate instead of ethyl (3*S*)-3-[2,5-bis(trifluoromethyl)phenyl]-3-
 15 [[4-[3-[2-[*tert*-butoxycarbonyl]-3-(1,3-dioxoisindolin-2-yl)propyl]amino]ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate. White solid. MS *m/z*: 961.5 [M+H]⁺, ESI pos.

Step 4: (3*S*)-3-[[4-[3-[2-[3-Aminopropyl(*tert*-butoxycarbonyl)amino]ethyl]phenyl]sulfonyl-2'-fluoro-spiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid 2,2,2-trifluoroacetic acid salt

 20

The title compound was prepared in analogy to **Example 5, step 9** starting with 2-[3-[*tert*-butoxycarbonyl]-2-[3-[6-[[(1*S*)-2-carboxy-1-[2-chloro-5-(trifluoromethyl)phenyl]ethyl]carbamoyl]-2'-fluoro-spiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-4-yl]sulfonylphenyl]ethyl]amino]propylcarbamoyl]benzoic acid instead of 2-
 25 [[(4*R*)-4-(*tert*-butoxycarbonylamino)-5-[3-[6-[[(1*S*)-2-carboxy-1-[2-chloro-5-(trifluoromethyl)phenyl]ethyl]carbamoyl]spiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-4-yl]sulfonylphenoxy]pentyl]carbamoyl]benzoic acid and stirring at 80 °C for 6 hours instead of 16 hours. White solid. MS *m/z*: 813.2 [M+H]⁺, ESI pos.

Step 5: *tert*-Butyl (17*S*)-17-[2-chloro-5-(trifluoromethyl)phenyl]-2'-fluoro-2,2,15,19-tetraoxo-spiro[24-oxa-2λ⁶-thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.023,27]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-25,1'-cyclopropane]-10-carboxylate (mixture of TRANS diastereomers) and *tert*-butyl (17*S*)-17-[2-chloro-5-(trifluoromethyl)phenyl]-2'-fluoro-2,2,15,19-

 30

tetraoxo-spiro[24-oxa-2 λ ⁶-thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.0^{23,27}]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-25,1'-cyclopropane]-10-carboxylate (mixture of CIS diastereomers)

To a solution of (3*S*)-3-[[4-[3-[2-[3-aminopropyl(*tert*-

- 5 butoxycarbonyl)amino]ethyl]phenyl]sulfonyl-2'-fluoro-spiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid 2,2,2-trifluoroacetic acid salt (140 mg, 0.15 mmol) in *N,N*-dimethylformamide (7 ml) was added DIPEA (78 mg, 0.6 mmol) and HATU (69 mg, 0.18 mmol), the mixture was stirred at 20 °C for 2 hours. The reaction mixture was concentrated.

- 10 The CIS and TRANS mixtures of diastereomers were separated using prep-HPLC (Phenomenex Luna C18 150 x 25 mm x 10 μ m, gradient water with 1 % 2,2,2-trifluoroacetic acid and acetonitrile). The CIS/TRANS stereochemistry was arbitrarily assigned and could not be confirmed by NMR analysis.

- 15 First eluting diastereomer (TRANS mixture, *R*_t = 0.663 min): *tert*-butyl (17*S*)-17-[2-chloro-5-(trifluoromethyl)phenyl]-2'-fluoro-2,2,15,19-tetraoxo-spiro[24-oxa-2 λ ⁶-thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.0^{23,27}]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-25,1'-cyclopropane]-10-carboxylate (mixture of TRANS diastereomers) (40 mg, 30 % yield). White solid. MS *m/z*: 695.3 [M-BOC+H]⁺, ESI pos.

- 20 Second eluting diastereomer (CIS mixture, *R*_t = 0.668 min): *tert*-butyl (17*S*)-17-[2-chloro-5-(trifluoromethyl)phenyl]-2'-fluoro-2,2,15,19-tetraoxo-spiro[24-oxa-2 λ ⁶-thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.0^{23,27}]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-25,1'-cyclopropane]-10-carboxylate (mixture of CIS diastereomers) (45 mg, 35 % yield). White solid. MS *m/z*: 695.3 [M-BOC+H]⁺, ESI pos.

- 25 Step 6: (17*S*)-17-[2-Chloro-5-(trifluoromethyl)phenyl]-2'-fluoro-2,2-dioxo-spiro[24-oxa-2 λ ⁶-thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.0^{23,27}]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-25,1'-cyclopropane]-15,19-dione 2,2,2-trifluoroacetic acid salt, mixture of 2 TRANS diastereomers (Example 57)

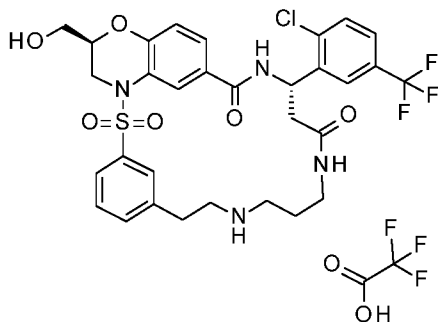
- 30 To a solution of *tert*-butyl (17*S*)-17-[2-chloro-5-(trifluoromethyl)phenyl]-2'-fluoro-2,2,15,19-tetraoxo-spiro[24-oxa-2 λ ⁶-thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.0^{23,27}]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-25,1'-cyclopropane]-10-carboxylate (mixture of TRANS diastereomers) (40.0 mg, 0.05 mmol) in dichloromethane (0.5 ml) and 2,2,2 trifluoroacetic acid

(0.5 ml, 0.05 mmol) was stirred at 25 °C for 1 hour. The reaction mixture was concentrated, water and acetonitrile were added and the solution was lyophilized to give (17*S*)-17-[2-chloro-5-(trifluoromethyl)phenyl]-2'-fluoro-2,2-dioxo-spiro[24-oxa-2λ⁶-thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.0.23,27]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-25,1'-cyclopropane]-15,19-dione 2,2,2-trifluoroacetic acid salt, mixture of 2 TRANS diastereomers (17.9 mg, 44 % yield) as a yellow solid. MS *m/z*: 695.1 [M+H]⁺, ESI pos.

Step 7: (17*S*)-17-[2-Chloro-5-(trifluoromethyl)phenyl]-2'-fluoro-2,2-dioxo-spiro[24-oxa-2λ⁶-thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.0.23,27]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-25,1'-cyclopropane]-15,19-dione 2,2,2-trifluoroacetic acid salt, mixture of 2 CIS diastereomers (Example 58)

To a solution of *tert*-butyl (17*S*)-17-[2-chloro-5-(trifluoromethyl)phenyl]-2'-fluoro-2,2,15,19-tetraoxo-spiro[24-oxa-2λ⁶-thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.0.23,27]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-25,1'-cyclopropane]-10-carboxylate (mixture of CIS diastereomers) (50.0 mg, 0.06 mmol) in dichloromethane (0.5 ml) and 2,2,2-trifluoroacetic acid (0.5 ml, 0.05 mmol) was stirred at 25 °C for 1 hour. The reaction mixture was concentrated, water and acetonitrile were added and the solution was lyophilized to give (17*S*)-17-[2-chloro-5-(trifluoromethyl)phenyl]-2'-fluoro-2,2-dioxo-spiro[24-oxa-2λ⁶-thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.0.23,27]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-25,1'-cyclopropane]-15,19-dione 2,2,2-trifluoroacetic acid salt, mixture of 2 CIS diastereomers (34 mg, 68 % yield) as a yellow solid. MS *m/z*: 695.1 [M+H]⁺, ESI pos.

Example 59: (17*S*,25*S*)-17-[2-Chloro-5-(trifluoromethyl)phenyl]-25-(hydroxymethyl)-2,2-dioxo-24-oxa-2λ⁶-thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.0.23,27]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-15,19-dione 2,2,2-trifluoroacetic acid salt



Step 1: Ethyl (3*S*)-3-[2-chloro-5-(trifluoromethyl)phenyl]-3-[[*(2R)*-4-[3-[2-[3-(1,3-dioxoisindolin-2-yl)propylamino]ethyl]phenyl]sulfonyl-8-fluoro-2-methyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate

The title compound was prepared in analogy to **Example 11, step 3** starting with ethyl (3*S*)-3-[[*(2S)*-4-[3-(2-aminoethyl)phenyl]sulfonyl-2-(hydroxymethyl)-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate (**Intermediate D13**) and 3-(1,3-dioxoisindolin-2-yl)propanal (CAS 2436-29-5) instead of methyl 4-[3-[(3*S*)-3-aminopyrrolidin-1-yl]-5-methyl-phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate and N-BOC-2-aminoacetaldehyde. Yellow solid. MS *m/z*: 857.4 [M+H]⁺, ESI pos.

Step 2: Ethyl (3*S*)-3-[[*(2S)*-4-[3-[2-[*tert*-butoxycarbonyl]-3-(1,3-dioxoisindolin-2-yl)propyl]amino]ethyl]phenyl]sulfonyl-2-(hydroxymethyl)-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate

The title compound was prepared in analogy to **Example 51, step 2** starting with ethyl (3*S*)-3-[2-chloro-5-(trifluoromethyl)phenyl]-3-[[*(2R)*-4-[3-[2-[3-(1,3-dioxoisindolin-2-yl)propylamino]ethyl]phenyl]sulfonyl-8-fluoro-2-methyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate instead of ethyl (3*S*)-3-[2-chloro-5-(trifluoromethyl)phenyl]-3-[[*(2R)*-4-[3-[2-[3-(1,3-dioxoisindolin-2-yl)propylamino]ethyl]phenyl]sulfonyl-8-fluoro-2-methyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate. Colorless oil. MS *m/z*: 957.5 [M+H]⁺, ESI pos.

Step 3: Ethyl (3*S*)-3-[[*(2S)*-4-[3-[2-[*tert*-butoxycarbonyl]-3-(1,3-dioxoisindolin-2-yl)propyl]amino]ethyl]phenyl]sulfonyl-2-(hydroxymethyl)-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate

The title compound was prepared in analogy to **Example 49, step 3** starting with ethyl (3*S*)-3-[[*(2S)*-4-[3-[2-[*tert*-butoxycarbonyl]-3-(1,3-dioxoisindolin-2-yl)propyl]amino]ethyl]phenyl]sulfonyl-2-(hydroxymethyl)-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate instead of ethyl (3*S*)-3-[2,5-bis(trifluoromethyl)phenyl]-3-[[4-[3-[2-[*tert*-butoxycarbonyl]-3-(1,3-dioxoisindolin-2-yl)propyl]amino]ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate. White solid. MS *m/z*: 947.5 [M+H]⁺, ESI pos.

Step 4: (3*S*)-3-[[[(2*S*)-4-[3-[2-[3-aminopropyl(*tert*-butoxycarbonyl)amino]ethyl]phenyl]sulfonyl-2-(hydroxymethyl)-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid

The title compound was prepared in analogy to **Example 5, step 9** starting with 2-[3-[*tert*-butoxycarbonyl-2-[3-[[[(2*S*)-6-[(1*S*)-2-carboxy-1-[2-chloro-5-(trifluoromethyl)phenyl]ethyl]carbamoyl]-2-(hydroxymethyl)-2,3-dihydro-1,4-benzoxazin-4-yl]sulfonyl]phenyl]ethyl]amino]propylcarbamoyl]benzoic acid instead of 2-[[[(4*R*)-4-(*tert*-butoxycarbonylamino)-5-[3-[6-[(1*S*)-2-carboxy-1-[2-chloro-5-(trifluoromethyl)phenyl]ethyl]carbamoyl]spiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-4-yl]sulfonylphenoxy]pentyl]carbamoyl]benzoic acid and stirring at 80 °C for 6 hours instead of 16 hours. Yellow solid. MS *m/z*: 799.5 [M+H]⁺, ESI pos.

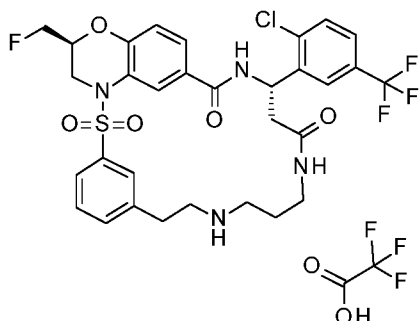
Step 5: : *tert*-Butyl (17*S*,25*S*)-17-[2-chloro-5-(trifluoromethyl)phenyl]-25-(hydroxymethyl)-2,2,15,19-tetraoxo-24-oxa-2λ⁶-thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.0^{23,27}]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-10-carboxylate

The title compound was prepared in analogy to **Example 51, step 5** starting with (3*S*)-3-[[[(2*S*)-4-[3-[2-[3-aminopropyl(*tert*-butoxycarbonyl)amino]ethyl]phenyl]sulfonyl-2-(hydroxymethyl)-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid instead of (3*S*)-3-[2-chloro-5-(trifluoromethyl)phenyl]-3-[[[(2*R*)-4-[3-[2-[3-aminopropyl(*tert*-butoxycarbonyl)amino]ethyl]phenyl]sulfonyl-8-fluoro-2-methyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoic acid. White solid. MS *m/z*: 799.5 [M+H]⁺, ESI pos.

Step 6: (17*S*,25*S*)-17-[2-chloro-5-(trifluoromethyl)phenyl]-25-(hydroxymethyl)-2,2-dioxo-24-oxa-2λ⁶-thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.0^{23,27}]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-15,19-dione 2,2,2-trifluoroacetic acid salt

The title compound was prepared in analogy to **Example 51, step 6** starting with *tert*-butyl (17*S*,25*S*)-17-[2-chloro-5-(trifluoromethyl)phenyl]-25-(hydroxymethyl)-2,2,15,19-tetraoxo-24-oxa-2λ⁶-thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.0^{23,27}]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-10-carboxylate instead of *tert*-butyl (17*S*,25*R*)-17-[2-chloro-5-(trifluoromethyl)phenyl]-22-fluoro-25-methyl-2,2,15,19-tetraoxo-24-oxa-2λ⁶-thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.0^{23,27}]nonacosa-3,5,7(29),20(28),21,23(27)-hexaene-10-carboxylate. White solid. MS *m/z*: 681.2 [M+H]⁺, ESI pos.

Example 60: (17*S*,25*S*)-17-[2-chloro-5-(trifluoromethyl)phenyl]-25-(fluoromethyl)-2,2-dioxo-24-oxa-2λ⁶-thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.0.23,27]nonacos-3(29),4,6,20(28),21,23(27)-hexaene-15,19-dione 2,2,2-trifluoroacetic acid salt



5 Step 1: Ethyl-(3*S*)-3-[2-chloro-5-(trifluoromethyl)phenyl]-3-[[2*S*)-4-[3-[2-[3-(1,3-dioxoisindolin-2-yl)propylamino]ethyl]phenyl]sulfonyl-2-(fluoromethyl)-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate 2,2,2-trifluoroacetic acid salt

The title compound was prepared in analogy to **Example 11, step 3** starting with ethyl-(3*S*)-3-[2-chloro-5-(trifluoromethyl)phenyl]-3-[[2*S*)-4-[3-(2-aminoethyl)phenyl]sulfonyl-2-(fluoromethyl)-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate 2,2,2-trifluoroacetic acid salt (**Intermediate D20**) and 3-(1,3-dioxoisindolin-2-yl)propanal (CAS 2436-29-5) instead of methyl 4-[3-[(3*S*)-3-aminopyrrolidin-1-yl]-5-methyl-phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate and *N*-BOC-2-aminoacetaldehyde. Yellow solid. MS *m/z*: 859.5 [M+H]⁺, ESI pos.

15 Step 2: Ethyl (3*S*)-3-[[2*S*)-4-[3-[2-[*tert*-butoxycarbonyl]-3-(1,3-dioxoisindolin-2-yl)propyl]amino]ethyl]phenyl]sulfonyl-2-(fluoromethyl)-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate

The title compound was prepared in analogy to **Example 51, step 2** starting with ethyl-(3*S*)-3-[2-chloro-5-(trifluoromethyl)phenyl]-3-[[2*S*)-4-[3-[2-[3-(1,3-dioxoisindolin-2-yl)propylamino]ethyl]phenyl]sulfonyl-2-(fluoromethyl)-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate 2,2,2-trifluoroacetic acid salt instead of ethyl (3*S*)-3-[2-chloro-5-(trifluoromethyl)phenyl]-3-[[2*R*)-4-[3-[2-[3-(1,3-dioxoisindolin-2-yl)propylamino]ethyl]phenyl]sulfonyl-8-fluoro-2-methyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate. Colorless oil. MS *m/z*: 859.5 [M-BOC+H]⁺, ESI pos.

Step 3: 2-[3-[*tert*-butoxycarbonyl]-2-[3-[[*(2S)*-6-[[*(1S)*-2-carboxy-1-[2-chloro-5-(trifluoromethyl)phenyl]ethyl]carbamoyl]-2-(fluoromethyl)-2,3-dihydro-1,4-benzoxazin-4-yl]sulfonyl]phenyl]ethyl]amino]propylcarbamoyl]benzoic acid

The title compound was prepared in analogy to **Example 49, step 3** starting with ethyl (*3S*)-3-
 5 [[*(2S)*]-4-[3-[2-[*tert*-butoxycarbonyl]-3-(1,3-dioxoisindolin-2-yl)propyl]amino]ethyl]phenyl]sulfonyl-2-(fluoromethyl)-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate instead of ethyl (*3S*)-3-[2,5-bis(trifluoromethyl)phenyl]-3-[[4-[3-[2-[*tert*-butoxycarbonyl]-3-(1,3-dioxoisindolin-2-yl)propyl]amino]ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate. Yellow solid. MS *m/z*: 849.5 [M-BOC+H]⁺, ESI pos.

Step 4: (*3S*)-3-[[*(2S)*]-4-[3-[2-[3-aminopropyl(*tert*-butoxycarbonyl)amino]ethyl]phenyl]sulfonyl-2-(fluoromethyl)-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid

The title compound was prepared in analogy to **Example 5, step 9** starting with 2-[3-[*tert*-
 15 *butoxycarbonyl*]-2-[3-[[*(2S)*]-6-[[*(1S)*]-2-carboxy-1-[2-chloro-5-(trifluoromethyl)phenyl]ethyl]carbamoyl]-2-(fluoromethyl)-2,3-dihydro-1,4-benzoxazin-4-yl]sulfonyl]phenyl]ethyl]amino]propylcarbamoyl]benzoic acid instead of 2-[[*(4R)*]-4-(*tert*-*butoxycarbonylamino*)-5-[3-[6-[[*(1S)*]-2-carboxy-1-[2-chloro-5-(trifluoromethyl)phenyl]ethyl]carbamoyl]spiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-4-yl]sulfonylphenoxy]pentyl]carbamoyl]benzoic acid and stirring at 80°C for 6 hours instead of 16 hours. Colorless liquid. MS *m/z*: 801.5 [M+H]⁺, ESI pos.

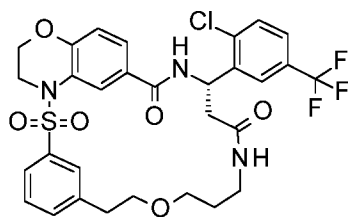
Step 5: : *tert*-Butyl (17*S*,25*S*)-17-[2-chloro-5-(trifluoromethyl)phenyl]-25-(hydroxymethyl)-
2,2,15,19-tetraoxo-24-oxa-2λ⁶-thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.0^{23,27}]nonacosa-
3(29),4,6,20(28),21,23(27)-hexaene-10-carboxylate

The title compound was prepared in analogy to **Example 51, step 5** starting with (*3S*)-3-[[*(2S)*]-4-
 25 [3-[2-[3-aminopropyl(*tert*-butoxycarbonyl)amino]ethyl]phenyl]sulfonyl-2-(fluoromethyl)-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid instead of (*3S*)-3-[2-chloro-5-(trifluoromethyl)phenyl]-3-[[*(2R)*]-4-[3-[2-[3-aminopropyl(*tert*-butoxycarbonyl)amino]ethyl]phenyl]sulfonyl-8-fluoro-2-methyl-2,3-dihydro-
 30 1,4-benzoxazine-6-carbonyl]amino]propanoic acid. White solid. MS *m/z*: 683.3 [M-BOC+H]⁺, ESI pos.

Step 6: (1*S*,25*S*)-17-[2-Chloro-5-(trifluoromethyl)phenyl]-25-(fluoromethyl)-2,2-dioxo-24-oxa-2λ⁶-thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.023,27]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-15,19-dione 2,2,2-trifluoroacetic acid salt

The title compound was prepared in analogy to **Example 51, step 6** starting with *tert*-butyl (1*S*,25*S*)-17-[2-chloro-5-(trifluoromethyl)phenyl]-25-(hydroxymethyl)-2,2,15,19-tetraoxo-24-oxa-2λ⁶-thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.023,27]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-10-carboxylate instead of *tert*-butyl (1*S*,25*R*)-17-[2-chloro-5-(trifluoromethyl)phenyl]-22-fluoro-25-methyl-2,2,15,19-tetraoxo-24-oxa-2λ⁶-thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.023,27]nonacosa-3,5,7(29),20(28),21,23(27)-hexaene-10-carboxylate. White solid. MS *m/z*: 683.4 [M+H]⁺, ESI pos.

Example 61: (1*S*)-17-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-10,24-dioxa-2λ⁶-thia-1,14,18-triazatetracyclo[18.6.2.13,7.023,27]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-15,19-dione



Step 1: *tert*-butyl *N*-(3-vinyloxypropyl)carbamate

To a solution of 3-ethenoxypropan-1-amine (CAS 66415-55-2, 1 g, 9.89 mmol) in dichloromethane (25 ml) were added DIPEA (5.1 ml, 29.7 mmol) and di-*tert*-butyldicarbonate (3.24 g, 14.83 mmol). The reaction mixture was stirred at 25 °C for 16 hours. The mixture was poured into water and extracted three times with ethyl acetate. The combined organic layer was dried with Na₂SO₄, filtered and evaporated. The crude product was purified via flash chromatography (silica gel, ethyl acetate/petroleum ether 0-60 %) to give *tert*-butyl *N*-(3-vinyloxypropyl)carbamate (2.0 g, 100 % yield) as a colorless oil.

¹H NMR (400 MHz, DMSO-*d*₆) δ ppm 1.37 (s, 9 H) 1.69 (quin, *J*=6.4 Hz, 2 H) 2.93 - 3.03 (m, 2 H) 3.65 (t, *J*=6.4 Hz, 2 H) 3.95 (dd, *J*=6.4, 1.6 Hz, 1 H) 4.15 (dd, *J*=14.4, 1.6 Hz, 1 H) 6.48 (dd, *J*=14.4, 6.8 Hz, 1 H) 6.83 (br s, 1 H)

Step 2: Methyl 4-[3-[2-[3-(*tert*-butoxycarbonylamino)propoxy]vinyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate

To a solution of methyl 4-(3-bromophenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate (1 g, 2.43 mmol) in N,N-dimethylformamide (20 ml) were added PPh₃ (254 mg, 0.97 mmol), triethylamine (2.7 ml, 19.4 mmol), palladium acetate (218 mg, 0.97 mmol) and tert-butyl *N*-(3-vinyloxypropyl)carbamate (732 mg, 3.64 mmol) at 100 °C for 16 hours under nitrogen. The mixture was poured into water and extracted three times with ethyl acetate. The combined organic layer was dried with Na₂SO₄, filtered and evaporated. The crude product was purified via flash chromatography (silica gel, ethyl acetate/petroleum ether 0-60 %) to give methyl 4-[3-[2-[3-(*tert*-butoxycarbonylamino)propoxy]vinyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate (400 mg, 25 % yield, 81 % purity) as a light yellow oil. MS *m/z*: 555.3 [M+Na]⁺, ESI pos.

Step 3: 4-[3-[2-[3-(*tert*-butoxycarbonylamino)propoxy]vinyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid

The title compound was prepared in analogy to **Example 49, step 3** starting with methyl 4-[3-[2-[3-(*tert*-butoxycarbonylamino)propoxy]vinyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylate instead of ethyl (3*S*)-3-[2,5-bis(trifluoromethyl)phenyl]-3-[[4-[3-[2-[*tert*-butoxycarbonyl-[3-(1,3-dioxoisindolin-2-yl)propyl]amino]ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate and stirring at 25 °C for 18 hours instead of 2 hours. White solid. MS *m/z*: 541.3 [M+H]⁺, ESI pos.

Step 4: Ethyl (3*S*)-3-[[4-[3-[2-[3-(*tert*-butoxycarbonylamino)propoxy]vinyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate

The title compound was prepared in analogy to **Intermediate 20, step 2** starting with 4-[3-[2-[3-(*tert*-butoxycarbonylamino)propoxy]vinyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid instead of (2*S*)-4-[3-[2-(*tert*-butoxycarbonylamino)ethyl]phenyl]sulfonyl-2-(fluoromethyl)-2,3-dihydro-1,4-benzoxazine-6-carboxylic acid. White solid. MS *m/z*: 698.3 [M-BOC+H]⁺, ESI pos.

Step 5: Methyl (3*S*)-3-[[4-[3-[2-[3-(*tert*-butoxycarbonylamino)propoxy]ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate

To a solution of ethyl (3*S*)-3-[[4-[3-[2-[3-(*tert*-butoxycarbonylamino)propoxy]vinyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate (300 mg, 0.38 mmol) in

methanol (6 ml) and aqueous ammonia (0.2 ml, 0.38 mmol) was added Raney Ni (150 mg, 1.75 mmol). The reaction was stirred at 30 °C for 2 hours under a hydrogen atmosphere (1 bar). The suspension was filtered, the filtrate was evaporated and purified by prep-HPLC (Phenomenex Luna C18 150 x 25 mm x 10 µm, gradient water with 2.25 % formic acid and acetonitrile) to afford methyl (3*S*)-3-[[4-[3-[2-[3-(*tert*-butoxycarbonylamino)propoxy]ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate (80.0 mg, 27 % yield). MS *m/z*: 627.3 [M+H]⁺, ESI pos.

Step 6: (3*S*)-3-[[4-[3-[2-[3-(*tert*-butoxycarbonylamino)propoxy]ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid

The title compound was prepared in analogy to **Example 49, step 3** starting with methyl (3*S*)-3-[[4-[3-[2-[3-(*tert*-butoxycarbonylamino)propoxy]ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate instead of ethyl (3*S*)-3-[2,5-bis(trifluoromethyl)phenyl]-3-[[4-[3-[2-[*tert*-butoxycarbonyl-3-(1,3-dioxoisindolin-2-yl)propyl]amino]ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate. White solid. MS *m/z*: 770.2 [M+H]⁺, ESI pos.

Step 7: (3*S*)-3-[[4-[3-[2-(3-aminopropoxy)ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid 2,2,2-trifluoroacetic acid salt

To a solution of (3*S*)-3-[[4-[3-[2-[3-(*tert*-butoxycarbonylamino)propoxy]ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid (80 mg, 0.1 mmol) in dichloromethane (5 ml) was added 2,2,2-trifluoroacetic acid (1.0 ml, 0.1 mmol). The reaction mixture was stirred at 25 °C for 1 hour. The solvent and excess of TFA were evaporated. The residue was purified by reverse phase flash chromatography (Flash Spherical C18, 20-35 µm; 100 Å; gradient water with 0.1 % TFA and acetonitrile) to give (3*S*)-3-[[4-[3-[2-(3-aminopropoxy)ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid 2,2,2-trifluoroacetic acid salt (60 mg, 63 % yield) as a white solid. MS *m/z*: 670.4 [M+H]⁺, ESI pos.

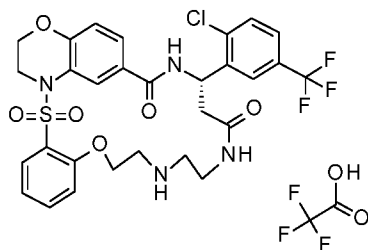
Step 8: (17*S*)-17-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-10,24-dioxo-2λ⁶-thia-1,14,18-triazatetracyclo[18.6.2.13,7.0.23,27]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-15,19-dione

The title compound was prepared in analogy to **Example 53, step 5** starting with (3*S*)-3-[[4-[3-[2-(3-aminopropoxy)ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid 2,2,2-trifluoroacetic acid salt

instead of (3*S*)-3-[[4-[[6-[2-[3-aminopropyl(tert-butoxycarbonyl)amino]ethyl]-2-

5 pyridyl]sulfonyl]-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid 2,2,2-trifluoroacetic acid salt. White solid. MS *m/z*: 652.3 [M+H]⁺, ESI pos.

Example 62: (18*S*)-18-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-9,25-dioxa-2λ⁶-thia-
1,12,15,19-tetrazatetracyclo[19.6.2.03,8.024,28]nonacosa-3(8),4,6,21(29),22,24(28)-hexaene-
10 16,20-dione 2,2,2-trifluoroacetic acid salt



Step 1: Ethyl (3*S*)-3-[[4-[2-[2-(*tert*-butoxycarbonyl)amino]ethoxy]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate

To a solution *N*-BOC-ethanolamine (CAS 26690-80-2, 85 mg, 0.53 mmol) and ethyl (3*S*)-3-[2-chloro-5-(trifluoromethyl)phenyl]-3-[[4-(2-hydroxyphenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate (250 mg, 0.41 mmol) in tetrahydrofuran (5 ml) were added triphenylphosphine (5.26 g, 20.1 mmol) and diethyl azodicarboxylate (4.37 g, 25.1 mmol) at 0 °C (dropwise addition). The reaction mixture was allowed to reach 25 °C and was stirred for 16 hours. The mixture was poured into water and extracted three times with ethyl acetate. The combined organic layer was dried with Na₂SO₄, filtered and evaporated. The crude product was purified by reverse phase flash chromatography (Flash Spherical C18, 20-35 μm; 100 Å; gradient water with 0.05 % TFA and acetonitrile) to give ethyl (3*S*)-3-[[4-[2-[2-(*tert*-butoxycarbonyl)amino]ethoxy]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate (230 mg, 71 % yield) as a white solid. MS *m/z*: 656.1 [M+H]⁺, ESI pos.

Step 2: Ethyl -(3*S*)-3-[[4-[2-(2-aminoethoxy)phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate 2,2,2-trifluoroacetic acid salt

The title compound was prepared in analogy to **Intermediate D11, step 3** using ethyl (3*S*)-3-[[4-[2-[2-(*tert*-butoxycarbonylamino)ethoxy]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate instead of ethyl (3*S*)-3-[2,5-bis(trifluoromethyl)phenyl]-3-[[4-[3-[2-(*tert*-butoxycarbonylamino)ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate. Yellow solid. MS *m/z*: 656.3 [M+H]⁺, ESI pos.

Step 3: Ethyl (3*S*)-3-[2-chloro-5-(trifluoromethyl)phenyl]-3-[[4-[2-[2-[2-(1,3-dioxoisindolin-2-yl)ethylamino]ethoxy]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate.

The title compound was prepared in analogy to **Example 28 and Example 29, step 3** starting from ethyl -(3*S*)-3-[[4-[2-(2-aminoethoxy)phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate 2,2,2-trifluoroacetic acid salt instead of ethyl (3*S*)-3-[[4-[3-(2-amino-1-methyl-ethoxy)phenyl]sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate. White solid. MS *m/z*: 829.1 [M+H]⁺, ESI pos.

Step 3: Ethyl (3*S*)-3-[[4-[2-[2-[*tert*-butoxycarbonyl]-2-(1,3-dioxoisindolin-2-yl)ethyl]amino]ethoxy]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate

The title compound was prepared in analogy to **Example 51, step 2** starting with ethyl (3*S*)-3-[2-chloro-5-(trifluoromethyl)phenyl]-3-[[4-[2-[2-[2-(1,3-dioxoisindolin-2-yl)ethylamino]ethoxy]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate instead of ethyl (3*S*)-3-[2-chloro-5-(trifluoromethyl)phenyl]-3-[[[(2*R*)-4-[3-[2-[3-(1,3-dioxoisindolin-2-yl)propylamino]ethyl]phenyl]sulfonyl-8-fluoro-2-methyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate. Off-white solid. MS *m/z*: 828.1 [M-100+H]⁺, ESI pos.

Step 4: 2-[2-[*tert*-Butoxycarbonyl]-2-[2-[6-[(1*S*)-2-carboxy-1-[2-chloro-5-(trifluoromethyl)phenyl]ethyl]carbamoyl]-2,3-dihydro-1,4-benzoxazin-4-yl]sulfonyl]phenoxy]ethyl]amino]ethyl]carbamoyl]benzoic acid

The title compound was prepared in analogy to **Example 49, step 3** starting with ethyl (3*S*)-3-[[4-[2-[2-[*tert*-butoxycarbonyl]-2-(1,3-dioxoisindolin-2-yl)ethyl]amino]ethoxy]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-

chloro-5-(trifluoromethyl)phenyl]propanoate instead of ethyl (3*S*)-3-[2,5-bis(trifluoromethyl)phenyl]-3-[[4-[3-[2-[*tert*-butoxycarbonyl]-[3-(1,3-dioxoisindolin-2-yl)propyl]amino]ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate and stirring at 25 °C for 16 hours instead of 2 hours. White solid.

5 MS m/z: 919.4 [M+H]⁺, ESI pos.

Step 5: (3*S*)-3-[[4-[2-[2-[2-aminoethyl(*tert*-butoxycarbonyl)amino]ethoxy]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid

The title compound was prepared in analogy to **Example 5, step 9** starting with 2-[2-[*tert*-
10 butoxycarbonyl]-2-[2-[[6-[[[(1*S*)-2-carboxy-1-[2-chloro-5-(trifluoromethyl)phenyl]ethyl]carbamoyl]-2,3-dihydro-1,4-benzoxazin-4-yl]sulfonyl]phenoxy]ethyl]amino]ethylcarbamoyl]benzoic acid instead of 2-[[[(4*R*)-4-(*tert*-butoxycarbonylamino)-5-[3-[6-[[[(1*S*)-2-carboxy-1-[2-chloro-5-(trifluoromethyl)phenyl]ethyl]carbamoyl]spiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-4-yl]sulfonyl]phenoxy]pentyl]carbamoyl]benzoic acid and stirring at 80 °C for 4 hours instead of
15 16 hours. White solid. MS m/z: 771.2 [M+H]⁺, ESI pos.

Step 6: *tert*-Butyl (18*S*)-18-[2-chloro-5-(trifluoromethyl)phenyl]-2,2,16,20-tetraoxo-9,25-dioxo-2λ⁶-thia-1,12,15,19-tetrazatetracyclo[19.6.2.03,8.024,28]nonacosa-3(8),4,6,21(29),22,24(28)-hexaene-12-carboxylate

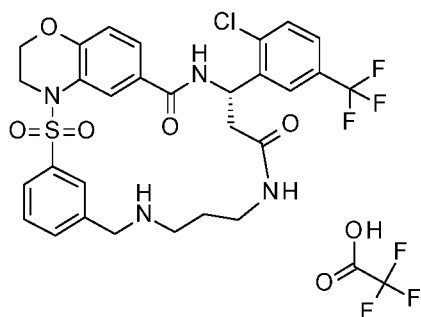
20 The title compound was prepared in analogy to **Example 8, step 8** starting with (3*S*)-3-[[4-[2-[2-[2-aminoethyl(*tert*-butoxycarbonyl)amino]ethoxy]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid instead of (3*S*)-3-[[4-[3-[[[(1*R*,4*R*)-5-(2-aminoethyl)-2,5-diazabicyclo[2.2.1]heptan-2-yl]phenyl]sulfonyl]spiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid hydrochloride salt and stirring at 25 °C for 1 hour instead
25 of 16 hours. White solid. MS m/z: 653.2 [M-BOC+H]⁺, ESI pos.

Step 7: (18*S*)-18-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-9,25-dioxo-2λ⁶-thia-1,12,15,19-tetrazatetracyclo[19.6.2.03,8.024,28]nonacosa-3(8),4,6,21(29),22,24(28)-hexaene-16,20-dione 2,2,2-trifluoroacetic acid salt

30 The title compound was prepared in analogy to **Example 51, step 6** starting with *tert*-butyl (18*S*)-18-[2-chloro-5-(trifluoromethyl)phenyl]-2,2,16,20-tetraoxo-9,25-dioxo-2λ⁶-thia-

1,12,15,19-tetrazatetracyclo[19.6.2.03,8.024,28]nonacosa-3(8),4,6,21(29),22,24(28)-hexaene-12-carboxylate instead of *tert*-butyl (17*S*,25*R*)-17-[2-chloro-5-(trifluoromethyl)phenyl]-22-fluoro-25-methyl-2,2,15,19-tetraoxo-24-oxa-2 λ^6 -thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.023,27]nonacosa-3,5,7(29),20(28),21,23(27)-hexaene-10-carboxylate. Yellow solid. MS *m/z*: 653.1 [M+H]⁺, ESI pos.

Example 63: (16*S*)-16-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-23-oxa-2 λ^6 -thia-1,9,13,17-tetrazatetracyclo[17.6.2.13,7.022,26]octacosa-3(28),4,6,19(27),20,22(26)-hexaene-14,18-dione 2,2,2-trifluoroacetic acid salt



10 Step 1: Ethyl (3*S*)-3-[[4-[3-[(*tert*-butoxycarbonylamino)methyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate

The title compound was prepared in analogy to **Intermediate C13, step 2** starting with ethyl (3*S*)-3-[[4-(3-bromophenyl)sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate (**Intermediate D5**) instead of methyl 4-[(6-bromo-2-pyridyl)sulfonyl]-2,3-dihydro-1,4-benzoxazine-6-carboxylate. Colorless oil. MS *m/z*: 726.4 [M+H]⁺, ESI pos.

Step 2: Ethyl (3*S*)-3-[[4-[3-(aminomethyl)phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate 2,2,2-trifluoroacetic acid salt

The title compound was prepared in analogy to **Intermediate D11, step 3** using ethyl (3*S*)-3-[[4-[3-[(*tert*-butoxycarbonylamino)methyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate instead of ethyl (3*S*)-3-[2,5-bis(trifluoromethyl)phenyl]-3-[[4-[3-[2-(*tert*-butoxycarbonylamino)ethyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate. White solid. MS *m/z*: 626.3 [M+H]⁺, ESI pos.

Step 3: Ethyl (3*S*)-3-[2-chloro-5-(trifluoromethyl)phenyl]-3-[[4-[3-[[3-(1,3-dioxoisindolin-2-yl)propylamino]methyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate 2,2,2-trifluoroacetic acid salt

To solution of ethyl (3*S*)-3-[[4-[3-(aminomethyl)phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate 2,2,2-trifluoroacetic acid salt (300 mg, 0.41 mmol) in methanol (10 ml) was added NaBH₃CN (255 mg, 4.05 mmol) and 3-(1,3-dioxoisindolin-2-yl)propanal (82 mg, 0.41 mmol). The reaction mixture was stirred at 25 °C for 2 hours. The reaction mixture was evaporated and the crude product was purified by reverse phase flash chromatography (Flash Spherical C18, 20-35 µm; 100 Å; gradient water with 0.1 % TFA and acetonitrile) followed by lyophilisation to give ethyl (3*S*)-3-[2-chloro-5-(trifluoromethyl)phenyl]-3-[[4-[3-[[3-(1,3-dioxoisindolin-2-yl)propylamino]methyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate 2,2,2-trifluoroacetic acid salt (260 mg, 67 % yield) as a white solid. MS m/z: 813.4 [M+H]⁺, ESI pos.

Step 4: Ethyl (3*S*)-3-[[4-[3-[[*tert*-butoxycarbonyl]-3-(1,3-dioxoisindolin-2-yl)propyl]amino]methyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate

The title compound was prepared in analogy to **Example 51, step 2** starting with ethyl (3*S*)-3-[2-chloro-5-(trifluoromethyl)phenyl]-3-[[4-[3-[[3-(1,3-dioxoisindolin-2-yl)propylamino]methyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate 2,2,2-trifluoroacetic acid salt instead of ethyl (3*S*)-3-[2-chloro-5-(trifluoromethyl)phenyl]-3-[[4-[3-[[3-(1,3-dioxoisindolin-2-yl)propylamino]ethyl]phenyl]sulfonyl-8-fluoro-2-methyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate. Colorless oil. MS m/z: 913.4 [M+H]⁺, ESI pos.

Step 5: 2-[3-[*tert*-Butoxycarbonyl]-[3-[[6-[(1*S*)-2-carboxy-1-[2-chloro-5-(trifluoromethyl)phenyl]ethyl]carbamoyl]-2,3-dihydro-1,4-benzoxazin-4-yl]sulfonyl]phenyl]methyl]amino]propylcarbamoyl]benzoic acid

The title compound was prepared in analogy to **Example 49, step 3** starting with ethyl (3*S*)-3-[[4-[3-[[*tert*-butoxycarbonyl]-3-(1,3-dioxoisindolin-2-yl)propyl]amino]methyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoate instead of ethyl (3*S*)-3-[2,5-bis(trifluoromethyl)phenyl]-3-[[4-[3-[2-[*tert*-butoxycarbonyl]-3-(1,3-dioxoisindolin-2-yl)propyl]amino]ethyl]phenyl]sulfonyl-

2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]propanoate and stirring at 25 °C for 18 hours instead of 2 hours. White solid. MS m/z: 903.5 [M+H]⁺, ESI pos.

Step 6: (3*S*)-3-[[4-[3-[[3-aminopropyl(*tert*-butoxycarbonyl)amino]methyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid 2,2,2-trifluoroacetic acid salt

The title compound was prepared in analogy to **Example 5, step 9** starting with 2-[3-[*tert*-butoxycarbonyl-[[3-[[6-[[[(1*S*)-2-carboxy-1-[2-chloro-5-(trifluoromethyl)phenyl]ethyl]carbonyl]-2,3-dihydro-1,4-benzoxazin-4-yl]sulfonyl]phenyl]methyl]amino]propylcarbonyl]benzoic acid instead of 2-[[[(4*R*)-4-(*tert*-butoxycarbonylamino)-5-[3-[6-[[[(1*S*)-2-carboxy-1-[2-chloro-5-(trifluoromethyl)phenyl]ethyl]carbonyl]spiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-4-yl]sulfonylphenoxy]pentyl]carbonyl]benzoic acid. White solid. MS m/z: 755.4 [M+H]⁺, ESI pos.

Step 7: *tert*-Butyl (16*S*)-16-[2-chloro-5-(trifluoromethyl)phenyl]-2,2,14,18-tetraoxo-23-oxa-2λ⁶-thia-1,9,13,17-tetrazatetracyclo[17.6.2.13,7.022,26]octacos-3(28),4,6,19(27),20,22(26)-hexaene-9-carboxylate

The title compound was prepared in analogy to **Example 8, step 8** starting with (3*S*)-3-[[4-[3-[[3-aminopropyl(*tert*-butoxycarbonyl)amino]methyl]phenyl]sulfonyl-2,3-dihydro-1,4-benzoxazine-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid 2,2,2-trifluoroacetic acid salt instead of (3*S*)-3-[[4-[3-[(1*R*,4*R*)-5-(2-aminoethyl)-2,5-diazabicyclo[2.2.1]heptan-2-yl]phenyl]sulfonylspiro[3*H*-1,4-benzoxazine-2,1'-cyclopropane]-6-carbonyl]amino]-3-[2-chloro-5-(trifluoromethyl)phenyl]propanoic acid hydrochloride salt and stirring at 25 °C for 1 hour instead of 16 hours. White solid. MS m/z: 737.4 [M+H]⁺, ESI pos.

Step 8: (16*S*)-16-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-23-oxa-2λ⁶-thia-1,9,13,17-tetrazatetracyclo[17.6.2.13,7.022,26]octacos-3(28),4,6,19(27),20,22(26)-hexaene-14,18-dione 2,2,2-trifluoroacetic acid salt

The title compound was prepared in analogy to **Example 51, step 6** starting with *tert*-butyl (16*S*)-16-[2-chloro-5-(trifluoromethyl)phenyl]-2,2,14,18-tetraoxo-23-oxa-2λ⁶-thia-1,9,13,17-tetrazatetracyclo[17.6.2.13,7.022,26]octacos-3(28),4,6,19(27),20,22(26)-hexaene-9-carboxylate instead of *tert*-butyl (17*S*,25*R*)-17-[2-chloro-5-(trifluoromethyl)phenyl]-22-fluoro-25-methyl-2,2,15,19-tetraoxo-24-oxa-2λ⁶-thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.023,27]nonacos-

3,5,7(29),20(28),21,23(27)-hexaene-10-carboxylate. White solid. MS m/z: 637.3 [M+H]⁺, ESI pos.

MET Inhibitory Activity

CellTiter-Glo®

- 5 2500 cells per well of EBC-1 and 2000 cells per well of Ba/F3 were seeded in a 384-well plate and let attach overnight. Cells were treated with MET inhibitors. Compound dilution was prepared using the Tecan D300e dispenser with T8+ dispense cassette (10 µM highest concentration, 8-times 1:3 dilution). Normalization to ≤ 1 % DMSO final concentration was applied. Cells were incubated for 3 days at 37°C RT, 100% RH, and 5% CO₂. CellTiter-Glo®
- 10 Luminescent Cell Viability Assay (Promega G7572) was performed to access cell viability. IC₅₀ values were calculated using GraphPad Prism software.

Cell line work

- The EBC-1 cell line, derived from human lung tissue, is associated with squamous cell lung carcinoma. This epithelial cell line, known to harbor MET gene amplifications, is typically
- 15 grown in Eagle's Minimum Essential Medium (EMEM) supplemented with 10% fetal bovine serum at 37°C in a 5% CO₂ atmosphere. It has been authenticated via short tandem repeat (STR) profiling and is deposited in the Japanese Collection of Research Bioresources (JCRB) Cell Bank (accession number JCRB0126).

Nucleofection

- 20 The optimization of nucleofection conditions was crucial to maximize transfection efficiency, ensure reproducibility and specificity of results, and promote cost-effectiveness by minimizing reagent usage. This optimization involved adjusting the nucleofection solution, pulse code, and the ratio between Cas9, sgRNA, and ssODN. The nucleofection was performed according to the manufacturer's protocol using a 4D-Nucleofector and Amaxa 96-well shuttle Device (Lonza).
- 25 Cells were nucleofected with TrueCut™ Cas9 Protein v2 (ThermoFisher A36499), a guide RNA specific to the insertion site (sgRNA 5'-ttctttatcatacatgtctc-3' according to SEQ ID NO:1) and single-stranded oligodeoxynucleotides with homology arms to the insertion site (ssODN2SNPMUT 5'-
- TTGTCCTTTCTGTAGGCTGGATGAAAAATTCACAGTCAAGGTTGCTGATTTTGGTCTC
- 30 GCGAGAGTCATGTATGATAAAGAATACTATAGTGTACACAACAAAACAGGTGCAAA

GCTGCCAGTGAAGTGGATGGCTTTGG-3' according to SEQ ID NO:2) adapted from Collie et al ¹.

The ssODN2SNPMUT harbors the desired D1228V mutation, a mutation in the PAM site, as well as a silent mutation.

- 5 Eight days post nucleofection, EBC-1 cells were grown for ten days in the presence of Capmatinib (10 - 30 nM). Capmatinib is designed to bind to the ATP-binding pocket of the MET receptor tyrosine kinase, inhibiting the kinase activity of the MET protein and preventing the downstream signaling that promotes cell growth and division. Since D1228V impairs capmatinib binding, MET D1228V mutant cell lines can grow under capmatinib pressure, while all non-
10 transfected MET wild-type cells will be eliminated.

Following the growth period, the pool of nucleofected wells was singularized via serial dilution to obtain single-cell clones, following the protocol from Corning ². The cells were then subjected to single-cell sorting

New Generation Sequencing

- 15 Following the generation of the D1228V cell line as described above, the mutation was to be confirmed using Next-Generation Sequencing (NGS). The NGS was to be performed by an external vendor, MicroSynth, who would utilize deep allele sequencing.

- Prior to sending the samples for sequencing, DNA extraction was performed on all the cell lines, including EBC-1 WT (MET-WT), EBC-1 D1228V (MET-Mutant)). The extraction was
20 conducted using the DNeasy Kit by Qiagen, following the manufacturer's protocol.

The extracted DNA samples were then sent to MicroSynth for deep allele sequencing, a method chosen for its ability to detect even low-frequency variants, making it suitable for confirming the presence of the D1228V mutation.

MET D1228H Biochemical TR-FRET IC₅₀ Assay

- 25 Reaction reagents were prepared in 25mM HEPES pH 7.5, 10mM MgCl₂, 1mM EGTA, 0.005% BSA and 0.01% NP40. Enzymatic reactions were ran for 60 minutes in an 8μL volume consisting of 10nM human dephosphorylated MET D1228H (aa1048-1348), 30μM ATP (at K_M) and 1μM biotin gastrin peptide. After 60mins reactions were terminated by the addition of 4μL of Stop and Detect reagents (25mM HEPES pH7.5, 60mM EDTA, 900mM KF, 0.005% BSA,
30 1.5nM Europium labeled anti-phosphotyrosine antibody and 75nM streptavidin conjugated

allophycocyanin). Time-resolved fluorescence energy transfer (TR-FRET) then determined the amount of phosphorylated biotin gastrin.

To determine compound IC₅₀'s reactions were incubated for 60minutes at room temperature in the presence of a 12-point concentration response curve of compound (starting concentration
5 50μM; 1 in 3 dilution between each point; 2% DMSO). TR-FRET was then plotted against the compound concentration to obtain an IC₅₀ as fitted via non-linear regression.

Results of the MET D1228H Biochemical TR-FRET IC₅₀ assay are provided for compounds of formula (I) in Table 1.

Compounds of the present invention are MET inhibitors. Thus, in one aspect, the present
10 invention provides the use of compounds of formula (I) as described herein for inhibiting MET kinase in a human.

In a further aspect, the present invention provides compounds of formula (I) as described herein for use in a method of inhibiting MET kinase in a human.

In a further aspect, the present invention provides the use of compounds of formula (I) as
15 described herein for the preparation of a medicament for inhibiting MET kinase in a human.

In a further aspect, the present invention provides a method for inhibiting MET kinase in a human, which method comprises administering an effective amount of a compound of formula (I) as described herein to the human.

Table 1

Example No.	MET D1228H TR-FRET biochemical assay, IC₅₀ (μM)	MET D1228V EBC-1 CellTiter-Glo® cell viability assay, IC₅₀ (μM)	MET EBC-1 CellTiter-Glo® cell viability assay, IC₅₀ (μM)
1	0.045	0.487	0.194
2	0.106	1.110	0.517
3	0.013	0.301	0.096
4	0.012	0.211	0.081
5	0.026	0.468	0.093
6	0.012	0.120	0.011

7	0.083	1.040	0.535
8	0.105	0.603	0.131
9	0.028	0.151	0.094
10	0.011	0.252	0.033
11	0.011	0.363	0.048
12	0.012	0.133	0.030
13	0.014	0.503	0.036
14	0.039	0.266	0.030
15	0.017	1.995	0.371
16	0.013	0.161	0.044
17	0.025	0.373	0.101
18	0.025	0.260	0.068
19	0.075	0.724	0.239
20	0.082	0.843	0.286
21	0.015	0.161	0.103
22	0.140	1.494	0.510
23	0.015	0.351	0.056
24	0.052	0.573	0.191
25	0.012	0.144	0.038
26	0.014	0.128	0.062
27	0.028	0.530	0.199
28	0.015	0.056	0.017
29	0.023	0.264	0.086
30	0.118	2.393	1.443
31	0.012	0.119	0.042
32	0.016	0.711	0.025
33	0.014	0.044	0.015
34	0.016	0.050	0.015
35	0.019	0.177	0.055
36	0.050	1.434	1.041
37	0.008	0.312	0.099
38	0.026	0.747	0.195
39	0.009	0.329	0.091

40	0.015	0.426	0.118
41	0.122		
42	0.025	0.099	0.033
43	0.015	0.281	0.064
44	0.015	0.090	0.022
45	0.025	0.687	0.148
46	0.015	0.236	0.051
47	0.012	0.037	0.013
48	0.007	0.330	0.080
49	0.013	0.216	0.075
50	0.012	0.301	0.111
51	0.017	0.053	0.020
52	0.008	0.160	0.065
53	0.055	>10.000	3.543
54	0.012	0.645	0.164
55	0.008	0.022	0.008
56	0.013	0.565	0.229
57	0.011	0.099	0.032
58	0.026	0.481	0.207
59	0.015	1.786	0.731
60	0.014	0.099	0.034
61	0.052	2.722	1.071
62	0.026	1.696	0.482
63	0.050	4.986	0.760

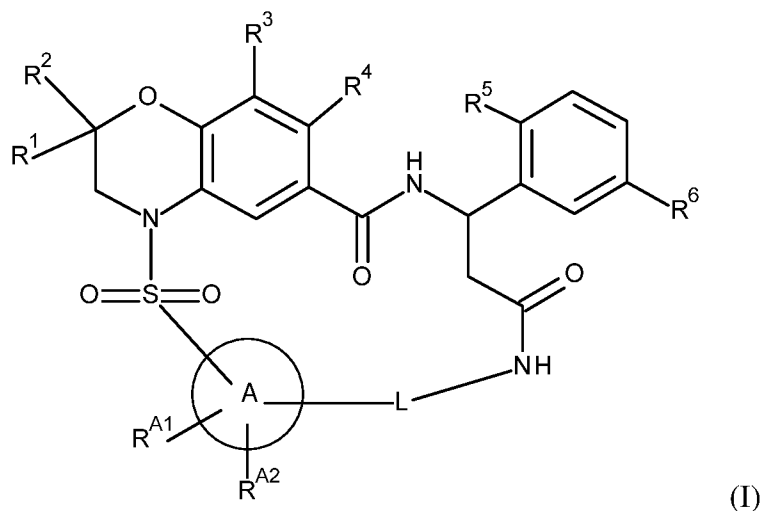
In one aspect, the present invention provides compounds of formula (I) and their pharmaceutically acceptable salts or esters as described herein, wherein said compounds of formula (I) and their pharmaceutically acceptable salts or esters have IC₅₀'s for MET inhibition
5 below 25 μ M, particularly below 10 μ M, more particularly below 5 μ M as measured in the MET inhibitory assay described herein.

In one embodiment, compounds of formula (I) and their pharmaceutically acceptable salts or esters as described herein have IC₅₀ (MET inhibition) values between 0.000001 μ M and 25 μ M,

particular compounds have IC_{50} values between 0.000005 μ M and 10 μ M, further particular compounds have IC_{50} values between 0.00005 μ M and 5 μ M, as measured in the MET assay described herein.

CLAIMS

1. A compound of formula (I)



5 or a pharmaceutically acceptable salt thereof, wherein:

ring A is C₆₋₁₀-aryl, 5- to 10-membered heteroaryl comprising 1 to 3 heteroatoms independently selected from N, O, S and the remaining atoms being carbon, or 3- to 10-membered heterocyclyl comprising 1 to 3 heteroatoms independently selected from N, O, S and the remaining atoms being carbon;

10 R^{A1} and R^{A2} are independently hydrogen, halogen, C₁₋₆-alkyl, cyano, C₁₋₆-alkoxy, halo-C₁₋₆-alkyl;

R¹ is hydrogen, halogen, C₁₋₆-alkyl, halo-C₁₋₆-alkyl, or hydroxy-C₁₋₆-alkyl; and

R² is hydrogen or C₁₋₆-alkyl; or

R¹ and R² together with the carbon to which they are attached form C₃₋₄-cycloalkyl

15 optionally substituted with hydroxy or halogen;

R³ is hydrogen or halogen;

R⁴ is hydrogen, halogen, or C₁₋₆-alkoxy;

R⁵ is halogen, C₁₋₆-alkyl, or halo-C₁₋₆-alkyl;

R⁶ is halogen or halo-C₁₋₆-alkyl;

20 L is

(i) a group *-X¹-L¹¹-Y¹-L¹²-; wherein:

X¹ is -O-, -NR⁷-;

Y^1 is $-\text{CH}_2-$, $-\text{O}-$, $-\text{NR}^8-$, or $-\text{CH}=\text{CH}-$, $-\text{C}\equiv\text{C}-$;

L^{11} is C_{1-3} -alkylene optionally substituted with R^9 ;

L^{12} is C_{1-3} -alkylene optionally substituted with R^{10} ;

R^7 and R^8 are independently hydrogen, C_{1-6} -alkyl, C_{1-6} -alkyl- $\text{C}(\text{O})-$

5 R^9 and R^{10} are independently hydroxy, amino, halogen, C_{1-6} -alkyl, or hydroxy-

C_{1-6} -alkyl, amino- C_{1-6} -alkyl; and

the asterisk indicates the point of attachment of L to ring A; or

(ii) a group $^*\text{-X}^2\text{-L}^{21}\text{-Y}^2\text{-L}^{22}\text{-}$; wherein:

X^2 is $-\text{C}\equiv\text{C}-$, $-\text{CH}=\text{CH}-$

10 Y^2 is $-\text{O}-$, $-\text{NR}^{11}-$;

L^{21} and L^{22} are independently C_{1-3} -alkylene optionally substituted with hydroxy, amino, halogen, C_{1-6} -alkyl, hydroxy- C_{1-6} -alkyl, or amino- C_{1-6} -alkyl;

R^{11} is hydrogen, C_{1-6} -alkyl, or C_{1-6} -alkyl- $\text{C}(\text{O})-$; and

the asterisk indicates the point of attachment of L to ring A; or

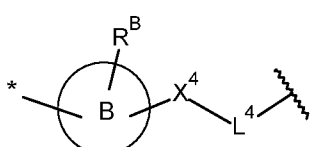
15 (iii) a group $^*\text{-L}^{31}\text{-X}^3\text{-L}^{32}\text{-}$; wherein:

X^3 is $-\text{O}-$, $-\text{NR}^{12}-$;

L^{31} and L^{32} are independently C_{1-3} -alkylene optionally substituted with hydroxy, amino, halogen, C_{1-6} -alkyl, hydroxy- C_{1-6} -alkyl, or amino- C_{1-6} -alkyl;

R^{12} is hydrogen, C_{1-6} -alkyl, or C_{1-6} -alkyl- $\text{C}(\text{O})-$; and

20 the asterisk indicates the point of attachment of L to ring A; or

(iv) a group ; wherein

X^4 is a covalent bond, $-\text{O}-$, $-\text{NR}^{13}-$, or $-\text{CH}(\text{OH})-$, $-\text{CH}(\text{NH}_2)-$,

L^4 is C_{1-3} -alkylene;

R^{13} is hydrogen, C_{1-6} -alkyl, or C_{1-6} -alkyl- $\text{C}(\text{O})-$;

25 R^B is hydrogen, hydroxy, amino, halogen, C_{1-6} -alkyl;

B is C_{3-10} -cycloalkyl, or 3- to 10-membered heterocyclyl comprising 1 to 3 heteroatoms independently selected from N, O, S and the remaining atoms being carbon; and

the asterisk indicates the point of attachment of L to ring A.

30 2. The compound of formula (I) according to claim 1, or a pharmaceutically acceptable salt thereof, wherein ring A is phenyl, 5- to 6-membered heteroaryl comprising 1 to 3

heteroatoms independently selected from N, O, S and the remaining atoms being carbon, or 5- to 6-membered heterocyclyl comprising 1 to 3 heteroatoms independently selected from N, O, S and the remaining atoms being carbon.

3. The compound of formula (I) according to claim 1, or a pharmaceutically acceptable salt thereof, wherein ring A is C₆-C₁₀-aryl.
4. The compound of formula (I) according to claims 1 or 2, or a pharmaceutically acceptable salt thereof, wherein ring A is phenyl, pyridyl, or piperidyl.
5. The compound of formula (I) according to any one of claims 1 to 4, or a pharmaceutically acceptable salt thereof, wherein ring A is phenyl.
- 10 6. The compound of formula (I) according to any one of claims 1 to 5, or a pharmaceutically acceptable salt thereof, wherein R^{A1} and R^{A2} are independently hydrogen, halogen, C₁₋₆-alkyl.
7. The compound of formula (I) according to any one of claims 1 to 6, or a pharmaceutically acceptable salt thereof, wherein R^{A1} is hydrogen, halogen, C₁₋₆-alkyl; and R^{A2} is hydrogen.
- 15 8. The compound of formula (I) according to any one of claims 1 to 7, or a pharmaceutically acceptable salt thereof, wherein R^{A1} is hydrogen, fluoro, or methyl; and R^{A2} is hydrogen.
9. The compound of formula (I) according to any one of claims 1 to 8, or a pharmaceutically acceptable salt thereof, wherein R^{A1} and R^{A2} are hydrogen.
10. The compound of formula (I) according to any one of claims 1 to 9, or a pharmaceutically acceptable salt thereof, wherein R¹ is hydrogen, halogen, C₁₋₆-alkyl, halo-C₁₋₆-alkyl, or hydroxy-C₁₋₆-alkyl; and R² is hydrogen; or R¹ and R² together with the carbon to which they are attached form C₃₋₄-cycloalkyl optionally substituted with hydroxy or halogen.
- 20 11. The compound of formula (I) according to any one of claims 1 to 10, or a pharmaceutically acceptable salt thereof, wherein R¹ is hydrogen, C₁₋₆-alkyl, or halo-C₁₋₆-alkyl; and R² is hydrogen.
- 25 12. The compound of formula (I) according to any one of claims 1 to 10, or a pharmaceutically acceptable salt thereof, wherein R¹ is hydrogen, fluoro, methyl, hydroxymethyl, or

fluoromethyl, and R^2 is hydrogen; or R^1 and R^2 together with the carbon to which they are attached form cyclopropyl or fluorocyclopropyl.

13. The compound of formula (I) according to any one of claims 1 to 12, or a pharmaceutically acceptable salt thereof, wherein R^1 is hydrogen, methyl or fluoromethyl; and R^2 is hydrogen.
14. The compound of formula (I) according to any one of claims 1 to 13, or a pharmaceutically acceptable salt thereof, wherein R^3 is hydrogen or fluoro.
15. The compound of formula (I) according to any one of claims 1 to 14, or a pharmaceutically acceptable salt thereof, wherein R^4 is hydrogen or halogen.
16. The compound of formula (I) according to any one of claims 1 to 14, or a pharmaceutically acceptable salt thereof, wherein R^4 is hydrogen, fluoro, or methoxy.
17. The compound of formula (I) according to any one of claims 1 to 16, or a pharmaceutically acceptable salt thereof, wherein R^4 is hydrogen or fluoro.
18. The compound of formula (I) according to any one of claims 1 to 17, or a pharmaceutically acceptable salt thereof, wherein R^5 is halogen or C_{1-6} -alkyl.
19. The compound of formula (I) according to any one of claims 1 to 17, or a pharmaceutically acceptable salt thereof, wherein R^5 is chloro, fluoro, methyl, or $-CF_3$.
20. The compound of formula (I) according to any one of claims 1 to 19, or a pharmaceutically acceptable salt thereof, wherein R^5 is chloro or methyl.
21. The compound of formula (I) according to any one of claims 1 to 20, or a pharmaceutically acceptable salt thereof, wherein R^6 is halo- C_{1-6} -alkyl.
22. The compound of formula (I) according to any one of claims 1 to 20, or a pharmaceutically acceptable salt thereof, wherein R^6 is chloro or $-CF_3$.
23. The compound of formula (I) according to any one of claims 1 to 22, or a pharmaceutically acceptable salt thereof, wherein R^6 is $-CF_3$.
24. The compound of formula (I) according to any one of claims 1 to 23, or a pharmaceutically acceptable salt thereof, wherein L is

(i) a group $*-X^1-L^{11}-Y^1-L^{12}-$; wherein:

X^1 is $-O-$, $-NR^7-$;

Y^1 is $-CH_2-$, $-O-$, $-NR^8-$, or $-CH=CH-$;

L^{11} is C_{1-3} -alkylene optionally substituted with R^9 ;

5 L^{12} is C_{1-3} -alkylene optionally substituted with R^{10} ;

R^7 and R^8 are independently hydrogen, C_{1-6} -alkyl;

R^9 and R^{10} are independently hydroxy, amino, C_{1-6} -alkyl, or hydroxy- C_{1-6} -alkyl;

and the asterisk indicates the point of attachment of L to ring A; or

(ii) a group $*-X^2-L^{21}-Y^2-L^{22}-$; wherein:

10 X^2 is $-C\equiv C-$;

Y^2 is $-O-$, $-NR^{11}-$;

L^{21} and L^{22} are independently C_{1-3} -alkylene;

R^{11} is hydrogen, or C_{1-6} -alkyl- $C(O)-$; and

the asterisk indicates the point of attachment of L to ring A; or

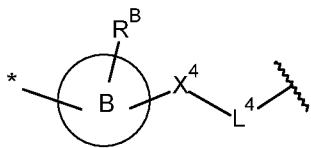
15 (iii) a group $*-L^{31}-X^3-L^{32}-$; wherein:

X^3 is $-O-$, $-NR^{12}-$;

L^{31} and L^{32} are independently C_{1-3} -alkylene;

R^{12} is hydrogen, C_{1-6} -alkyl; and

and the asterisk indicates the point of attachment of L to ring A; or

20 (iv) a group ; wherein

X^4 is a covalent bond, $-O-$, $-NR^{13}-$, or $-CH(OH)-$;

L^4 is C_{1-3} -alkylene;

R^{13} is hydrogen;

R^B is hydrogen or hydroxy;

25 B is C_{5-6} -cycloalkyl, or 5- to 7-membered heterocyclyl comprising 1 to 2 heteroatoms independently selected from N and O and the remaining atoms being carbon; and

the asterisk indicates the point of attachment of L to ring A.

25. The compound of formula (I) according to any one of claims 1 to 24, or a pharmaceutically acceptable salt thereof, wherein L is

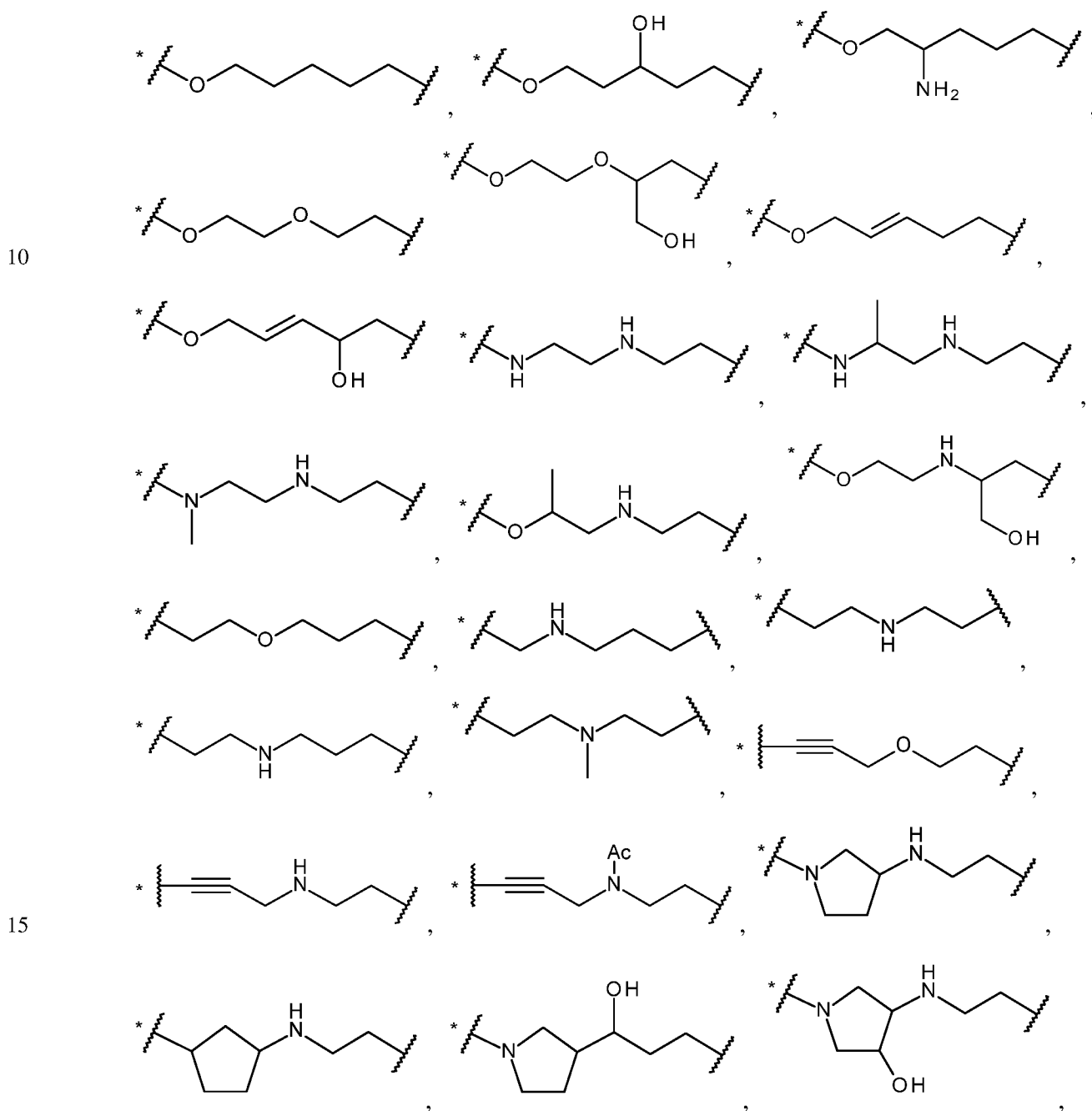
a group $*-L^{31}-X^3-L^{32}-$; wherein:

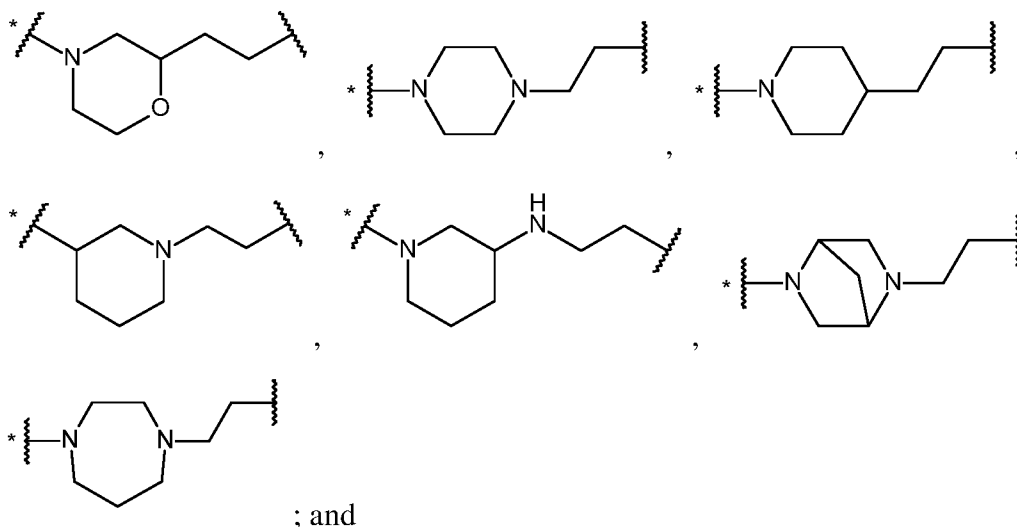
X^3 is $-NH-$;

5 L^{31} and L^{32} are independently C_{1-3} -alkylene;

the asterisk indicates the point of attachment of L to ring A.

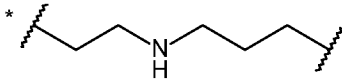
26. The compound of formula (I) according to any one of claims 1 to 24, or a pharmaceutically acceptable salt thereof, wherein L is





the asterisk indicates the point of attachment of L to ring A.

- 5 27. The compound of formula (I) according to any one of claims 1 to 26, or a pharmaceutically

acceptable salt thereof, wherein L is ; and the asterisk indicates the point of attachment of L to ring A.

28. The compound of formula (I) according to claim 1, or a pharmaceutically acceptable salt thereof, wherein:

- 10 ring A is phenyl, 5- to 6-membered heteroaryl comprising 1 to 3 heteroatoms independently selected from N, O, S and the remaining atoms being carbon, or 5- to 6-membered heterocyclyl comprising 1 to 3 heteroatoms independently selected from N, O, S and the remaining atoms being carbon;
- R^{A1} and R^{A2} are independently hydrogen, halogen, C_{1-6} -alkyl;
- 15 R^1 is hydrogen, halogen, C_{1-6} -alkyl, halo- C_{1-6} -alkyl, or hydroxy- C_{1-6} -alkyl; and R^2 is hydrogen; or R^1 and R^2 together with the carbon to which they are attached form C_{3-4} -cycloalkyl optionally substituted with hydroxy or halogen;
- R^3 is hydrogen or halogen;
- R^4 is hydrogen, halogen, or C_{1-6} -alkoxy;
- 20 R^5 is halogen, C_{1-6} -alkyl, or halo- C_{1-6} -alkyl;
- R^6 is halogen or halo- C_{1-6} -alkyl;
- L is

- (i) a group $*-X^1-L^{11}-Y^1-L^{12}-$; wherein:

X^1 is $-O-$, $-NR^7-$;

Y^1 is $-CH_2-$, $-O-$, $-NR^8-$, or $-CH=CH-$;

L^{11} is C_{1-3} -alkylene optionally substituted with R^9 ;

L^{12} is C_{1-3} -alkylene optionally substituted with R^{10} ;

5 R^7 and R^8 are independently hydrogen, C_{1-6} -alkyl;

R^9 and R^{10} are independently hydroxy, amino, C_{1-6} -alkyl, or hydroxy- C_{1-6} -alkyl;

and the asterisk indicates the point of attachment of L to ring A; or

(ii) a group $*-X^2-L^{21}-Y^2-L^{22}-$; wherein:

X^2 is $-C\equiv C-$;

10 Y^2 is $-O-$, $-NR^{11}-$;

L^{21} and L^{22} are independently C_{1-3} -alkylene;

R^{11} is hydrogen, or C_{1-6} -alkyl- $C(O)-$; and

the asterisk indicates the point of attachment of L to ring A; or

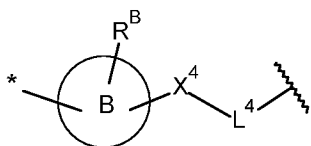
(iii) a group $*-L^{31}-X^3-L^{32}-$; wherein:

15 X^3 is $-O-$, $-NR^{12}-$;

L^{31} and L^{32} are independently C_{1-3} -alkylene;

R^{12} is hydrogen, C_{1-6} -alkyl; and

and the asterisk indicates the point of attachment of L to ring A; or

(iv) a group ; wherein

20 X^4 is a covalent bond, $-O-$, $-NR^{13}-$, or $-CH(OH)-$;

L^4 is C_{1-3} -alkylene;

R^{13} is hydrogen;

R^B is hydrogen or hydroxy;

B is C_{5-6} -cycloalkyl, or 5- to 7-membered heterocyclyl comprising 1 to 2

25 heteroatoms independently selected from N and O and the remaining atoms being carbon; and

the asterisk indicates the point of attachment of L to ring A.

29. The compound of formula (I) according to claim 28, or a pharmaceutically acceptable salt thereof, wherein:

30 ring A is phenyl;

R^{A1} and R^{A2} are hydrogen;

R^1 is hydrogen, C_{1-6} -alkyl, or halo- C_{1-6} -alkyl; and

R^2 is hydrogen;

R^3 is hydrogen or halogen;

5 R^4 is hydrogen or halogen;

R^5 is halogen or C_{1-6} -alkyl;

R^6 is halo- C_{1-6} -alkyl;

L is a group $*-L^{31}-X^3-L^{32}-$; wherein:

X^3 is $-NH-$;

10 L^{31} and L^{32} are independently C_{1-3} -alkylene; and

the asterisk indicates the point of attachment of L to ring A.

30. The compound of formula (I) according to claim 28, or a pharmaceutically acceptable salt thereof, wherein:

ring A is phenyl, pyridyl, or piperidyl;

15 R^{A1} and R^{A2} are hydrogen;

R^1 is hydrogen, fluoro, methyl, hydroxymethyl, or fluoromethyl, and R^2 is hydrogen; or

R^1 and R^2 together with the carbon to which they are attached form cyclopropyl or fluorocyclopropyl;

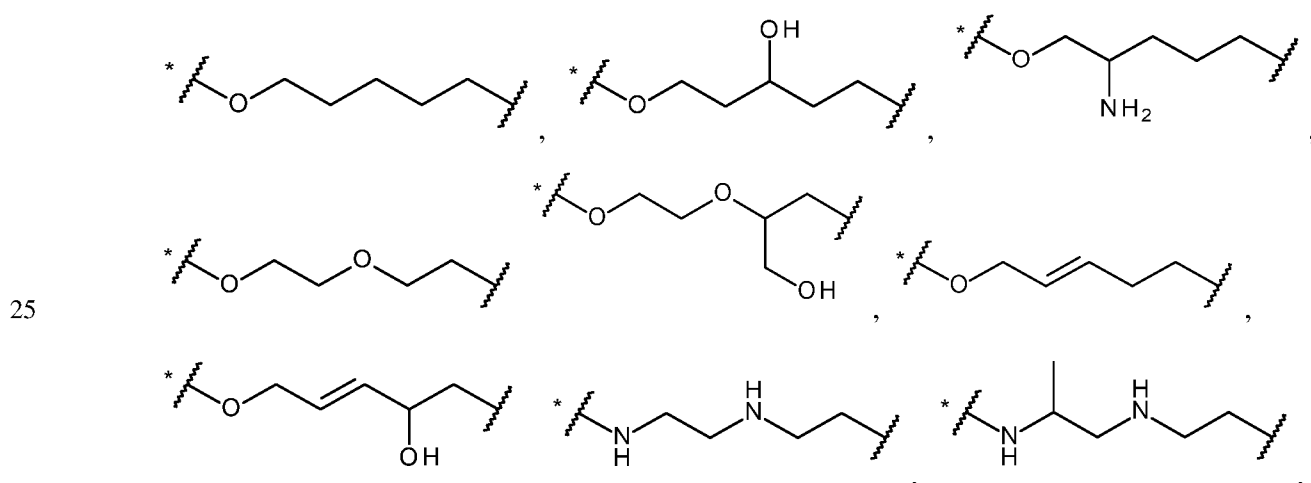
R^3 is hydrogen or fluoro;

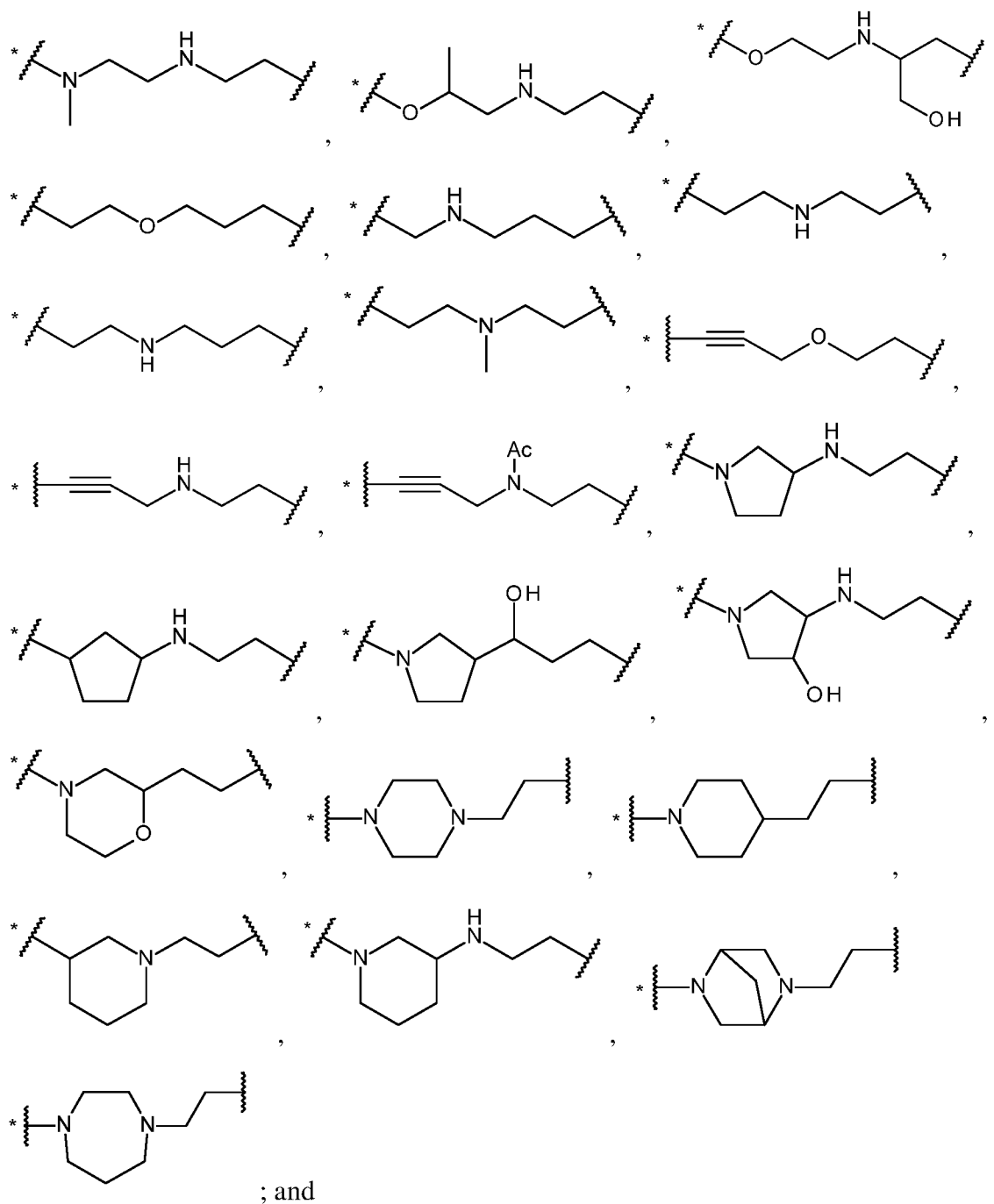
20 R^4 is hydrogen, fluoro, or methoxy;

R^5 is chloro, fluoro, methyl, or $-CF_3$;

R^6 is chloro or $-CF_3$;

L is





the asterisk indicates the point of attachment of L to ring A.

- 10 31. The compound of formula (I) according to claim 29, or a pharmaceutically acceptable salt thereof, wherein:

ring A is phenyl;

R^{A1} and R^{A2} are hydrogen;

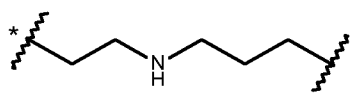
R¹ is hydrogen, methyl or fluoromethyl; and R² is hydrogen;

15 R³ is hydrogen or fluoro;

R⁴ is hydrogen or fluoro;

R⁵ is chloro or methyl;

R⁶ is -CF₃;

L is ; and

5 the asterisk indicates the point of attachment of L to ring A.

32. The compound of formula (I) according to claim 1, or a pharmaceutically acceptable salt thereof, wherein said compound of formula (I) is:

10 (17S)-17-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-diketo-8,11,24-trioxa-2λ⁶-thia-1,14,18-triazatetracyclo[18.6.2.13,7.023,27]nonacosa-3(29),4,6,20,22,27-hexaene-15,19-quinone;

(17S)-17-[2-fluoro-5-(trifluoromethyl)phenyl]-2,2-dioxo-8,11,24-trioxa-2λ⁶-thia-1,14,18-triazatetracyclo[18.6.2.13,7.023,27]nonacosa-3,5,7(29),20(28),21,23(27)-hexaene-15,19-dione;

15 (17S)-17-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-spiro[8,11,24-trioxa-2λ⁶-thia-1,14,18-triazatetracyclo[18.6.2.13,7.023,27]nonacosa-3,5,7(29),20(28),21,23(27)-hexaene-25,1'-cyclopropane]-15,19-dione;

(17S)-17-[2-chloro-5-(trifluoromethyl)phenyl]-5-methyl-2,2-dioxo-8,11,24-trioxa-2λ⁶-thia-1,14,18-triazatetracyclo[18.6.2.13,7.023,27]nonacosa-3,5,7(29),20(28),21,23(27)-hexaene-15,19-dione;

20 (10R,17S)-10-Amino-17-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-spiro[8,24-dioxa-2λ⁶-thia-1,14,18-triazatetracyclo[18.6.2.13,7.023,27]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-25,1'-cyclopropane]-15,19-dione;

25 (18S)-18-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-25-oxa-2λ⁶-thia-1,8,12,15,19-pentazapentacyclo[19.6.2.28,12.13,7.024,28]dotriaconta-3,5,7(32),21(29),22,24(28)-hexaene-16,20-dione;

(17S)-17-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-24-oxa-2λ⁶-thia-1,8,11,14,18-pentazapentacyclo[18.6.2.28,11.13,7.023,27]hentriaconta-3,5,7(31),20(28),21,23(27)-hexaene-15,19-dione;

(10R,12R,20S)-20-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-spiro[27-oxa-2 λ^6 -thia-1,8,14,17,21-pentazahexacyclo[21.6.2.13,7.08,12.010,14.026,30]dotriaconta-3(32),4,6,23(31),24,26(30)-hexaene-28,1'-cyclopropane]-18,22-dione;

5 (17S)-17-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-spiro[24-oxa-2 λ^6 -thia-1,8,14,18-tetrazapentacyclo[18.6.2.28,11.13,7.023,27]hentriaconta-3,5,7(31),20(28),21,23(27)-hexaene-25,1'-cyclopropane]-15,19-dione;

(11S,18S)-18-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-25-oxa-2 λ^6 -thia-1,8,12,15,19-pentazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-3,5,7(31),21(29),22,24(28)-hexaene-16,20-dione;

10 (11S,18S)-18-[2-chloro-5-(trifluoromethyl)phenyl]-5-methyl-2,2-dioxo-25-oxa-2 λ^6 -thia-1,8,12,15,19-pentazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-3,5,7(31),21(29),22,24(28)-hexaene-16,20-dione;

15 (11S,18S,26R)-18-[2-chloro-5-(trifluoromethyl)phenyl]-26-methyl-2,2-dioxo-25-oxa-2 λ^6 -thia-1,8,12,15,19-pentazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-3(31),4,6,21(29),22,24(28)-hexaene-16,20-dione;

(11S,18S)-18-[2-chloro-5-(trifluoromethyl)phenyl]-5-fluoro-2,2-dioxo-25-oxa-2 λ^6 -thia-1,8,12,15,19-pentazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-3,5,7(31),21(29),22,24(28)-hexaene-16,20-dione;

20 (11S,18S)-18-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-spiro[25-oxa-2 λ^6 -thia-1,8,12,15,19-pentazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-3(31),4,6,21(29),22,24(28)-hexaene-26,1'-cyclopropane]-16,20-dione;

(12S,19S)-19-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-26-oxa-2 λ^6 -thia-1,8,13,16,20-pentazapentacyclo[20.6.2.13,7.18,12.025,29]dotriaconta-3,5,7(32),22(30),23,25(29)-hexaene-17,21-dione;

25 (17S)-17-[2-chloro-5-(trifluoromethyl)phenyl]-8-methyl-2,2-dioxo-24-oxa-2 λ^6 -thia-1,8,11,14,18-pentazatetracyclo[18.6.2.13,7.023,27]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-15,19-dione;

30 (12R,18S)-18-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-spiro[11,25-dioxa-2 λ^6 -thia-1,8,15,19-tetrazapentacyclo[19.6.2.13,7.18,12.024,28]hentriaconta-3(31),4,6,21(29),22,24(28)-hexaene-26,1'-cyclopropane]-16,20-dione;

(12S,18S)-18-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-spiro[11,25-dioxa-2 λ^6 -thia-1,8,15,19-tetrazapentacyclo[19.6.2.13,7.18,12.024,28]hentriaconta-3(31),4,6,21(29),22,24(28)-hexaene-26,1'-cyclopropane]-16,20-dione;

5 (12R,18S)-18-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-11,25-dioxa-2 λ^6 -thia-1,8,15,19-tetrazapentacyclo[19.6.2.13,7.18,12.024,28]hentriaconta-3(31),4,6,21(29),22,24(28)-hexaene-16,20-dione;

(12S,18S)-18-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-11,25-dioxa-2 λ^6 -thia-1,8,15,19-tetrazapentacyclo[19.6.2.13,7.18,12.024,28]hentriaconta-3(31),4,6,21(29),22,24(28)-hexaene-16,20-dione;

10 (17S)-17-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-11,24-dioxa-2 λ^6 -thia-1,14,18-triazatetracyclo[18.6.2.13,7.023,27]nonacosa-3,5,7(29),20(28),21,23(27)-hexaene-8-yne-15,19-dione;

15 rac-(11S,18S)-18-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-25-oxa-2 λ^6 -thia-1,3,8,12,15,19-hexazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-21(29),22,24(28)-triene-16,20-dione;

(17S)-17-[2-chloro-5-(trifluoromethyl)phenyl]-6-fluoro-2,2-dioxo-24-oxa-2 λ^6 -thia-1,8,11,14,18-pentazatetracyclo[18.6.2.13,7.023,27]nonacosa-3,5,7(29),20(28),21,23(27)-hexaene-15,19-dione;

20 rac-(10E,17S)-17-[2-chloro-5-(trifluoromethyl)phenyl]-12-hydroxy-2,2-dioxo-8,24-dioxa-2 λ^6 -thia-1,14,18-triazatetracyclo[18.6.2.13,7.023,27]nonacosa-3(29),4,6,10,20(28),21,23(27)-heptaene-15,19-dione;

(17S)-17-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-8,24-dioxa-2 λ^6 -thia-1,11,14,18-tetrazatetracyclo[18.6.2.13,7.023,27]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-15,19-dione;

25 (17S)-17-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-8,24-dioxa-2 λ^6 -thia-1,14,18-triazatetracyclo[18.6.2.13,7.023,27]nonacosa-3,5,7(29),20(28),21,23(27)-hexaene-15,19-dione;

30 (17S)-17-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-8,24-dioxa-2 λ^6 -thia-1,14,18-triazatetracyclo[18.6.2.13,7.023,27]nonacosa-3,5,7(29),10,20(28),21,23(27)-heptaene-15,19-dione;

(9R,17S)-17-[2-chloro-5-(trifluoromethyl)phenyl]-9-methyl-2,2-dioxo-spiro[8,24-dioxa-2 λ^6 -thia-1,11,14,18-tetrazatetracyclo[18.6.2.13,7.023,27]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-25,1'-cyclopropane]-15,19-dione;

5 (9S,17S)-17-[2-chloro-5-(trifluoromethyl)phenyl]-9-methyl-2,2-dioxo-spiro[8,24-dioxa-2 λ^6 -thia-1,11,14,18-tetrazatetracyclo[18.6.2.13,7.023,27]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-25,1'-cyclopropane]-15,19-dione;

(12S,17S)-17-[2-chloro-5-(trifluoromethyl)phenyl]-12-(hydroxymethyl)-2,2-dioxo-8,11,24-trioxa-2 λ^6 -thia-1,14,18-triazatetracyclo[18.6.2.13,7.023,27]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-15,19-dione;

10 (11S,17S)-17-[2-chloro-5-(trifluoromethyl)phenyl]-11-hydroxy-2,2-dioxo-8,24-dioxa-2 λ^6 -thia-1,14,18-triazatetracyclo[18.6.2.13,7.023,27]nonacosa-3,5,7(29),20(28),21,23(27)-hexaene-15,19-dione;

15 (17S)-17-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-spiro[24-oxa-2 λ^6 -thia-1,11,14,18-tetrazatetracyclo[18.6.2.13,7.023,27]nonacosa-3,5,7(29),20(28),21,23(27)-hexaen-8-yne-25,1'-cyclopropane]-15,19-dione;

(17S)-11-acetyl-17-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-spiro[24-oxa-2 λ^6 -thia-1,11,14,18-tetrazatetracyclo[18.6.2.13,7.023,27]nonacosa-3,5,7(29),20(28),21,23(27)-hexaen-8-yne-25,1'-cyclopropane]-15,19-dione;

20 (19S)-19-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-26-oxa-2 λ^6 -thia-1,12,16,20-tetrazapentacyclo[20.6.2.13,7.18,12.025,29]dotriaconta-3(32),4,6,22(30),23,25(29)-hexaene-17,21-dione;

(11S,18S,26R)-18-[2-chloro-5-(trifluoromethyl)phenyl]-26-fluoro-2,2-dioxo-25-oxa-2 λ^6 -thia-1,8,12,15,19-pentazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-3(31),4,6,21(29),22,24(28)-hexaene-16,20-dione;

25 (12S,18S)-18-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-spiro[11,25-dioxa-2 λ^6 -thia-1,8,15,19-tetrazapentacyclo[19.6.2.13,7.18,12.024,28]hentriaconta-3(31),4,6,21(29),22,24(28)-hexaene-26,1'-cyclopropane]-16,20-dione;

30 (11S,18S,26S)-18-[2-chloro-5-(trifluoromethyl)phenyl]-26-fluoro-2,2-dioxo-25-oxa-2 λ^6 -thia-1,8,12,15,19-pentazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-3(31),4,6,21(29),22,24(28)-hexaene-16,20-dione;

(10R,11R,18S)-18-[2-chloro-5-(trifluoromethyl)phenyl]-10-hydroxy-2,2-dioxo-25-oxa-2 λ^6 -thia-1,8,12,15,19-pentazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-3(31),4,6,21(29),22,24(28)-hexaene-16,20-dione;

5 (11S,12R,18S)-18-[2-chloro-5-(trifluoromethyl)phenyl]-12-hydroxy-2,2-dioxo-25-oxa-2 λ^6 -thia-1,8,15,19-tetrazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-3(31),4,6,21(29),22,24(28)-hexaene-16,20-dione

(11S,12S,18S)-18-[2-chloro-5-(trifluoromethyl)phenyl]-12-hydroxy-2,2-dioxo-25-oxa-2 λ^6 -thia-1,8,15,19-tetrazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-3(31),4,6,21(29),22,24(28)-hexaene-16,20-dione;

10 (11R,12R,18S)-18-[2-chloro-5-(trifluoromethyl)phenyl]-12-hydroxy-2,2-dioxo-25-oxa-2 λ^6 -thia-1,8,15,19-tetrazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-3(31),4,6,21(29),22,24(28)-hexaene-16,20-dione;

15 (11R,12S,18S)-18-[2-chloro-5-(trifluoromethyl)phenyl]-12-hydroxy-2,2-dioxo-25-oxa-2 λ^6 -thia-1,8,15,19-tetrazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-3(31),4,6,21(29),22,24(28)-hexaene-16,20-dione;

(18S)-18-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-25-oxa-2 λ^6 -thia-1,12,15,19-tetrazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-3(31),4,6,21(29),22,24(28)-hexaene-16,20-dione;

20 (8S,11S,18S)-18-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-25-oxa-2 λ^6 -thia-1,12,15,19-tetrazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-3(31),4,6,21(29),22,24(28)-hexaene-16,20-dione;

(8R,11S,18S)-18-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-25-oxa-2 λ^6 -thia-1,12,15,19-tetrazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-3(31),4,6,21(29),22,24(28)-hexaene-16,20-dione;

25 (8S,11R,18S)-18-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-25-oxa-2 λ^6 -thia-1,12,15,19-tetrazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-3(31),4,6,21(29),22,24(28)-hexaene-16,20-dione;

30 (8R,11R,18S)-18-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-25-oxa-2 λ^6 -thia-1,12,15,19-tetrazapentacyclo[19.6.2.13,7.18,11.024,28]hentriaconta-3(31),4,6,21(29),22,24(28)-hexaene-16,20-dione;

(17S)-17-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-24-oxa-2 λ^6 -thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.023,27]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-15,19-dione;

(16S)-16-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-23-oxa-2 λ^6 -thia-1,10,13,17-tetrazatetracyclo[17.6.2.13,7.022,26]octacosa-3(28),4,6,19(27),20,22(26)-hexaene-14,18-dione;

(17S)-17-[2,5-bis(trifluoromethyl)phenyl]-2,2-dioxo-24-oxa-2 λ^6 -thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.023,27]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-15,19-dione;

(17S)-17-[2-chloro-5-(trifluoromethyl)phenyl]-10-methyl-2,2-dioxo-24-oxa-2 λ^6 -thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.023,27]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-15,19-dione;

(17S,25R)-17-[2-chloro-5-(trifluoromethyl)phenyl]-22-fluoro-25-methyl-2,2-dioxo-24-oxa-2 λ^6 -thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.023,27]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-15,19-dione;

(17S)-17-[2-methyl-5-(trifluoromethyl)phenyl]-2,2-dioxo-24-oxa-2 λ^6 -thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.023,27]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-15,19-dione;

(17S)-17-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-24-oxa-2 λ^6 -thia-1,10,14,18,29-pentazatetracyclo[18.6.2.13,7.023,27]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-15,19-dione;

(17S)-17-[2-chloro-5-(trifluoromethyl)phenyl]-21-methoxy-2,2-dioxo-24-oxa-2 λ^6 -thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.023,27]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-15,19-dione;

(17S)-17-[2-chloro-5-(trifluoromethyl)phenyl]-21-fluoro-2,2-dioxo-24-oxa-2 λ^6 -thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.023,27]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-15,19-dione;

(17S)-17-(2,5-dichlorophenyl)-2,2-dioxo-24-oxa-2 λ^6 -thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.023,27]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-15,19-dione;

(17S)-17-[2-chloro-5-(trifluoromethyl)phenyl]-2'-fluoro-2,2-dioxo-spiro[24-oxa-2 λ^6 -thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.023,27]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-25,1'-cyclopropane]-15,19-dione;

(17S)-17-[2-chloro-5-(trifluoromethyl)phenyl]-2'-fluoro-2,2-dioxo-spiro[24-oxa-2 λ^6 -thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.023,27]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-25,1'-cyclopropane]-15,19-dione;

(17S,25S)-17-[2-chloro-5-(trifluoromethyl)phenyl]-25-(hydroxymethyl)-2,2-dioxo-24-oxa-2 λ^6 -thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.023,27]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-15,19-dione;

(17S,25S)-17-[2-chloro-5-(trifluoromethyl)phenyl]-25-(fluoromethyl)-2,2-dioxo-24-oxa-2 λ^6 -thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.023,27]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-15,19-dione;

(17S)-17-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-10,24-dioxa-2 λ^6 -thia-1,14,18-triazatetracyclo[18.6.2.13,7.023,27]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-15,19-dione;

(18S)-18-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-9,25-dioxa-2 λ^6 -thia-1,12,15,19-tetrazatetracyclo[19.6.2.03,8.024,28]nonacosa-3(8),4,6,21(29),22,24(28)-hexaene-16,20-dione;

(16S)-16-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-23-oxa-2 λ^6 -thia-1,9,13,17-tetrazatetracyclo[17.6.2.13,7.022,26]octacosa-3(28),4,6,19(27),20,22(26)-hexaene-14,18-dione.

33. The compound of formula (I) according to claim 1, or a pharmaceutically acceptable salt thereof, wherein said compound of formula (I) is:

(17S)-17-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-24-oxa-2 λ^6 -thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.023,27]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-15,19-dione 2,2,2-trifluoroacetic acid salt;

(17S,25R)-17-[2-chloro-5-(trifluoromethyl)phenyl]-22-fluoro-25-methyl-2,2-dioxo-24-oxa-2 λ^6 -thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.023,27]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-15,19-dione 2,2,2-trifluoroacetic acid salt

(17S)-17-[2-methyl-5-(trifluoromethyl)phenyl]-2,2-dioxo-24-oxa-2 λ^6 -thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.023,27]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-15,19-dione;

(17S)-17-[2-chloro-5-(trifluoromethyl)phenyl]-21-fluoro-2,2-dioxo-24-oxa-2 λ^6 -thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.023,27]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-15,19-dione;

(17S,25S)-17-[2-chloro-5-(trifluoromethyl)phenyl]-25-(fluoromethyl)-2,2-dioxo-24-oxa-2 λ^6 -thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.023,27]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-15,19-dione;

34. The compound of formula (I) according to claim 1, or a pharmaceutically acceptable salt thereof, wherein said compound of formula (I) is:

(17S)-17-[2-chloro-5-(trifluoromethyl)phenyl]-2,2-dioxo-24-oxa-2 λ^6 -thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.023,27]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-15,19-dione;

(17S,25R)-17-[2-chloro-5-(trifluoromethyl)phenyl]-22-fluoro-25-methyl-2,2-dioxo-24-oxa-2 λ^6 -thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.023,27]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-15,19-dione;

(17S)-17-[2-methyl-5-(trifluoromethyl)phenyl]-2,2-dioxo-24-oxa-2 λ^6 -thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.023,27]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-15,19-dione;

(17S)-17-[2-chloro-5-(trifluoromethyl)phenyl]-21-fluoro-2,2-dioxo-24-oxa-2 λ^6 -thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.023,27]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-15,19-dione;

(17S,25S)-17-[2-chloro-5-(trifluoromethyl)phenyl]-25-(fluoromethyl)-2,2-dioxo-24-oxa-2 λ^6 -thia-1,10,14,18-tetrazatetracyclo[18.6.2.13,7.023,27]nonacosa-3(29),4,6,20(28),21,23(27)-hexaene-15,19-dione;

35. A compound of formula (I) according to any one of claims 1 to 34, or a pharmaceutically acceptable salt thereof, for use as a therapeutically active substance.

36. A pharmaceutical composition comprising a compound of formula (I) according to any one of claims 1 to 34, or a pharmaceutically acceptable salt thereof, and a pharmaceutically acceptable excipient
37. The use of a compound of formula (I) according to any one of claims 1 to 34, or a
5 pharmaceutically acceptable salt thereof, for the treatment and/or prophylaxis of disease or disorder mediated by MET kinase.
38. The use according to claim 37, wherein said disease or disorder mediated by MET kinase is cancer.
39. The use according to claim 38, wherein said cancer is lung cancer.
- 10 40. The use according to claim 39, wherein said lung cancer is non-small-cell lung cancer (NSCLC).
41. The use of a compound of formula (I) according to any one of claims 1 to 34, or a pharmaceutically acceptable salt thereof, for the preparation of a medicament for the treatment and/or prophylaxis of disease or disorder mediated by MET kinase.
- 15 42. The use according to claim 41, wherein said disease or disorder mediated by MET kinase is cancer.
43. The use according to claim 42, wherein said cancer is lung cancer.
44. The use according to claim 43, wherein said lung cancer is non-small-cell lung cancer (NSCLC).
- 20 45. A compound of formula (I) according to any one of claims 1 to 34, or a pharmaceutically acceptable salt thereof, for use in the treatment and/or prophylaxis of disease or disorder mediated by MET kinase.
46. The compound for use according to claim 45, wherein said disease or disorder mediated by MET kinase is cancer.
- 25 47. The compound for use according to claim 46, wherein said cancer is lung cancer.
48. The compound for use according to claim 47, wherein said lung cancer is non-small-cell lung cancer (NSCLC).

49. A method for the treatment or prophylaxis of disease or disorder mediated by MET kinase, which method comprises administering a therapeutically effective amount of a compound of formula (I) according to any one of claims 1 to 34, or a pharmaceutically acceptable salt thereof, to a human.
- 5 50. A method according to claim 49, wherein said disease or disorder mediated by MET kinase is cancer.
51. A method according to claim 50, wherein said cancer is lung cancer.
52. A method according to claim 51, wherein said lung cancer is non-small-cell lung cancer (NSCLC).
- 10 53. The invention as described hereinbefore.

INTERNATIONAL SEARCH REPORT

International application No
PCT/EP2025/080568

A. CLASSIFICATION OF SUBJECT MATTER		
INV.	C07D513/08	C07D513/18
	A61P35/00	A61P31/00
ADD.	C07D513/22	C07D515/08
		A61K31/5415
According to International Patent Classification (IPC) or to both national classification and IPC		
B. FIELDS SEARCHED		
Minimum documentation searched (classification system followed by classification symbols) C07D		
Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched		
Electronic data base consulted during the international search (name of data base and, where practicable, search terms used) EPO-Internal, CHEM ABS Data		
C. DOCUMENTS CONSIDERED TO BE RELEVANT		
Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
A	WO 2017/040963 A1 (FORMA THERAPEUTICS INC [US]; ZHENG XIAOZHANG [US] ET AL.) 9 March 2017 (2017-03-09) page 155 - page 156; example 21; compounds I-87 -----	1-53
A	SPENCER JONATHAN A. ET AL: "Design and Development of a Macrocyclic Series Targeting Phosphoinositide 3-Kinase [delta]", ACS MEDICINAL CHEMISTRY LETTERS, vol. 11, no. 7, 3 June 2020 (2020-06-03), pages 1386-1391, XP093287996, US ISSN: 1948-5875, DOI: 10.1021/acsmmedchemlett.0c00061 the whole document ----- -/-	1-53
☒ Further documents are listed in the continuation of Box C. ☒ See patent family annex.		
* Special categories of cited documents : "A" document defining the general state of the art which is not considered to be of particular relevance "E" earlier application or patent but published on or after the international filing date "L" document which may throw doubts on priority claim(s) or which is cited to establish the publication date of another citation or other special reason (as specified) "O" document referring to an oral disclosure, use, exhibition or other means "P" document published prior to the international filing date but later than the priority date claimed "T" later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention "X" document of particular relevance;; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone "Y" document of particular relevance;; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art "&" document member of the same patent family		
Date of the actual completion of the international search 3 December 2025		Date of mailing of the international search report 15/12/2025
Name and mailing address of the ISA/ European Patent Office, P.B. 5818 Patentlaan 2 NL - 2280 HV Rijswijk Tel. (+31-70) 340-2040, Fax: (+31-70) 340-3016		Authorized officer Hallen, Lukas

INTERNATIONAL SEARCH REPORT

International application No
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C(Continuation). DOCUMENTS CONSIDERED TO BE RELEVANT		
Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
A	WO 2023/179600 A1 (MINDRANK AI LTD [CN]) 28 September 2023 (2023-09-28) pages 25, 119; examples C2-28 -----	1-53
A	WO 2015/171610 A2 (LYCERA CORP [US]) 12 November 2015 (2015-11-12) page 132 - page 337; examples 1-7, 18, 29-97 -----	1-53

INTERNATIONAL SEARCH REPORT

Information on patent family members

International application No

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WO 2015171610 A2	12-11-2015	AU 2015256190 A1 CA 2947290 A1 EP 3140291 A2 JP 6728061 B2 JP 2017514855 A US 2017183331 A1 US 2018208587 A1 US 2018265502 A1 WO 2015171610 A2	08-12-2016 12-11-2015 15-03-2017 22-07-2020 08-06-2017 29-06-2017 26-07-2018 20-09-2018 12-11-2015